Lecture Notes in Networks and Systems 1171

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Preface

Education is the cornerstone of any society, and it serves as one of the foundations for many of its social values and characteristics. State-of-the-art and novel methodologies and technologies allow researchers, designers, and domain experts to pursue Technology Enhanced Learning (TEL) solutions targeting not only cognitive processes but also motivational, personality or emotional factors. Nowadays, we can identify two main legs, providing necessary and complementary strengths to a TEL oriented design process: appropriate technologies should be applied, and appropriate methods should guide such application. Technologies in TEL can deliver smart, personalized, tailored, and motivating learning solutions. Methods come from different fields, such as psychology, medicine, computer science, and from diverse communities, where collaboration and co-working is used, such as maker communities and participatory design communities. In addition, learning analytics can help manage (big) data to enhance learning opportunities for learners and educators alike, for instance, by supporting self-regulated learning or adaptation of the learning material.

As to these topics, the annual appointment of MIS4TEL established itself as a consolidated fertile forum where scholars and professionals from the international community, with a broad range of expertise in the TEL field, share results and compare experiences. The calls for papers of the 14th edition of the conference welcomed novel research in TEL, even from new research fields (ranging from artificial intelligence and agent-based systems to robotics, virtual reality, Internet of Things and wearable solutions, among others) concerning methods and technological opportunities, and how they serve to create novel approaches to TEL, innovative TEL solutions and valuable TEL experiences.

The result of the call for papers is that both the main track of MIS4TEL 2024 and its four related workshops: *Artificial Intelligence for Education (Ai4Ed)*, *Integration of emerging technologies into education and training (ETELT)*, *Technology Enhanced Learning in Nursing Education (NURSING)*, and *Technology-Enhanced Learning for Future Citizens (TEL4FC)*, contribute to novel research in TEL and expand on the topics of the previous editions. This volume presents the papers that were accepted for the main track of MIS4TEL 2024.

All papers underwent a peer-review selection: each paper was assessed by at least three different reviewers, from an international panel composed of about 71 members from 21 countries. From a total of 43 articles, the program of MIS4TEL 2024 counts 29 contributions (18 full papers and 12 short papers) from diverse countries.

This edition of the conference is organized by the BISITE Research Group of the University of Salamanca (Spain). We would like to thank all the contributing authors, the members of the Program Committee, the reviewers, the sponsors and the Organizing

Committee for their hard and highly valuable work. Thanks for your help – MIS4TEL 2024 would not exist without your contribution.

Christothea Herodotou
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Teaching Neural Networks Using Comic Strips

Guido Camerlingo[1], Paolo Fantozzi[2], Luigi Laura[1(✉)], and Marco Parrillo[1]

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Abstract. There is a rich variety of information in comic strips: dialogues, images, face expressions and body postures. In this paper we describe our experience in using this variety to help students learning how to use neural networks. In particular, the scope is to generate dialogues between chosen characters, and this is achieved by a sequence of steps needed to build the proper dataset: face detection to recognize the characters in the comic, text recognition, and assigning texts to the characters. Finally, we use a Transformer network to generate a dialogue.

Keywords: Neural Networks · Comics · Natural Language Processing · Image Recognition

1 Introduction

Comic strips, with their rich tapestry of dialogues, images, facial expressions, and body postures, present a unique and engaging medium. This distinct form of visual narrative art offers a multifaceted view into storytelling, making it an ideal candidate for educational purposes, especially in the realm of artificial intelligence (AI) and machine learning. In this paper, we explore the use of comic strips as a resourceful tool to assist students in learning how to apply and manipulate neural networks, specifically focusing on the generation of dialogues between characters depicted in the comics.

The convergence of traditional comic strip elements with cutting-edge AI technologies presents a novel approach to tackling complex machine learning algorithms in an accessible manner. By integrating visual and textual data from comics, we can create dynamic datasets that are not only fun and engaging but also challenging and informative for AI training purposes. This method not only enhances the learning experience but also introduces students to practical applications of neural network technologies in processing and generating human-like dialogues.

Our methodology involves several crucial steps designed to construct a comprehensive dataset suitable for training an AI model. The initial phase consists of face detection technology, which serves to identify and differentiate characters

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within the comic strips. This process is critical as it ensures that the subsequent dialogue generation is accurately attributed to the correct characters, maintaining the narrative's integrity.

Following character recognition, we employ a vision API [1] for text recognition to extract spoken words and textual elements from the comic strips. This stage is imperative for gathering the raw data needed to train the neural networks, as it converts visual text into a digital format that can be processed by AI algorithms.

The extracted text is then meticulously assigned to the respective characters identified in the first step. This association between character and dialogue is what allows for the contextual training of our AI model, providing a foundation for the generated interactions to reflect the personalities and story arcs of the characters involved.

Finally, we utilize a Transformer [2] network, a type of model renowned for its effectiveness in handling sequence-to-sequence tasks and its ability to manage long-range dependencies in text data. The Transformer network is trained with the data prepared in the preceding steps, enabling it to generate dialogues that are not only coherent and contextually appropriate but also creative and true to the characters' original depictions in the comic strips.

Through this paper, we aim to demonstrate the educational potential of using comic strips as a basis for teaching AI and machine learning concepts. We discuss the challenges encountered, the solutions implemented, and the outcomes observed while employing this innovative teaching approach. This exploration not only contributes to academic knowledge but also paves the way for future research and application of AI in educational tools and content creation.

This paper is organized as follow: next section discusses related work, whilst Sect. 3 describes the project phases; Sect. 4 provides the results and some ideas for further development.

2 Related Works

This paper is in the wider contest of comic automatizing generation starting, for example, from a contest or illustration descriptive sentence. 'Comic Gan' [3] is one of the works which deal with the use of Generative Adversarial Network to generate illustrations. However it does not state with certainty if a GAN can generate comics even if the network is applied to pictures, generation with great realism but starting from "photos" and not from illustrations. The problem of comics generation concerns the semantic connection between text and illustration which, mostly in the training phase, can hardly be applied as a comic text is not descriptive enough. A possible option to generate corresponding cartoons starting from the text, Text-to-Image Synthesis, could be recent evolutions of the GAN models, for example StyleGAN [4] or GigaGAN [5] being able to arrive at the automatic generation of complete comics. In pictures classification a useful instrument may be the use of ResNet in order to opt for the eventual classification of the contest also according to the text. Another aspect is the assignment

of text to the characters, although many methods use distance measurements, there are works based on deep learning, Scene Graph Generation Models [6], specifically related to Comics Speaker Detection [7]; for such a task it might be possible to use also the approach of Dalla Torre et al. [8]. The generation of comics is complex and there are numerous aspects to consider and possible techniques that can be used through the use of different AI tools. The ultimate goal, for the development of comics dataset able to be used to train neural network would be something that would allow automatic classification of people such as in Di Mascio et al. [9]. As always, when we deal with neural networks, the explainability [10] is an important issue, also due to recent laws such as the European AI act.

3 Project Scheme

The project is made up of four different phases divided into as many repositories which are identified in: The project consists of four different phases which, due to study methods, have been divided into four repositories which are identified in:

- **Dilbert - Cascade Classifier**[1]: the goal of this repository is to practically do the training of a haar cascade classifier with opencv, using dilbert's comic balloons as an example.
- **Dilbert and Boss - Cascade Classifier**[2]: the goal of this repository is to merge the two classifiers created (Dilbert and Boss) to locate them together in a single panel.
- **Dilbert - Text Extractor**[3]: the main purpose of this repository is through the recognition of the text of the panels (identified in the first part of the Dilbert and Boss - Cascade Classifier) using the Vision OCR
- **Dilbert and Boss - Dialogue Generator Transformer**[4]: the main purpose of this repository is the generation of dialogues, the training and practical implementation of the Transformer architecture and the prediction script. Through the use of the Python language and the Tensor Flow and Karas libraries. This is the fourth phase of the project following the construction of the dataset

Figure 1 shows the project flow, schematizing all the phases and steps for the production of the various outputs and inputs. The first point, Dilbert - Cascade Classifier, regards the creation of a classifier, face detection, and therefore with its training in order to produce two cascade classifiers which are basically two xml files. At point two this production is exploited to identify the panels that contains the characters trained for recognition. This output feeds point three that is the text extraction phase, Text Extractor. This phase extracts the panel texts

[1]url: https://github.com/GuiZ88/dilbert-cascade-classifier/.
[2]url: https://github.com/GuiZ88/dilbert-boss-cascade-classifier.
[3]url: https://github.com/GuiZ88/dilbert-text-extractor.
[4]url: https://github.com/GuiZ88/dilbert-boss-dialogue-generator-transformer.

to move on to the dataset build phase, which also includes a data cleaning phase. A short manual verification phase has been necessary as the panel are 'noisy' [11] in terms of text recognition. It generally involves fictitious words addition, as it has been noted, at the end of the sentence. The fourth phase exploits the dataset created to train two Transformer [2] models, one for each character in question respectively, which in this case are two. In Fig. 1 model dilbert.h5 and model boss.h5 are used for the last phase of dialogue generation. The dialogues have as input a topic or a starting sentence, as mentioned above. The output is composed of n passages and obviously it is possible to manipulate the various sentences that can be passed as input to shape the speech and establish a further long-term memory at the input level.

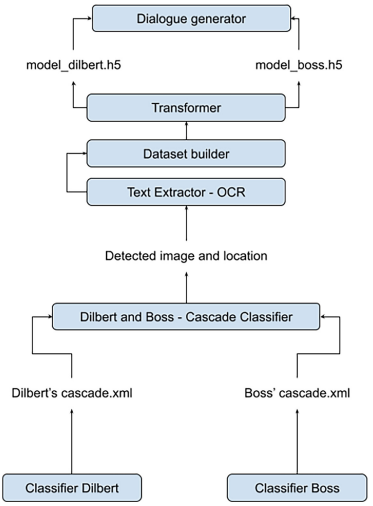
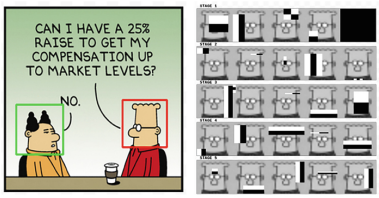


Fig. 1. Scheme and phases of the project

3.1 Dataset

A consistent dataset is not needed in the entire strip but substantially only in the single panel in the question and in the answer, as in characterizing these entities it is necessary to produce a question that would make Dilbert, an answer that would give the Boss and vice versa. For the identification of the characters, the use of algorithms, derived from Viola-Jones Object detection framework [12], was chosen, specifically the implementation of OpenCv: Haar Cascades Classifier [13, 14], which requires a small initial effort in the preparation and in the manual identification of the samples but which returns a work clearly proportional to the care of the proposed samples. Once the two classifiers have been trained, the two characters in the single panel can be detected by combine this two classifiers. An interesting point would be the adoption of the most recent *YOLO* [15], You



(a) (b)

Fig. 2. OpenCV Detection (a) Dilbert & Boss detected. (b) Features application during detection

Only Look Once, which is highly efficient even if more oriented for real-time applications even in the absence of GPU (Fig. 2).

During this procedure, the next step is extracting the text using OCR recognition. The Google APIs [16] are used relative to Cloud Vision [1] allowing text extraction with excellent precision by grouping it into blocks, where, using the *boundingBox* property, it is possible to identify the centroid of the block. This procedure makes it possible to use a measurement distance for the matching with the corresponding character. An intermediate json is therefore generated containing the extracted text and its centroid, which, in a subsequent processing, is assigned to the character for the creation of the final dataset.

3.2 Dialogue Generation

This phase concerns the generation of dialogues, the training and practical realization of the Transnformer architecture [2] and of the prediction script[5] by using TensorFlow and Keras libraries. Preliminary steps are performed in order to prepare the dataset:

- Extract the questions and answers and place them in two lists, the answers and the questions;
- Preprocessing for the elimination of contracted forms;
- Phrase tokenizer and addition of START and END tokens;
- Phrase pad at maximum length.

In Figure ?? the description of the final Transformer. For the training phase the cost function *SparseCategoricalCrossEntropy* is used. It computes/calculates the crossentropy loss between the labels and predictions. It is defined where the y are considered the labels and y_i the predictions as:

$$CE = -\sum_{i=1}^{i=N} y_i \cdot log(\widehat{y}_i) \tag{1}$$

[5] Available at https://github.com/GuiZ88/dilbert-boss-dialogue-generator-transformer.

For the learning rate (l_r), the one expected from the paper [2] is reported and therefore

$$l_r = d_{model}^{-0.5} \cdot min(step_num^{-0.5}, step_num \cdot warmup_steps^{-1.5}) \quad (2)$$

It is essentially linearly increasing the learning rate for the first warm up steps ($warmup_steps$), that is the a real warm up and decreasing proportionally the step num to the inverse of the square root of $step_num$.

4 Results

In the different phases of this project different aspects can be evaluated. Starting from the face detection it has been possible to verify the excellent result of the classification obtained by the Haar cascade classifier and therefore the algorithms of Viola and Jones through features application. To verify the correct functioning of the classifier, a test sample of 411 panels has been taken randomly from strips from 2004 to 2023. In the classifier relating to the'Ponty Haired Boss' trained with only 171 images carefully selected from panels from 2018 to 2023 with factors such as angle of the character, character from behind, etc. . . it has been possible to verify that on the test sample there have been only 6 false positives and 3 false negatives. For Dilbert, a less 'characteristic' character than the Boss who has prominent tip hair, 368 panels taken from years 2018 to 2023 strips have been used for recognition. It has been possible to verify that on the test sample there have been only 11 false positives and 0 false negatives.

For the dataset extraction therefore for the combined recognition of Dilbert and the Boss out of 25327 panel relating from 2004 to 2023, 4470 panel have been obtained, about 17%, where both characters are represented. Reasonably by manually verifying the presence of Dilbert and the Boss on the test sample, it has been noted their presence in 64 cartoons and therefore about 15%. It is an expected percentage. Some examples in Fig. 3.

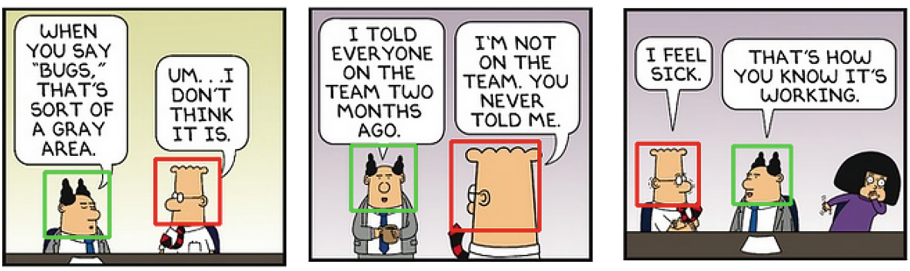


Fig. 3. Characters detected in different panel

For what regards OCR extraction, a dataset of 1425 'question' and 'answer' composed dialogues have been created. It can be deduced that out of the 4470

panels where both Dilbert and the Boss appear only 1425 contain a complete dialogue. In fact, this could be an improving but also a quite complex aspect, which consists in understanding the dialogues on several panels within the same strip. Transformers training has been performed on 90 epochs and has produced good results, considering the limited number of available lines. Over the 90 epochs it has been noted that the loss function reaches its limit while precision/accuracy, which evaluates the performance does not get great improvements and stands at 0.56 for both models. Even if this is not an excellent number, it has been experimented that with an expanded dataset, therefore with more lines assigned to Dilbert and the Boss, this increases but obviously losing the main characterization factor. An amount around 10,000 to 15,000 lines can achieve superior accuracy. So for the training of the model related to Dilbert, in detail, as in Fig. 4, after 90 epochs a 0.0378 for the loss function while a 0.57 for the accuracy have been obtained. Equally for the Boss, in the same Fig. 4, an equivalent result, 0.0338, and a precision of 0.5578 can be seen for the loss function.

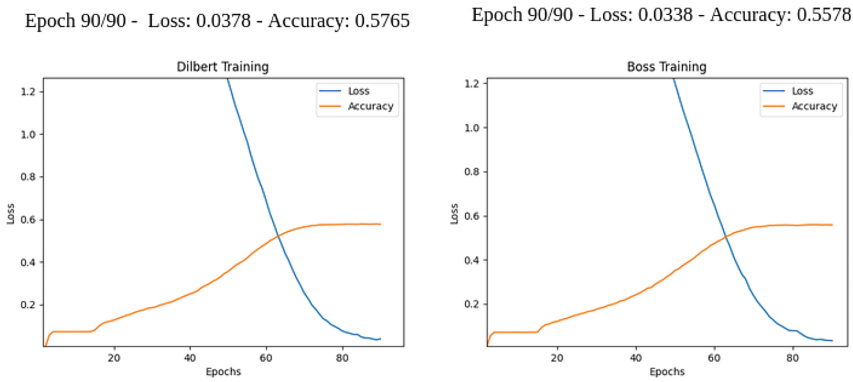


Fig. 4. Training trend for loss & accuracy

The choice to create two separate models for the generations is also due to problems inherent to validation as well as to the elimination of any problems of overfitting and therefore of its effects. It is therefore necessary to find a method of evaluating the prediction using a qualitative method since it is clear when a prediction is successful but it is still a must to give a quantification that we have tried to summarize in three parameters for each prediction under the aspect of congruence:

- of a logical type
- grammatical
- functional related to the characterization

The one-to-ten evaluation for each congruence aspect is personal. It does not refer to the exchange but to the dialogue, which in this case is made up of three

exchanges. A total of twenty predictions have taken place, respectively ten where Dilbert starts the speech and ten where the Boss starts it. The start sentence is taken randomly from among the respective ones in the dataset because of its limited terminology and in any case the discursive nature, composed of exchanges, can be widely evaluated in the following speeches and exchanges. Positive and less positive dialogues were obtained. Starting from initial random sentences, starter sentences, taken from the dataset, less sensible speeches are obtained, since a random sentence can also be an answer or a sarcasm. Otherwise writing a sentence 'manually' such as 'it's friday!' better results are obtained as in the following example:

BOSS: it's friday!
DILBERT: so . . . what are do some feedback
BOSS: or maybe we need to act fast.
DILBERT: i can read it ?
BOSS: why can't you ?
DILBERT: so it was very wise of you.

In sentence construction the training has been effective, there are not big mistakes and the sentences are mostly correct. A further more complex example, below, demonstrates its particular effectiveness:

BOSS: are you trying to kill us?
DILBERT: thats a real thing.
BOSS: but seriously, we can finish the project ?
DILBERT: it might be all of those.
BOSS: well, the bad data to support that claim.
DILBERT: well, we have been getting some inconsistent results when users try to log in . sometimes it works, for a little bit.

A logic and coherent talking is sometimes lacking but this is mainly due to the lack of occurrences in the dataset, as more widely treated themes in it do not decline this problem:

BOSS: when do you expect to finish your project?
DILBERT: maybe we should have no way to find my staff to the software for them.
BOSS: what are you doing that?
DILBERT: i think you are a great job?
BOSS: hmm, i can ask him not have a feature to you.
DILBERT: well, i was thinking, can you create an ai that can make me funny?

Grammar usage is equivalent to sentence construction as there are not great problems. The characterization, the central point of the paper, is absolutely valid and the characters behave and discuss like the original actors, often focusing on sarcastic dialogues. There are some problems always due to the lack of dataset such as some repeated sentences. Although these are strictly personal evaluations, the results are illustrated in Table 1 and 2 for Dilbert and the Boss relating to ten sentences, each based on who starts the dialogue (Fig. 5).

Table 1. Results for prediction for Dilbert

Sentences	Logical	Grammatical	Characterization
1	6	8	8
2	5	8	7
3	8	4	8
4	6	8	10
5	4	8	7
6	7	10	8
7	9	9	6
8	6	7	8
9	3	6	9
10	9	8	9
Average	6.3	7.6	8

Table 2. Results for prediction for Boss

Sentences	Logical	Grammatical	Characterization
1	6	8	8
2	5	8	7
3	8	4	8
4	6	8	10
5	4	8	7
6	7	10	8
7	9	9	6
8	6	7	8
9	3	6	9
10	9	8	9
Average	6.3	7.6	8

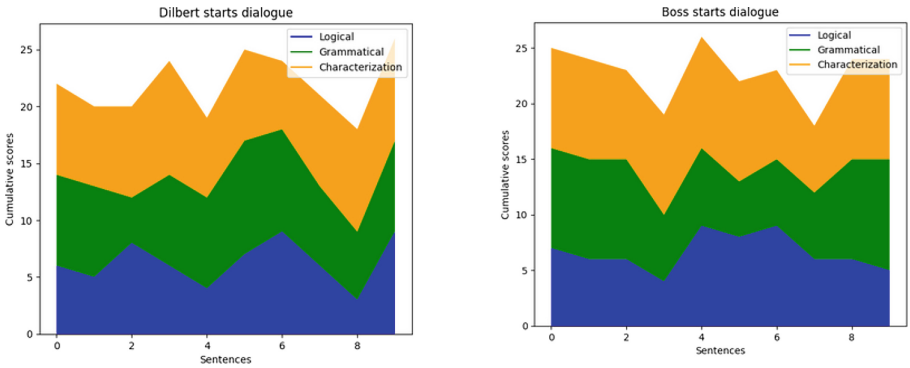


Fig. 5. Graph of the scores obtained in the three categories

5 Conclusions

Keeping in mind that the objective is the characterization of a generative model, it was possible to verify how different combined techniques produced the final result, a good final result. The "sum" of errors, deriving from all phases, was therefore low and therefore the extraction techniques produced valid results for generative purposes, in addition to the training itself. By placing the code produced by the repositories as the basis of use, it is easily applicable in other fields such as the characterization of a character using previous chats. The dialogues individuation in several consecutive panels between the two characters considered can identify further exchanges that do not appear in a single panel. Another important aspect is the evaluation of performance during training, the impact on the CPU and the possible optimization techniques such as quantization [17].

This would basically require identifying all the characters in every single panel in order to identify the presence of a corresponding text. Another methodology could be the identification of the belonging of the text, a sort of character text

detection, that is no longer identifying the characters through face detection but directly through the text. So understanding who the 'speaking' character is and through the characterization with whom he is speaking. The Sentiment Analysis models [18] could be used, so substantially a training which takes as a positive label the texts relating to the character to be identified while all the others are negative. A further development could be to find a best way for the identification of multiple exchanges within the same panel, which, although rare, are often configured in the ABA scheme because of space requirements.

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Exploring the Impact of LLM-Generated Feedback: Evaluation from Professors and Students in Data Science Courses

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Abstract. Providing students with exercises and timely feedback is pivotal in education, particularly in data science. This paper aims to assess the quality of three different types of feedback, one generated through traditional artificial intelligence methods called "schematic feedback", and the other two with different prompts passed to a Large Language Model (LLM), called "discursive feedback" and "combined feedback", the latter also exploiting the "schematic feedback". The three types of feedback are evaluated from both instructors' and students' perspectives, in terms of the feedback's ability to provide helpful suggestions on how to solve an exercise correctly. Instructors considered the "combined feedback" the most helpful feedback, compared to the "schematic feedback" and "discursive feedback", which were not considered substantially different. Students used the "schematic feedback" and "combined feedback" during their formative assessment activities: (i) they experienced the feedback better than expected, (ii) students of the current academic year (who used the "combined feedback" in addition to the "schematic feedback") reported an improved experience compared to students of the previous academic year (who used only the "schematic feedback"). In conclusion, we observed an improved quality of the feedback using LLM (reported by both instructors and students), as long as the prompt includes specific context, in our case, achieved by the "combined feedback". This study underscores the importance of leveraging LLMs in educational settings to provide tailored and helpful feedback, ultimately fostering a more effective learning environment.

Keywords: Generative AI · Large Language Models · Automated feedback · Data science

1 Introduction

Several courses at the Department of Life, Health and Environmental Sciences of the University of L'Aquila (Italy) have data science as a subject and teach how to manage datasets and carry out descriptive and inferential statistics, using R as the scripting language. The courses have a practical approach and are

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based mainly on solving exercises related to the topics above. Given the practical nature of the courses, formative assessment activities occur regularly, in the form of exercises focusing on a specific topic, whose solutions comprise a list of commands, outputs and possible comments. Formative assessment was initially characterised as a technique for offering feedback crucial for grasping the advancement of learning [9]. Research indicates that the implementation of formative assessment appears to enhance learning and improve outcomes [6]. For formative assessment, timely and elaborated feedback is pivotal [4]. Accordingly, the authors started developing an adaptive learning system [1] that includes a module [11] able to provide automated feedback to students about the solutions given to exercises. A recent evaluation of the automated feedback highlighted the need for improving it [12]. In the research reported in this paper, we explore the possibility of using Generative AI (GAI, i.e., the set of computational techniques that are capable of generating meaningful content from training data) to improve the feedback for students in the aforementioned scenario. The research questions we address are:

- RQ_1: How can we exploit Large Language Models (LLMs, i.e., the specific category of GAI models specialised on text-based data) to provide helpful feedback to students in our educational scenario?
- RQ_2: What are students' opinions of the provided feedback?

To answer these research questions, we first introduce the related work, focusing on feedback and the current applications of GAI/LLMs for feedback. Then, we present three different types of feedback we developed, namely: the "schematic feedback" (the initially available feedback), the "discursive feedback" (a LLM generated feedback), and the "combined feedback" (a LLM generated feedback that also exploits the "schematic feedback"). Hence, to understand the capability of each type of feedback to give the necessary suggestions to correct a mistake, we report on the following evaluations:

- two professors evaluated the three different types of feedback, as applied to a sample of real mistakes made by students of the previous academic year;
- forty-four students evaluated the LLM-generated feedback chosen by the professors during their formative assessment activities and compared it with the "schematic feedback".

The paper continues by discussing the findings and ends with short conclusions that summarise the results and sketch future work.

2 Related Work

A recent literature review [3] underlines the importance of automated feedback for learners, as well as its effectiveness due to the recent advances of computer technologies achieved in the last decades. In particular, GAI and LLMs have given an additional push, showing abilities and performances that seem to open

new educational opportunities [14]. Accordingly, researchers started to explore the possibility of using GAI in education from different perspectives. In [2], the authors present an LLM capable of generating more detailed feedback about students' performance than human instructors and of providing feedback on the process of students completing a task. In [7], the author evaluated LLM-generated feedback on writing tasks: LLM-generated feedback increased revision performance and task motivation and increased positive emotions compared to revising without feedback. Finally, Jacobsen et al. [5] conducted a study on the effectiveness of LLMs in generating feedback in higher education contexts: they found that the quality of generated feedback depends on the quality of the initial prompt, and highlighted that, although LLMs outperformed novice feedback and rivalled expert feedback, human expertise remains indispensable in particular for relationship-building and character definition.

3 Feedback

This section presents the different types of feedback evaluated in the paper. The three types of feedback are discussed using an actual exercise given by the system and an actual wrong solution submitted by a student.

The exercise is:

> *Let us consider dataset sbp. The dataset includes data on 10 patients, and we know their gender (variable gender) and systolic blood pressure before and after the administration of an antihypertensive drug (variables before and after). For the male subjects, calculate the median systolic blood pressure after the drug administration.*

The (wrong) solution given by a student was:

```
> median(sbp$after)
   [1] 179
```

The solution is not correct because the median is calculated on all observations, and not for the male subjects. The correct solution is instead:

```
> median(sbp$after[sbp$gender=="m"])
   [1] 196
```

Given this sample case, the following subsections show how each feedback is built. What is returned to the student, due to page space limitation constraints, is available at URL https://vittorini.univaq.it/?page_id=3904.

3.1 Schematic Feedback

The "schematic feedback" is built using an ad-hoc toolchain, briefly summarised below (see [11,13] for a more detailed description). Given an exercise:

- the toolchain parses both the correct solution given by the professor and the solution given by a student, and returns two lists of triples (i.e., the command, the output, and the optional comment)

– the two lists are compared, and (i) a distance is calculated as an estimate of the final grade; (ii) a static code analysis tool evaluates the differences between the commands/outputs. It returns feedback about the missing commands, the commands with different outputs and the correct commands; (iii) a classifier asserts if the comment is correct or not, and returns simple feedback about the correctness of the comment.

3.2 LLM-Generated Feedback

In this section, we describe two types of feedback generated through ChatGPT-4 (version 1106-preview). The first is produced from a prompt that includes (i) the student model, (ii) the exercise, (iii) the provided solution, and (iv) a sentence asking ChatGPT to explain the mistakes (if any). The second feedback enriches the prompt with the "schematic feedback". For both types of feedback, we instructed ChatGPT to generate explanations based on the following system's role:

> *You are a university professor of data science. Your background is in statistics, computer science and R programming. You need to help students solve simple data analysis problems by presenting the basics of statistics and then the solution in R with exercises. You must present simple examples. You need to be as concise as possible. Do not explain how to load the dataset, assume it is already in the environment.*

3.2.1 Discursive Feedback

The "discursive feedback" is produced from a prompt built with the same information available to the "schematic feedback", i.e., the student model, the exercise, and the submitted solution[1]. The prompt goes as follows:

Student Model. In our system, the student model contains (among the many) the student's general ability in the course, as well as the ability in every concept taught in the course. The abilities are estimated through the Rasch model [8], and may potentially range from $[-\infty, \infty]$, in practice between $[-3, 3]$. See [10] for details. To clarify to ChatGPT the student's general ability in the course, we use the following prompt:

> *I am a student with [qualitative description] competence in data science.*

The "qualitative description" is an adjective that translates the student's ability θ into the conversational prompt as follows:

– if $\theta < -2.5$, then translate the ability as the word "no";
– if $-2.5 \leq \theta < -1.5$, then translate the ability as the word "poor";
– if $-1.5 \leq \theta < -0.5$, then translate the ability as the word "mediocre";
– if $-0.5 \leq \theta < 0.5$, then translate the ability as the word "average";
– if $0.5 \leq \theta < 1.5$, then translate the ability as the word "good";

[1] Note that the prompts are produced in the Italian language, but they are translated into English in the paper for straightforward comprehension.

– if $1.5 \leq \theta < 2.5$, then translate the ability as the word "very good";
– if $\theta \geq 2.5$, then translate the ability as the word "excellent".

For instance, for a student with an estimated ability of $\theta = -0.65$, the initial declaration of the student model is:

> *I am a student with mediocre competence in data science.*

A similar approach is followed for translating the abilities a student has in every course concept. So, for instance, we declare a complete student model to ChatGPT as follows:

> *I am a student with mediocre competence in data science. I have mediocre competence in central tendency. I have average competence in dispersion. I have average competence in distribution. I have medium proficiency in confidence intervals. I have average proficiency in normality. I have average proficiency in hypothesis testing.*

Exercise. ChatGPT is then notified of the exercise proposed to the student. This part of the prompt consists of the following sentence:

> *I need to solve the following exercise: [exercise description]*

where, obviously, "exercise description" is the exercise the student must solve. Therefore, the following sentence is an example of a typical exercise given in the course:

> *I need to solve the following exercise:*
> *"Let us consider dataset sbp. The dataset includes data on 10 patients, and we know their gender (variable gender), as well as the systolic blood pressure before and after the administration of an antihypertensive drug (variables before and after). For the male subjects, calculate the median of the systolic blood pressure after the drug administration."*

Submitted Solution. This part of the prompt consists of an initial sentence:

> *I provided this solution: [student's solution]*

An example of a (wrong) solution given by a student, as exploited in this prompt:

> *I provided this solution:*
>
> ```
> > median(sbp$after)
> [1] 179
> ```

Concluding sentence. The prompt is then concluded by asking ChatGPT to explain the error:

> *Explain the mistake and how to solve the exercise correctly.*

3.2.2 Combined Feedback

The "combined feedback" is produced from a prompt built like in the previous section, plus a part exploiting the "schematic feedback" introduced right before the concluding sentence. The "schematic feedback" is, therefore, exploited in this prompt as follows:

I received the following feedback: [schematic feedback]

Therefore, for the sample case we are using, we have the following:

I received the following feedback:

Command:

```
> median(sbp$after)
   [1] 179
```

The command seems correct, but the output differs from the right solution. The boolean predicate doesn't seem correct. The after parameter with boolean predicate gender=="m" seems missing. The correct command is:

```
> median(sbp$after[ sbp$gender=="m" ])
```

The correct output is:

```
[1] 196
```

4 Evaluation by Teachers

Objective. This study aims to answer RQ_1. The research question is addressed with a twofold evaluation. The first evaluation compares the three different types of feedback presented above ($RQ_{1.1}$). This study aims to identify the approach that builds the best feedback for students. The second evaluation investigates how the AI-generated feedback is "sensible" to different student models ($RQ_{1.2}$). In our adaptive learning system, the ability to provide tailored content and feedback is an important feature that is, therefore, worth investigating.

Materials and Methods. For $RQ_{1.1}$, we randomly selected 60 solutions with mistakes (taken from actual solutions submitted by students), generated the three different types of feedback, and asked two professors with expertise in data science and R to assess if the different types of feedback were clear enough to explain to a student how to solve the exercise correctly. For $RQ_{1.2}$, we randomly selected 30 solutions with mistakes (taken from actual solutions submitted by students), and generated the feedback for two categories of students (i.e., a student with significant gaps in the subject, a student very proficient in the subject) using the best method identified in $RQ_{1.1}$, and asked two professors with expertise in data science and R to assess if the first feedback was easier to understand than the second one. For both evaluations, the experts came to an agreement before providing their opinion. To analyse the data of both evaluations, we used frequencies for descriptive statistics; confidence intervals and McNemar's χ^2 test for inferential statistics.

Results. The results regarding $RQ_{1.1}$ are summarised in Table 1. The feedback that was considered the most useful is CF, i.e., the AI-generated feedback built with a prompt that incorporates the "schematic feedback". It provided suitable feedback for 70% of the cases, followed by DF (i.e., the feedback provided by

ChatGPT) and SF (i.e., the "schematic feedback") with 41.7% and 38.3% of the cases, respectively. The statistically significant differences were found when comparing CF with SF and DF, whereas no statistically significant difference exists between SF and DF. Concerning $RQ_{1.2}$, the evaluators considered the feedback proper to the two different types of students only in 19 cases (63.3%, $CI_{95\%}$ = [45.5%, 78.1%]). In addition, both professors reported difficulties in appreciating the differences. They also noticed that when one feedback was longer than the other, it included more explanations for the "weaker" student or additional details/commands for the "stronger" student.

Table 1. Comparison of the different types of feedback (SF = Schematic feedback, DF = Discursive feedback, CF = Combined feedback)

Feedback	*N*	%	$CI_{95\%}$
SF	23	38.3%	[27.1%, 51.0%]
DF	25	41.7%	[30.1%, 54.3%]
CF	42	70.0%	[57.5%, 80.1%]

Comparison	χ^2	*p-value*
$SF - DF$	0.029	n.s.
$SF - CF$	12.96	0.0003
$DF - CF$	6.919	0.0085

5 Evaluation by Students

Objective. This study focuses on RQ_2. In particular, we investigated both the usefulness of the feedback generated by the tool, as evaluated by the students ($RQ_{2.1}$) and the preference of the students between "schematic feedback" and "combined feedback" ($RQ_{2.2}$).

Materials and Methods. For $RQ_{2.1}$, we asked the students: (i) at the beginning of the course, their expectations about the usefulness of the AI-generated feedback, (ii) before the exam, their experience with the AI-generated feedback. We asked them to rate the ability of feedback to help them understand and solve their mistakes, using a 5-point Likert scale, where 1 = "very difficult" and 5 = "very easy". At the same time, we compared these answers to the answers given by the students of the previous academic year to the same question, when the only available feedback was the schematic one. For $RQ_{2.2}$, we asked the students about their experience with the "schematic feedback" and the "combined feedback". Their experience was collected with two questions. The first asked to select among the following statements:

- The "schematic feedback" suffices;
- The "schematic feedback" suffices, but sometimes the "combined feedback" was useful;
- Both types of feedback were equally useful;
- The "schematic feedback" was not enough, I often referred to the "combined feedback";
- The "combined feedback" is essential.

The second question was an open-ended request for suggestions about how to improve the feedback. To analyse the data, we used means and frequencies for descriptive statistics, confidence intervals and Wilcoxon rank-sum or signed-rank tests (for independent or paired samples, respectively) for inferential statistics.

Results. Table 2 summarises the results regarding $RQ_{2.1}$. Eighty-eight students answered the expectation question, and forty-four the experience question[2]. The average expectation was 3.03, i.e., not easy nor difficult; the average experience was 3.39, i.e., more easy than difficult. The difference is statistically significant (p-value = 0.032). In the previous academic year, seventy students reported an average experience with the (schematic) feedback of 3.11. Despite the lower experience, the difference between the two academic years is not statistically significant. The results regarding $RQ_{2.2}$ are presented in Table 3. All possible answers are shown in the table rows with their absolute frequencies, respective percentages and 95% confidence intervals. The majority of the students considered the detailed feedback either useful or essential (54.5%), a small part considered the two types of feedback equally useful (27.3%), and a minimal part considered the "schematic feedback" sufficient (18.2%). Finally, the open-ended answers mainly reported that sometimes the "combined feedback" included solutions using commands not illustrated during the course (e.g., to use the rowSums function instead of the easier "+" operator adopted during the course) and that the "combined feedback" could be simplified (anyway, without specifying how).

Table 2. Students evaluation of the feedback quality: pre/post results on the top, comparison between academic years on the bottom

	N	Avg.	$CI_{95\%}$	*p-value*
Pre	88	3.03	[2.72, 3.34]	0.032
Post	44	3.39	[3.08, 3.71]	
AA 2023	70	3.11	[2.91, 3.31]	n.s.
AA 2024	44	3.39	[3.08, 3.71]	

6 Discussion

The differences between the two types of feedback returned by ChatGPT are worth noting. The "discursive feedback" contains additional sentences about assumptions on the dataset (e.g., that gender for males could be indicated with "M") and more complex commands for solving the exercise (e.g., the two commands in sequence `sbp_males = sbp[sbp$gender == "M", ]` and `median_males = median(sbp_males$after)`). On the other hand, the "combined feedback" exploited that gender for males was expressed with "m" (in lowercase, instead of uppercase) and also the use of a more straightforward syntax (i.e., `median_males_after = median(sbp$after[ sbp$gender == "m"`

Table 3. Usefulness of the schematic feedback (SF) and the combined feedback (CF)

	N	%	$CI_{95\%}$
SF	0	0.0%	[0.0%, 16.0%]
SF plus CF	8	18.2%	[4.5%, 34.6%]
Same	12	27.3%	[13.6%, 43.7%]
CF plus SF	11	25.0%	[11.4%, 41.4%]
CF	13	29.5%	[15.9%, 46.0%]

[2] At the time of writing, we could only collect data about the students that started preparing for the first round of exams.

`]))`. These differences clearly result from the additional context provided by the "schematic feedback" and the LLM's ability to exploit it properly.

The results from the instructors' evaluation suggest that the most proper feedback for students is "combined feedback", i.e., the LLM-generated feedback that exploits the "schematic feedback". Moreover, the "discursive feedback" was not better than the "schematic feedback". At the same time, the ability of LLM to provide adaptive feedback appears more limited. On the other hand, the students experienced the feedback better than expected and better than the previous academic year (even if the latter comparison is not statistically significant). Moreover, the majority of students considered the "combined feedback" helpful/essential to solving the exercise. In addition, the students of this academic year (who used the "combined feedback") reported a higher quality of feedback than those of the previous academic year (who used the "schematic feedback"). This point is particularly relevant because it suggests a positive influence of using LLMs to provide student feedback. At the same time, we also collected suggestions on how to improve the "combined feedback", which requires an improvement and refinement of the prompt used to query the LLM.

The two evaluations have limits. The first is that only two professors evaluated the different types of feedback. The second regards the comparison between students from two academic years (even if the involved degree courses, the teaching methods and topics remained the same). By taking this into account, we can conclude that adopting LLMs for building students' feedback seems useful as long as the prompt is crafted by providing the necessary context and information (in our case, by exploiting the "schematic feedback"). Moreover, this finding appears true both from a professor's and a student's viewpoint.

7 Conclusions

The paper initially presented three different types of feedback. The first (called the "schematic feedback") is crafted from a toolchain with static code analysis and a short-sentence classification module. The other two types of feedback (called "discursive feedback" and "combined feedback") exploit LLMs using two similar prompts, the first using the same information as the "schematic feedback", the second including the "schematic feedback" as additional context. We evaluated all types of feedback from both the instructors' and students' perspectives, with a general satisfaction regarding the quality of the feedback powered by LLMs. These findings suggest the positive impact of adopting LLMs to support the generation of student feedback, as similar studies—discussed in the related work – highlight.

Finally, the possible impact of "combined feedback" on the learning outcomes is part of the future work and will be evaluated as soon as the students complete the final exam. In addition, to support adaptation, we plan to improve the prompt engineering describing the student model to better inform the LLM about the differences between the different students.

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A Systematic Literature Review on Applicability of Robot Assistants in Higher Education

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Abstract. As higher education navigates the shift toward digital and remote learning, innovative technological solutions, including robot assistants, are becoming increasingly important. This systematic literature review explores the potential use of robot assistants in higher education. The research method involves a systematic literature review using the Kitchenham approach, focusing on studies conducted from January 2019 to February 2024. The review identifies 30 relevant articles, discovering the utilization of robot assistants in supporting lectures, teaching practices, and improving learning processes. The discussion emphasizes the utilization of robot assistants in higher education, suggesting their transformative potential in providing personalized support. Ongoing exploration is essential for effectively utilizing robot assistants in higher education. However, the findings suggest that addressing the technical limitations of robots, alongside implementing clearer policies and enhancing teacher training, is essential for the effective adoption of these technologies.

Keywords: robot assistants · educational robots · robot-assisted education · robotic teaching assistants · robot teaching roles · higher education

1 Introduction

Due to the recent global pandemic's unprecedented disruption of traditional in-person learning, the higher education (HE) environment has faced a significant challenge in adapting to remote teaching settings [1]. This challenge highlights the urgent need for advancements in technology-enhanced learning, demanding innovative solutions [2], and calls for innovative approaches in HE to ensure that all students, regardless of their location, would receive a comprehensive and immersive education [3]. While robot-assistants have been actively used in several sectors like manufacturing, healthcare and transportation [2], robot-assisted education supported by artificial intelligence (AI) has been identified as an innovative approach that not only addresses the immediate challenges posed by the pandemic but also holds promise for transforming teaching and learning practices [4]. Moreover, robot-assisted education enhances teaching by making it more interactive, responsive, and inclusive, accommodating everyone from students

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with special needs to remote students and teachers, and thereby expands accessibility of education [5]. This systematic literature review explores how robot assistants are changing educational practices in HE with particular focus on teaching practices, student engagement, learning outcomes, and educational challenges.

To guide the systematic literature review and address the research objectives effectively, specific research questions have been formulated, serving as the foundation for the review and providing a structured approach to evaluating the existing body of knowledge. The research questions, which encompass the goal of the review, are as follows:

- RQ1: What types of robots have been utilized for robot-assisted education within higher education over the past 5 years, and what specific educational purposes have they served?
- RQ2: What are the key benefits and challenges associated with integrating robot assistants in higher education, with a focus on their impact on learning outcomes?
- RQ3: How have technological advancements, such as artificial intelligence contributed to the development and effectiveness of robot assistants in higher educational settings?

These research questions show diverse applications of robot assistants for HE and the promising potential of this technology.

The rest of the paper is structured as follows. In Sect. 2, the research method applied for literature review is discussed, Sect. 3 presents the results and answers the research questions, and in Sect. 4, discussion and draw conclusions are provided.

2 Research Method

The systematic literature review was conducted to compile and analyze evidence regarding the applicability of robot assistants in HE. As suggested by Beecham et al. [6], Kitchenham [7] approach, which provides structured guidelines by focusing on essential steps such as developing a protocol, defining research questions, search strategy and maintaining rigorous data extraction and reporting practices, was taken, supplementing it with general guidelines from Baheer et al. [8] for the method of this literature review to include important aspects of robot-assisted education in HE, such as choosing studies that closely relate to this topic and methods for analyzing their information.

2.1 Search Strategy

After forming the research questions, a plan and a focused search strategy were established to identify studies relevant to the use of robot assistants in HE aligned with the goals of the review. A set of keywords: "`robot assistants`", "`robotic assistant`", "`educational robots`", "`robot-assisted learning`", "`robotic teaching assistants`", "`ai-based robots`", "`social robots`", "`humanoid robots`", "`telepresence robots`" and "`robot teaching roles`" in combination with "`higher education`" were identified to search through well-known and respected electronic databases such as IEEE Xplore, Web of Science, Scopus and SpringerLink. The selected terms

were combined in the search process to ensure that all significant research on the utilization of robotic assistance in educational settings, particularly within the context of HE was included. The review specifically targeted journal articles and conference papers, given their relevance and contribution to advancing knowledge in the field. Studies were limited to those published in English to ensure the clarity and accessibility of findings for the international audience. The publication period was set from January 2019 to February 2024, focusing on the most recent developments and addressing the constantly progressing features of emerging technologies in educational environments. The details of the search strategy are summarized in Table 1.

Table 1. Search strategy.

Electronic databases	IEEE Xplore Web of Science Scopus SpringerLink
Type of searched literature	Journal and Conference Papers
Search string	("robot assistants" OR "robotic assistant" OR "educational robots" OR "robot-assisted learning" OR "robotic teaching assistants" OR "ai-based robots" OR "social robots" OR "humanoid robots" OR "telepresence robots" OR "robot teaching roles") AND "higher education"
Language of the study	English
Publication period	From January 2019 to February 2024

2.2 Inclusion and Exclusion Criteria

Search inclusion and exclusion criteria were established to decide which search results to consider for this literature review. The abstract, keywords, introduction, and conclusion of each result, if applicable, were reviewed to filter the number of findings in each database (Table 1).

Five **inclusion criteria** were formulated to assist in the selection of relevant studies for this review. Namely, the study had to be (a) published in a peer-reviewed journal or conference proceedings; (b) published in English language; (c) an empirical study; (d) published not earlier than January 2019; and (e) align with research on HE robot assistants, verified by analyzing abstract, keywords, introduction, and conclusion.

Additionally, two **exclusion criteria** were integrated to strengthen the process of searching for evidence when assessing the relevance of each screened article. Namely, studies: (a) representing working papers, conference abstracts, and organization websites; and (b) not addressing robot assistants within the context of HE, were excluded.

2.3 Study Search and Selection

The initial search, conducted on 26 February 2024, identified 537 articles as follows: IEEE Xplore (n = 51), Web of Science (n = 86), Scopus (n = 111) and SpringerLink (n = 289). After a preliminary review based solely on the titles of the papers, titles not falling within the scope of this research were excluded, reducing the number of articles to 127. Removing duplicates left 114 records for screening. The screening process excluded 73 articles due to these reasons: 28 were not aimed at robot-assisted education and 45 did not use robots within the context of HE. This left 41 articles for full-text review. Of these, 11 records were found suitable after reviewing their titles and abstracts in mentioned databases, however, they were further excluded as full access to the papers was restricted due to subscription limitation, resulting in 30 journal and conference papers. The literature selection process, including formulation of inclusion and exclusion criteria, and resulting number of studied articles, is illustrated in Fig. 1.

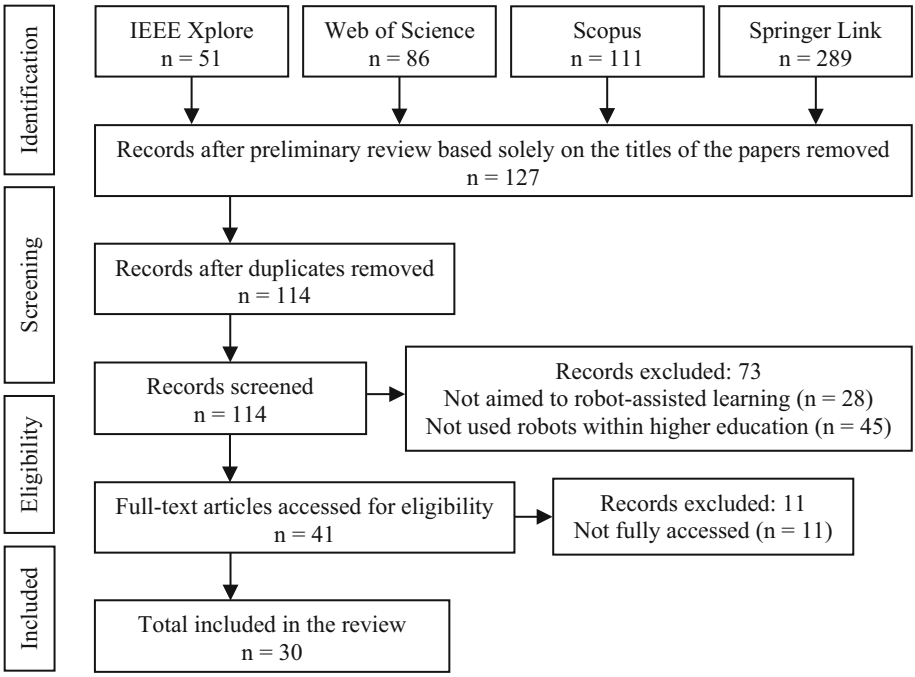

Fig. 1. Literature selection process flowchart.

3 Results

This section presents the findings of the systematic literature review by analyzing recent studies conducted between January 2019 and February 2024 on the utilization of robots in HE, and answers the set research questions RQ1–RQ3.

3.1 Types of Robots and Their Educational Purpose in HE

RQ1 is addressed by examining what types of robots have been utilized for robot-assisted education within HE over the past 5 years, and what specific educational purposes have they served. Recent studies show the use of various types of robots in HE over the past five years, including telepresence robots like Double [9–14], Kubi [9, 14], Beam [13–15], Ohmni [10], and AV1 [16]. The Pepper robot [17–23], noted for its semi-humanoid design, ranks as another frequently utilized robot, facilitating interactive learning experiences. The custom-developed Cognito robot [24] integrates advanced technologies for personalized student recognition through facial and fingerprint features. The Nao robot [18, 25, 26], an autonomous programmable humanoid, is distinguished for promoting group collaboration. In addition, the Reeti robot [27], known for its expressive communicative abilities with 15 degrees of freedom across its face, head, and eyes, and Baxter [28], a robot that supports hybrid human-robot collaboration teams, have been adopted in teaching settings. Moreover, innovative AI-integrated software robots [29, 30] have emerged, providing gamified learning experiences [31] and acting as professors and assistants, further enriching the educational environment. These findings show how robots, from telepresence to AI software, are making real changes in HE by supporting teachers, helping students, and improving how we learn and teach. Various types and models of robots used in robot-assisted education identified by the literature review for HE are detailed in Table 2.

Table 2. Types and models of robots used in robot-assisted learning in higher education.

Type of robot	Model of robot	Study
Telepresence robot	Double, Kubi, Beam, Ohmni, AV1	[9–16]
Semi-humanoid robot	Pepper	[17, 19–23, 32]
Custom-developed robot integrating various technologies	Cognito	[24]
Autonomous, programmable humanoid robot	Nao	[18, 25, 26]
AI-based software intelligent robot	Proprietary, called Professor, or Assistant	[29, 30]
Expressive and communicative Robot	Reeti	[27]
Industrial robot	Baxter	[28]
Gamified AI educational robot	(n/a)	[31]

Next, how robot assistants have been utilized in HE for various educational purposes are examined. Specifically, for the teaching process, robot assistants are used to facilitate remote attendance for both students and teachers [9–16, 29, 30], monitor attendance through facial and fingerprint recognition [24], provide real-time feedback [21], enhance engagement through gamification and robotics [27, 31], and automate administrative tasks [30]. For lecturers, robots act as examiners [24, 29, 33], supervise exams [10], detect plagiarism [19], support tutoring and assessment via Learning Management Systems (LMS) [21], and improve psychological and social communication during oral

assessments [34]. For students, robot assistants offer support for those with learning difficulties and disabilities [9, 14], assist with exam preparation [17, 18, 23, 32], transcribe faculty lectures and detect emotions to motivate [24], and enhance group collaboration [25, 28]. This variety of purposes highlights how robot assistants can significantly change education by improving learning outcomes, simplifying administrative tasks to free up more time for teaching, and promoting a more inclusive and active learning environment that engages all students more effectively. The aims of using robot assistants in HE identified by the literature review are shown in Table 3.

Table 3. Aims of using robot assistants in higher education.

Category	Aim of robot assistant	Study
For the teaching process	Allows students and teachers to attend classes remotely	[9–16, 29, 30]
	Attendance monitoring through facial/fingerprint recognition	[24]
	Providing real-time feedback	[21]
	Enhancing engagement with gamification and robotics	[27, 31]
	Automating administrative tasks	[30]
For the lecturer	Robot as examiner	[24, 29, 33]
	Exam supervision	[10]
	Plagiarism detection	[19]
	Tutoring and assessment support via LMS	[21]
	Improve psychological and social communication during oral assessment	[34]
For the student	Provides support for learning difficulties and disabilities	[9, 14]
	Exam preparation	[17, 18, 23, 32]
	Transcribing faculty lectures	[24]
	Emotion detection of students to provide motivation	[24]
	Enhance group collaboration	[25, 28]

3.2 Benefits and Challenges of Robot Assistants in HE

In addressing RQ2, the key benefits, challenges, and impacts associated with integrating robot assistants in HE with a focus on learning outcomes are explored.

The benefits identified in literature review include enhanced engagement, confidence, and motivation [32], improved exam scores [17], collaboration fostering [35], real-time evaluations [33], increased reliability through automated tasks [24], enhancing social presence [11], bias elimination and fairer evaluations [29], increased confidence

[22], and fewer bureaucratic obstacles [30]. These improvements make education more dynamic and fairer. Challenges encompass technical and operational issues such as connectivity [13], navigation difficulties [11], limited field of view [12], battery life [9], audio/video quality concerns [36], concerns like privacy and legalities [37], high costs [35], knowledge gaps in robot usage [19], potential reductions in human interaction [24], and decreased social connectivity [25]. These challenges highlight the need for careful implementation and ongoing support to mitigate potential disadvantages. The impact on education appears significant, with improvements in remote accessibility and flexibility [10], positive perceptions of technology use [36], enhanced engagement and presence [15], development of teamwork skills [35], time savings [24], improved learning efficiency [38], administrative efficiency [37], objective evaluations [29], and addressing the robomorphism challenge [16]. This suggests that, despite challenges, robot assistants can significantly enhance educational experiences and results. Benefits, challenges and impact of robot assistants in HE are described in Table 4.

Table 4. Benefits, challenges and impact of using robot assistants in higher education.

Benefits	Std.	Challenges	Std.	Impact	Std.
Engagement, confidence, motivation improvement	[32]	Connectivity issues	[13]	Remote accessibility	[10]
Eliminating bias	[29]	Navigating the robot	[11]	Positive perception	[36]
Fostering collaboration	[35]	Narrow field of view	[12]	Improved engagement and presence	[15]
Evaluations in real-time	[33]	Audio/video issues	[36]	Teamwork skills	[35]
Enhances reliability through automation tasks	[24]	Privacy and legal concerns	[37]	Saving time	[24]
Enhancing social presence	[11]	Battery life	[9]	Flexibility	[10]
Enhanced exam scores	[17]	Cost	[35]	Learning efficiency	[38]
Increasing fairness in evaluation	[29]	Lack of usage knowledge	[19]	Administrative efficiency	[37]
Increase confidence	[22]	Reduction in human capability	[24]	Objective evaluation	[29]
Reducing bureaucratic hurdles	[30]	Reducing social connectivity	[25]	Robomorphism	[16]

3.3 Contribution of Technological Advancements to Robots in HE

As the last research question (RQ3), how have technological advancements, such as AI, contributed to the development and effectiveness of robot assistants in HE settings are

explored. The literature review highlights those technological advancements, particularly in the field of AI, have played an essential and transformative role in significantly enhancing both the functionality and the overall impact of robot assistants within educational environments. AI-driven robot examiners [33] offer fair and efficient assessment reviews, while robot moderators [33] utilize AI to store, analyze, and verify content accuracy. Advanced recognition technologies and speech-to-text conversion [24] enable personalized interaction, with machine learning [24] further applied for accurate exam corrections and sensing students' emotions. AI significantly improves the Pepper robot's ability to communicate through speech, gestures, and tablet use [20, 22], making educational experiences more interactive. Autonomous teaching robots [29], powered by sophisticated software, operate with minimal human intervention, illustrating the move towards independent educational facilitators. Virtual reality (VR) and smart robots [28] transform learning into a more interactive and engaging experience. ChatGPT [30] illustrates AI robots' capacity for customizing learning materials, interactive teaching, and automating administrative tasks. Moreover, the integration of AI with gamification [31] introduces dynamic and motivating learning environments, particularly in safety education. Finally, specialized algorithms [34] enable Assistance Humanoid Robots (AHRs) to adapt their social interactions during oral assessments, addressing specific needs of students. All these AI improvements make learning with robots easier, more fun, and better adapted to each student's needs. These developments highlight AI's crucial contribution to making robot assistants more flexible, effective, and supportive in educational contexts. Table 5 presents the discovered technological advancements and AI's influence on HE robot assistants, detailing their roles and effectiveness.

Table 5. Technological advancements – AI influence on robot assistants in higher education.

Technological advancements – AI role	Study
Robot examiners ensure fair, efficient AI-driven answer review	[33]
Robot moderators store, analyze, and fact-check content using AI	[33]
Facial, fingerprint recognition and speech-to-text conversion	[24]
Machine learning for exam correction and emotion detection	[24]
AI enhances Pepper robot's speech, gesture, and tablet interaction functionalities	[20, 22]
Software enables autonomous teaching robots with minimal human intervention	[29]
VR and intelligent robots make education more interactive and engaging	[28]
ChatGPT showcases AI robots' ability to customize learning, teach interactively, and automate tasks	[30]
AI and gamification enhance educational tools, with AIER using AI for voice interaction and gamification to engage students in safety courses	[31]
Algorithm enhances AHRs for effective, adaptive social interaction in oral assessments	[34]

4 Discussion and Conclusion

In this study, the role of robot assistants in HE, particularly in the context of recent increased attention towards digital and remote learning was examined. Via a systematic literature review, the utilization of robot assistants in the HE context, aiming to understand their potential benefits, challenges, and impact on teaching and learning outcomes was investigated.

Results suggest that the variety of robot assistants utilized in HE shows their potential to enhance and supplement traditional teaching methods and make learning more interactive and responsive [9]. Via supporting lectures, teaching practices and learning processes, robot assistants can offer personalized support and contribute to making the educational experience more dynamic and immersive [24]. Robot assistants can enhance engagement, collaboration, and real-time evaluations, facilitating a more inclusive learning environment [21]. Improvements in accessibility, flexibility, efficiency, and fairness in evaluations contribute to enhancing learning outcomes and promoting student success [29]. The impact of robot assistants on learning outcomes can be substantial (as demonstrated in [10]), with improvements observed in various areas such as accessibility, engagement, efficiency, and fairness in evaluations [21].

However, despite the promising potential of this technology, several challenges and limitations, such as technical and operational issues, privacy concerns [38], and high costs [36] need to be addressed for the effective implementation of robot assistants in HE settings. In addition, in the interests of fluent integration, clearer policies, ongoing support, and teacher training are essential [19].

Technological advancements, particularly in the field of AI, seem to be crucial in shaping the effectiveness of robot assistants in the near future. AI-driven innovations such as robot examiners, moderators [33], and advanced recognition technologies [25] could enhance the functionality and adaptability of robot assistants, making learning more interactive, personalized, and adaptive to individual student needs.

Limitations and Future Direction. The study examined the papers published after January 2019 listed in several major academic databases. Although recent technological advancements have made robot assistants more relevant to the needs of HE, earlier works could offer significant theoretical contributions to framing the field of robot-assisted education. Potential future directions include studies on the development of advanced AI-driven functionalities for robot assistants, potential integration of virtual reality applications, and assessment of long-term impacts on educational practices in order to understand the potential of robot assistants in HE fully.

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Generative AI: Simplifying Text for Cognitive Impairments and Non-native Speakers

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Abstract. It is well known that people with cognitive disabilities, older people, and immigrants face barriers in understanding technical texts. Some organisations makes efforts in developing guidelines for the creation of Easy-to-Read texts, although these have limitations, such as a lack of specificity and a focus on intertextual cohesion. There have been several recent papers that put forward new techniques for simplification like the addition of notes and Q&A sections while others emphasise a balance between simplification and quantity of information. They also criticise that conventional simplification methods are dependent on expertise and resources that may not be available and that there is a lack of focus on the individual needs and different target groups, so artificial intelligence presents an opportunity to expedite these processes. The results of the use case and the comparison between the resulting LLM's yielded correct components but also shortcomings. These give room for further exploration to consider optimising prompts for better results.

Keywords: Textual simplification · Generative AI · Dyslexia

1 Introduction

Reading, technical texts, is a regular part of daily life for many individuals, but there is a specific demographic for whom this poses more difficulties. On the one hand, one must consider those people with different cognitive impairments. Within this group, there is a variety of impairments, such as dyslexia, autism spectrum disorder (ASD), and Down syndrome, among others. However, these do not always have to be disabilities; also, the older population and immigrants, who lack the necessary language skills, are affected. All they have a point in common: the fact that they have difficulties understanding specific and more difficult texts. For these reasons, these groups are affected and even marginalised within society. To offer better inclusion to these citizens, some institutions have advocated for development and implementation of different tools for text simplification. The International Federation of Library Associations and Institutions (IFLA) developed the Guidelines for Easy-to-Read Materials [1], and

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the European Association ILSMH (International League of Societies of Persons with Intellectual Disabilities) established Make it Simple - European Guidelines for the Production of Easy-to-Read [2] texts. Most of these regulations propose simplification using easier synonyms, shorter sentences, and the representation of only one idea per sentence. Conventional simplification guidelines have been previously studied and implemented, but the results obtained cannot be adequately "quantified" [3]. Therefore, the aim of this study is to determine the degree of simplification that each target group need. The study focuses on promoting inclusion through text simplification, applying an integrated methodological approach that combines qualitative and quantitative analyses to assess the effectiveness of simplifying interventions [4]. Simplification strategies and their outcomes will be investigated to verify compliance with comprehension and accessibility objectives. Multilingual texts of varying degrees of complexity will be examined with a view to establishing a framework for the potential application of generative artificial intelligence. Finally, the article will present the following structure: Sect. 1 will identify the target groups and their needs, i.e., catalogue the different target groups, recognising and detailing their specific needs in relation to the comprehension of medical and legal texts. Section 2 will conduct a critical review of current text simplification methods, identifying their strengths and weaknesses, to better understand the limitations and opportunities in the field. Section 3 will examine the technological contributions that have impacted text simplification, highlighting significant achievements and areas that still require development. In the last section, a case study will be presented to illustrate how artificial intelligence can be applied in practice to simplify texts, evaluate its effectiveness, and explore the implications of its use in real-life scenarios.

2 Related Works

In the last few decades, text simplification methods, such as Easy Reading, Easy-to-Read and Plain Language, have been widely studied and discussed as necessary tools to facilitate text comprehension. Some of those methods, mentioned above, were created by activists and institutions, but authors of guidelines such as Nomura et al. [1] claim that further research is needed to provide empirical support, which allow to consider better the needs of readers with cognitive disabilities. Similarly, Fajardo et al. [3] indicate that the guidelines proposed by IFLA and ILSMH are too general, which makes their application difficult. Within the superficial readability formulas, it is suggested that textual difficulty can be measured by variables such as the length of words and sentences, as well as the frequency of words, according to the Flesh Index [5]. However, there are authors who show that the frequency of words does not influence literal or inferential comprehension of texts [6]. A limitation observed in these methods is the omission of intertextual cohesion, which refers to the relationship between different sections of a text. Crossley and collaborators emphasise that greater presence of cohesive elements simplifies text comprehension. Simplification of texts can be done in several ways, one of which was suggested in a study [7], in which

they sought to make clinical documents more accessible to patients by means of annotations that provided additional information, a simplified version, a section of possible questions and answers related to the documents, and a to-do list.

A further limitation of these methods is the difficulty of determining the optimal degree of simplification. Maaß and Rink [8] point out that simplification can generate overly simple texts that omit crucial information from the original text. Their approach suggests that one should seek to balance the amount of information needed to produce accessible texts without jeopardizing essential content. In addition, Bott and Saggion [9] mention another weakness of the Easy-to-Read simplification method, this method would be very dependent on the knowledge of professionals and requires a lot of resources. So, the amount of simplified textual content is restricted. To speed up this process, Bott and Saggion propose an automatic simplification system to make the texts more understandable for the target population. Another approach, such as lexical simplification, attempts to solve the problem by using simpler alternatives. This work began to be implemented in 1998 [10]. As previously mentioned, simplified texts are also aimed at people who have a low level of language knowledge, such as immigrants. However, Baker [11] criticises that immigrants are seen by the receiving societies as a problem, because they can not understand the language, and this require solutions such as easy reading, but also criticises the action of institutions that promote glotopolitics such as the IOM that seek to contribute to ensure a humane management of migration the problem is "that they are acting as entrepreneurs, capitalising on their malleability and reputation for efficiency," is the observation that Bradley [12] (p. 97) offers. However, within the related state of the art, no approaches have been found on the possibility of elaborating simplifications with the help of generative AI.

3 The Different Target Groups

In the context of the present study, the objective of which has been outlined above, it is imperative to establish the identity and characteristics of the target group. The elucidation of the criteria delimiting membership in this group is essential for understanding the intrinsic need for textual accessibility. A detailed characterization of the profile of the target group, including but not limited to their reading comprehension skills, educational background, and possible cognitive or linguistic barriers, is therefore required to substantiate the demand for simplified texts and to justify the proposed textual adaptations. Among the group of people who need simplified texts are people with dyslexia. Dyslexia affects between 10 and 17.5% of Spaniards, causing problems in word recognition and comprehension, as well as in writing, due to a phonological deficit [13]. Those affected experience difficulties with reading that limit the development of their vocabulary and knowledge [14]. In addition, they face challenges with words of similar spelling and morphological and syntactic errors. Text formatting can also complicate comprehension [15]. Another disability that hinders understanding is autism spectrum disorder (ASD), according to Autismo España. In 2015,

Autismo España reported a total of 2,116 ASD diagnoses in the country, implying a substantial increase of 119% over the previous year [16]. The capacity for understanding among those affected by ASD varies significantly, reflecting a heterogeneity of needs. One segment of this population shows above-average cognitive development, while another segment experiences pronounced deficits in text comprehension, particularly in the recognition and understanding of meaning [17]. The difficulties of the latter group are accentuated in inferring contextual information and understanding long sentences with complex syntactic structures [18]. Research indicates that the main challenges for reading comprehension in individuals with ASD are related to syntactic complexity and coordination and subordination mechanisms [19]. However, not all people who have difficulties in understanding complex texts are people with cognitive disabilities, the elderly population and foreign language learners also have difficulties. Factors such as age-related cognitive decline and limited literacy in the target language compromise their ability to process complex texts without external assistance. Hirsh and Nation [20] argue that comprehending non-simplified literature, such as novels, requires a vocabulary of at least 5000 terms. This challenge is amplified for immigrants who are confronted with specialised texts in their new cultural and linguistic context, which may hinder their integration. The competence to infer the meaning of unfamiliar words from context requires a language proficiency of 97–98% [21], a threshold that many immigrants do not reach. This comprehension deficit affects their social participation and adaptation to their environment.

3.1 Analysis of Traditional Text Simplification Methods

Baker's [11] analysis of easy-to-read texts, originally designed for individuals with intellectual disabilities and later applied to immigrants in Spain, revealed that such texts often carry stigmatizing implications. The assumption that these groups cannot understand the "correct" language or have limited comprehension, as described by Bredel and Maaß [22], also highlights a possible negative bias inherent in the creation of these texts. The text-makers, who are not the end recipients of the easy-to-read versions, may unintentionally create texts that resemble xenolects, and this would contribute to social stigmatization. Criticism of the easy-to-read format intensified when German documents were presented in this format for a vote, raising concerns about the simplification of language potentially diminishing citizen-government engagement [23]. Baker underscores that the intent behind providing simplified texts is not to homogenize difficulty levels but to promote inclusion by offering texts that cater to varying needs. Further, she explores the adaptation of the Easy-to-Read method to Spanish, initially formulated by the IFLA, addressing not only those with disabilities but also individuals with neuropsychiatric conditions, sensory impairments, aphasia, dementia, limited education, and children. In contrast, the *Dilofácil* project led by García Muñoz [24] includes similar target groups but further distinguishes subcategories within immigrant populations. This makes sense, because, within the group of immigrants, there are people with different needs, also some with

cognitive impairments. So, at this point it is crucial to ask how much simplification is needed for each group? And how to simplify texts for people who are facing more than one barrier? Baker's examination of online newspaper articles indicates a troubling trend where immigrants, regardless of cognitive abilities, are frequently equated with those having disabilities, a conflation that undermines inclusive practices.

Rink [8] addresses the translation of simplified texts, particularly legal and administrative texts, and the balance between providing enough information for comprehension without overwhelming the reader. Simplified texts should not be excessively short or superficial, as this can lead to under-informing the reader and perpetuating stigmatization. An ideal text strikes a balance, offering a point of entry into the discourse and establishing a foundation for deeper understanding. Bredel and Maaß [22] emphasize that simplified texts should be accessible, understandable, and actionable. In this context, simplified texts involve linguistic, global, and medial strategies, where linguistic strategies are geared towards enhancing word, sentence, and text-level comprehension, and conceptual strategies deal with managing cognitive complexity and content structure. The investigation into text simplification strategies, especially within the realm of complex legal and economic documents, underscores the inadequacy of relying solely on Easy Language guidelines to foster accessibility. Rink's analysis [8] points out that such simplification does not fully address the participation challenges faced by individuals with disabilities, as envisioned by the UN CRPD ambitions. It is apparent from this insight that a more nuanced approach is necessary to bridge the accessibility gap in these hybrid texts that blend informative and instructive elements. The specific study titled "Simplify or Help? Text Simplification Strategies for People with Dyslexia" delves into alternative strategies beyond the conventional substitution of complex terminology with easier synonyms [25]. Focusing on dyslexic readers, the study examines the effectiveness of linguistic simplifications that cater to difficulties with long or complex words [26]. The use of the LexSiS automatic simplification tool is central to this study. LexSiS employs a semantic model alongside web-based resources to provide context-appropriate synonyms while consciously maintaining the original term's frequency and semantic integrity [27]. Comparative analysis within the study pitted two simplification strategies against each other: SubsBest, replace the complex word with the simplest synonym, and SHOWSYNS, which offers a choice of synonyms, implemented through the DysWebexia [28]. The study's methodology involved measuring the impact of these strategies on readability and comprehension distinctly. Findings from this investigation were enlightening; texts modified using the SHOWSYNS strategy outperformed those processed using SubsBest or not at all in terms of readability and comprehension for dyslexic readers. Instead, providing a range of synonym options could enhance the accessibility and usability of texts for this demographic [25]. Given these findings, the SHOWSYNS strategy holds particular promise for the adaptation of electronic legal and economic texts, which often prove challenging for individuals with reading disabilities.

3.2 Text Simplification with AI

To understand how textual simplification is carried out with the help of generative AI, it is important to get an introduction to what promts and promt engineering are. The effectiveness of Large Scale Language Models (LLMs) "such as GPT-4" on specific tasks depends on prompt engineering, a technique that involves formulating detailed and clear instructions to guide the model towards accurate and relevant answers. This discipline requires a balance between the technical and creative knowledge of the LLM and the domain in question [29]. Strategies such as "Zero-shot Prompting" [30] and "Chain-of-thought" [31] are key to improving interaction with LLMs and overcoming challenges such as variability in responses and the production of incorrect data. The prompt engineering technique in text simplification consists of two parts: the task prompt, which defines the action to be performed, and the consideration prompt, which details specific aspects to be taken into account. To explore the use case, we take as a reference the study [32], in which a five-step methodology using AI for textual simplification for learning purposes was used. Only step two, which involves simplification with the help of a large language model, specifically ChatGPT-4, will be considered in this use case. In this context, an ethical concern arises, since the texts that are simplified with the help of these models contain sensitive and confidential data that must not be stored or passed on to third parties. How do LLMs handle this data?

4 Results and Discussion

4.1 Case Study

Since some of the guidelines or rules of textual simplification have been detailed [1,2,24] different research works have been analysed that criticise the deficiencies or shortcomings of these modalities [8,9,11], and a research study has been detailed that implements a methodology to simplify legal texts by implementing a LLM, we explore the possibility of simplifying a legal text to make it suitable for the target group, in this case people with dyslexia. For this purpose, a text from the legal field, a template for a declaration of horizontal division, will be used. The text was translated in a legal translation class from German into Spanish. The selected use case is closely aligned with the methodology used, as the design of the case study is based on user-centred design. The first stage of the methodology proposed refers to a process of empathy and knowledge of the initial context.

4.2 Criteria for Text Simplification Without AI

After exploring optimised simplification methods, a list of criteria was created. Those norms were considered necessary for simplification for people with dyslexia and for this specific type of text. The aim of this is to make conventional criteria and the criteria of LLMs comparable (Table 1).

Table 1. Criteria for Textual Simplification

• Adding context that makes implicit content explicit
• Inclusion of definitions to enable understanding of terms that cannot be substituted
• Lexical simplification of complex words by simpler or more audience-friendly synonyms
• Use of clear and consistent typography and a text format that provides an easy way to access information
• Concrete and concise writing, so that a short sentence only contains a single action
• Logical writing and use of textual connectors so that the text has clear continuity
• Use of symbols or some method of visualization, if possible, that allows the content to be understood
• If the text appeals to the reader to perform an action, this should be highlighted and clearly outlined (e.g. using brackets), without details that are not crucial

4.3 Criteria for Text Simplification with AI

The prompt for the simplification was made as follows:

prompt1 + prompt2 + [Original Text].

< Simplify the text > < considering that the target group is people with dyslexia > < Horizontal division statement.>

By applying this structure, ChatGPT-4 is guided to process and reformulate the information according to the defined instructions, maintaining the coherence and essence of the original text while reducing its complexity. It is crucial that the

Table 2. Strategies Taken into Account by ChatGPT-4 in Textual Simplification

• Clarity and conciseness: I used short, direct sentences to present information in a simple way, which helps to avoid cognitive overload
• Logical structure: I presented the information in a logical and sequential order to make it easier to follow
• Simple vocabulary: I chose words that are common and easy to understand rather than complicated legal terminology, reducing the difficulty of the text
• Avoid jargon and technicalities: I eliminated or explained technical and legal terms so that anyone, regardless of their familiarity with the subject, can understand the content
• Use of examples and explanations: Where possible, I provided additional examples or explanations to clarify complex concepts
• Clear formatting: Although visual formatting cannot be presented in text, I mentioned that attachments are organised and are an important part of the document, suggesting that attention should be paid to how these are physically presented
• Numbered points and lists: I used numbered lists to break down the information into manageable points, making it easier to keep track of important parts of the document
• Minimal repetition: I avoided repeating unnecessary information that could cause confusion or distraction
• Focus on the essentials: I focused on the most important information that needs to be understood, omitting details that are not crucial to the overall comprehension of the text

Table 3. Original Spanish text

Sección	Contenido
Al Registro de propiedad (Lugar)	Declaración de división horizontal
	Para el establecimiento de la propiedad de viviendas y propiedad parcial, según el artículo 8 de la Ley alemana.
Artículo 1 Descripción del inmueble	(1) El firmante consta como propietario del inmueble en el registro de propiedad de (lugar) en el tomo ... y hoja (número): (Descripción del terreno)
	(2) Sobre este está construido/se construirá un edificio con (Descripción del edificio)
Artículo 2 División del inmueble	(1) A partir de ahora el propietario divide la propiedad de propiedad inmueble en cuotas de copropiedad de forma que vincula por cada cuota de copropiedad, la propiedad privativa de un apartamento (propiedad de viviendas) o de un espacio no destinado a la vivienda, de la siguiente forma:
	1. Cuota de copropiedad de (fracción)/1000, vinculada a la propiedad privativa al apartamento no. 1 que está indicado en el plan de repartición en la planta baja a la derecha del sótano número 1.
	2. La propiedad de vivienda y la propiedad no destinada a la vivienda son independientes conforme al artículo 3 del párrafo 2 de la Ley de propiedad horizontal. Tanto el plano de distribución como el certificado de delimitación suficiente de la autoridad competente se encuentran adjuntos y forman parte integrante de esta escritura. Se ha referencia a estos/remítase a ellos.
Articulo 7 Venta de la propiedad horizontal	(1) La propiedad de vivienda se puede enajenar y dejar en herencia.
	(9) En caso de venta de una de las propiedades de viviendas y espacios no destinados a la vivienda lo siguiente:
	1. Todos los pagos efectuados a la comunidad (reserva para mantenimiento u otros pagos a la comunidad) por el vendedor, se transmitirán al adquiriente.
	2. El adquiriente responde como deudor solidario por todas las obligaciones de pago o del enajenador frente la comunidad, esto implica también los atrasos en los pagos de la comunidad y otros.

task prompt and the consideration prompt are aligned with the capabilities of the model and with the specific goal of simplification to ensure optimal results. The effectiveness of this methodology lies in its simplicity and its ability to provide the model with clear direction, which facilitates the generation of a simplified text that meets the user's expectations. This approach is consistent with best practices in prompt engineering, which seek to optimise the functionality of LLMs for specific natural language processing tasks.

ChatGPT-4 was asked to explain what guidelines and strategies it considered when generating the simplified text (Table 2):

Table 4. Simplified versión by ChatGPT-4 [33]

Sección	Contenido
Registro de la Propiedad	Lugar
División de un Edificio	Este documento explica cómo se divide un edificio para que diferentes partes pertenezcan a diferentes personas. Esto se hace siguiendo las reglas de una ley especial
Información del Edificio	– El propietario registrado del terreno está en los documentos del Registro de la Propiedad del lugar, en el libro y página que se mencionan – El terreno se describe y hay o habrá un edificio construido en él
Cómo se Divide el Edificio	– Ahora el propietario ha dividido el edificio en partes. Cada parte tiene un número y corresponde a un apartamento o un espacio que no es para vivir – Cada apartamento o espacio tiene un número y está en un plano. Este plano y otros documentos importantes están adjuntos y forman parte de este documento
Venta de los Apartamentos o Espacios	– Se pueden vender o dejar en herencia los apartamentos o espacios – Cuando se vende, cualquier dinero que el vendedor haya pagado para el mantenimiento del edificio o cualquier otro pago, pasará al nuevo dueño – El nuevo dueño también será responsable de cualquier deuda o pago atrasado que el vendedor tenía con el edificio

4.4 Simplified Text Using Other LLM

To make a comparative work between the LLM ChatGPT-4 and other models, exactly the same instructions were given to the Code-Llama-34b model and the Llama2-70b model in order to compare the simplifications generated by these models (Table 5).

Table 5. Simplified version by Code-Llama-34b

Este es un contrato de compraventa de una propiedad horizontal en Alemania. El comprador adquiere una parte de la propiedad de un inmueble, que incluye un apartamento y un espacio no destinado a la vivienda. La propiedad se divide en dos partes: la propiedad privativa, que pertenece al comprador, y la propiedad común, que es compartida con los demás propietarios del inmueble
El contrato establece que el comprador es responsable de todos los pagos efectuados a la comunidad, incluyendo los atrasos en los pagos. Además, el comprador es solidario con el vendedor en caso de venta de la propiedad
Este contrato es importante porque establece los derechos y obligaciones de los propietarios de la propiedad horizontal, así como las responsabilidades de cada uno en caso de venta o herencia de la propiedad. Es importante que los propietarios lean y entiendan bien este contrato antes de firmarlo

5 Conclusions

After reviewing recent research proposing advanced methods for text simplification and taking advantage of the opportunities offered by new technologies, one of these advanced technologies was tested. The objective was to simplify a legal document using a simple prompt that would generate a preliminary output. This result complied with certain pre-established guidelines, although areas for improvement were identified. In the comparison made, discrepancies between the results provided by the different large language models are recognised. The question of how LLMs treated personal data was not answered in this paper as the document that was simplified did not contain personal data and was a template to be filled in. However, this issue should be developed in future work. Future research also should consider in-depth analysis of the creation of more accurate prompts and define parameters that could lead to superior results. The idea of creating a simplification that is suitable for people who fit into more than one target group at the same time was not explored. However, it is believed that a simplification suitable for people with dyslexia is also suitable someone with little knowledge of the language, because the criteria for dyslexics overlap with the criteria for lower language levels. This could be analysed in future work that focuses only on dyslexics.

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A Study on Analyzing Classroom Teaching Questions Supported by Generative Artificial Intelligence

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Abstract. Classroom questioning can promote students' learning to happen and is an important means of classroom teaching. In order to analyze the target direction of questions raised by teachers in current classroom teaching, clarify the difference between questions on different topics, and discover the problems of classroom teaching, the article takes the mathematics discipline competence as the direction of classroom questions, constructs a question coding framework based on the analysis of the connotation and behavioral performance of mathematics discipline competence, and analyzes the 20 examples of elementary school mathematics classroom teaching in collaboration with generative artificial intelligence. The results of the study found that the current classroom teaching questions are focused on learning comprehension and application practice, and less on creative transfer. The teaching questions of different teaching topics vary, with the teaching questions of number and algebra, graphing and geometry focusing on generalization and comprehension, and illustration and argumentation, and the questions of probability and statistics focusing on speculation and explanation. Finally, the article suggests that the design of questions should take into account the teaching objectives, teaching content, higher-order cognitive questions, and scientific expression of teaching questions based on the idea of cue engineering.

Keywords: Classroom Teaching Questions · Generative Artificial Intelligence · Human-Computer Collaboration · Mathematics Discipline Competencies

1 Introduction

Classroom questioning is an important way to promote student learning, and has been called "the most influential teaching behavior" in classroom teaching [1]. Enlightening wisdom and promoting learning through dialogue and question-and-answer sessions can be traced back to the time of Confucius and Socrates to optimize or change students' learning behaviors by constantly stimulating their cognition [2]. Through questioning and answering, teachers guide students to master basic knowledge and skills, and students construct knowledge and develop cognition in the process of problem solving. Some studies have shown that students' question-answering behavior in the classroom accounts for 80% of all language behaviors [3]. For this reason, researchers at home

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and abroad have carried out a lot of studies on classroom questions. Some studies have analyzed the characteristics of higher-order questions in primary and secondary school classrooms, and quantitatively characterized the proportion of higher-order questions and styles in the classroom [4]. Some studies have categorized questions into literacy questioning, prompting questioning, analytical questioning, application questioning, and evaluative questioning based on Bloom's categorization of objectives [5]. Or, based on the authenticity and openness of the questions, the types of questions are categorized into demonstrative, authentic, and clarifying questions [6]. For the interactivity of classroom question-and-answer (Q&A), some studies have proposed different Q&A interaction structures, such as "Inspire-Respond-Feedback" [7] and "Ask-Answer-Evaluate" [8]. However, the current teachers' classroom questioning can seldom pay attention to the occurrence of students' in-depth thinking, and most of the questions raised are based on memorization and practice, and divergent questions and higher-order questions are rarely seen [9]. The goal of classroom questioning is also vague [10], which is ineffective in supporting the achievement of teaching goals. Since classroom questions can promote the achievement of various teaching goals, the design of teaching questions should be diversified and directional. As the goal of students' learning, discipline competence contains three dimensions: learning and understanding, application and practice, and transfer and innovation, which are the key competencies gradually formed by students through discipline learning [11]. The formation of discipline competence is the key path to promote the realization of core literacy, and it also provides a clear direction for students' learning and teachers' teaching. However, current research is less involved in classroom questioning directed at subject competence. Therefore, this study takes the cultivation of mathematics subject competence as the goal of classroom questioning, and explores the types of questions asked in classroom teaching, the goal and the specific aspects of teaching improvement by analyzing high-quality lesson examples.

With the arrival of the intelligent era, generative artificial intelligence has become the focus of attention, and its superb natural language processing and analyzing ability makes it have excellent performance in human-computer dialogue, classroom teaching [12], and scientific research [13]. However, there are still fewer studies on the use of generative AI to optimize dialogic learning [14], and it is unknown whether generative AI can support teachers' classroom questioning. To summarize, this study adopts a human-computer collaborative research method, using generative artificial intelligence as a research assistant. On the one hand, it is used to verify whether the generative AI can support human research work, and on the other hand, its high level of intelligence is fully utilized to assist in classifying and analyzing classroom teaching questions.

Based on this, this study asks the following two research questions:

RQ1: What disciplinary competencies do the classroom instructional questions point to?
RQ2: How do the disciplinary competencies to which the classroom teaching questions belong differ across themes?

2 Method

This study used a collaborative human-computer content analysis approach as the research methodology. The researcher and the generative AI technology worked together to code and analyze the questions posed in the classroom instructional videos and examine the results. Then, based on the definition of mathematics subject competence connotations and exoteric behaviors, the sub-dimensions of subject competence to which the questions belonged were coded in order to characterize the classroom teaching questions. The specific research methodology of this study is described below:

2.1 Subjects of the Study

In order to guarantee the quality of the research object, 20 high-quality teaching videos of the first book of the fifth grade of elementary school mathematics of the Renmin Jiao version in China's National Wisdom Education Platform for Primary and Secondary Schools were selected for this study, The content covers four major areas: number and algebra, graphing and geometry, probability and statistics, and practical and integrated applications. The main contents include multiplication and division of decimals, location, probability, and tree planting problem solving. Meaningful questions raised by teachers during teaching were extracted from the 20 teaching videos, and a total of 202 math teaching questions were eventually extracted.

2.2 Question Coding Framework Construction

Teaching questions should have a clear goal orientation. This study uses the sub-competency points of the nine dimensions of mathematics discipline competence to one-to-one correspondence between teaching questions and discipline competence. Mathematics discipline competence is nine sub-competence points constructed and formed on the basis of the three dimensions of learning and understanding, application and practice, and creativity and transfer, as shown in **Table 1**.

Studies have been conducted to describe in detail the connotations of competence in the subject of mathematics and the specific manifestations of the sub-competencies [15]. This study analyzes the outward manifestations of competence in the subject of mathematics based on the conceptual descriptions.

The learning comprehension level contains three competency points of observation and memorization, generalization and comprehension, and illustration and argumentation, which require students to initially perceive and filter information, and then process the input knowledge through information extraction, comparison, and inductive analogy. The practical application level contains three competency points of analyzing and calculating, speculating and interpreting, and simple problem solving, which mainly reflects the simple application of mathematical knowledge and problem solving, in which students select what they have learned to match the information and relationships in the problem context, and exhibit external behaviors such as extracting, analyzing, interpreting, representing, calculating, and relating. The creative transfer dimension of the mathematics discipline includes three competence points: complex problem solving, conjectural inquiry, and discovery and innovation. According to the four cognitive stages of complex

Table 1. Mathematics Subject Competency Types

Discipline Competency	Dimensions of Mathematics Discipline
Learning and Understanding	Observation and Memorization
	Generalizing and Understanding
	Explanation and Argumentation
Application and Practice	Analyze and Calculate
	Speculate and Explain
	Simple Problem Solving
Creativity and Transfer	Comprehensive Problem Solving
	Conjecture and Exploration
	Discovering and Innovation

problem solving, the problem solving process goes through the process of observing the problem situation, discovering the problem point, characterizing the problem in mathematical language, designing the investigation plan, and carrying out the investigation. The learners have previously established a new schema, have a new cognitive structure, and have a certain degree of critical cognition of unfamiliar phenomena and problems, so they will also show behaviors such as questioning problems, phenomena, and solutions.

Based on the above analysis, this study constructs a question coding framework pointing to mathematics discipline competencies, as shown in Table 2 in which the specific behavioral manifestations, and representative question types of each type of subject competency are recorded to provide a basis for coding the subsequent questions.

2.3 Questions Coding

In this study, in order to give full play to the role of generative artificial intelligence in scientific research and to analyze and study teaching questions in a human-computer collaborative manner, we used a generative artificial intelligence software product developed by a Chinese company, the Starfire Big Model, to code and analyze questions.

Considering the research theme and scientific nature of this study, firstly, the connotative definitions of the dimensions related to mathematics subject competence, competence externality behaviors, and corresponding typical types of questions in Table 2 were provided to the big model for training and fine-tuning until the big model was able to identify the mathematics subject competence and the types of teaching and learning questions with a higher degree of accuracy. Secondly, in order to reduce the impact of errors on the coding results caused by the material text, unclear prompts, etc., the extracted question text was cleaned and standardized by removing irrelevant characters, punctuation, etc., and converting all the text to lowercase in order to reduce the variability of the model's processing, and the questions were provided to the large model for processing in batches to avoid exceeding the processing limit of the large model and to safeguard the analysis and coding quality. In this study, we adopt the research results

formed by the existing research on cue language engineering, and strictly use the cue language form of "character + instruction + background + output indicator".

Table 2. A Coding Framework for Classroom Questions Pointing to math subject competencies

Code	Discipline Competency	Behavior	Typical question statement
OM	Observation and Memorization	Observe, identify, select, describe, etc	"What information did you find by looking at the pictures?"
GU	Generalizing and Understanding	Comparing, searching, extracting, sifting, etc	"What are the similarities between the two methods?"
EA	Explanation and Argumentation	Generalizing, analogizing, associating, interpreting, etc	"Can you explain why you use departures?"
AC	Analyze and Calculate	Calculate, draw, extract, analyze, etc	"What is the answer to 6.4 ÷ 2?"
SE	Speculate and Explain	Guessing, interpreting, reasoning, judging, etc	"Look at the vertical equation and answer where the 0 after the remainder comes from?"
SPS	Simple Problem Solving	Characterization, association, calculation, etc	"Is there enough money left over to buy a $20 carton of eggs?"
CPS	Comprehensive Problem Solving	Characterizing, designing, solving, etc	"What are some problem solving strategies that can be applied"
CE	Conjecture and Exploration	Discovery, conjecture, inquiry, etc	"How can we verify the pattern of decimal division?"
DI	Discovering Innovation	Creating, questioning, etc	"Are there any other new solution ideas?"

In order to ensure the reliability of the question coding, after the researcher and the big model coded the same batch of questions in a double-blind manner, the consistency test of the two coding results was conducted, and the results are shown in Table 3, with the Kappa coefficient reaching 0.818, which is a high level of consistency. For the different coded items in the coding results, one by one calibration was performed after analyzing the dialog with the big model.

Table 3. Coding consistency test results

	Kappa	*p*
Human Coding & AI Coding	0.818	0.000**

* *: $p < 0.01$, this indicates a significant correlation*

3 Analysis and Results

3.1 Distribution of Mathematics Discipline Competencies Belonging to Classroom Teaching Questions

Figure 1 visualizes the percentage of questions in mathematics classroom teaching on the nine discipline competency dimensions. As can be seen from the figure, the higher percentage of questions posed in mathematics classroom teaching were in the two categories of illustrative-argumentative questions and generalization and comprehension questions, with 36.8% and 22.22%, respectively, and analytical-computational questions were the next highest (12.87%). Since there is a probability and statistics theme content in the math teaching theme, which involves students' speculation, inquiry and interpretation of uncertain events, the speculative inquiry type of teaching questions also has a certain percentage (8.77%). Overall, the classroom instructional questions were skewed toward the learning comprehension level.

The distribution of questions pointing to the competence of the mathematical discipline shows that the questions posed by teachers focus on the cultivation of students' learning comprehension and application and practical competence, and that teachers are accustomed to guiding students to make observations of mathematical information and to provide simple descriptions and explanations of situational questions. By analyzing the content of specific questions, it can be seen that the main functions of these two main types of teaching questions are to consolidate students' a priori knowledge, to stimulate students' recall through classroom questions, to activate students' cognitive structure, to construct meanings through verbal language and pictorial information dissemination, and to promote the connection of old and new knowledge in students' cognitive structure. Specifically, the learning comprehension level is dominated by the categories of generalization and comprehension and illustration and argumentation. Teachers present learning situations in books and ask questions to guide students to generalize and extract features of mathematical concepts with the same attributes, for example, asking students to identify the commonalities of mathematical concepts and methods of different computational processes as well as the differences between them. On the basis of memorizing and summarizing the relevant information, students integrate what they have learned and transform the knowledge into individual cognition through rational reasoning and judgment. For example, in the theme of "multiplying decimals by decimals", the teacher asks the question "Is there anything we should be reminded of?". By answering this question, students not only extract knowledge points, but also explain and justify why they are important, how they can be applied, and common errors they may make. At the level of innovation and discovery ability, there is a lack of higher-order questions that synthesize both problem solving and discovery and innovation. Such questions focus

on guiding students to synthesize and apply a variety of knowledge and methods, and through keywords such as designing, answering, questioning, creating, etc., students are guided to make the appropriate learning behaviors to solve the problems of the integrated and complex contexts, so as to contribute to the attainment of the higher-order teaching goals.

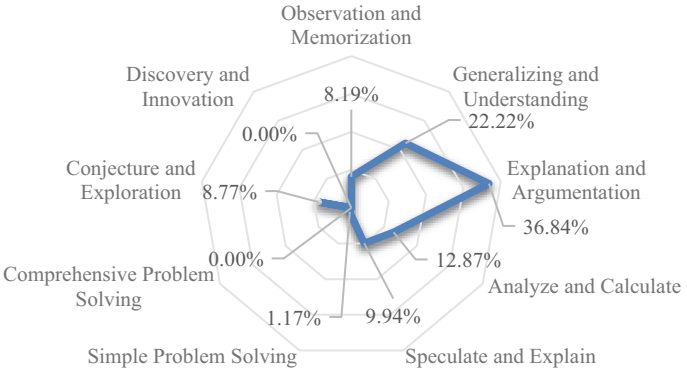

Fig. 1. Radar Chart of Overall Distribution of Types of Classroom Teaching Questions

3.2 Distribution of Mathematics Discipline Competencies to Which Different Thematic Classroom Questions Belonged

Fig. 2 shows the distribution of the types of instructional questions for each topic in the selected 20 instructional videos in the form of a bubble chart. The size of the bubbles in the graph represents the percentage of each type of question, the larger the bubble the greater the percentage of the number of questions. The distribution of instructional questions varies across content topics.

Horizontally, the same theme classroom instruction was dominated by learning comprehension and application practice types of questions (the bubbles corresponding to OM, GU, EA, AC, and SE in the figure), and they relatively accounted for a large proportion of the questions. Relatively few questions were designed for higher-order subject competencies. Conjectural Exploration questions (CE) were covered in all topics of instruction, but the number of questions in the synthesized problem solving and discovery of innovations categories was zero. This is the same as the overall distribution results.

Longitudinally, there were relatively more questions in the illustrative and argumentative categories of instruction for the topics of Number and Algebra and Graphics and Geometry. The speculative explanation category of questions in the Statistics and Probability theme was the most prevalent of the four themes' instructional questions. The Practical and Integrated Applications theme had more analytical and computational questions and conjectural investigations than any other theme. It is worth noting that the percentage of Simple Problem Solving (SPS) is very low and the percentage of Comprehensive Problem Solving (CPS) is zero. This may be related to factors such as the

design of classroom questioning language and the design of teaching activities. This is analyzed in the "Discussion" section.

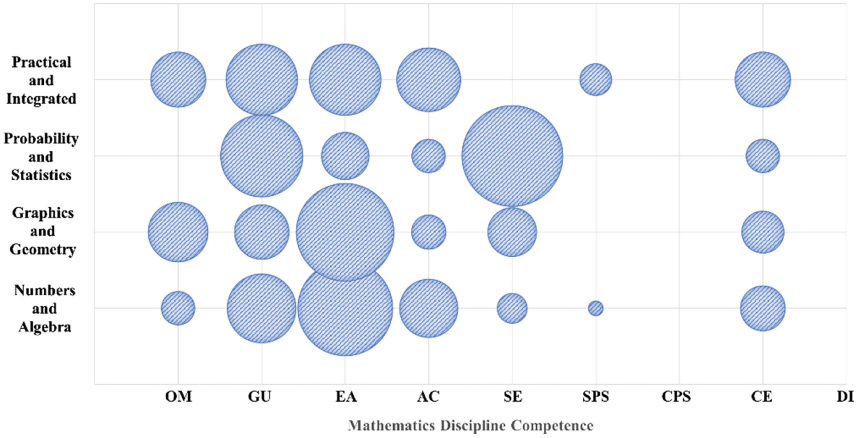

Fig. 2. Bubble Chart of the Distribution of Types of Teaching Questions for Different Topics on the Subject

4 Discussion

In this study, the teaching questions in the high-quality teaching videos were analyzed, and the distribution of question types and discipline competence was visualized and presented. From the analysis results, it can be seen that the distribution of classroom teaching questions types mainly focuses on the learning and comprehension level and the practice and application level of mathematics discipline competence, which is dominated by illustrative-argumentative, generalization and comprehension, and conjecture-inquiry questions. Other types of questions accounted for a relatively small proportion. Neither the comprehensive problem solving category nor the discovery and innovation category of questions were involved. This section will explain the possible causes of these problems, discuss their impact on student learning processes and outcomes, and suggest strategies on how to improve the current situation.

(1) Mathematics teaching themes and content have an effect on the distribution of question types in the classroom. The results show that different teaching topics have different distribution of question types. Specific analysis of the teaching contents also shows that most of the contents of the theme of "Number and Algebra" are related to identifying and explaining quantitative relationships and calculating quantities, while most of the contents of the theme of "Graphics and Geometry" are related to drawing graphs, calculating the area of graphs, explaining relationships and thinking methods of calculations. "Graphics and Geometry" covers mostly drawing shapes, calculating the area of shapes, illustrating relationships between shapes and thinking about related calculations. Teachers should fully study the curriculum standards,

analyze students' learning characteristics and fully understand students' learning problems. Ask targeted questions in classroom teaching. This can avoid the teaching mode fixed single.

This requires students to demonstrate judgment and understanding of quantitative and graphical relationships through verbalization. Therefore, most of the questions in these two themes are of the illustrative and argumentative type. In contrast, the theme of "Probability and Statistics" requires students to make conjectures or explanations about various events and to explain the likelihood of their occurrence. Students are required to experience the randomness of the results after going through the process of "observation-conjecture-practice-verification". Therefore, the largest proportion of questions in the teaching of this theme is of the conjectural inquiry type.

(2) Teachers' question design ability is a key factor influencing the distribution of question types in classroom teaching. From the results of the analysis, most of the current teachers' design of questions are factual questions that drive the pace of the classroom, such as "What information is found?" "What is the quantitative relationship?" "Is the event likely to happen?" etc., where students are only required to perform a simple information retrieval process without much in-depth thinking. Teachers' weak question design ability is mainly manifested in the following aspects. Firstly, it is difficult to achieve the teaching objectives in the design of teaching questions. Teaching objectives are hierarchical from high to low, and the corresponding teaching questions should also be hierarchical, and the achievement of higher-order objectives requires the design of higher-order questions or tasks. Second, the structure of teaching questions is single. Good questions often have appropriate question words and clear objectives. Due to the weak ability of teachers, the question words of teaching questions are usually "what", "whether", etc., and lack of "how", "why", etc., which require deep thinking and understanding.

Therefore, in classroom teaching, teachers not only need to pay attention to the appropriateness of teaching questions and teaching objectives and content, but also need to design the questions carefully. Teachers should flexibly apply a variety of questioning techniques, based on the idea of cue engineering, and adopt scientific question formulation methods to reduce the impact of meaningless questions on students' learning. In addition, the occurrence of students' learning not only depends on the guidance of teachers' questions, but also needs to promote the occurrence of cognition in the learning activities of knowledge construction. Teachers' teaching questions should be embedded in specific teaching activities, guiding students in the process of inquiry and providing them with cognitive scaffolding.

(3) The percentage of question types needs to be determined based on specific instructional objectives. The current imbalance in the proportion of question types in teaching may be related to the setting of teaching objectives. Most of the teaching has a large proportion of lower-order objectives, while the proportion of higher-order objectives is very small. This, to some extent, leads to the lack of higher-order questions in classroom teaching. On the one hand, teachers need to set teaching objectives reasonably according to the course content and learning requirements in teaching, and increase the cultivation of higher-order objectives appropriately. On the other

hand, teachers need to rationally design teaching questions in combination with the objectives to ensure the solution of low-order and high-order questions.

Learning is a process of knowledge exploration and generation, from the overall distribution of questions and the distribution of questions on some topics, the types of questions vary. The high and low cognitive levels of the questions are also different, which is corroborated by the results of existing research, dialogic teaching, lecture-type teaching teacher-student verbal exchanges in more than 80%, while higher-order questions only account for 13% [16], and the path of learning is therefore different. Although teaching advocates that teachers should cultivate students' higher-order cognitive abilities [17] and emphasize the use of higher-order cognitive questions to narrow the gap between students [18] so as to achieve teaching goals. However, study has shown that lowering the proportion of teachers' behavior has a positive effect on the improvement of students' academic performance [19], and teaching should not pursue the formulation of higher-order questions. Lower-order questions are equally important as the basis for students' understanding and application of scientific knowledge. The appropriateness of question formulation is also very important, as appropriate, multiple questions can stimulate students' interest in learning, while poor questioning can limit the learning process and have a negative impact on student learners [20].

5 Limitations

Although this study reveals the distribution of competencies directed by instructional questions in the current classroom and identifies areas of instruction that still need to be improved. However, the study still has a little limitation:

(1) This study only coded and analyzed the teaching questions, lacked to analyze the correspondence between specific teaching objectives and teaching questions, and has not yet explored whether the distribution of teaching questions is related to the setting of teaching objectives. Further in-depth analysis of the relationship between the two may be needed in subsequent studies to obtain more patterns.
(2) Only 20 videos of mathematics classroom instruction were selected for coding and quantitative analysis in this study, which is a small and disciplinary sample size. There is a subsequent need to increase the sample size and expand the disciplinary scope of the study.
(3) This study has only quantitatively analyzed the distribution of classroom teaching questions and suggested possible influencing factors. However, it has not yet explained how to design different types of questions in teaching, and further methods or strategies for designing teaching questions need to be proposed in subsequent studies to provide a basis for teachers to teach.

6 Conclusions

In this study, we took the mathematics discipline competence as the direction of teaching questions, used the content analysis method of human-computer collaboration as the research method, and coded and quantitatively analyzed the teaching questions of 20

mathematics classrooms together with generative artificial intelligence. Although teachers have been able to design teaching questions flexibly according to the teaching content and teaching objectives, the distribution of questions is still dominated by learning comprehension and application practice, and the design of questions for creative transfer needs to be further strengthened. Although this study reveals the questions existing in the current classroom teaching questioning, it has not yet carried out in-depth excavation of the relationship between teaching objectives and teaching questions, and the research on the teaching of other disciplines needs to be further expanded. In the future research, we will focus on these limitations and continue to follow up on teachers' ability to ask questions, and make full use of the advantages of generative AI technology to assist in classroom teaching questioning.

In addition, the article uses generative AI to code instructional questions with humans with the aim of trying to validate the effectiveness of AI in teaching research assistance through this study. However, there is still a large amount of training to achieve a certain level of accuracy in the use process, and it cannot be used blindly.

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Practical Use of AI-Based Learning Recommendations in Higher Education

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Abstract. A Learning Experience Platform (LXP) is an extension of a Learning Management System (LMS) with the emphasis on personalized learning support and learner motivation. In this paper extensions for the well established LMS Moodle are proposed, that include, beyond others, AI supported learning recommendations and user based feedback mechanisms to transform Moodle to Moodle LXP. The architecture of this Moodle LXP provides plug-ins for three types of recommendations, i.e. content-based and two collaborative recommendations. Furthermore, a separate module was developed that is responsible for data storage and calculation of recommended learning resources. Various metadata such as description, number of clicks of learning resources, rating, and feedback of the resources are used for the calculation. The extensions have been integrated into several lectures for three semesters. The system operates effectively and the students use the plug-ins for learning. To learn more about the quality of the recommendations, a mechanism for evaluating the results of the content-based recommendation using feedback from the lecturers was developed.

Keywords: Recommender component · learning experience platform · recommended resources evaluation

1 Introduction

LMSs are widely deployed in Higher Education Institutions [11]. In practice, the main focus of LMS lies on the presentation of learning materials and administrative functions, such as course announcements [13]. In recent years, a new generation of platforms, called LXPs have been proposed firstly for workplace environments, but subsequently for higher education. LXPs are designed with the emphasis on personalized learning support and learner motivation by a better user experience similar to social media applications. Functions of LXPs comprise the integration of OER (open educational resources), social interaction by rating and recommendation of learning resources, gamification, and micro learning etc. [23].

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For the LMS Moodle, AI supported learning recommendations and user based feedback mechanisms were developed and implemented in order to enhance Moodle to an LXP. The integration of these new features is done in a way that they can be used optionally. Each lecturer can decide for themselves whether they want to incorporate the extensions into their courses. Furthermore, the extensions are designed with a modular approach so that these new plug-ins can be integrated in any Moodle system and used by other universities as well.

2 Related Work

The LXP market is dominated by proprietary LXP vendors, like Degreed, Valamis, Learning Pool, etc. [9], which are mostly dedicated to workplace learning. In higher education LMS have already been widely deployed for many years. However, switching to another platform is not only costly but also involves substantial amount of effort. Especially in Germany, the LMS market share of Moodle, an open source platform, is over 50% [12] and thus, the widespread use of Moodle leads to the question, whether it is possible to enhance Moodle with the functions of an LXP. Extending Moodle with the features of an LXP has the advantage of a smooth transition to new features and gives lecturers the flexibility to rely on the classical LMS and provide high-quality and comprehensive learning support to their students.

Learners experience difficulties finding learning resources similar to what they are currently learning, due to the large number of learning resources in online learning platforms [25]. However, AI-driven learning recommendations, a fundamental feature of LXPs, present a solution to this issue by finding learning resources relevant to their competencies. A distinction can be made between content-based, collaborative and hybrid e-learning recommendations [1]. Content-based recommendations rely on textual information that describes the content of learning resources, e.g. tags, keywords or short descriptions of the content. Collaborative recommendations, on the other hand, are based on the behavior of learners. The system generates recommendations based on an assessment of the learners' past behavior, represented by ratings, review comments, and the selected resources [19]. Hybrid recommendations combine collaborative and content-based recommendations and may also integrate special knowledge about the learners, e.g. their personal interests and preferences [3].

The literature extensively covers proposal, documentation and evaluation of recommendation system for e-learning content. However, their integration into typical e-learning environments or LMS are rare. An exception is MoodleRec, a recommendation system integrated as a Moodle plug-in [8]. The plug-In recommends external content, otherwise called learning objects to the lecturers. Conversely, the approach described in this paper recommends internal resources to the learners.

3 Design and Implementation

To enhance Moodle with functions of an LXP, the architecture shown in Fig. 1 was designed. The LMS Server (Moodle) is extended by recommendation plug-ins which present the learning recommendations to students. A separate recommender component server is responsible for the computationally intensive AI-based operations. The separation of servers can reduce processing load on the LMS server and keep the server working separately without dependency. The recommender component server consists of a web service module and recommender modules. The detailed explanation of the individual components can be found in the following sections.

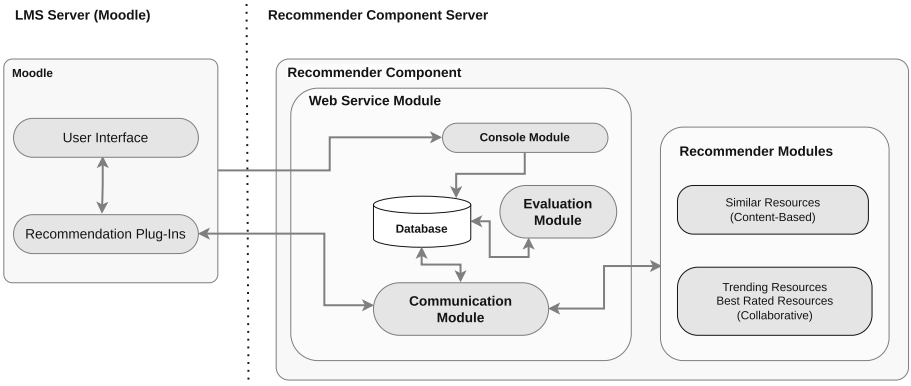

Fig. 1. General Architecture of the Moodle LXP with Recommender Modules.

3.1 Moodle Recommendation Plug-Ins

Many OERs are publicly available from many different OER pools such as LinkedIn learning videos, Youtube videos, interactive exercises, etc. They can be used as additional learning resources by the lecturers. However, with the increasing amount of learning resources in a single course, students face the challenge of finding those learning resources that align best with their current learning needs. LXPs solve this problem by offering recommendations of learning resources. Moodle as an LMS lacks in recommending various learning resources to the learners. Therefore, to extend Moodle and to add the feature of recommending learning resources to learners, three plug-ins were developed.

Rating Plug-In. Moodle offers a rating system for forum topics, database items, and glossary items, which could not be used for the rating of learning resources. Hence, a new Moodle plug-in was developed that allows users to rate

various learning resources using a one-to-five-star system. Additionally, learners can provide brief reviews for resources, such as "This book covers the topic extensively and is easy to understand". The plug-in calculates and displays the average ratings of each resource anonymously. Learners can benefit from accessing resources rated by others, fostering a collaborative learning environment.

Similar Items Plug-In. The plug-in displays similar learning resources to the resource that the learner has selected. It communicates with the recommender component server through the communication module to fetch a list of learning resources which are similar to the selected one. Additional information on the process can be found in the subsequent section.

Trending Items Plug-In. This plug-in lists the current popular resources by utilizing a trend calculation algorithm detailed in the subsequent section. It identifies and lists the most popular learning resources, primarily by tracking learner clicks on specific resources.

All three plug-ins are independent and are implemented separately. Each plug-in can be added to a course and is presented in a separate block to support modularity.

3.2 Recommender Modules

In the first step of the project, three different recommender modules were designed and implemented. They are trending learning resources, similar learning resources, and the best rated resources.

Trending Learning Resources. Identifying learning materials that are gaining popularity among learners helps them in discovering suitable resources for their needs. These kinds of systems have been explored for a long period, yielding insights into their effectiveness, constraints, and benefits. Numerous technologies and algorithms have been explored and suggested for real-world implementation [10].

There are various methods which can be used to calculate trending items such as Slope analysis (Mann-Kendall and Sen's slope analysis), z-score, and chi squared tests [15]. Nevertheless, at present, z-score is selected to calculate the trending items. Currently, the sample size taken is fourteen, though it is flexible and is subject to change in the future.

The z-score is particularly helpful since it calculates the probability of a score falling within a normal distribution and allowing to compare two scores from different distributions. The z-score accomplishes this by transforming (or standardizing) scores from a normal distribution to z-scores, resulting in a standard normal distribution [22].

Supposing that the number of clicks received follows normal distribution having mean value (μ) and standard deviation (σ), the time series data (x_i) can be standardized by:

$$z = \frac{x_i - \mu}{\sigma} \tag{1}$$

The standardized z will follow standard normal distribution.

Values of z can be used to identify trends in the distribution. The higher the positive z value, the greater the popularity of the resource. Conversely, if the z-score is negative, it indicates that the resource was once popular but is now declining in popularity.

The number of clicks of a particular learning resource is recorded for a defined period. This period is called pane, which is currently set as a day. There are 14 panes which constitute a window. The number of panes is configurable. The trending items plug-in records the number of clicks of the resource for the duration of the window, effectively recording a time series data. The window slides daily, removing the last pane and adding a new pane for the current day, thereby making it a sliding window.

The window is divided into two parts. The first part is termed as history panes which is used to calculate mean and standard deviation of the time series. In the second part, termed as current panes, z-scores are calculated for each pane, and averaged. This score is compared with the other learning resources and those resources with the highest score are listed in the trending items plug-in.

Similar Learning Resources. The similarity between several learning resources can be calculated in the context of content-based recommendations by using the metadata related to these resources. The metadata could include descriptive text, tags, and keywords. Different methods are used in content-based recommendations to compute the similarity of the metadata of learning resources. One such method is Term Frequency - Inverse Document Frequency (TF-IDF) [5,7,26,28]. TF-IDF represents a text processing method utilized in machine learning approaches for Natural Language Processing (NLP), a subset of Artificial Intelligence (AI) [5,7,16]. The TF-IDF features are extracted and cosine similarity using the sklearn Python API libraries is computed [20].

The recommendation algorithm uses inputs such as course ID, resource ID, resource name, resource type, resource description, and other metadata from Moodle. It then analyzes these to estimate the similarity of learning resources within a course.

The following steps are performed to process the textual descriptions of learning resources. Initially, the descriptions are divided into individual words, a process known as tokenization. Subsequently, the resulting text undergoes analysis with a NLP tool called a lemmatizer, which identifies the lemma or base form of each word. Stopwords, which are common words in a language that often have no significant meaning or do not contribute much to the analysis, are excluded from further analysis. The resulting text is then used to compute the TF-IDF scores of the learning resources represented by vectors in a high-dimensional space. The vectors serve as input for the cosine similarity algorithm to calculate the similarity between learning resources.

The AI-based recommender module, finally, sends the computed result in JSON format (JavaScript Object Notation) to the recommender component's web service module, which is then stored in the database of the module.

Best Rated Learning Resources. Best rated learning resources typically refers to those resources that have received the highest level of emotional impact and are likely to recommend to others [2,21]. These resources are often considered to be high quality and effective resources for the learners. Such resources can have various benefits, such as access to high quality learning materials to make learning more effective and to contribute to successful learning experiences. The recommender module calculates which resources are rated highly by the cohorts and recommends these resources.

3.3 Web Service Module

The web service module is the essential technical component of the project. It is developed using the Symfony framework. Symfony is a web framework which aims to speed up the development and maintenance of web applications, with a set of reusable components [24]. The Symfony framework was chosen for the web service module because of its modularity and proven track record. In the beginning basic Symfony components were used and later other, more advanced components were integrated as the project grew.

The web service module has varied functions such as receiving data and instructions from Moodle and Moodle plug-ins, storing them in the internal database and executing tasks, such as synchronizing courses and learning resources information from Moodle, and re-calculating the TF-IDF similarity matrix.

The web service has three different sub-modules namely, evaluation, communication and console. The main function of the evaluation sub-module is to display similar resources of each learning resource calculated by algorithms and to offer the opportunity to evaluate the results to the lecturers. The communication sub-module provides bi-directional communication and data transfer between Moodle and the recommender component using REST (Representational State Transfer) endpoints. It also provides bi-directional communication between itself and the recommender modules. The third sub-module provides interfaces between linux console commands and the web service module. These commands perform various tasks such as synchronizing and updating data from Moodle and calculating the TF-IDF similarity matrix.

The MySQL database serves as the storage backend for the web service module. There are two fields in the database namely, Scores and Evaluation which hold significance. The Scores field stores the result of the calculation of similarity between two learning resources. Alternatively, the evaluation field stores the result of assessment of similarity between various learning resources provided by the lecturers.

4 Results and Discussion

Over the course of three semesters, the project team has added in total eight courses with the described recommendation plug-ins. Students have positively accepted the new plug-ins and have rated and provided feedback on the learning resources constructively. These comments give lecturers an insight into learning resources that are helpful for students and also list where problems occur, which, in many cases, can be easily solved by the lecturers.

Figure 2 shows the user interaction to the courses and the list of resources chosen by the user. When a learner selects a learning resource (by clicking on it or by moving the mouse pointer over it), the learner will now see a list of resources relevant to the chosen resource in the similar items block plug-in. Learners will also see a list of trending items and a list of top rated items in the block plug-ins.

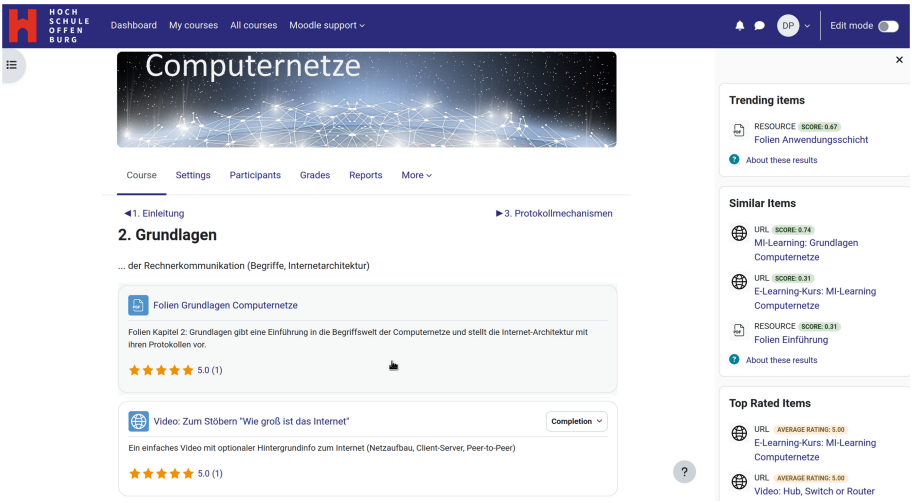

Fig. 2. The Similar Items, Trending Items and Top-Rated Plug-Ins.

In addition, the lecturer provides a description for each learning resource to give students a brief information about the learning resource that they will learn. These descriptions are also utilized as input for the content-based recommendation algorithm to find similarities between learning resources. The result of the content-based recommendation was evaluated by using the Jaccard coefficient [27].

The Jaccard Index, also known as the Jaccard similarity coefficient [4], is used to measure the overall accuracy of recommendations by comparing the evaluation results from lecturers with the calculation results using TF-IDF and cosine similarity. The Jaccard index is calculated as the size of the intersection of the set EL (evaluation results of similarity conducted by lecturers) and set EA (the set of scores computed using TF-IDF and cosine similarity) divided by

the size of the union of these sets. The Jaccard Index is calculated using the following equation and ranges from 0% to 100%.

$$JaccardIndex = \frac{|E_L \cap E_A|}{|E_L \cup E_A|} \times 100\% \tag{2}$$

Two courses were evaluated which are Software Engineering and Computer Networks, each of which has 79 and 40 learning resources, respectively. The higher the similarity value between the learning resources, the better the result that the recommendation system provides for the students. The average of similarity values between 40 learning resources in the Computer Networks course was 82.58%. While, the Software Engineering course which has 78 learning resources was 72.50%.

Currently, these outcomes seem to be suitable for the specific scenario. To gain a deeper understanding of the evaluation findings and make well-informed choices, it is intended to incorporate these results with those of other algorithms that will be implemented in the upcoming phase of the project.

5 Future Work

The next steps in our project include a hybrid recommendation component that combines content-based recommendations with ratings and trending topics to create personalized recommendations. This type of recommendation is intended to improve the quality, coverage, diversity, and adaptability of the recommendation system, so that the system can contribute to an effective learning experience. We will integrate further Artificial Intelligence algorithms for recommendations. Similarly, evaluation from the perspective of users who use this recommendation system will be conducted by using the ResQue (Recommender systems' Quality of user experience) framework [17].

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Exploiting Collaboration to Engage High-School Learners in Experiencing Sensors and Actuators

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Abstract. Due to the rapid smartification of cities and houses we have experienced over the years, (future) citizens are urged to familiarize themselves with sensors and actuators as elementary components of the Internet-of-Things (IoT) paradigm. Learning experiences must pivot towards engaging mechanisms and electronic kits to empower learners to take on the active role of designers. This paper investigates the possibility of engaging 230 high-school learners by enabling them explore professional technologies of STMicroelectronics, relying on hands-on activities, and verifying the impact of working in collaboration. To do so, the protocol distinguishes collaborative and individual phases, and participants auto-assessed the perceived level of engagement in each phase by an anonymous post-questionnaire administered immediately after the in-person workshop. Our findings reveal statistically significant differences, indicating a strong preference among participants for collaborative work-a trend that aligns with existing literature. This study sheds light on the positive effect of collaborative learning approaches in engaging young minds in the realm of emerging technologies.

Keywords: Sensor · Actuator · Hands-on · Electronic · Making · Programming · Engagement · High-school learners · Collaboration

1 Introduction

1.1 Background and Motivation

We are witnessing a rapid smartification of our homes and cities due to the proliferation of the Internet-of-Things (IoT) paradigm. IoT is an ecosystem of

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devices equipped with sensors, computing capabilities, and networking technologies that create an autonomous environment where components work collaboratively. While sensors can be assimilated into human senses to perceive external inputs, actuators are analogous to human limbs, such as legs and arms, to react accordingly to the registered input and transform signals into visible evidence [20]. For example, smart curtains can be provided with light sensors and react by opening or closing themselves according to the external light.

Every day, we interact with an increasing number of digital and autonomous devices that replace traditional human interactions. This shift necessitates that all citizens become acquainted with systems based on IoT, robots, and smart things with sensors and actuators, mastering the mechanisms to provide clear input signals and interpret responses accurately. Our primary focus is on young learners up to 18 years old, who should be not just passive recipients of IoT technologies, but potential designers and innovators. They represent the future generations who must be made aware of these technologies, equipped to effectively navigate them, and empowered to control and predict their behavior.

In recent years, researchers and educators have proposed a range of initiatives [3,13,15,16,22] that focus not just on introducing young learners to the world of sensors and actuators but investigate how to empower them to become designers of these technologies. By adopting a white-box approach, learners can transition from passive users to active creators, engaging in the design process from conceptualization to prototyping. Engaging young learners throughout the entire design process demands a significant investment of time [14], but is rewarding as it enriches their comprehension [12], fosters their creativity, and instills a sense of ownership in the technologies they use. These initiatives often rely on in-person collaboration [7,10,11] and primarily focus on practical hardware-centered projects [1]. Hand-on activities require a set of sensors and actuators for prototype development and often rely on Arduino, Raspberry Pi, and Micro-bit because of their broad community backing, cost-effectiveness, accessibility of open-source code, and plentiful sample projects [1].

1.2 Contribution and Organization

This paper discusses a workshop organized by STMicroelectronics for high school scholars, which took place as an in-person activity. During the workshop learners explore sensors and actuators in the direction of demystifying the IoT paradigm and actively learning how individual components must be connected to communicate effectively. While STMicroelectronics primarily focuses on industry applications, this workshop is part of a broader initiative called the "STEM your way" program[1] which aims to let educators, scholars, and enthusiasts familiarize themselves with STMicroelectronics components. The motivation behind this initiative stems from the company's commitment to sustainability and the desire to allow learners to experience in the first person components widely used across various industries, ranging from electric vehicles to aerospace. Through

[1] STEM your way: www.st.com/STEMyourway.

this program, participants not only gain exposure to future-relevant topics but also acquire practical skills applicable to their future job careers.

The workshop reported in this paper was held in February 2024 involving 230 high-school learners and teachers from the Campania region. Participants were tasked with working with the Nucleo board, exploring different sensors and actuators that can be embedded with the board to develop a prototype which assembles individual components in a manner that addresses a simplified yet real-world challenge. The workshop alternates individual and collaborative stages. Participants' engagement was assessed using a standard self-report questionnaire administered online. The aim is to verify the role of collaboration in perceived engagement by comparing the level of engagement reached during individual commitment with that achieved during collective collaboration. According to the results, participants are highly engaged during the collaborative hands-on, coherently with last year's edition of the same activity [4]. Also, they preferred to work collaboratively, in accordance with the literature.

The paper is structured as follows. Section 2 overviews similar initiatives to enable learners up to 18 years old to experience sensors and actuators. Section 3 overviews the structure of the workshop in terms of participants and settings, the performed protocol, and the data-gathering approach. Section 4 reports results which are discussed in Sect. 5.1. The paper concludes by outlining the limitations of the study and future directions.

2 Related Work

There are different approaches and initiatives to enable learners younger than 18 years old to explore sensors and actuators, and develop smart solutions with them, e.g., smart things or IoT.

Some approaches utilize digital and making tools across the entire design process, so that partitcipants can acquire programming skills by building programs starting from the initially conceived ideas. Examples are SNaP [9] and DigiSNaP [21] that focus on the smart object design via micro:bit boards. Similarly, RapidIoT which enables IoT application prototyping with Arduino boards. Another approach use the IoTgo toolkit that is an adaptable toolkit for human-centred engineering with micro-electronics. It guides young learner from exploring the workings of sensors, actuators, and wireless communication to the development of prototypes with them. It also encourages reflection in the design process [6,7]. Another example is the GAIA education lab kit which aims to teach students about basic energy consumption aspects using a hands-on approach, in which they get to use IoT components and electronics [18].

Other approaches take a more electronic-centric approach and offer tools to assist learners in developing and prototyping with electronic components. Recognizing that constructing electronic circuits can be challenging besides error-prone for novice users, these approaches strive to provide support and mitigate potential issues. An example of such an approach involves exploring the use of Augmented Reality (AR) technology to tackle the challenges associated with

novice users building electronic circuits [2]. Similarly, [19] adopt an AR software to combine real-time IoT data from a sensing infrastructure inside a fleet of school buildings as companions to a set of educational lab activities aimed at exploring IoT components, sensors, and actuators and promoting energy awareness in a STEM context. On the same trend, many other initiatives are based on problem-based learning in real-life scenarios that have the potential to spur learners' engagement [17]. In this direction, Machet and Lowe propose educational activities challenging participants to use IoT-enabled wireless sensors to share laboratory activities based on real-world sensed data [15]. Similarly, The EDUCABOT3D project focuses on environmental awareness while letting participants interact with sensors and actuators via Arduino [22].

Besides focusing on real-world scenarios, some initiative also exploit collaboration. For instance, the AYO! project engage high-school learners in ideating, programming, and prototyping air quality monitoring systems by experiencing sensors and actuators [3]. It is based on a familiarization stage to introduce participants to Python and the Raspberry Pi board, followed by a collaborative implementation stage, and a public presentation of the achieved outcome. Similarly to the workshop presented in this paper, participants relied on already assembled components, that should be correctly linked relying on tutors' support, if needed, and had the possibility to customize an initial version of publicly available code. Learners had the possibility to acquire technical skills, useful for their labor world, and increase soft skills mainly relying on collaboration. Similarly, the GREEN project challenges participants to design educational smart city keeping an eye of road signs able to contextually deal with traditional and autonomous vehicles, while experiencing air quality sensors [11]. It leverages on the STMicroelectronics components wrapped into Roobopoli [5], an educational smart city where high-school learners have to program autonomous vehicles and smart cities, and air sensors designed and developed by FBK. Participants first familiarize themselves with programming and sensors, then have to solve programming challenges collaboratively, and finally publicly present their outcome. Participants were highly engaged and successfully exploited collaboration. On this path, the workshop in this article explores the role of collaboration in programming sensors and actuators and assembling them in a prototype to address a simplified but real-world challenge.

3 Workshop Design

3.1 Participants and Setting

A total of 230 participants joined the workshops, belonging to 27 high-school institutes in the Campania region, 40 (17%) females. Participants' ages range between 15 and 19, mainly from the 11th and 12th grades. Schools who are interested in participating in free workshops offered by STMicroelectronics were invited to nominate up to 8 students to attend. Attendance is mandatory for selected students. All participating schools received four boards each that became their property after the activity. The workshop was organized in two turns, each

turn involving up to 15 institutes per time. The workshops took place in February 2024 as in-person activity at the STMicroelectronics headquarters in Arzano.

3.2 Goal of the Workshop and Protocol

The overall purpose of the workshops is to empower young learners to delve into sensors and actuators and program their behaviors accordingly to meet specific challenges. The ultimate challenge entails participants designing and programming a motor, symbolizing a vehicle, which must respond accurately (e.g., stopping) upon detecting an obstacle. The same protocol is repeated for each group, per two consecutive days. Each workshop lasts 7 h plus a break for lunch, starting from 9 am to half past 4 pm. While the introductory phase takes 4.5 h, the hands-on phase takes 2.5 h. The break splits the two phases. More details about the protocol follow.

- **Step-by-step warm-up.** This phase represents the core part of the introductory stage and aims to let participants experience STMicroelectronics on their own, familiarize themselves with protocols to let components communicate, and the programming environment to program STMicroelectronics boards. An expert in the field working at STMicroelectronics behaved as a moderator. They introduced all the concepts as a theoretical lesson, always sharing the screen, welcoming questions, and providing participants with any required clarification. First, the moderator overviewed ready-to-use exercises accessible via a public repository and comments on the code by interacting with participants. Then, all participants were invited to assemble the circuit, supported by their professors or STMicroelectronics tutors if needed, and test the code independently. This phase is designed to be an individual activity, requiring participants to share some material, e.g., a board for the couple. Participants explored the code, set up the circuit, and tested the program. Each exercise focused on sensors and actuators as isolated components, covering LED, joystick, infrared, and motor, as visible in Fig. 1. Learners were encouraged to apply changes to the code and observe the reaction of the prototype. All the participants were free to seek support from the moderators, tutors, and their peers, if needed.
- **Collaborative hands-on.** Each school was invited to collaboratively solve a simplified but real-world challenge that combines sensors and actuators explored in isolation during the step-by-step warm-up. They were required to implement a motor, simulating a vehicle that moves continuously, stopping at any time the infrared detects an obstacle, simulated by hands put close to the infrared. An LED should report the status of the motor, distinguishing between stop and in movement. This stage requires participants to design, program, and prototype the prototype able to solve the challenge, working in groups and assisted by tutors if needed. Participants could modify codes explored during the step-by-step warm-up without starting from scratch.

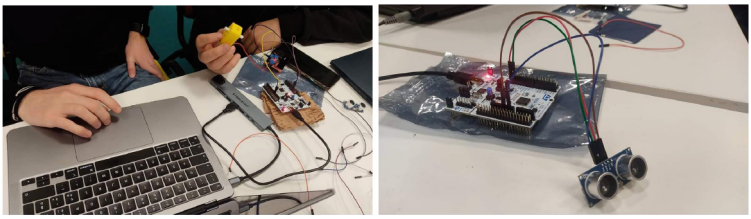

Fig. 1. Material and setting experienced by high-school learners in the warm up phase.

3.3 Data Collection Approaches

A self-report questionnaire is administered at the end of the workshop via Google Form to assess learners' engagement levels. All the participants have been informed that the questionnaire is anonymous and that the collected results will be used in scientific contributions. While observers have been authorized by an STMicroelectronics responsible to join the activity, the school professors dealt with parental authorization. The questionnaire asks participants to rate the engagement obtained during the step-by-step warm-up and the collaborative hands-on. The questionnaire adapted inquires grounded on the literature and similar to those previously experienced by the same research group, such as [11,21], customizing adjectives used to characterize the designed activity. For each phase, the questionnaire asks participants to rate to what extent they consider that stage *stimulating*, *clearly explained*, *easy*, *trivial*, and *engaging*. Ratings were given with a 5-point Likert scale, ranging from not at all (1) to very much (5). Additionally, two open-ended questions asked participants about the aspects of the activity they liked most and those they liked least.

4 Results

154 participants out of 230 (20% females) joined the questionnaire. Participants' scores are analyzed via Python using the scipy.stats package to identify if scores reported to the individual warm-up are statistically different from those used to rate the collaborative hands-on.

As the Shapiro-Wilk test return that the variables are not normally distributed, we used the non-parametric Wilcoxon-Mann-Whitney test to verify if and to what extent scores used to grade different aspects concerning the warm-up and the hands-on stages are statistically different in the two workshop stages. Since the warm-up phase was designed as an individual activity, while the hands-on phase involves collaboration, this analysis seeks to verify whether collaboration influences engagement. Table 1 reports descriptive statistics in terms of mean value, standard deviation (st. dev.), and quartiles. Table 2 reports results of the Wilcoxon-Mann-Whitney test in terms of Z-score and p-value obtained by using two-sided as alternative hypotheses. The results of the Wilcoxon signed-rank test indicate that collaboration leads to a statistically significant increase

Table 1. Descriptive statistics

	N	Mean	St.Dev.	Min	Percentiles 25th	Median	75th	Max
Stimulating								
Warm up	154	3.61	1.07	1.00	3.00	4.00	4.00	5.00
Hands-on	154	3.79	1.18	1.00	3.00	4.00	5.00	5.00
Clearly presented								
Warm up	154	3.46	1.23	1.00	3.00	4.00	4.00	5.00
Hands-on	154	3.53	1.10	1.00	3.00	4.00	4.00	5.00
Easy								
Warm up	154	3.49	1.05	1.00	3.00	4.00	4.00	5.00
Hands-on	154	3.33	1.09	1.00	3.00	3.00	4.00	5.00
Trivial								
Warm up	154	3.46	1.23	1.00	1.00	2.00	3.00	5.00
Hands-on	154	2.46	1.25	1.00	1.00	2.00	3.00	5.00
Engaging								
Warm up	154	3.52	1.12	1.00	3.00	4.00	4.00	5.00
Hands-on	154	3.77	1.15	1.00	3.00	4.00	5.00	5.00

Table 2. Wilcoxon Signed Ranks Test statistics

	Z	Asympt. Sig. (2-tailed)	
warm up-hands-on *stimulating*	902.5	0.013	*
warm up-hands-on *clarity*	884.0	0.37	
warm up-hands-on *easiness*	988.5	0.06	
warm up-hands-on *trivial*	782.0	0.41	
warm up-hands-on *engaging*	1,089.5	0.002	**

in stimulation (Z = 902.5, p = 0.013) and engagement (Z = 1,089.5, p = 0.002) among participants when programming sensors and actuators, graphically represented in Fig. 2. However, collaboration does not elicit statistically significant changes in improving the clarity of the assigned tasks (Z = 884.0, p = 0.37), the perceived easiness (Z = 988.5, p = 0.06) and in making the task trivial (Z = 782.0, p = 0.41).

Results from the open-ended questions are as follows and are graphically represented in Figs. 3 and 4. In response to the question "which activity did you like most?", out of 154 total respondents, 33 either provided irrelevant answers or did not respond at all. Overall, 121 answers were considered. 3% of the respondents reported enjoying the entire experience without specifying any activity. 17% of the respondents preferred the practical activity done in the warm-up phase, which involved connecting components and assembling circuits. 21% of

the respondents, declared that they liked the hands-on phase activities the most. Some respondents specified specific hands-on activities they preferred most, such as programming (13%), using a joystick (7%), working with LED lights (2%), and mechanical components (8%). 9% explicitly declared that the most engaging part was when they worked in group and collaborate.

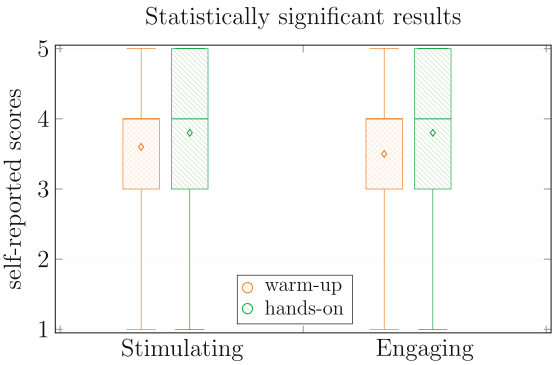

Fig. 2. Box plots focusing on statistically significant results according to the Wilcoxon Signed Ranks Test.

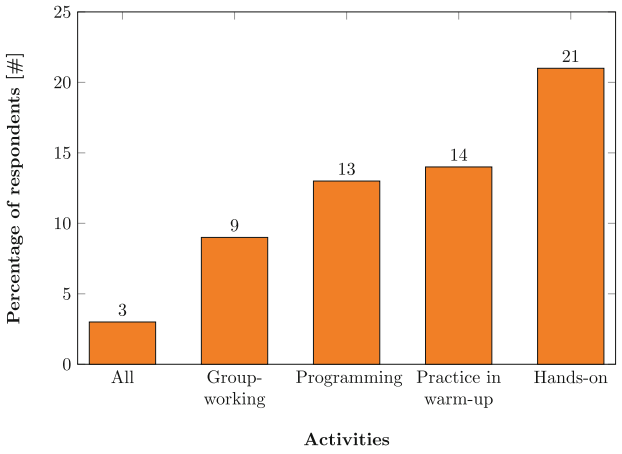

Fig. 3. Preferred activities reported by 121 questionnaire respondents.

Regarding the activities that were least liked, we considered 107 responses. Out of all the respondents, 36% stated that they did not dislike any activity, while 5% said that they did not like the entire experience. The warm-up phase

was the least liked activity, as indicated by 31% of the respondents. Among them, 21% liked least the theoretical part of the warm-up phase, while 10% liked least some practical aspect of the warm-up activity like connecting wires, or embedding circuits with specific sensors. As part of the hands-on phase, 7% of respondents reported disliking the programming activity.

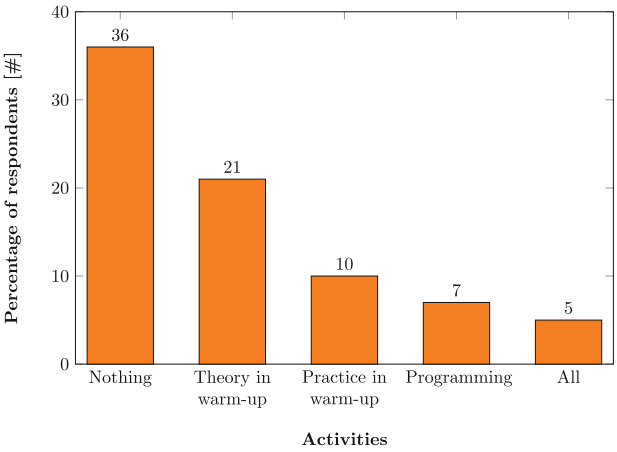

Fig. 4. Least preferred activities reported by 107 questionnaire respondents.

5 Discussion and Conclusion

This paper reports educational workshops involving 230 high school learners. The workshop, proposed by STMicroelectronics, aims to enable high-school learners to explore STMicroelectronics components via a step-by-step warm-up, mainly conceived as an individual activity, and collaborative hands-on. This paper presents the findings of a self-report anonymous online questionnaire featuring both closed-format and open-format questions. Participants were asked to rate each phase on a 5-point Likert scale, indicating the extent to which they found it stimulating, clearly presented, easy, trivial, and engaging.

5.1 Discussion of the Workshop Results

By comparing the scores reported for rating the warm-up and hands-on phases, participants implicitly assess their preference for working individually or collaboratively, which is the key difference between the two stages. Generally, the hands-on phase is preferred over the warm-up phase regarding stimulation and engagement. However, the hands-on activities are not considered easy overall.

The Wilcoxon signed-rank test showed that collaboration elicited a statistically significant change in stimulating and engaging participants when programming sensors and actuators.

Looking at the mean values registered in the two different stages and used to rate the level of stimulation, the clarity in presenting the task, the perceived easiness and triviality, and the overall engagement, the collaborative hands-on is generally attached to a higher value with respect to the warm-up stage, but not in the easiness and triviality aspect. The median confirms a similar trend, which is the same value in both stages, but when we focus on the perceived ease. Surprisingly, the hands-on is perceived as harder than the warm-up stage. This can be justified by the fact that during the warm-up stage, components are explored individually, while the collaborative hands-on required is to design and implement a complex circuit based on three sensors and an actuator. Hence, the hands-on task is objectively more complex than those in the warm-up stage. Results are coherent with previous editions of the same workshop, where programming STMicroelectronics boards is perceived as funny, but not easy [4].

The qualitative results regarding the most liked and disliked activities confirmed the aforementioned trend. Specifically, the activities participants liked the most were those carried out during the hands-on phase, even though they were perceived as the most challenging. Additionally, a notable percentage of respondents (8%) explicitly stated that working in groups and collaborating with their peers was the most engaging activity of the workshop. While collaboration requires dealing with divergent ideas and making cooperation work [11], it has the potential to let participants co-design the solution and join forces to solve tasks together, as confirmed in similar initiatives levering on collaboration [8,11].

The theoretical lesson of the warm-up phase was reported as the least engaging activity. As for the practical aspects of the warm-up, which encompass activities such as connecting wires, building electronic circuits, and exploring the functioning of specific sensors and actuators, the responses were divided. Approximately 13% of respondents found the practical aspects of the warm-up engaging, while another 13% did not enjoy them. This aligns with existing literature, which indicates that prototyping with electronic circuits and activities such as connecting wires are error prone and challenging for novice users [2].

5.2 Limitations and Future Work

While 230 high-school learners joined that activity, 67% filled in the questionnaire. Further effort should be invested in engaging them more in the conclusive questionnaire, such as minimizing the required time needed to compile it or leading it with class-level tasks, such as via Mentimeter.

Collaboration was likely an influencing factor in the result. However, we acknowledge the presence of some confounding issues. The warm-up started with a theory lecture, which is objectively less fun than working on a problem. The warm-up might have been perceived as more challenging as this was new to most of the learners. By the time they reached the collaborative hands-on, they

would have been more familiar with the materials. The first phase lasted 4.5 h before lunch, and the second phase lasted 2.5 h after lunch. The third factor in itself may have contributed to differential reactions between the phases. Those observations go in the direction of considering results almost trivial in that the hands-on is more engaging than the warm-up stage by design. On the other hand, since the hands-on session followed mid-day work, participants might have been physiologically more tired. Furthermore, as during the warm-up, they spent most of the time fixing issues, they might become discouraged by the hands-on. Moreover, if any did not solved some of the challenges raised during the warm-up, they were not in the favorable condition to join the hands-on. Finally, while during the warm-up novel participants might have been engaged by the curiosity of the novelty, it might have been mitigated during the hands-on stage. In this direction, results are even more encouraging than they have been presented.

This contribution focus on the engagement dimension, mainly via a quantitative questionnaire. Self-report questionnaires should be compared with qualitative results reported by external observers. We will explore it in the future. Moreover, once verified that the participants considered the proposed workshop engaging, with a mean score of 3.77 out of five and a standard deviation of 1.15, we need to triangulate those results with learning outcomes.

While this contribution focuses on a specific commercial brand of electronic components, the protocol can be easily adapted to any other brand, which should be carefully selected according to the moderators' knowledge, the available equipment at the school, and any requirement imposed by the workshop settings. Similar initiatives should not be targeted for technicians, but might be considered as promising opportunities to democratize the access to technical concepts that are even and even more common in our daily tasks due to the unprecedented spread of concepts connected to internet of things and smart cities.

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Identity Shield: Cultivating Privacy Awareness Through AR for Young Adults

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Abstract. This research focuses on young adults' vulnerability to cyber threats and develops an *Identity Shield* AR application to enhance privacy awareness through challenges. The game consists of three main challenges: Facebook Auth, Email Phishing, and Malware Hunter. These challenges focus on four main cyber attacks: identity theft, brute force, phishing, and malware. To test whether the awareness of the participants increased, an experiment was undertaken on 30 university students. Results indicate a significant impact on participants' awareness ($p < 0.05$), though limitations include a small sample size and basic AR features. Future work entails expanding AR capabilities, broadening the audience, and improving game mechanics and UI for enhanced educational impact.

Keywords: Privacy Awareness · AR · Security

1 Introduction

The increasing digital interconnectedness of our lives has raised concerns about online privacy and personal data protection. With the proliferation of online social networks, the risk of cyber threats such as identity theft, social engineering, and malware attacks has escalated [15,24]. Information shared on these platforms becomes vulnerable, exposing users to data leakage risks [22]. Young adults aged 17 to 30 are particularly susceptible to these threats due to their heavy use of social networks [1]. Research on AR for privacy awareness includes various mechanisms such as browser plugins for data leakage detection [12], security monitoring platforms [27], and recommendation systems [31]. Privacy concerns are prevalent across multiple domains, with studies highlighting data accessibility issues on social platforms like Facebook [18] and privacy control shortcomings on major platforms [11]. Cyber hygiene methods have been used to enhance employee privacy awareness in healthcare [3], and surveys show persistent privacy concerns among technical users [5]. Mobile app security risks have been

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analyzed through user reviews [28], with calls for secure development practices [9]. Research also explores privacy and security in mobile applications [16], and location information disclosure in mobile AR applications [14]. AR and VR technologies have effectively raised awareness in various educational fields, such as an epidemiology video game [13] and VR headsets in military applications [23]. Despite these advancements, there remains a gap in AR applications targeting privacy awareness among young adults, especially in Egypt.

Focusing on Egyptian society, few studies have tested the privacy awareness of young adults. This paper addresses this gap by examining the cybersecurity behaviors of students at the German International University in the New Capital, Egypt. There is a crucial need to understand and mitigate cyber threats among young adults. One proposed solution is raising privacy awareness using Augmented Reality (AR) technologies.

This paper presents the design and development of *Identity Shield*, an AR game built using the Unity Game Engine, aimed at fostering privacy awareness among users. To the best of our knowledge, this is the first AR game designed to raise awareness of the selected attacks. The game incorporates scenario-based simulations, puzzles, role-playing scenarios, and a game score system to engage users in learning about privacy and cybersecurity. The game was developed for Android devices and tested on 30 participants. Surveys before and after gameplay assessed privacy-related concepts, and a t-test analyzed the changes. The study found a significant increase in privacy awareness based on pre-and post-game survey mean scores.

This work is part of the larger Private-Eye project, which examines privacy perception and awareness between Egypt and Germany. Its goal is to improve privacy practices through education on privacy engineering techniques and to bridge cultural gaps. This paper contributes by developing an AR game to raise privacy awareness among young adults and testing its effectiveness in an Egyptian context.

2 Approach

This section outlines the research objectives and theoretical framework of the AR game, emphasizing privacy breaches such as identity theft, social engineering, phishing, brute force, and malware and their importance. Discussions dive into the game's design elements, including scenario-based simulations, puzzles, role-playing scenarios, and a game score system, all aimed at enhancing privacy awareness among users.

2.1 Privacy Attacks and Their Importance

The five critical privacy attacks and the emphasis on the importance of awareness to mitigate these threats to user data privacy, particularly for young adults, are discussed.

1) *Identity Theft:* Involves unauthorized use of personal information for financial gain or fraudulent activities, such as accessing bank accounts or creating fake social media profiles [6].
2) *Social Engineering:* Manipulates individuals by exploiting human behavior and trust to gain unauthorized access or extract sensitive information, including phishing and pretexting [29].
3) *Phishing:* A deceptive practice where attackers send fraudulent communications, often emails, to trick individuals into revealing sensitive information like login credentials or financial details [26].
4) *Brute Force Attacks:* Involves systematic attempts to access user accounts by trying numerous username and password combinations until the correct one is found [30].
5) *Malware:* Malicious software that compromises user systems, steal information, or disrupt normal operations, including viruses, ransomware, and spyware [19].

These privacy breaches are critical for raising awareness among young adults because they are prevalent, directly target user data, and can cause significant harm, including financial loss or reputational damage. By focusing on these specific attacks and providing tangible examples, our initiative aims to equip young adults with the knowledge and skills necessary to effectively safeguard their user data. This strategic focus ensures that privacy awareness is comprehensive and tailored to address the most common threats in the digital landscape.

2.2 Game Design and Mechanics

The game aims to raise privacy awareness among young adults through immersive cyber threat scenarios, utilizing four main design elements: scenario-based simulations, puzzles/challenges, role-playing scenarios, and a game scoring system, each enhancing engagement and learning.

1) *Scenario-based Simulations:* it simulations replicate real-world experiences for learning and problem-solving by mimicking situations individuals may encounter, enhancing engagement in education [21].
2) *Puzzles and Challenges:* Puzzles test knowledge, creativity, and problem-solving, while challenges are tasks that test abilities and provide opportunities for goal achievement in games and learning [2].
3) *Role-playing Scenarios:* it involve participants assuming specific roles within simulated situations to explore interactions and decisions, fostering learning and interpersonal skills [20]. They allow users to understand different attack perspectives. For instance, users may play the role of an attacker to comprehend potential threats or face phishing and malware cases to learn to detect and avoid attacks, enhancing awareness and decision-making skills.
4) *Game Score:* it quantifies a player's performance, achievements, or progress within a game, representing success or proficiency based on specific actions or challenges completed during gameplay [8].

In the phishing challenge, users earn 20 points for correct answers and lose 10 points for incorrect ones. Total email scores are determined by the number of targeted phishing signs, providing a metric for user performance.

3 Identity Shield Overview

This section outlines the implementation of the game, which tests privacy breaches like identity theft, brute force, social engineering, phishing, and malware through three separate challenges, each focusing on different privacy concepts. The game aims to boost privacy awareness through an engaging user experience. It's developed using Unity with Vuforia SDK for AR features. To smoothly integrate Figma designs into Unity, the 'FigmaUnityBridge' plugin[1] is utilized. The game's structure comprises three distinct challenges.

3.1 Facebook Auth Challenge

In the *Facebook Auth* challenge, players scan a Facebook profile to collect publicly shown information, aiming to simulate hacking. This challenge addresses identity theft, brute force, and social engineering, based on findings from [18] regarding participants' privacy awareness on Facebook. After scanning the Facebook profile, the player encounters an OTP page simulating a confirmation code sent to highlight the use of time passwords for secure logins. This aims to raise awareness about security mechanisms. Finally, the player views a loading page displaying the targeted Facebook profile's basic information, followed by the actual feed where they can post content (Fig. 1).

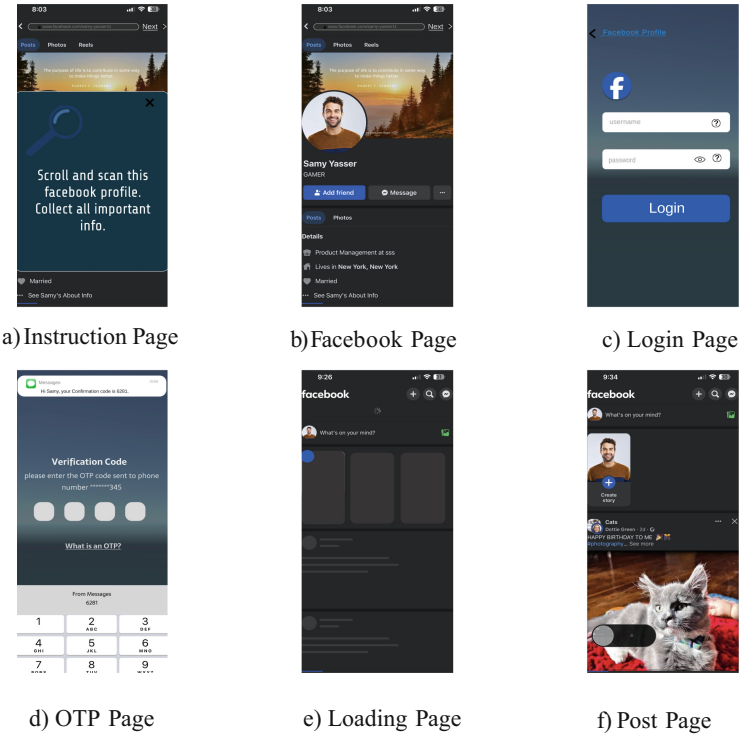
3.2 *Email Phishing* Challenge

The *Email Phishing* challenge educates users on phishing signs by scanning emails and identifying them, with correct selections earning 20 points and incorrect ones resulting in a loss of 10 points. Inspired by "Phishing Wars[2]" by the Cyber Security Agency of Singapore, this challenge aims to teach users about email spam and avoid falling into phishing traps. Given the prevalence of email as a confirmation method for online processes, particularly among young adults [17], this challenge addresses common threats. The scoring system, serving as a motivational tool [10], encourages user engagement and competition, enhancing the challenge's dynamism and effectiveness in improving awareness and detection skills (Fig. 2).

3.3 *Malware Hunter* Challenge

In the *Malware Hunter* challenge, users navigate a block maze to help a character named Dino reach a blueberry while avoiding bombs. Movement is based

[1] https://github.com/simonoliver/UnityFigmaBridge.git.
[2] https://www.csa.gov.sg/programmes/sg-cyber-safe-students/games/phishing-wars/index.html.



a) Instruction Page b) Facebook Page c) Login Page

d) OTP Page e) Loading Page f) Post Page

Fig. 1. *Facebook Auth* Challenge a) Instruction Page b) Facebook Page c) Login Page d) OTP Page e) Loading Page f) Post Page

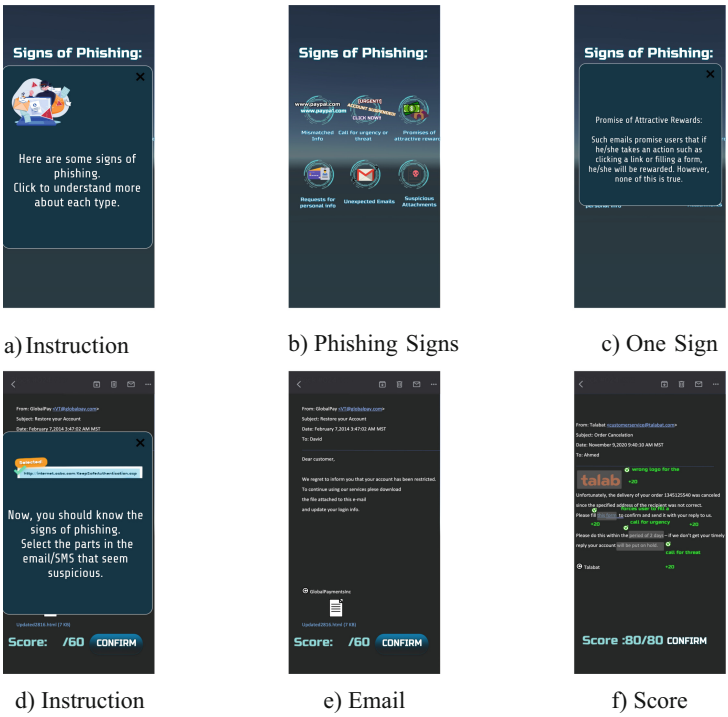
on selecting between two images, one of which contains malware. Choosing the malware-free image allows Dino to progress while selecting the wrong one results in collision with a bomb and moving backward. This setup emphasizes the careful selection of malware-free images to pass the challenge. The selection of malware as the challenge theme is motivated by its prevalence and significance in cybersecurity. By focusing on malware, the challenge aims to raise awareness of associated risks and preventive measures, highlighting its importance within the cybersecurity domain (Fig. 3).

4 Experiment and Results

This section evaluates the game using the G/P/S model [7], detailing the experiment and its results to address the research question.

4.1 G/P/S Model

Following the G/P/S classification model [7] used to identify the key elements of serious games, *Identity Shield* can be summarized as follows. In terms of



a) Instruction b) Phishing Signs c) One Sign

d) Instruction e) Email f) Score

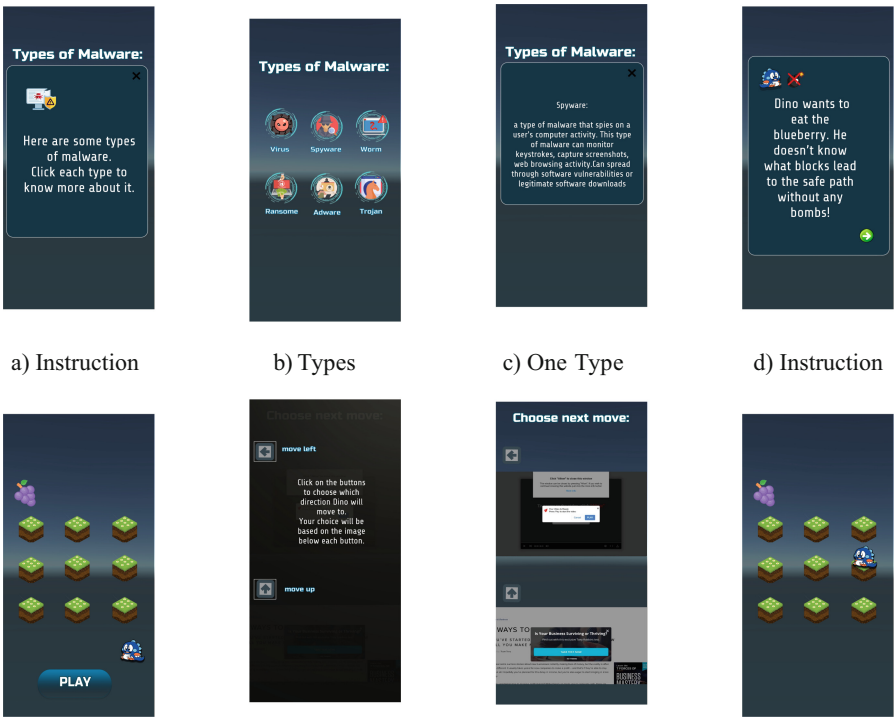
Fig. 2. *Email Phishing* Challenge a) Instruction b) Phishing Signs c) One Sign d) Instruction e) Email f) Score

gameplay, it is a mobile AR game featuring scenario-based simulations, puzzles, challenges, and role-playing scenarios designed to enhance players' critical thinking and problem-solving skills in privacy awareness. These components enrich the overall gameplay experience. The purpose of *Identity Shield* is to educate users, particularly young adults, teens, and students, about privacy concerns and cyber threats using AR technology. It actively engages users and motivates them to understand the importance of privacy awareness.

The scope of the game includes the three challenges, providing a comprehensive learning experience in privacy and cybersecurity. Developed using Unity and Vuforia SDK for AR, the game ensures an informative experience. Overall, the game aligns with the Gameplay/Purpose/Scope model by offering engaging challenges, an educational purpose focused on privacy awareness, and a well-defined scope covering various aspects of cybersecurity.

4.2 Experiment

The experiment involved 30 participants aged 17 to 27, consisting of 18 females and 12 males from various fields such as Computer Science, Business Informatics, Architecture, Business Administration, and Engineering. Informed consent was



a) Instruction b) Types c) One Type d) Instruction

e) Play f) Button Explanation g) Next Move h) Game

Fig. 3. *Malware Hunter* Challenge a) Instruction b) Types c) One Type d) Instruction e) Play f) Button Explanation g) Next Move h) Game

obtained from all participants. The experiment began with a pre-test questionnaire [25], followed by a gameplay session lasting 30 to 40 min where participants played the three challenges, and concluded with post-test and SUS questionnaires [4]. Participants scanned QR codes and virtually clicked the instruction box to play each challenge as shown in Figure. The pre- and post-tests included 13 identical questions (7 true/false and 6 multiple choice) focused on privacy concepts explored in the game and its usability. Example questions include: *"Request for personal info" is a sign of phishing* and *A spyware is a computer program that spies on a user's computer activity.* After the post-test, participants answered 3 additional demographic questions to assess engagement with the AR features. These included a Five-point Likert scale item (*Scanning QR code is engaging*) and two "Yes" or "No" items (*I prefer playing without the AR feature*).

4.3 Results

To determine whether awareness increased among participants, an analysis of both pre-game and post-game testing surveys was conducted. The number of

correct answers increased after playing the game. As a result, the following hypotheses were formed:

- *H0: The mean scores of before and after-game testing are equal.*
- *H1: The mean scores of before and after-game testing are not equal.*

To test the hypothesis, a paired sample t-test was conducted on the pre and post scores of the participants. The α was set to 0.05. The t-statistic is –5.719, and the p-value is 0.000003 which is <0.05. This indicates that the post-game score is higher and that the awareness of the participants has increased. Thus, there is a significant difference between the mean scores of pre-game and post-game testing surveys. In other words, the null hypothesis was rejected, ensuring that the scores were not equal. Table 1 shows the number of participants who answered correctly pre and post the gameplay session.

Table 1. Pre-test and Post-test Results

# of question	Pre-test (# of participants with correct answers)	Post-test (# of participants with correct answers)	Pre-test (% of participants with correct answers)	Post-test (% of participants with correct answers)
1	24/30	28/30	80%	93.33%
2	10/30	14/30	33.33%	46.67%
3	26/30	29/30	86.67%	96.67%
4	8/30	8/30	26.67%	26.67%
5	22/30	30/30	73.33%	100%
6	23/30	27/30	76.67%	90%
7	27/30	29/30	90%	96.67%
8	17/30	23/30	56.67%	76.67%
9	15/30	20/30	50%	66.67%
10	15/30	21/30	50%	70%
11	11/30	24/30	36.67%	80%
12	9/30	24/30	30%	80%
13	17/30	22/30	56.67%	73.33%

Moreover, 63.3% of participants found scanning QR codes engaging and preferred playing with the AR feature. To assess usability, a System Usability Scale (SUS) survey was administered to 26 participants after testing. The average SUS score was 77.1, indicating a high level of acceptability [4].

5 Conclusion and Future Work

The *Identity Shield* app educates young adults about cyber risks through scenario-based simulations, puzzles, challenges, and role-playing, covering identity theft, phishing, and malware. A significant difference in pre- and post-game survey scores (p-value = 0.000003) demonstrates its effectiveness in boosting

privacy awareness. High user engagement with the AR feature and a SUS score of 77.1 confirm its usability. The app serves as a foundation for further research on cyber threats and the integration of advanced AR features. Despite limitations like small sample size and time constraints, future plans include enhancing AR features, broadening the audience, improving the UI, and extending security coverage.

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Exploring an Artificial Intelligence Model Based on Integer Linear Programming for Allocating Students to Industry Placements

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Abstract. Allocating students to industry placements is a necessary task in certain educational contexts, both in Vocational Educational Training and at the University level. This refers to the process of assigning students to specific work placements within companies, where they can gain practical experience related to their field of study. In such scenarios, preferences may play a crucial role in the allocation process. This article proposes a multi-objective integer linear programming model for allocating students to industry placements. The proposal takes into consideration competence vectors, which allow the expression of aspects such as the current competences that a student has developed, the competences a student wishes to develop in a placement, or the competences required by a company for a specific placement. In this paper we study the scalability of the proposed model and what factors may affect the scalability and time performance of the model.

Keywords: Artificial intelligence · education · optimization

1 Introduction

In educational contexts, students undertake placements in companies during their training. This type of training is usually seen as beneficial for both companies and students, as companies do not incur significant costs, and students can gain valuable professional experience for their future [3]. In this regard, there is fairly widespread consensus on the positive effects of these internships, such as improved job prospects, better salaries, or higher satisfaction [8]. Traditionally,

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educational institutions manually assign students to industry placements, which may be time-consuming and cognitively complex for decision-makers. We refer to the problem of assigning students to industry placements or internships as the student-industry placement allocation (SIPA) problem.

The proposal presented in this paper extends preliminary work [10] by evaluating the scalability and performance of a multi-objective integer linear programming (ILP) model for the SIPA problem. The model takes into account different types of preferences associated with students and companies. Unlike other preference models used in similar allocation problems in education, we model preferences as three competence vectors: the current competences that a student has developed, the competences the student would like to develop in a placement, and the competences that are required by a company in a specific placement. Additionally, in this model, we incorporate other factors that are also relevant, such as the distance to the companies and the interdependence of placements. The main objective and contribution of this paper is to understand how the model scales under different scenarios and to understand what factors may affect the most the time performance of the model. The rest of the paper is organized as follows. Section 2 presents related work on assignment problems in educational contexts, with a special focus on works addressing SIPA problems. In Sect. 3, we detail the integer linear programming model under study. Section 4 describes various experiments to validate our proposal. Finally, in Sect. 5, we present some concluding remarks and outline future lines of work.

2 Related Work

In the literature, we can find proposals for assignment problems in educational contexts. For instance, partitioning students into teams [12], assigning a student(s) to a project, as well as assigning a lecturer to supervise a student's project. Although these problems are different from SIPA problems, they share common actors such as students, projects, or lecturers.

One of the problems that has generated significant interest in the scientific community is the student-project allocation problem (SPA), where the goal is to optimally assign students to projects. Typically, these problems use to define students' preferences for various offered projects, with the optimization objective being to maximize student satisfaction based on their preferences. For example, in [5] a genetic algorithm is presented to solve this allocation problem by considering the preferences of students for the projects. Other works introduce additional complexity by involving lecturers who will supervise student mentoring. In these cases, some works such as [6], also introduce preference rankings associated with each available lecturer for mentoring students. The work [9] considers departmental preferences when accepting students, while [2] takes into account the professors' capacity to supervise projects, and [15] aims to balance the workload among different departments. Another variant of these problems considers students' preferences for being mentored by a specific lecturer and the capacity (number of students) of the lecturers [11]. A similar problem is addressed in [13],

where a genetic algorithm is proposed while considering a range of supervision capacities between a minimum and maximum, in addition to student preferences. The proposal of [14] also integrates lecturer preferences for mentoring students.

Other proposals address the group-project allocation problem (GPA). Optimization in these cases involves assigning a group of students to a specific project. Proposals like [4] consider student preferences based on project topics and propose group assignments of up to three students based on these preferences. Moreover, incorporating a group of students in the allocation process leads to considering other requirements. In this regard, works like [7] take into account the skills needed for a project and the skills that a student wants to develop while working on that project, which can be related to preferences.

The proposal addressed in this paper is focused on the SIPA problem. In this type of problem, students are assigned to industry placements, that are offered by different companies. Related to this, in [16] the authors tackle the issue of allocating students to industry placements. In the model proposed in this work, each company can offer one or more placements, and companies interview students to define their preferences. The model aims to optimize company satisfaction but does not take into account student preferences. The work [1] presents a greedy algorithm for assigning students to student placements. In this case, real data is used to assign 127 students to placements offered by companies located in 23 different towns. These placements can be supervised by tutors from the students' university and lecturers from other universities closer to the companies. This work also does not take into account aspects related to preferences. Instead, the optimization problem focuses on minimizing the distance between assigned professors and the internship locations.

While many of the above works have successfully addressed various aspects of student allocation problems, they present some limitations. Preferences have typically been addressed as indications of a student's interest in one or more specific projects or a lecturer's interest in supervising one or more specific students. However, this way of modeling preferences requires defining direct evaluations that can introduce biases, especially when there are a large number of possible assignments. In this regard, our model aims to advance the state of the art in SIPA problems by incorporating preferences through competence vectors. Specifically, we consider the current level of competence development of a student, the competences that a student would like to develop during a company placement, and the competences demanded by a company for a particular placement. Hence, our method aims to optimise both student and company satisfaction, providing a more balanced and realistic solution to the SIPA problem.

3 Integer Linear Programming Model

We start by formalizing the SIPA problem. Let us define the set of students to be allocated by an educational institution as $\mathcal{S} = \{1, \ldots, s\}$. Let $\mathcal{P} = \{1, \ldots, p\}$ denote the placements that the education center has available for students. Please, note that, in this particular model, it is assumed that $p \geq s$ so that

it is possible to assign every single student to an industry placement. This is a common setting in Spanish Vocational Educational Training (VET) programs, where industry placements are compulsory for final-year students. Therefore, the educational institution should offer at least one possibility to each student. In addition to this, $\mathcal{B} = \{1, \ldots, b\}$ represents the set of companies that offer any placement for the educational center. Each placement $j \in \mathcal{P}$ is associated with a given company $k \in \mathcal{B}$. Without losing any generality let us define by $P_k \subset \mathcal{P}$ as the set of placements that are offered by company k. Each company $k \in \mathcal{B}$ has agreed with the educational center a minimum number of students to be allocated q_k and a maximum number of students to take as placement students, Q_k. In practice, some companies offer placements that have interdependencies between them. This is typically the case when these placement positions are bound to be working on different parts of the same project. Thus, if any of these placement positions are not covered, the company's project would not be completed. Let us define $\mathcal{I} = \{1, \ldots, n\}$, with each $d \in \mathcal{I}$ being a set of placement positions that have interdependence. That is, $d \subset \mathcal{P}$.

From the perspective of an academic institution, the problem of allocating students to placements consists of successfully allocating a placement for each student, considering both the preferences of students and the preferences of companies, given the requirements formulated in the previous paragraphs. Following, we show the details of the multi-objective integer linear programming model for tackling this problem based on competences. This multi-objective integer linear programming model is aimed to be used by a single academic institution or a centralized institution to match students to available placements. The main features of this model are: (i) its ability to consider both the students' and the companies' preferences using a multi-objective approach; (ii) the use of competences as a key element when matching students to placements; (iii) the inclusion of distance as a factor for considering the utility reported by an industry placement to a student; (iv) and the ability to model interdependencies between placements.

We employ $\mathcal{C} = \{1, \ldots, c\}$ to refer to the set of competences that are relevant for the assignment of students to placements. Competences are important as they provide a common ontology for businesses and students to describe both the capabilities needed by an industry placement and the students' preferences to develop or use certain capabilities. In addition to this, education and assessment based on competences is nowadays a core part of the European Higher Education Area. Hence, education institutions can also provide a factual and multifaceted evaluation of the level of acquisition of competences by students, thus acting as an objective evaluator. These competences could include course-specific competences (e.g., ability to develop network programs in C++, ability to design basic linear programming models, etc.) and also generic competences related to teamwork, oral communication, etc.

The proposed model assumes that each placement $j \in \mathcal{P}$ has associated a vector of competence relevance $\Phi_j =< \phi_{j1}, \ldots, \phi_{jc} >$, where each ϕ_{jl} represents the importance of competence of $l \in \mathcal{C}$ for the development of placement j. Similarly, a student i has two vectors associated with competences. $\Psi_i =< \psi_{i1}, \ldots, \psi_{ic} >$, where each ψ_{il} represents the current development of competence $l \in \mathcal{C}$ by stu-

dent i, and $\Theta_i =< \theta_{i1}, \ldots, \theta_{ic} >$, where each θ_{il} represents the preference of student i for developing competence l during her placement. For the sake of simplicity, it is assumed that $\sum \phi_{jl} = 1$ with $0 \leq \phi_{jl} \leq 1$, and $0 \leq \theta_{ir}, \psi_{ir} \leq 1$. It should be highlighted that, in practice, we expect the number of relevant competences to be general and limited. Thus, these vectors can be easily elicited from stakeholders. For instance, Φ_j can be elicited from the company that offers the placement when agreeing placements with schools, Ψ_i can be elicited from the academic institutions, and Θ_i can be elicited from a simple questionnaire given to students before being allocated their placements.

We formally define the utility reported by placement j to student i as $\tau_{ij} = w_{id} \times (1 - d_{ij}) + w_{ic} \times (\Phi_j \cdot \Theta_i)$ where w_{id} is the importance given by student i to the distance from her home to the allocated placement, w_{ic} is the importance given by the student to the competences to be developed in the placement, and d_{ij} is the distance between the student's home and the workplace of the placement. This distance is normalized between 0 and 1 with a min-max scaler for each student. Distance is included as a factor to be considered since it was highlighted by consulted experts as a major element in allocating students to placements. The main rationale is that many students have a young age and many do not have a driving license. Therefore, very distant placements may be unreachable by students or result in long commuting times using public transportation. The formula also includes the dot product of Φ_j and Θ_j as an average of the preferences of students with respect to competences weighted by the importance of each competence for the placement. The utility reported by student i to placement j, is defined as $\eta_{ij} = \Phi_j \cdot \Psi_i$ where the formula represents an average of the student's competences weighted by the importance of each competence for the placement. Given this, we formalize a multi-objective ILP model.

$$\max \quad \frac{\alpha}{|\mathcal{S}|} \sum_{i=1}^{s} \sum_{j=1}^{p} \tau_{ij} \times x_{ij} + \frac{\beta}{|\mathcal{S}|} \sum_{i=1}^{s} \sum_{j=1}^{p} \eta_{ij} \times x_{ij} \tag{1a}$$

s.t.

$$\sum_{j=1}^{p} x_{ij} = 1 \qquad \forall i \in \mathcal{S}, \tag{1b}$$

$$\sum_{i=1}^{s} x_{ij} \leq 1 \qquad \forall j \in \mathcal{P}, \tag{1c}$$

$$\sum_{i=1}^{s} \sum_{j \in P_k} x_{ij} \leq Q_k \qquad \forall k \in \mathcal{B}, \tag{1d}$$

$$\sum_{i=1}^{s} \sum_{j \in P_k} x_{ij} \geq q_k \qquad \forall k \in \mathcal{B}, \tag{1e}$$

$$\left(\sum_{i=1}^{s} \sum_{j \in I} x_{ij} \right) - |I| \times \delta_I = 0 \qquad \forall I \in \mathcal{I} \tag{1f}$$

where $x_{ij} \in \{0,1\}$ are binary decision variables that represent if student i is allocated to placement j, $\delta_I \in \{0,1\}$ are binary decision variables that indicate if interdependent placements $I \in \mathcal{I}$ are allocated or not. The multi-objective aspect of the problem is modeled by means of a weighted sum of the two objectives. Each objective is represented by one of the two terms in the function. The first term represents the average utility of allocations from the students' perspective, and it is related to the overall satisfaction of students with the allocation. The second term represents the average utility reported to businesses by the proposed allocations, and it is related to the overall satisfaction of companies with the allocation. These two terms are weighted by α and β which represent the importance given by the education institution to the students' and companies' satisfaction respectively, with $\alpha + \beta = 1$.

In addition to this, constraints 1b guarantee that each student is allocated to exactly one placement, constraints 1c ensure that each placement is allocated at most once, and constraints 1d and 1e guarantee that each company is allocated a number of students within the minimum and maximum agreed number of students. Constraint 1f is designed to force placements with interdependencies to be allocated all together in case that they are allocated.

As the reader may observe, the model also allows us to tackle the problem from a single objective perspective. By setting $\alpha = 1$ and $\beta = 0$ one can solve the problem by just considering the objective associated with the students' satisfaction. Conversely, by setting $\alpha = 0$ and $\beta = 1$ we can solve the problem by just considering the objective associated with the companies' satisfaction. Both objectives are measured on a 0–1 scale. Therefore, the weights α and β are related to the importance given by the educational institution to each objective. Depending on its criteria, they can set both accordingly.

4 Experiments

As suggested, one of the objectives of this article is to analyze the scalability of the proposed model and gain an understanding of what factors may affect the time performance of the model. To make this analysis, it is necessary to generate varied and different scenarios and test the proposed model under those different conditions. We start by describing how different problem instances with different characteristics were generated. Then, we describe the experiments that have been carried out and analyze the results to understand the scalability of the model and the factors that affect its scalability.

4.1 Instance Generation

We created several problem instances by changing the type and distribution of the different inputs of our linear integer programming model. Next, we specify what characteristics were taken into consideration when generating instances.

– **Number of students:** The number of students plays a major role in the size of the problem. The more students one has to allocate, the more time it

will take. In order to study the scalability of the model with the number of students, we generated problem instances with 100, 200, 300, and 400.

- **Number of placements:** Another factor that may affect the problem size is the number of placements available. The more placements, the more decision variables will be necessary, as the number of decision variables is exactly equal to $|\mathcal{S}| \times |\mathcal{P}|$. Any problem with $|\mathcal{S}| > |\mathcal{P}|$ is non-feasible. We will use the term surplus to refer to the difference between the number of placements and the number of students $|\mathcal{P}| - |\mathcal{S}|$. For each number of students, we generated problem instances that have a 0%, 10% and 20% of surplus.
- **Students' homes and companies' locations:** Students' homes are randomly sampled from points in 21 small cities in the metropolitan area of Valencia (Spain), and Valencia. Companies' locations are randomly sampled from 200 real company locations in Valencia and surrounding areas.
- **Student competence development Ψ_i:** We aim to create instances that are distinct from each other, regardless of the institution. Therefore, we have chosen to consider six competencies for these experiments. We generated students with a high, intermediate and low competence development. To match realistic proportions, when generating a student its competence development is assigned to high, medium and low competence development with 5%, 45% and 50% probability. For students with a high development, two random competences follow a normal distribution with $N(\mu = 0.8, \sigma = 0.2)$, while the rest follow a normal distribution with $N(\mu = 0.4, sigma = 0.4)$. In the case of students with a medium development one competence follows a normal distribution with $N(\mu = 0.8, \sigma = 0.2)$ and the rest follow a normal distribution with $N(\mu = 0.2, \sigma = 0.4)$. For students with a low development, all the competences follow a normal distribution with $N(\mu = 0, \sigma = 0.1)$. All of random numbers generated were truncated to the [0,1] interval. Thus, students with a high competence development will tend to have 2 strong competences, students with an intermediate competence development will tend to have one strong competence, and students with a low competence development will tend to have no strong competence.
- **Student competence preference Θ_i:** The preferences of each student with respect to the development of competences during the placement is generated in a way that one random θ_{il} has a value of 1 (i.e., the most preferred competence), one has a value of θ_{il} of 0.5 (i.e., the second most preferred competence, and the rest are generated from a uniform distribution $U(0, 0.3)$, being options with a low preference for students.
- **Student preferences for distance** In this case, the preference of student i with respect to distance w_{id} is generated from a uniform distribution $U(0, 1)$ and w_{ic} is calculated as the complement $1 - w_{id}$.
- **Placement competence importance Φ_j:** We understand that different placement options may require of one important or several important competences. We generated some scenarios where l random competences are the primary ones for the placement, and another competence acts as a secondary competence for the placement. We refer to these scenarios as l primary competence scenarios. In these scenarios, Φ_j is generated with l random competences

taking $\phi_{jl} = 1$ (i.e., the primary competence), and a random competence taking $\phi_{jl} = 0.5$ (i.e., the secondary competence), and the rest being uniformly generated from $U(0, 0.3)$. We generated scenarios with one, two and three primary competences.
- **Placement interdependence:** A company with at least 2 placements may state interdependence between some of their placements (30% probability).

For each possible configuration we generated 5 different problem instances. Thus, in total we have generated $4 \times 3 \times 2 \times 5 = 120$ problem instances. These instances will be employed to study the performance and scalability of the model.

4.2 Experimental Design

As any integer linear programming model, it can be solved via a variety of solvers. In this case we are interested on studying how the solver scalers both with commercial and free solvers. While the former represent the state-of-the-art software for solving mathematical programming problems, we recognize that many schools and institutions may only have access to free or open solutions. Hence, we include both types of solvers in the experiments. More specifically, we selected two non-commercial (SCIP and CBC) and one commercial solver (Gurobi). All of the solvers were configured to have a time limit of 15 min per problem and to use a single thread.

We employed the 120 problem instances described in the previous subsection in our experimental setting. Each problem instance was solved with three configurations for α and β: $(\alpha = 0.5, \beta = 0.5)$, $(\alpha = 1, \beta = 0)$, and $(\alpha = 0, \beta = 1)$. The first configuration represents a balanced scenario where the same importance is given to the students' preferences and the companies' preferences for the allocation. The second and third configuration represent scenarios where only the students' preferences are considered, and only the companies' preferences are considered, respectively. These different configurations are employed to assess the conflict between the different objectives of the model. Each problem instance and configuration was solved 30 different times to capture stochastic variations in the computation time. Thus, this gives a total of $120 \times 3 \times 3 \times 10 = 10,800$ problem resolutions. We collected information about the execution time and the value of the objective function for both the students and companies. The experiments were run on 2.2 GHz processors 16 GB of RAM.

4.3 Results

We aggregated all of the repetitions into mean values for the same problem instance, configuration, and solver. Then, with the resulting dataset we conducted an ANOVA analysis using the execution time as response variable and the rest of the variables (i.e., number of students, surplus, number of primary skills, configurations for α and β, and the solver) as factors. The main goal of this analysis is understanding how the different variations of the problem and the solver type may affect the execution time of the model and, thus, its resolution in practice.

Table 1. ANOVA type III ($\alpha = 0.05$) analysis taking the execution time as a response variable. A bold font is employed to highlight significant p-values.

	Df	Sum squares	Mean Squared	F value	p-value
A: Configuration for alpha and beta	2	1.68E+11	8.43E+10	1.68	0.1869
B: Primary competences	1	5.91E+9	5.91E+9	0.12	0.7315
C: Solver type	2	3.12E+13	1.56E+13	311.42	**0.00**
D: Number of students	3	7.58E+13	2.52E+13	503.61	**0.00**
E: Surplus	2	6.69E+12	3.34E+12	66.69	**0.00**
Interaction AB	2	7.44E+10	3.72E+10	0.74	0.4769
Interaction AC	4	1.50E+11	3.76E+10	0.75	0.5579
Interaction AD	6	3.28E+11	5.48E+10	1.09	0.3658
Interaction AE	4	8.11E+10	2.02E+10	0.40	0.8059
Interaction BC	2	4.21E+10	2.79E+10	0.42	0.6576
Interaction BD	3	8.37E+10	1.74E+10	0.56	0.6442
Interaction BE	2	3.48E+10	5.19E+12	0.35	0.7006
Interaction CD	6	3.11E+13	4.29E+12	103.48	**0.00**
Interaction CE	4	1.71E+13	8.00E+11	85.50	**0.00**
Interaction DE	6	4.80E+12	5.02E+10	15.94	**0.00**
Residuals	1030	5.17E+13	3,97E+11		

The ANOVA table can be found in Table 1. The table contains the study of the effect of the aforementioned factors on the mean execution time, as well as the effect of all possible interactions of the second degree. In addition to this, we have included plots with the means and the Tukey intervals ($\alpha = 0.05$) for those factors that were found to have a significant effect on the mean execution time (Fig. 1). Finally, we include interaction plots for those second-order interactions that were found significant. From the main factors, we found the number of students, the surplus, and the solver type to have a significant effect on the average execution time. As observed in Fig. 1, Gurobi is the fastest solver.

Let us start by analyzing the effect of the number of students. As we expected, Table 1 suggests that the number of students has a significant effect on the mean execution time. This is natural, as it controls the number of decision variables and the number of constraints of the model. Hence, as one could expect, the more decision variables and constraints the more execution time it takes to solve the problem, as they are related to the size of the problem. In Fig. 1 one can observe the increasing tendency of the mean execution time and the number of students. It can be highlighted that the mean execution time grows in a possibly exponential way. As a consequence, one conclusion that can be drawn from these experiments is the necessity for alternative methods to solve the problem

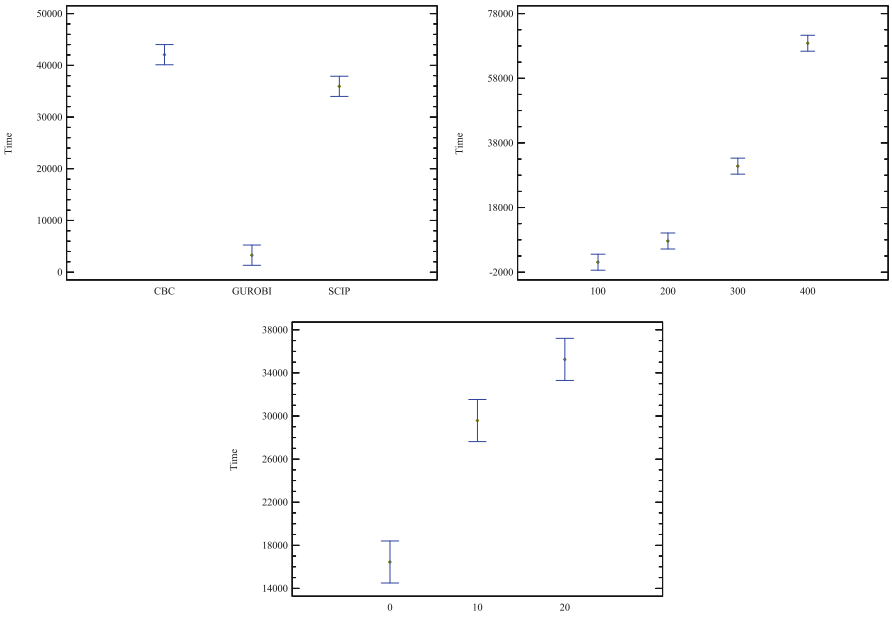

Fig. 1. Means Plot and Tukey HSD intervals (95% confidence) for the factor type of solver (left), the number of students (right), and the surplus (bottom)

when one faces larger problem instances. The interaction between the number of students and the solver type was found significant (see Fig. 2, suggesting a different growth for the execution time when increasing the number of students attending to the solver type. Effectively, Gurobi has a slower growth compared to CBC and SCIP, being the fastest of the three solvers and showing consistent behavior across problem instances. Despite showing a good performance, not all schools and educational institutions may have access to commercial solvers. Hence, they may need to employ non-commercial solvers, which have a faster growth for the mean execution time and whose mean execution times may result in unacceptable for larger problem instances.

Some similar conclusions can be drawn by analyzing the effect of the surplus on the mean execution time (Fig. 1 and Table 1). The larger the surplus, the longer the mean execution time. This is also natural, as more placements mean more decision variables and, therefore, a larger problem size. What is interesting is the fact that not all of the solvers show the same behavior when facing problems with more placement opportunities than students, as shown in the interaction plot between solver and surplus (see Fig. 2). CBC shows very competitive execution times when the number of students is equal to the number of placements, being close to the times achieved by Gurobi, the fastest among the three solvers. However, this performance quickly deteriorates when the number of available placements arises, becoming the worst option among the three solvers. SCIP shows a more stable behavior as well as Gurobi. We can attribute

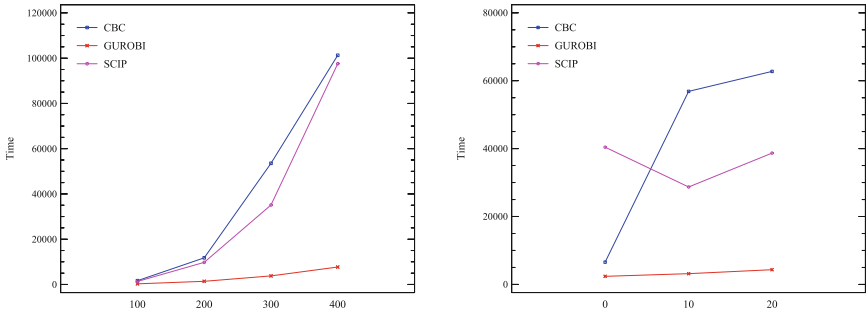

Fig. 2. Interaction plot of the number of students and surplus and the solver

Gurobi's better performance to the more sophisticated heuristic implementations it integrates, as well as other optimizations. This performance depends on the type of problem, but in our specific problem, the performance of the commercial software is clearly superior.

5 Conclusions

In this paper, we have introduced a multi-objective integer linear programming model for the SIPA problem, which is focused on matching students with industry placements. This proposal would be particularly suitable for environments that require the assignment of students to practical work, such as Vocational Educational Training or high school. This model incorporates the student and company preferences with a multi-objective approach. Preferences are defined through the competences associated with the current development level of students, their competence development preferences within the placement, and the competence requirements specified by each placement. In addition, the model also incorporates the distance between the student and the placement site, as well as interdependencies between placements.

We formally defined the SIPA problem and a model to address it. Then, we conducted an experimental comparison of various commercial and non-commercial solvers to analyze the scalability and performance of the model. To do this, we created different instances using real student and company data. According to the results, the non-commercial solvers do not scale as well as the Gurobi commercial solver. However, it is important to consider that many schools and educational institutions do not have access to commercial solvers and must rely on non-commercial software. Furthermore, while Gurobi shows fairly scalable and efficient behavior, it is important to note that its growth is exponential, which can be a drawback when dealing with larger instances than those used in these experiments. Therefore, as part of future work, we are working on defining a heuristic or metaheuristic method to solve instances of larger problems within a reasonable timeframe. It is worth noting that lecturers are another

relevant actor to consider in future versions of this model, as they need to supervise students. Hence, as part of future work, we aim to develop a model that also takes into account lecturer preferences for supervising student placements and, consequently, making visits to the companies. Finally, aim at conducting user evaluations to measure the effectiveness and pedagogical relevance of our approach.

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The Fading of F(AI)th: Tracing the Technological Promises of a Wellbeing App in K-12 Education

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Abstract. This paper explores the implementation of a wellbeing app in a Swedish Upper Secondary School. The aim is to understand how ideas of data driven school improvement underpinned by promises of artificial intelligence (AI) and learning analytics (LA) change the work of teachers. The study draws on video-ethnography from 17 meetings between five teachers/form tutors. The produced data is analysed using actor-network theory to focus on the various stages of the implementation process and the interactions between the learning analytics dashboard (LAD) and the teachers. To capture the complexity of the data, the empirical material is presented through cartoon-inspired illustrations grounded in a *Thinking through Cartoons* methodology. Findings show how teachers took on new roles and responsibilities in relation to the wellbeing app, most notably the role of collecting data from students. Teachers came to act as data analysts which imposed constant negotiations and uncertainties. To address the declining engagement of students over time, a student-facing LAD was introduced. The teachers shifted their focus to motivate students to engage with their own data in different ways. Despite no improvements in students' response rates teachers remained committed to the app, trusting that new AI and LA functionalities would compensate unsatisfactory outcomes. In conclusion, instead of improving teachers' capacity to identify at-risk students, the wellbeing app increased teachers' workload and led to different dilemmas related to teacher-student relations and teachers' professional judgement.

Keywords: Artificial Intelligence · K12 · data-driven schooling · actor-network theory (ANT) · learning analytics · EdTech · socio-materiality · PERMA-model

1 Introduction

What happens when emergent technologies are introduced in educational practice and how can these actions be understood in an AI era? This paper goes beyond the recent trends of research into generative AI and AI chatbots, by exploring areas where AI in K-12 education, in conjunction with learning analytics (LA), has been common for years [1]. The introduction of AI and LA is often grounded in the belief that all educational

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processes can be represented by data and that this data can provide teachers with valuable insights [2]. However, to materialise AI in school practice, data first needs to be collected, analysed and become a part of teachers' daily work. This study tries to understand the complexity and challenges of such materialisation processes by drawing on a Swedish case study.

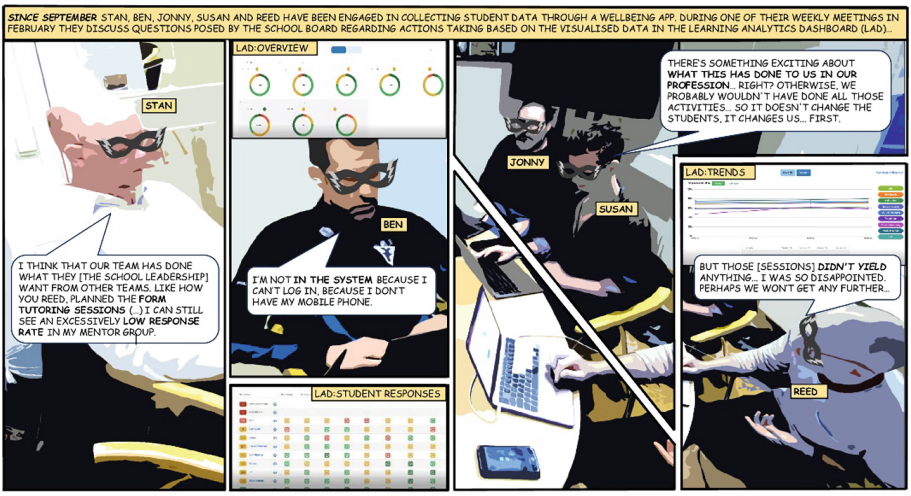

Fig. 1. Conversation between five form tutors during a weekly team meeting, when interacting with a learning analytics dashboard (LAD). Author's reconstruction.

The main actors in the paper, as introduced in the cartoon-inspired illustration (Fig. 1) are a commercial wellbeing app and a group of five upper secondary school teachers and form tutors (Stan, Ben, Jonny, Susan, and Reed). The wellbeing app was co-developed with an EdTech company specialising in AI and LA and is instrumental to understanding how a frequent monitoring of students' wellbeing might offer teachers/form tutors new insights about their students and prompt appropriate actions. Once a week, students were asked to respond to eight questions. The answers were then automatically converted into interactive pie charts and graphs and presented through the LAD. However, as reflected in the dialogue between Stan and Reed, despite various measures taken, for example, by planning the form tutor sessions differently, response rates were low and student behaviour remained largely unchanged. Thus, rather than helping teachers to support students' wellbeing, the app, underpinned by the promises of AI and LA, had implications for teachers' work. To quote Susan, "it doesn't change the students, it changes us, first". These changes will be further explored in this paper.

1.1 Data-Driven School Improvement

The wellbeing app provides a local context for global assumptions about *data-driven school improvement*. Such assumptions have, for the past two decades, become an essential component in the realms of education and have been accelerated by the recent rise

in Big Data and the recent development and implementation of data-driven technologies like AI and LA in K-12 [3]. The collected data, for example on student progress, is commonly communicated via advanced statistical techniques and personalised recommendations through a learning analytics dashboard (LAD) that can be directly faced at either teachers or learners [4]. LADs are often designed to help, or 'nudge', teachers to identify students who require individual attention [5]. However, research indicates that the effectiveness of LADs in enhancing teachers' capabilities is highly speculative and lacks scientific evidence [6].

1.2 Solving the Problem of Wellbeing with AI and LA

The correlation between wellbeing and academic achievement has been demonstrated in research and policy reports [7]. In recent years, the growing concern for student wellbeing, exacerbated by the pandemic, has led to the development of digital technology initiatives aiming to address this problem [8, 9]. Results from initiatives where wellbeing has been monitored through digital apps have shown low uptake and a lack of alignment with students' needs [10]. However, with the promise that AI can provide both automated and personalized feedback, software like The Mind Tutor for Higher Education [11] and the School Wellbeing Model for K-12 education [12] are hoped will meet these new expectations. The Swedish wellbeing app in this paper serves as an illustration of this trend. Its development reflects the growth of EdTech solutions with claimed AI and LA functionalities which address not yet comercially exploited educational problems through the intense collection of student data.

1.3 Aim

This paper explores the implementation of a wellbeing app in one Swedish Upper Secondary School with the aim of understanding how data-driven school improvement underpinned by promises of AI and LA changes the work of teachers. Here technologies like AI and LA are conceptualised *relationally* [13]. This conceptualisation considers what happens within a particular moment of interaction between a teacher and, in this case, a wellbeing app that is underpinned by the promises of AI and LA. A relational understanding adds to the fluent realities of AI and LA in school practice. Unmaterialised dimensions of AI also form specific social and material relations as well as the conditions that actively configure and shape teachers as well as students. Consistent with a relational understanding of AI and LA, this study understands the wellbeing app not as a neutral tool but as an active entity with specifically inscribed ideas and politics of teaching and learning that affects and shapes the work of teachers. Thus, the study focuses less on teachers as *users* of the app and more on how the teachers and the wellbeing app change each other and what emerges from these changes. For this reason, Actor-Network Theory (ANT), as a method for analytical exploration, has been found particularly productive, as outlined in the next section.

2 Actor-Network Theory

Actor-Network Theory is grounded in science and technology studies (STS) [14] and provides this study with a set of concepts that help it to explore the relational dynamics of technology in situated practice. In the context of education, this approach highlights the importance of examining both the social and material dimensions of technology to better understand what these mean for teaching and learning [15]. The ANT concepts *actors, networks* and *translations* are used in this study to unpack how data-driven school improvement, underpinned by the promises of AI and LA, changes the work of teachers and to understand the effects arising from these relations.

2.1 Actors and Networks

ANT seeks to describe technology as more or less stable and far-reaching alliances or networks of *actors* [16]. An actor can be a human (e.g. a teacher) or a thing (a wellbeing app) as long as it acts. Actors then come together, exert force, and change other in *networks* [14]. A *network* can be described as the "point-to-point connections" that are left behind by actors and can be traced [14 p.131]. The number of point-to-point connections between actors is essential as they render networks durable and make invisible, or "black box", their inner workings [16]. Hence, if actors are removed from the network, for example when students stop engaging with a wellbeing app, counteractions by the remaining actors are necessary to prevent the network from dissolving. Thus, networks are only durable as long as there is interaction between the mobilised actors [17].

2.2 Translations

Also described as a *sociology of translation* [18], ANT suggests that *translations*—the focus of this study – can explain how actors impose their interests onto other actors and how the network performs knowledge in specific ways. *Translations* can also describe various stages of a network's formation. The first stage, *problematisation,* occurs when an actor identifies and defines the nature of a problem that requires the potential involvement of other actors. Of specific importance for this stage is the *obligatory passage point* [18, p. 205] where interactions are particularly dense and where actors, independently of their objectives, converge on a specific aim and thus mediate all interactions within the network. The second stage, *interessement,* takes place when several actors are engaged in the process of addressing an identified problem. This stage is crucial for establishing a network of actors. The third stage, *enrolment,* involves the formation of alliances between actors to ensure that each actor is working towards a common goal. The final stage of network formation, *mobilisation,* occurs when the proposed solution gains broader acceptance within the network and leads to the emergence of new networks. The moments of translation which occur when the network progresses to the next stage are used in this study to understand the implementation of a wellbeing app and its change processes. More specifically, this study has focused on how a wellbeing app makes teachers act, (a question of methodology) and which educational realities are being enacted (a question of analysis) [19].

3 Method

This study builds on previously conducted ethnographical fieldwork that explored the co-design of a commercial Learning Management System (LMS) with AI and LA specifically tailored for K-12 education [20]. The wellbeing app was officially launched in June 2022 to further advance AI and LA in education. The development was carried out in close collaboration with a Swedish Upper Secondary school as part of their participation in a data-driven school improvement and professional development program. The implementation of the wellbeing app across the entire school also served as a way of understanding how data-driven wellbeing measures could help teachers identify early signs of potential problems.

Starting in August 2022, approximately 225 students aged 16–19 were asked to complete a weekly survey consisting of eight questions accessed through their mobile phones. The survey was based on the PERMA model [21] and was designed to collect data about student wellbeing, motivation, social relationships with teachers and other students, academic performance, workload, depression, and anxiety. The statistics generated from the surveys were visualised through a teacher-facing LAD in the form of interactive pie charts, line charts, and matrix charts. Individual answers were accessible to only to the students' form tutors, whereas anonymised group results were accessible by a wider group of teachers as well as the school leadership. Six months after implementation, students were also provided with a student-facing LAD, which showed their individually accumulated responses over time.

3.1 Data Production

The study took place at the Swedish Upper Secondary School mentioned earlier. The produced data consists of 17 video recordings of all the form tutor meetings that took place between November 2022 and June 2023 in real life on a weekly basis. These meetings consisted of a working team of five teachers who each were responsible as form tutors for 15 students and thus representing a total of 75 students. The meetings were intended to focus on the analysis presented through the LAD. Each recording lasts between 14–30 min and I was present during all 17 meetings via Zoom. During a subsequent visit to the school in March 2023, four individual in-person interviews were conducted with two teachers, a development teacher, and the school principal, and audio-recorded.

In addition, fieldnotes taken during the visits, together with the school's participation report, teachers' meeting agendas, and a student evaluation of the app, were gathered to provide an understanding of how the app was implemented across the entire school.

The study received approval from the Swedish Ethical Review Authority in August 2022. Informed consent was obtained from all participating teachers along with additional consent from the supervised students. Confidentiality considerations prevent the disclosure of websites, company names, reports, or individuals' names and affiliations.

3.2 ANT Analysis

The analysis focuses on the different stages of the translation process by drawing on a *method assemblage* approach. Law describes this as a way to *craft out-thereness by*

condensing particular patterns and repetitions whilst ignoring others [22, p. 113] and is an important part of an ANT approach [14, 15]. In practice, this involved seeking a profound qualitative understanding of how heterogeneous actors related, interacted, changed and were being changed during the implementation of the wellbeing app with a particular focus on teachers' work.

First, video captures were analysed by watching all the recordings in an open-minded way and only making brief annotations. A full transcription was not carried out at this stage. During a second screening all sequences where teachers were interacting with or relating to the wellbeing app were transcribed verbatim and accompanied by a related screen shot. In the final stage of the analysis, the translation concepts were 'plugged into the data', which allowed the actions within the that were most salient in relation to teachers' work to be traced [23]. The four audio-recorded interviews, field notes, school's report of their participation on data driven school improvement, teachers' meeting agendas, and a student evaluation provided a contextual backdrop to the entire analytical process.

3.3 Crafting the Cartoon Inspired Episodes

A recurring challenge with video data is how to absorb the multimodal density of what is happening and represent it in a meaningful and representative manner [24]. In this case, the video recording was affected by the school's video conference system, where the software automatically identified every participant in the room and split up the image into different parallel sections. This influenced how data was crafted and suggested that it could be usefully presented as cartoon-inspired episodes. This cartoon crafting was also part of a deliberate decision to adopt a *Thinking Through Comics methodology* (TTC) [25] because of its ability to show how each actor 'makes other actors do unexpected things [14, p.122] and to trace a set of relations between LAD and teachers based on the teachers' accounts. The TTC methodology involves integrating the process of comic-making into the doing of research at different stages. In this paper, the methodology has been used as part of the analytical work allowing for understanding the data like a system or network of heterogeneous actors, which allows for different readings. While the strips can be read sequentially like a text, their entire composition can also be taken in – or viewed – all at once, with potential associations stretching web-like across the page and weaving fragments into a cohesive whole [26, p. 62]. The strips also allow for an allegorical reading, inspired by the Marvel comic book series.

Each cartoon-inspired episode, including selected excerpts, comes from one meeting that could illustrate the everyday interactions between teachers and the wellbeing app. The dialogues have been taken f shortened slightly and translated from Swedish into English while trying to remain faithful to the original narratives. Corresponding screen-shots were then processed as individual panels. Faces were then anonymised. Speech bubbles and screenshots of the LAD of the wellbeing app were then added manually.

4 Findings in Three Episodes

The findings highlight specific moments in time of the implementation process of the wellbeing app. These moments illustrate the complexity of the network and show the negotiations taking place that were necessary to form it.

4.1 Episode I: Doing the Heavy Lifting

This episode depicts a conversation between Ben, Reed, Jonny, and Susan during an early stage of the implementation process. The conversation makes is possible to trace the different actors that associate in the early stages of a network formation (*problematization*): in this case, teachers, students, developers, and various technological components like computers screens, mobile phones, and data interfaces (Fig. 2).

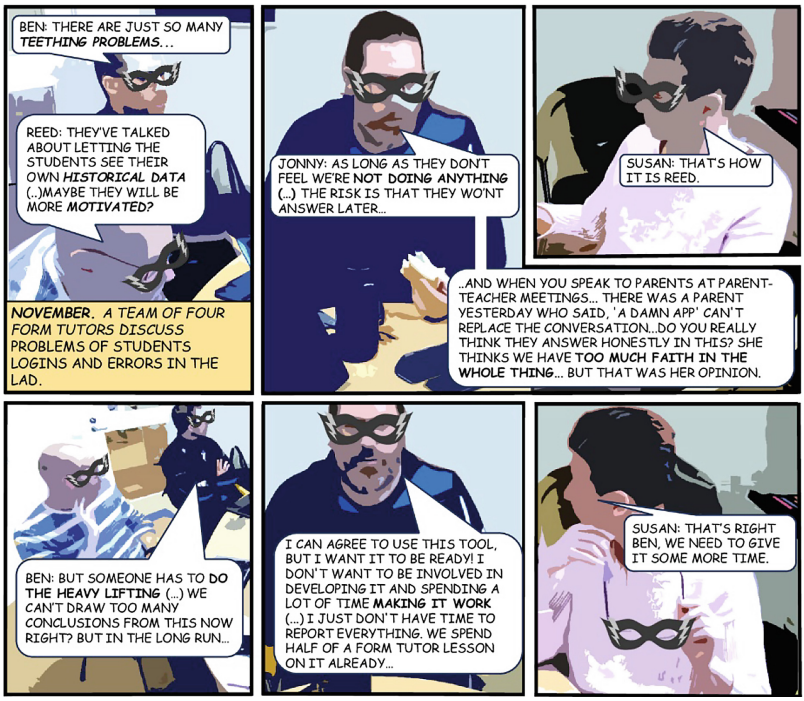

Fig. 2. Episode I in which teachers discuss the initial challenges when implementing the wellbeing app. Author's reconstruction.

In this network formation, the wellbeing app plays a central role both in defining the problem (the question statistically measuring the emotional state of students) and proposes itself as the solution to this problem, as the *obligatory passage point* through which all the other actors need to pass for data-driven education to emerge. However, as expressed by both Ben and Jonny, the wellbeing app is not translating these interests without the teachers contributing a substantial degree of work or doing the 'heavy

lifting'. This work includes tasks such as managing student logins and reporting erroneous LAD features to the EdTech company. While convinced that the app can be useful to address the problem of wellbeing, teachers show resistance to becoming part of the development process as required, which indicates a fragility in the network formation. The app malfunctions are also understood to affect students' motivation to generate new data through the surveys which in turn produces incomplete visual representations of students' wellbeing in the LAD. This becomes evident in the comment by Susan who does not know how to interpret the different lines in the LAD.

The episode also points to different dilemmas emerging from these interactions. The incomplete data generates statistics and visuals in the LAD that are flawed. This in turn makes it difficult for teachers to take effective action, while knowing that individual students might report that they are not doing well. Teachers' so-called (in)action risks jeopardising student-teacher relations, in which students might feel that their call for help is not taken seriously. Dilemmas also arise in relation to parents, as indicated by Jonny, who suggests that parents' scepticism risks undermining the authority of teachers.

During the research period, a solution to these teething problems was suggested. One way to sustain ongoing efforts and strengthen the relationships between teachers, students, and the wellbeing app is currently under development by the EdTech company: a student-facing LAD. This new network ally is expected to alter the dynamics of the app. It is expected to translate the interests of the wellbeing app for students and make them more motivated to complete the weekly surveys. Because they have faith in this promised technological solution, the teachers agree to give the wellbeing app more time. At this initial problematization stage, teachers' work with collecting data and handling technical problems, combined with their belief that students' wellbeing can be transformed into individual data points, is instrumental in promoting the idea of data-driven school improvement.

4.2 Episode II: Enacting a Tool for Self-reflection

The second episode illustrates how human and non-human actors jointly enact data-driven school improvement and how this leads to shifting the responsibility of data analysis from teachers to students. During this phase (*interessement*), the LAD attempts to stabilize the roles of the teachers and other actors. The app's visualizations compel teachers to consider students' wellbeing in specific ways, a process that is formalised by questions from the school board – distributed digitally– looking to find out what insights teachers have gained from the wellbeing app thus far (Fig. 3). The translation process here turns the teachers into data analysts. They are expected to translate the interests of the LAD into their own professional understanding and practice. In this context this necessitates seeing the visualised pie charts as probabilistic indicators of students' different dimensions of wellbeing.

As reflected in the line graph shown here, almost all of the parameters, with one exception, remained consistently positive. This should, according to the logic of the LAD, indicate that all of the students are doing fine. The one exception is feelings about workload, which peak higher near the autumn break and decrease as it approaches the Christmas holidays. This pattern, which teachers interpret as a reality for students and

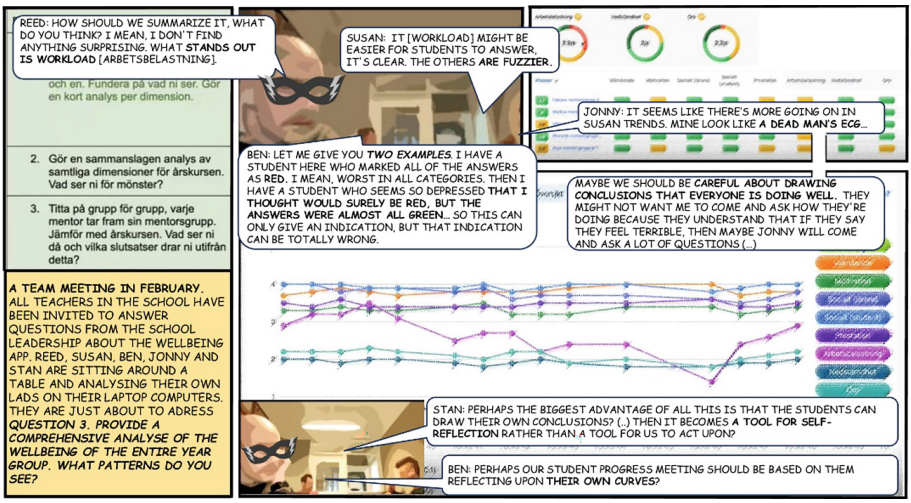

Fig. 3. Episode II illustrates how the ambiguity reflected by the LAD makes teachers shift the responsibility for data analysis to the students. Reconstruction by author.

as something that is that is less "fuzzy" to measure, seems to sustain the relevance of the LAD and gives it a particularly important role in the interessement phase.

While the teachers appear reassured by the statistics, the flat curves could suggest the opposite. As Ben points out, the curves could be deceptive and, in fact, indicate the opposite of what teachers know to be the case from their personal encounters with individual students. Could the positive answers simply reflect student indifference or a possible resistance to a process that they are obliged to engage with, but enables them to avoid having teachers following up on their results? This ambiguous insight, combined with the disappointment of not being able to 'see' anything unexpected in the data, highlights how the translated interests of the LAD force teachers to constantly negotiate the statistics with their own professional judgment. Accepting the interests of the LAD implies downplaying their own knowledge based on daily encounters with individual students inside and outside the classroom. Reed's comment about'not seeing anything surprising' shows that the humanity of the actors in the network is brittle and that contradictions risk dissolving the alliance the app is trying to create. In addition, the flat curves in the LAD, which are likened to a 'dead man's ECG,' seems to actually prevent the actors from acting. After all, what actions can be taken when everything looks fine at a first glance? At this point a shift of roles becomes necessary. Rather than the teachers needing to take action, the LAD is there to help students reflect on their own wellbeing. This transformation demands different actions from teachers and helps to sustain the wellbeing app as the obligatory passage point in the network. Faith in data-driven education can remain if the responsibility of data analysis shifts from teachers to students by letting students 'own their own curves'.

4.3 Episode III: Anticipating Seeing New Things

The third episode captures a moment during one of the last form tutor meetings. The conversation centres around two main points: students' response rates and their unwillingness to answer the questions in the app in a way that seems credible to teachers, and a new technical feature that could make the data presented in wellbeing app more relevant (Fig. 4).

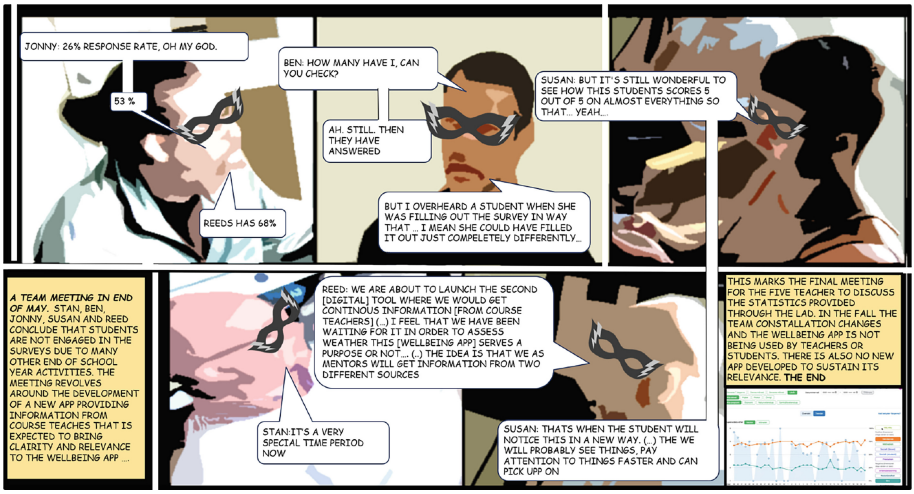

Fig. 4. Episode III illustrates how teachers, rather than giving up the idea of data-driven education anticipate a new and improved app. Reconstruction by author.

After an entire school year of implementation, with substantial efforts from teachers to motivate students by shifting the ownership of the wellbeing app, students' response rates remains low, indicating that they have not fully joined the network or been *enrolled* as anticipated. This again raises questions by teachers about whether any meaningful conclusions can be drawn from the LAD, in particular as they also notice an indifference when filling out the surveys. Susan confirms the difficulties of trusting the data by providing an example of a student who consistently gives the highest rating to everything even though this, for unknown reasons, is not accepted as a credible response.

Rather than abandoning the idea of data-driven school improvement and the wellbeing app once and for all, a new initiative to sustain the network is already on its way as the conversation shifts to discuss the development of a complementary app with AI functionality designed to collect data from subject teachers on how they perceive the progress of individual students. Through this app, form tutors are expected to receive statistics that can better support their interpretation of the wellbeing app and help them to be more proactive. The network is again temporarily sustained by faith in new human and non-human allies anticipated to join the network. However, as stated in the afterword, the final phase in which the network is expected to expand, *mobilisation,* never occurs. The new ally does not joint the network as planned and so it collapses. Resembling the incident when scallops detached themselves from the researchers' project on the

Normandy coast [20], the interests of the wellbeing app were not successfully translated onto the students in a Swedish upper secondary school. They resisted, either explicitly or implicitly, the ideal of a data-driven school. While changing teachers' practices was an effect, the problem of translation was eventually extended also to them who had, like the heroes in comic books, tried to overcome every possible imagined obstacle in order to save the day. Hopes and desires for future technological development, including the promises of AI and LA, were not sufficient to sustain the network and, in the end, the teachers at the school stopped using the wellbeing app.

5 Concluding Discussion

Many conversations about emergent technologies, including AI and LA, are dominated by promises and expectations of how technology alone can solve problems in education [27]. Such optimism is rarely promoted by teachers and educators or grounded in the realities of everyday educational life. However, as demonstrated in this paper, teachers' actions, here consisting of work behind the scenes combined with technology optimism, proved instrumental for AI and LA to materialise in school practice. The added work this created for teachers is problematic for at least two reasons: teacher workload is already a growing problem internationally; and the technology added little value for teachers and students. On the contrary, the wellbeing app introduced dilemmas related to trust, confidence and privacy between students and teachers and to teachers' relational 'knowing' about their students.

As illustrated in the comic strips, the collected data turned out to be inaccurate, incomplete and ultimately a very poor indicator of how students actually were doing. While teachers were aware that they were dealing with indicators that could be completely inaccurate, they tended to treat the data visualised through the LAD as true. This was particularly the case when the indicators could be interpreted as positive or explained what they already knew, such as stress levels relating to fluctuating workloads. These concerns stretch well beyond conventional notions of data validity and instead challenge the appropriateness of using data models to represent educational processes or emotional states of students. In practice the LAD "forced" teachers either to distrust their students' answers or to downplay their own professional judgment.

The study also provides a good empirical example of the significant efforts of teachers' work in constructing the infrastructure of data-driven practices in schools for the introduction of AI technologies. In this context, AI and LA in the role of augmenting teachers' insights, assumes an executive position where all elements converge. Therefore, at this early stage, it is crucial not only to focus on AI and LA applications but also to consider the intricate dynamics involved in setting up the preconditions for data-driven education at large. Furthermore, the case study illustrates how emergent technologies are beginning to expand into not yet highly commercialised areas, such as the realm of students' wellbeing. Thus, what can be considered as the most private, sensitive and non-measurable elements of education is becoming subject to commercialisation.

Acknowledging that the findings are based on a distinct, contextually bound case related to a development project on data-driven school improvement, this study exemplifies the complexity of technology implementation in schools, despite the best intentions

and measures taken by the school leadership and teachers. Using the ANT epistemology and focusing on relations between human and non-human actors, the three distinct episodes highlight the messiness, unpredictability, and failure in mobilising the network to support the specific interests of the wellbeing app. Understanding these dynamics as a translation process helps to illustrate the active roles of both humans and material artifacts in shaping educational realities. Focusing on translations can also help to illustrate how the trust in not yet materialised technologies like AI can provoke actions and sustain brittle network formations. Most importantly, the case illustrates that technology in education is not a determined and inevitable process, but one where what teachers and students *do* and do *not do* makes all the difference. Thus, examining these processes can provide with new perspectives of how digital technologies bring change in K-12 education and inform future design initiatives that serve the best interests of both teachers and students.

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Democratizing Gamification Strategy Design on Multitenant Platforms Through a Low-Code Approach

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Abstract. Gamification is a growing trend, driving behavioral shifts and nurturing user engagement across online platforms. Yet, a critical challenge persists in sustaining the underlying gamification engines that drive platform-oriented gamification. Typically, these systems are the domain of developers, but what if this capability were democratized for end users? This paper presents a case study wherein a gamification engine underwent evaluation within the dynamic ecosystem of the CAMPUS platforms. Given the technical complexity inherent in construction and the inadequacy of hard coding, the study's evaluation centered on a design-based research methodology. This approach encompassed four iterative cycles, testing various phases of prototype designs and culminating in the development of a low-code interface tailored for platform administrators' use. According to the results of this study, it was possible to validate the relevance and importance of implementing the low code interface since it proved useful in simplifying the gamification creation process and was suitable for the target audience, bringing novelty to the current gamification system.

Keywords: gamification · low-code · end-user development · interaction design

1 Introduction

In the ever-evolving landscape of digital platforms, gamification, as the application of elements and game mechanics to non-game related contexts [1], emerges as a potent tool, fostering engagement, promoting learning practices, and driving user behavioral changes [2]. This approach has proven to cultivate motivation, boost user retention, and enhance user experience [3]. However, despite the undeniable benefits of gamification, the challenge persists in maintaining and innovating the underlying gamification engines that power platform-oriented gamification strategies. The responsibility for developing and maintaining these systems lies with the developers, who possess the technical expertise to navigate gamification design and development [4]. But what if we could democratize this capability, empowering the end user to actively participate in creating gamification strategies tailored to their specific platform goals and needs? The present paper explores that notion, proposing a paradigm shift in how gamification strategies are designed and

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implemented on multitenant platforms. By leveraging a low-code approach, this study aims to dismantle entry barriers, enabling a broader audience to engage in this creative process.

This study was conducted on digital platforms that feature CAMPUS technology, where we can find an environment for resource sharing oriented towards online communities. The nature of CAMPUS technology is multi-tenant, it aggregates several independent platforms that consume the same code base hosted in the same development environment [5]. The CAMPUS gamification is platform-oriented, compelling users to use platform features to progress to next levels, boosting platform interaction. This requires well-defined gamification strategies to ensure that level progression is logical and missions are aligned with the appropriate difficulty level.

This work aims to evaluate a developed interface in these platforms to promote administrators' activity in implementing gamification strategies with a low-code approach. Evaluating the potential benefits of applying this new interface guarantees to ease the complexity of developing them. Applying the principles of design-based research, it was possible to evaluate the various prototypes of the designed low-code interface through four iterative cycles. Subsequently, the final iteration was implemented and submitted for final evaluation with platform administrators and specialists, either in the field of gamification or interaction design leading to positive results.

2 Designing Gamification Strategies with Low-Code Interfaces Having an End-User Development Approach

2.1 Designing Personalized Gamification Strategies

Understanding the motivations and behaviors of distinct types of users in gamification enables the creation of tailored gamification systems. Thus, as criticized by Orji et al. [3], gamification strategies often opt for one-size-fits-all approaches that generally fail rather than adopting tailor-made approaches that provide the potential for personalization. Applying this notion to gamification systems by understanding and incorporating diverse user motivations and behaviors can enhance engagement and effectiveness [2].

Gamification frameworks provide a set of recommendations and guidelines for creating gamified experiences [6], offering best practices to aid in understanding and implementing gamification strategies. To comprehend how to approach gamification systems design, the MDA framework, widely applied in game design, was explored. Many adaptations of this framework have emerged to encompass specific situations, aligning with the goals of gamification. In the MDA framework, mechanics (M) corresponds to game rules, dynamics (D) represents the user interaction with mechanics, and aesthetics (A) depicts emotions elicited through game interaction [7]. This framework emphasizes approaching game construction from mechanics to aesthetics and understanding player experiences from aesthetics to mechanics. However, it is argued that the unidirectional flow of the MDA framework may not apply to many game scenarios [8]. An alternative model proposed by Duarte and Battaiola [8] acknowledges the multi-directionality of MDA layers (M, D, A), suggesting a more fluid interpretation rather than a strict unidirectional relationship between player, game, and designer. For instance, in a game context,

the initial approach to learning game rules and dynamics can enhance the aesthetic experience [8]. Considering the perspective of game designers, it is crucial to recognize that players/designers can influence the mechanics, dynamics, and aesthetics of gamification strategies in diverse ways and directions. This is particularly relevant for gamified experiences where adherence to pre-established rules allows for the enjoyment of badges. Gamification tries to leverage familiar design constructs with recognized metaphors to align user behavior with the designer's intentions [9]. Therefore, designing gamification systems with known metaphors ensures alignment of behavior and recognition with the designer, so users can later construct these logics.

2.2 End-User Development

End-User Development (EUD), a field within Human–Computer Interaction (HCI), analyzes user interactions with low-code platforms, it evolved from spreadsheet-based development to creating functional designs and programs collaboratively [10]. Consequently, web design is evolving towards easy-to-develop interfaces, shifting from the late easy-to-use paradigm [11]. The fundamental premise of end-user development is to not only facilitate interaction interface usage, but also empower end-users with development capabilities. Furthermore, EUD promotes an adjusted level of complexity that is suited to the skills of different individuals [12]. An end-user is not perceived as a professional programmer but rather an individual who, for several reasons, needs to develop a system within a specific platform they need to use [13]. This way non-experienced programmers can develop software systems through visual interfaces without the need for additional effort in creating complex code logic [14]. Although process automation can save time, poorly applied automation can lead to the opposite of the expected outcome, which is not typically observed in manual implementations [13].

Designing for user-driven content creation requires not only implementing solutions that reduce cognitive load but also providing user suggestions [13]. This involves creating visually intelligent designs tailored to the new development paradigm where Blackwell [13] challenges traditional design methods, particularly questioning Shneiderman's concept of "Direct Manipulation" which involves implementing actions akin to physical tasks into visual interfaces. This allows users to perform actions without the need to recall them. Blackwell [13] argues that these principles are no longer the primary focus of constructing tacit interfaces for the use cases of end-user development. While these principles consider attention investment, they were crafted for manual processing rather than automation, where dealing with objects that emerge only after construction is a crucial step in development [13]. Therefore, for effectiveness in interface construction, particularly those involving tailoring, is imperative to comprehend how to expose future functionalities and components [13]. It is vital to emphasize that when providing tools to end users in which they can modify existing programs or create their own, the traditional working structure in the software development industry can be jeopardized. Therefore, end-user development is strongly influenced, both socially and technically, with socio-technical implications that need to be considered [11].

2.3 Low-Code Interfaces

Building visual and graphical interfaces designed to reduce implementation effort through lines of code represents one approach to End-User Development (EUD).

Low-code interfaces encompass platforms with graphical elements for constructing complex systems while still allowing code insertion and usage. These platforms offer end-users the opportunity to develop software at the level of a web developer within a given context [4]. Referred to as citizen developers by Di Ruscio et al. [15], these end-users, even without programming skills, are the target audience for low-code platforms, as they own the power to develop their desired applications. Understanding these platforms entails grasping how both programmers and non-programmers approach them. Low-code platforms enhance system quality by enabling customization, are perceptually intuitive, and offer enhanced usability and flexibility, thereby improving user experience [16]. Depending on whether data transformation or activity execution is emphasized, the puzzle metaphor of creating with building blocks aligns user intentions with appropriate levels of abstraction to indicate desired operations [17]. These platforms aim to significantly reduce the amount of required code, democratize application development, short time-to-market, and increase software development productivity [18]. This is supported by a study made by Forrester, which revealed that several companies reported these implementations accelerated the development process from five to ten times faster than usual [19]. While low-code platforms offer advantages, limitations include a learning curve, high pricing, restricted customization, perceived lower power compared to traditional programming, and the need for basic programming knowledge to comprehend digital product construction logic.

Over the past 40 years, user digital literacy has significantly increased, underscoring the relevance of teaching computer programming fundamentals as part of basic education [20]. The omnipresence of software calls for low-code techniques due to the increasing demand for implementation surpassing available skilled resources [16]. Ongoing research is crucial to clarify the future and current position of low-code platforms in the technological landscape [16]. Developing user-centered tools can play a crucial role in making programming more accessible, paving the way for universal software development [21]. Opinions on the benefits and limitations of low-code platforms vary, and current studies lack depth and consensus. Shridhar [22] noted that low-code platforms do not aim to replace conventional development approaches but to provide end-users with resources to build their applications autonomously.

3 Methods

3.1 Methodological Approach: Design-Based Research

Within the scope of design-based research [23] inspired by the DBR Process Model framework proposed by Hoadley & Campos [25], this methodology emphasizes the interrelationship between the design, the context, and deductions obtained to find different design hypotheses through the interpretation of the data collected. After adequate research and data collection for the theoretical background, this study presents findings derived from iterative evaluations with end-users (platform administrators) across different iterations of the low-code interface in the scope of CAMPUS platforms.

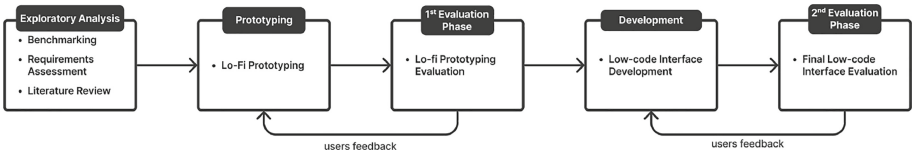

Fig. 1 Stages of the investigation.

The investigation included five stages depicted in Fig. 1: (i) Exploratory Analysis – initial exploration to understand similar platforms and user requirements; (ii) Prototyping – development of lo-fi prototype designs based on the findings and user feedback; (iii) 1st Evaluation Phase – cycles of usability testing to evaluate the lo-fi prototypes developed; (iv) Development – Implementation of the low-code interface on code base; (v) 2nd Evaluation Phase – Final round of usability testing the low-code interface developed, having in mind the found refinements. Each one of these stages was executed to ensure a systematic approach to design iteration and improvement based on user feedback and each cycle evaluation results.

3.2 Study Sample, Data Collection and Data Analysis

The first evaluation phase spanned three months. In the first usability testing session, nine participants from the CAMPUS platform's R&D team were tested to align the interface requirements with the expertise of technology specialists. The second cycle involved eight participants responsible for developing the platform's gamification system to validate the interface's accuracy. In the third iteration, five testers from previous cycles participated to validate prior feedback. The second evaluation phase included ten potential end-users who tested the developed interface to finally assess its attractiveness and usability.

Table 1 outlines the methods and tools used for data collection. Initially, the Exploratory Analyses encompassed benchmarking exercises comparing similar low-code platforms, a thorough literature review to identify common low-code features and a requirements assessment to determine which features would be integrated into the lo-fi prototype. Subsequently, following the development of the lo-fi prototypes, they underwent testing with the target audience via the Maze.co platform where tasks had to be completed by the participants. An observation grid was utilized to document user interactions and Likert scale questions were presented to gather user feedback. This evaluation process was repeated three times with end-users. Upon completion, the low-code interface was developed and integrated into the code base for testing within its authentic environment. The final testing phase involved further evaluation with end-users by their achievement of implementing a new gamification strategy with the low-code interface developed, this incorporated an observation grid, open-ended questions to solicit feedback, and the AttrakDiff scale [24] to assess the hedonic and pragmatic qualities of the final product.

Table 1. Techniques and Instruments to Collect Data.

Techniques	Methods	Instruments	Typology
Exploratory Analyses	Benchmarking Literature Review	Observation Grid	Qualitative
	Requirements Assessment		
Usability Testing	Lo-fi Prototypes Evaluation	Observation Grid	Qualitative
		Maze.co Platform	Mixed
		Likert Scale	Qualitative
	Low-code Interface Evaluation	Observation Grid	Qualitative
		AttrakDiff Scale	Quantitative
		Open-ended questions	Qualitative

4 Results

The gamification system features a hierarchical structure to create platform gamification. Levels contain missions, missions have triggers, and triggers include conditions, that can be created with a combination of events, operators and input values. All these elements must be created to set up specific missions completed by platform actions to build a new level. This study aimed to find a way not to use a JSON file to generate gamification, prototype and construct the low-code interface while reducing the learning curve associated with creating these gamification strategies. This makes sure to empower platform administrators to actively participate in crafting them to meet CAMPUS platform goals. This section presents key findings from both phases of usability testing through successive iterations of prototypes.

4.1 1st Evaluation Phase – Lo-fi Prototypes

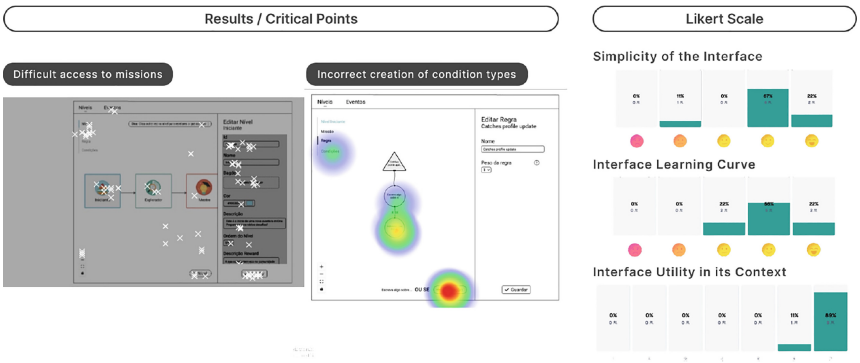

Fig. 2 First evaluation cycle results

In the first usability testing cycle, the first lo-fi prototype showed good results despite limitations. Two critical issues (Fig. 2) were identified: difficult access to missions and incorrect creation of condition's relationship types. This highlighted the inherent learning curve for users but showed positive user interest in the interface. Refinements included adding a button for direct access to the missions panel, changing the menu navigation, and removing the immediate condition type creation prompts to reduce user mistakes.

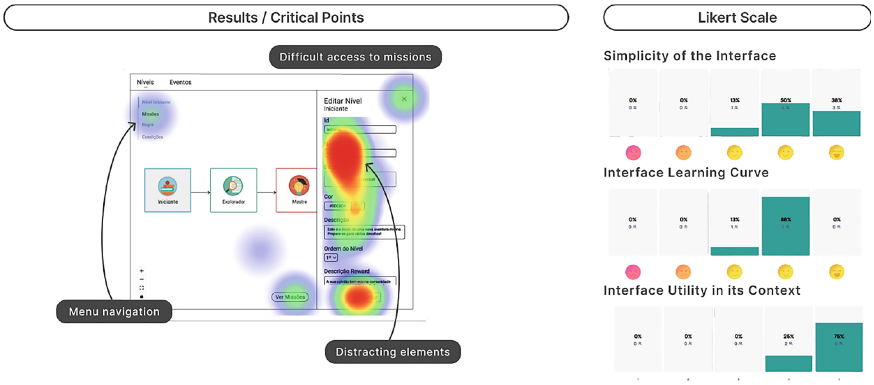

Fig. 3. Second evaluation cycle results.

The second usability testing cycle revealed a significant learning curve, but users completed tasks more efficiently. Redesigning the navigation system and layout was necessary to reduce initial interaction time and enhance usability. Accessing missions and sidebar distraction remained a problem (Fig. 3), but improvements were evident in creating correct relationships between conditions. It was noticeable an interaction improvement from the last cycle.

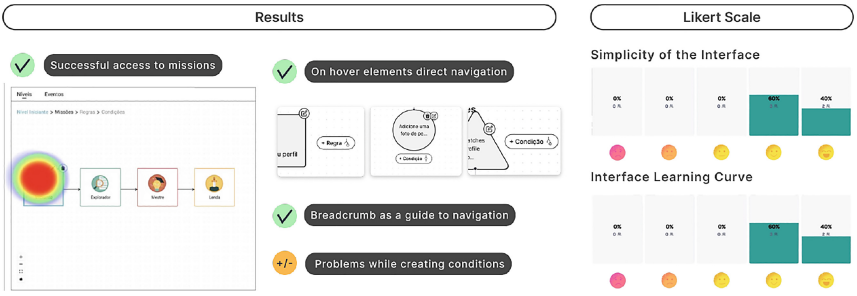

Fig. 4. Third evaluation cycle results.

The third usability testing cycle yielded the best results. Refining navigation and element creation tasks achieved significant improvements (Fig. 4). Mission access had an 80% success rate, with reduced task completion time and misclick rates. Breadcrumb navigation and hover elements enhanced the user experience. Feedback was the most

positive so far. However, preventing the errors in condition relationships still required attentive refinements. Participants suggested improvements in visual elements, addition of on-hover elements and item pre-selection to ease navigation, and finally a dedicated step by step condition creation modal process for the next iteration.

4.2 2nd Evaluation Phase – Low-Code Interface

In the final evaluation phase, the goal was to assess the attractiveness and usability of the interface with participants matching the profile of the end-users. Refinements from the previous cycle were applied to develop and test the low-code interface. Usability testing enabled the creation of a comprehensive gamification strategy, including a new gamified level. Users praised the interface for its simplicity, well-structured layout, cleanliness, and flexibility. These qualities were consistently highlighted, resulting in positive feedback about the platform structure. While the creation of relationships was generally well-received, questions arose about whether non-technical audiences would understand the differences between different types of relationships. The affordance of creating a new level was not immediately perceivable. Though users later accessed this functionality, making the action button more visible could improve this. Additionally, the learning curve has progressively decreased across testing cycles.

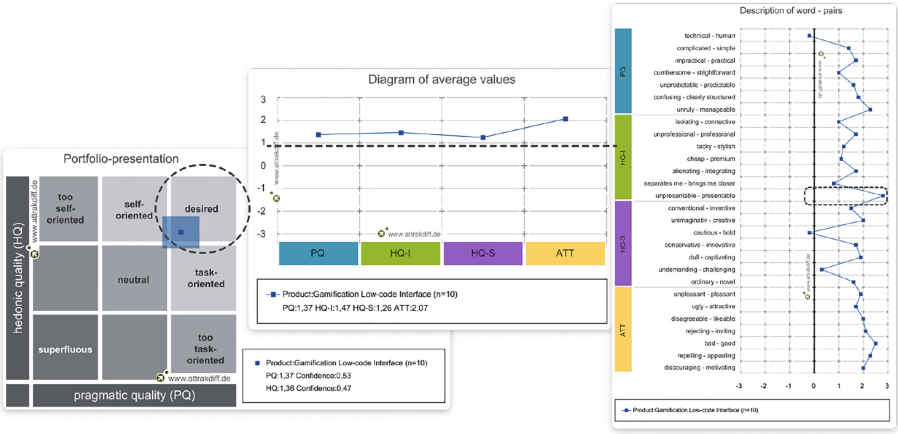

Fig. 5. AttrakDiff results.

The AttrakDiff results (Fig. 5) lie in the upper right quadrant of the "Portfolio-presentation" diagram, indicating an optimal balance of pragmatic and hedonic qualities. The "Diagram of average values" shows values above 1.26, with the highest value of 2.07 for attractiveness, surpassing negative values. The "Description of word-pairs" shows a positive trend, particularly in technical aspects of the interface. These results validate the overall attractiveness of the interface and the methodology used for its development. The AttrakDiff scale proved effective in assessing participants' experience with the gamification strategy-building interface. The findings affirm the importance of

the design-based research methodology for iterative development and the success in democratizing the capability of developing gamification strategies for end users.

5 Conclusions

Given that this interface aimed to simplify the process of gamification strategy construction while preventing hard coding, it is broadly concluded that a low-code interface could enhance the speed of conceptualization, user experience and development of customization processes for the gamification system across CAMPUS platforms. Implementing a low-code platform emerged as an ideal solution concerning the need to abstract complex logic for platform administrators, satisfying their needs and offering broad applicability by democratizing gamification system construction. As citizen developers, they may lack the expertise in coding required to build platform-oriented gamification. From this, it can be inferred that the outcome of this interface stems from the methodological process undertaken in this research, where iterative cycles of evaluation were employed to enhance usability. In conclusion, this interface demonstrates significant relevance in its application, as it enables universal access to a broader audience. The simplification of gamification construction was demonstrated by the ease of use for users without prior system or coding knowledge, as supported by the positive AttrakDiff results. Ultimately the interface drives innovation and novelty to the gamification system, which is increasingly prevalent in digital platforms, and it opens its access for maintenance and development to a broader audience.

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Intelligent Interface: Enhancing Lecture Engagement with Didactic Activity Summaries

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Abstract. Recently, multiple applications of machine learning have been introduced. They include various possibilities arising when image analysis methods are applied to, broadly understood, video streams. In this context, a novel tool, developed for academic educators to enhance the teaching process by automating, summarizing, and offering prompt feedback on conducting lectures, has been developed. The implemented prototype utilizes machine learning-based techniques to recognise selected didactic and behavioural teachers' features within lecture video recordings. Specifically, users (teachers) can upload their lecture videos, which are preprocessed and analysed using machine learning models. Next, users can view summaries of recognized didactic features through interactive charts and tables. Additionally, stored ML-based prediction results support comparisons between lectures based on their didactic content. In the developed application text-based models trained on lecture transcriptions, with enhancements to the transcription quality, by adopting an automatic speech recognition solution are applied. Furthermore, the system offers flexibility for (future) integration of new/additional machine-learning models and software modules for image and video analysis.

Keywords: Artificial intelligence in education · Lecture evaluation · Natural language processing · Machine learning

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1 Introduction

It is easy to notice that the majority of newly proposed solutions, adopting artificial intelligence (AI) in higher education, focus on supporting students. This involves student learning [6,10], measuring student performance [1], or assessing available learning sources [16]. Moreover, since approximately 2022, great interest in utilising and evaluating use of generative AI in education, for learning and teaching [5] has emerged.

A much less frequently addressed subject is the support that could be provided for the teachers, by observing them while they are conducting lessons, and providing personalized feedback. Only very few AI-based solutions have been developed to address this need, e.g. tools for observing students' behaviour and cooperation in teamwork during class [12], as well as other activities [2,14]. Another type of support – related to preparing lessons – includes AI-based tutors developed to support teachers in schools by directing them towards achieving educational goals [7]. One more recent tool, is described in [4] and is closely related to the approach proposed in this contribution. Here, the work describes a system that includes teaching evaluation indicators that combines traditional assessment scales with new values derived from the computer vision, audio speech recognition, and machine learning. while quite interesting, it provides only a comprehensive behavioural classification of the teacher, e.g. teacher's style (i.e. humorous, passionate, solemn), or teacher's media usage (i.e. multimedia vs blackboard). Research reported here proposes to take the teacher support, using already available AI-based methods, a step further. Specifically, the main goal is to improve the academic lecturing process by concentrating on lecturers' didactic behaviours, and other techniques that can be used to visualise the lecture topic. It is stipulated that this can be achieved by automating, summarising, and providing quick feedback to the teachers by using machine learning (ML) methods applied to the video recordings of their lectures.

The overarching research question is: is it possible to apply already existing ML-based methods to design a system that will have as its input a recording of a lecture, and as an output a set of recommendations for the teacher. To achieve this goal, the didactic features found in the lecture recordings are annotated, and machine learning models are trained to recognise them. Detailed experiments, dedicated to the selected ML models, are available in [15]. They have been used to set up the proposed approach, which can be summarized as follows:

1. establishing a set of didactic features feasible to annotate and detect automatically in lecture videos,
2. collecting a dataset of lecture video recordings (that contain the selected features),
3. designing a set of deep learning models trained to determine the presence of the selected didactic feature in the collected dataset – the video recordings.

This contribution more precisely describes the captured set of didactic features, and relates them to the teaching practices. Moreover, the reported research has been focused on designing and implementing a prototype, supported with

ML models, to automatically recognise the selected didactic features and to prepare summaries and visualisation for the lecturers. Another functionality of the implemented prototype is the possibility of collecting additional datasets and annotating them automatically. This, in turn, allows the proposed system to be further extended and fine-tuned.

The remainder of this work is structured as follows. Section 2 describes the recognized didactic features and the collected dataset. Next, Sects. 3, 4, and 5 introduce the system prototype and the results, including the interface for the feature visualisation. Finally, the work is summarized in Sect. 6. There the main findings, and limitations to be considered in the future research, are presented.

2 The Didactic Features and the Collected Dataset

Let us start the presentation by discussing the *didactic features* that have been selected and the reasons for their selection.

2.1 Didactic Features Captured in the Research

At first, the teaching practices have been thoroughly analysed, following the approaches described in [11,13]. This allowed to establish which features can be used as standards for, objectively, assessing the lessons. Next, a set of didactic features, particularly valuable from the teaching point of view have been selected, and adapted to the academic lecturing use case, considered in this work.

Obviously, it has to be possible to detect the selected didactic features, using the appropriately selected and adapted ML models, applied to the lecture video recordings. Figure 1 presents an example of frames that were extracted from lecture recordings (found in the collected dataset). Note that layouts of such frames often consist of two regions – one originates from the camera view focused on the lecturer, while the second one is a direct input from a computer screen (typically, presentation slides).

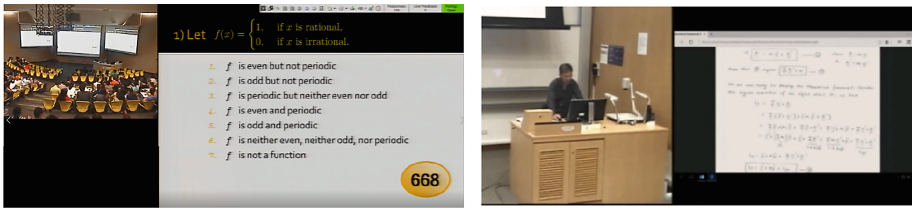

Fig. 1. Example frames from the recorded lectures in the collected dataset; this dataset has been also utilized in [15]

The results of analysis of pertinent literature have been summarized in Table 1. Here the selected features are presented and their correspondence to

the main categories of Singapore Teaching Practices taxonomy [13], e.g., "activating prior knowledge," "arousing interest", indicated. In particular the set of actions, performed by the lecturer, such as "giving class outline" or "writing on a whiteboard" has been selected. In what follows, these teachers' actions are also called the didactic features. Additionally, the slides' features that help understand the lecture, have been collected. Note that the qualitative and quantitative didactic features comprise (a) slide characteristics, and (b) different educational behaviours of teachers explaining scientific concepts, e.g. "plots on slides", "asking questions to the students", or "actively explaining on the board". Taking into account the goal of capturing didactic features using ML-based approaches, in the Table, the selected features have been divided into three categories: (1) the ones that can be automatically detected from a video stream, i.e. images or videos; (2) those that need to be detected only from the lecture audio stream; and (3) those features that can be detected based on both sources.

2.2 Collecting Dataset

After conceptualizing the didactic features, a dataset with those features has been collected and annotated. Preparing the annotation dataset is a challenging, complex, and time-consuming task. In the case reported here, the process started with designing a set of clear and understandable features (as described above), continued through annotator training and coding. Next, the annotation cleaning and verification ensued, and the process finished with selecting the gold standard for training and testing ML models. This required the combined efforts of multiple people and the application of programs and automation tools. In the annotation process, to facilitate annotations of life events (the features) occurring through time, BORIS – a free and open-source Behavioral Observation Research Interactive Software (BORIS is dedicated for video/audio coding and live observations) [3] has been used. (see, Fig. 2).

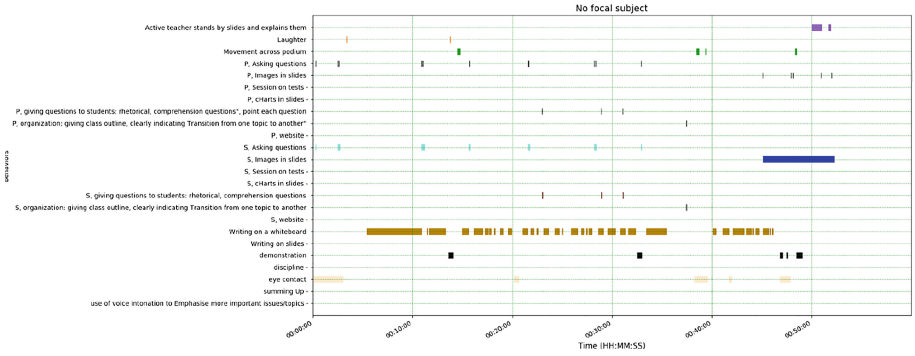

Fig. 2. Example annotations in the dataset, a view from BORIS; P – indicated point event, S – state event

Table 1. Set of didactic features implemented in this and the previous work [15]

Feature	Singapore Teaching Practices areas
Audio-based features	
Asking questions	Activating prior knowledge; Encouraging learner engagement; Using questions to deepen learning
Giving questions to students: rhetorical, comprehension questions	Using questions to deepen learning; Arousing interest; Providing clear explanation
Students are asking questions and generating their ideas	Facilitating collaborative learning; Empowering learners; Encouraging learner engagement
Laughter	Arousing interest
Discipline	Maintaining positive discipline
Students' discussion	Facilitating collaborative learning; Empowering learners; Encouraging learner engagement
Use of voice intonation to emphasize more important issues/topics	Encouraging learner engagement; Pacing and maintaining momentum
Visual features (that can be detected on slides or the view of the teacher)	
Films or animations in slides	Providing clear explanation; Arousing interest
Images in slides	Providing clear explanation; Arousing interest
Charts in slides	Providing clear explanation; Arousing interest
Showing websites	Providing clear explanation; Arousing interest
An active teacher stands by slides and explains them	Encouraging learner engagement
Movement across podium	Encouraging learner engagement
Writing on a whiteboard	Providing clear explanation; Arousing interest
Writing on slides	Providing clear explanation; Arousing interest
Demonstration	Arousing interest; Providing clear explanation
Eye contact	Encouraging learner engagement
Visual and/or audio-based features	
Referring to the bibliography, other research	Deciding on teaching aids and learning resources
Session on tests	Activating prior knowledge; Checking for understanding and providing feedback; Assessment
Giving hints on how to do something	Activating prior knowledge; Encouraging learner engagements
Organization: giving class outline, clearly indicating a transition from one topic to another	Setting expectations and routines; Establishing interaction and rapport, determining lesson objectives; Deciding on teaching aids and learning resources; Pacing and maintaining momentum
Assignments	Supporting self-directed learning; Setting meaningful assignments
Summing up	Concluding the lesson

Finally, the partner university (Nanyang Technological University in Singapore) provided 128 lectures related to the Physics discipline (delivered by multiple lecturers), recorded in English (on average, one lecture lasts an hour and a half).[1] At least three independent research assistants annotated every lecture. Each individual data element is a full-length annotation of didactic features present in a given lectures. Cumulatively, 380 observations were collected. Each feature was identified as either a state or point event. State events last a specific time, i.e., they have intervals with a precisely defined start time and end time. Point events occur at a given moment and show the changes in the behaviours or the occurrence of a feature, without its specific anchoring in a given time interval. An event means a single didactic feature in a given lecture, annotated by a research assistant. An observation is a set of events for one lecture annotated by one research assistant. An annotation is a set of all observations created for one lecture. The annotations were then used to train machine learning models.

In the next project phase, various methods how to pre-process and prepare the deep learning models to detect the selected features and how to design ML models to cope with them have been tried. It was necessary to prepare methods that allowed to split two sources of information within the recordings, i.e. the slide and the teacher views. Afterwards, both or separate views have been used to design ML models to apply to the visual features. Moreover, ML techniques needed to identify features in the audio streams have been selected. Moreover, transcription models have been applied to the collect texts – the audio stream of the lectures. Detailed results of tests run with all explored techniques have been reported in [15]. Here, the models have been improved with the latest state-of-the-art techniques, which have been utilize to build the system prototype. The overarching goal of the process was to design and implement an interface that can automatically help teachers to get feedback and to visualize their lectures so that they can derive ways to improve their performance.

3 System Design

In the proposed system design, it has been assumed that the automatically detected features will be used for the objective summary evaluation, to provide feedback for the academic teachers on their lecturing style and effectiveness. To achieve this goal the prototype's graphic interface allows to upload a lecture video and get a list of didactic features occurring within it. Overall, the developed application is organized in a way that does not require technical knowledge. The user should obtain the annotation of didactic features, view the visualisation of features detected by the model, and see which part of the lecture was automatically annotated. In addition, users should be able to see the visualisation of trends, based on all of the lectures uploaded and annotated in the previous sessions.

[1] We have an ethical acceptance from the NTU to use this dataset in the reported experiments.

The prototype's functional requirements were defined assuming that the developed application has two types of users: Lecturer (User) and Administrator. Without the need for ML knowledge, the lecturer uses the application to get feedback on their lecturers (see, Fig. 3). The Administrator, on the other hand, has machine learning knowledge and can change or upgrade the models, available in the system.

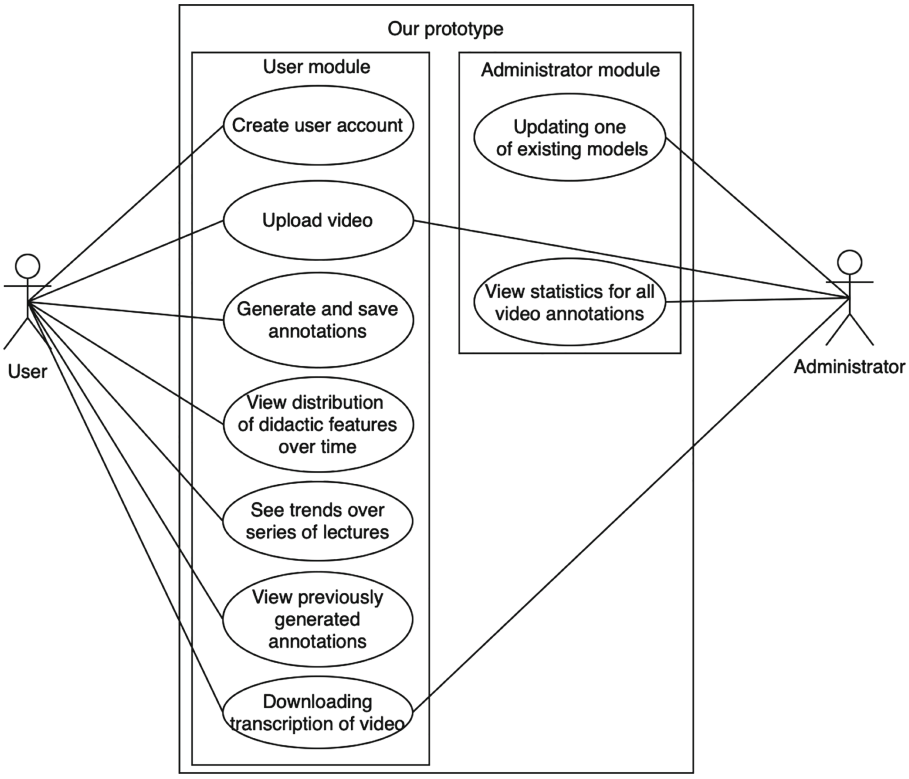

Fig. 3. System use case diagram

Moreover, the system was designed to fulfil the standard non-functional requirements for web applications, i.e. (i) usability – intuitive interface, widely-used programming standards to reuse the application source code, reliability; (ii) performance – short time of response for ML-models; and (iii) supportability – easy-to-change ML models, easy extending source code with up-to-date standards (e.g. Python, dash, plotly, pytorch, whisper, sqlite3).

The implemented prototype comprises six modules: *Controller, GUI, Preprocessing, ML Models, Database Connector* and *Database* (see, Fig. 4). The *Controller* module communicates with the other modules, and runs the application. The *Preprocessing* module transforms video into a format that can be passed to

the models that are being trained, e.g. a sequence of images, only audio stream. Preprocessing of the video consists of two steps: (A) the transcription is created with the `openai-whisper` package; (B) the JSON file with generated transcription is transformed into a data frame with the start and the end timestamps of the sentence. The *ML Models* module stores imported from pytorch binary files, with the model weights and other parameters, which are then used to create predictions. The *GUI* module is responsible for creating the graphic interface and handling input from the user. To make the interface, the `dash` framework,[2] was used, as it allows integration of interactive plots into the web application. The graphic interface allows the users to upload a new video, get transcription, and view the analysis of predicted didactic features. The *Database* module is where all datasets (including videos uploaded by the users) are stored.

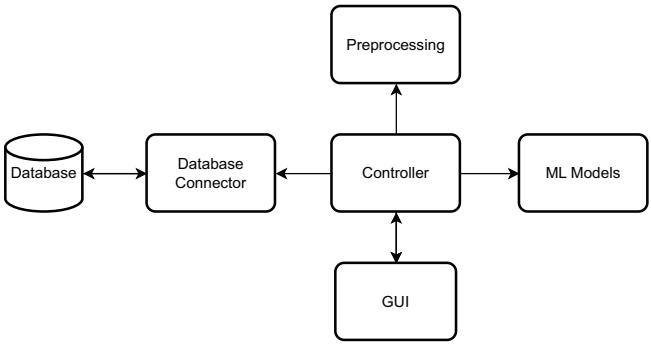

Fig. 4. Implemented prototype's design – module diagram

4 Machine Learning Module

In the developed prototype, the audio was used to create the transcription and, in subsequent steps, the transcription was used for building text-based models. Let us now describe it in some more detail.

4.1 Transcription Process

Even though the errors of the transcription models could be propagated to the classification model, the method based on transcription achieved generally better results than the method employing audio characteristics, and this transcription and training, or predicting, pipeline takes less computation power. The transcription results were tested on six short, manually transcribed lecture fragments. Ultimately, selected was the tool with the best word error rate (WER) metric results [9] – see Table 2.

[2] https://dash.plotly.com.

Table 2. WER on test dataset for lecture transcription

Deepspeech	Azure	Watson	Vosk	A-S-R	Jasper	GCP	Whisper
50.18	19.02	47.15	43.62	63.04	49.97	34.26	**16.81**

4.2 Text-Based Models

The transcriptions obtained from the existing transcriptions tools were used to train two machine learning models: for the binary classification task, noting whether the given sentence contains a question, and for the multiclass classification task, which marks all relevant text-based didactic features.

The state-of-the-art transfer learning with Transformer BERT-based models provided by HuggingFace [8][3], was used. The dataset classification tasks are selected as the downstream tasks each time the training procedure and the pretraining are performed on another, larger dataset, the BookCorpus dataset [17][4]. A feed-forward neural network head is added at the end of the Transformers, and predicts classes from the dataset. Default uncased BERT-Base model hyperparameters were not changed during the hyperparameter optimization process, and instead, the focus was on fine-tuning the last layer, the feed-forward network.

The models used for the recognition of features regarding questions achieved a precision of about 20%, recall of about 50%, and F1 score of about 30%. It is worth mentioning that the model recognised six features at once, so the random results would be about 16%. Hence, the obtained results seem reasonable, mainly because the dataset was relatively small, and the task was challenging to annotate.

5 System Interface

The implemented prototype automatically produced summaries of the detected didactic features. Figure 5 shows an interface with the results for a single lecture recording with text-based features, e.g. 'asking questions' and 'organizational issues'. This visualisation contains a number of occurrences of each feature during a given lecture and varying time spans for each feature. The first plot is a bar graph with the number of occurrences of every feature. The second plot's x-axis shows time in minutes; on the y-axis, there are didactic features. Every point represents a single occurrence of a feature in the lecture.

The user can also see and download the transcription results combined with the results of the text-based models, showing in which sentences any text-based features occurred. This functionality is available when the Text or the Questions model is selected. A dropdown list with features that have been recognized in the selected lecture(s) is displayed. After selecting one, the user can see a table

[3] https://huggingface.co/roberta-base.
[4] https://huggingface.co/datasets/bookcorpus.

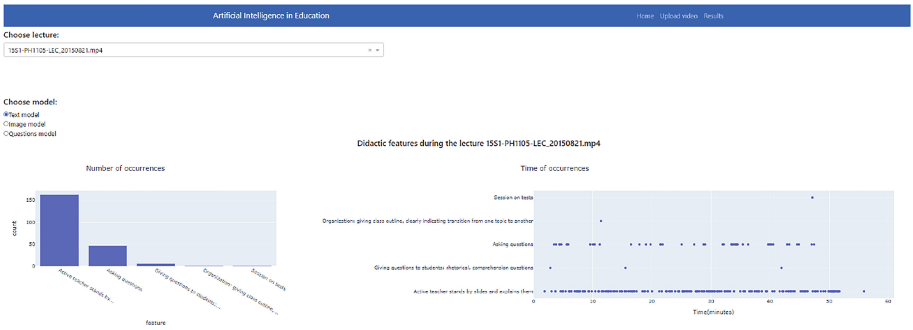

Fig. 5. Displaying plots with a summary of didactic features that appeared during the selected lecture: number and time of occurrences of features predicted by the ML model

with all moments where the model predicted the occurrence of the feature, with text, time of start, and time of the end of each occurrence. An example of a table look is shown in Fig. 6.

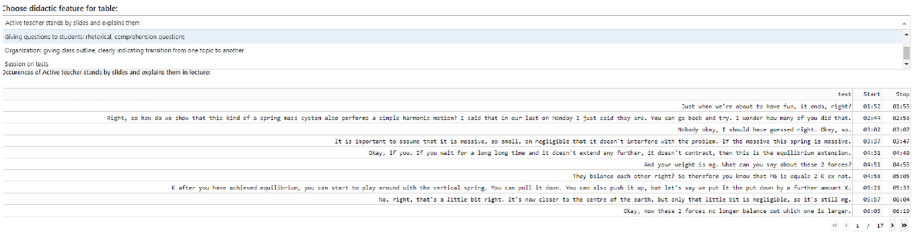

Fig. 6. Displaying table with text fragments connected to chosen didactic feature and time interval of its occurrence.

6 Concluding Remarks

The goal of the reported research was to develop and implement a prototype of an academic lecturer support system, based on application of ML-models to the lecture video recordings. To achieve this goal a set of didactic features was selected and adapted to objectively and quantitatively evaluate and compare academic lectures. Moreover, the implemented prototype supports automating the annotation process of lecture video recordings using pre-trained machine-learning models. Utilizing these annotations, summaries and visualizations of the didactic features observed within the lectures have been created. The developed system is anchored in in-depth analysis of required system's functionalities, which guided the design of its architecture, dataflow, and interfaces. The modularity of the designed system allows easy addition of ne ML models, if such

models will be found to be promising for the task at hand. Obtained results, validating the approach, are reasonable, taking into account the size of the dataset used for model training.

The primary target of future research has to be collection of a larger dataset, to be used for model training. Here, what is worthy repeating is that the developed prototype can be used to form a partially-supervised continual learning loop. Specifically, the existing system can be used for preliminary didactic feature extraction, helping annotate future videos (human-in-the loop). These videos can then be used to up-train the model(s).

The dataset and the software of the system is available upon request from the corresponding author. However, note that restrictions related to the GDPR (and other pertinent regulations) apply, and applicable documents will guide granting access to both the dataset and the code.

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Eliciting and Empowering Teachers' AI Literacy: The Devil is in the Detail

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Abstract. An emerging challenge from the rapid integration of Artificial Intelligence (AI) in society and education is how to prepare teachers for this evolving reality. Teachers need to integrate Generative AI (GenAI) into their teaching practices and educate students about AI within their specific teaching contexts. Although several frameworks have been published to guide AI literacy education, these frameworks often remain abstract and disconnected from the actual teaching environments where AI literacy needs to be applied. This disconnect makes it difficult to link the theoretical knowledge proposed in these AI literacy frameworks to the specific AI-related requirements that teachers face. Pinpointing AI literacy aspects relevant to teachers' specific teaching contexts and prior knowledge is needed for guiding teachers to relevant Continuous Professional Development (CPD) and supporting materials. Thus, to address these challenges we have developed a method that has recently been tested and piloted with teachers. The results of our efforts suggest that the tested method effectively triggers discussions about using GenAI in teaching contexts. Distinct patterns of interaction with the GenAI application emerged, which are discussed in the paper with the long-term goal of developing tools to identify misconceptions about AI and enhance understanding.

Keywords: AI literacy · Teacher education · Generative AI · AI education · AI readiness

1 Introduction

AI literacy is a concept that commonly refers to essential skills and knowledge needed in a world where AI is ubiquitously embedded [1–4]. AI is becoming integrated into many tools and services used in our daily lives, having a broad impact

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on society. This growing influence has led to calls for introducing AI literacy in K-12 education to prepare pupils for the demands of this new reality. Although AI literacy has been researched for quite some time it has so far remained an academic topic with little impact on teaching and learning [1,3–7]. However, the public release of Generative AI in the form of OpenAI's ChatGPT in November 2022 changed this since the implications for pedagogy suddenly became tangible. Challenges in dealing with the implications of AI for assessment and teaching practices suddenly became practical issues for teachers.

Frameworks outlining competencies and skills required to become AI literate have been widely proposed, along with suggested methods for how to teach these skills [8]. Recent studies, however, highlight the gap between the knowledge and skills suggested by these frameworks and teachers' actual knowledge and understanding of AI and their opportunities to achieve such knowledge [6,9,10]. Due to vague definitions of the broader concept of digital competence/literacy and lack of inclusion in teacher education and current policy and curricula, there is a lack of support to teach AI literacy (Örtegren & Velander, 2023). Furthermore, teachers are often not aware of what AI-related knowledge is relevant to their specific teaching. To complicate matters further there are many (pre)conceptions and misunderstandings about AI which can be partly linked to the recent AI hype in media and beyond which have allowed for strong emotional responses and attitudes to the topic [6].

As most AI-supported systems today are data-driven, the concept of AI is often used interchangeably with Machine Learning (ML) and Large Language Models (LLMs). Due to the immediate implications of ML and Natural language processing (NLP) for teaching [11], this study focus particularly on Generative AI to explore how these methods affect current research and frameworks for AI literacy.

This paper presents an exploratory study set in Sweden, investigating a scenario-based method for eliciting teachers' knowledge and knowledge gaps of AI related to different teaching contexts. Workshops were designed based on our previous research efforts to understand teachers' knowledge and knowledge gaps [12] and a questionnaire on teachers AI awareness and attitudes. This paper reports on the results of piloting the workshop design and contributes to two pressing calls regarding AI literacy specifically considering generative AI. The first is a methodological challenge of eliciting teachers' knowledge, understanding, and emotions related to ill-defined concepts such as AI. Secondly, our pilot study will reveal how teachers' knowledge and knowledge gaps using generative AI can be elicited, understood, and related to the current theoretical AI literacy frameworks. The following research question was formulated to guide this study:

> *What interactions and topics emerge when teachers are asked to create teaching scenarios using LLMs? Does this scenario creation task help teachers reflect critically on AI and its use in educational contexts?*

2 Related Research

In this section, research concerning the conceptual understanding of AI literacy is presented as we are trying to connect abstract conceptual understanding with its practical implications for teaching and learning. As the study ultimately concerns teachers and teacher education, we also review existing literature that explores AI literacy in teacher education.

2.1 AI Literacy - Conceptualizations in Research and Policy

There has been a flurry of activity in the research area related to AI and education [13]. Many publications propose guidelines and frameworks for AI literacy but often leave the concept of literacy unidentified [1,3,8]. In response, recent studies have investigated underlying assumptions and narratives present in AI literacy research. Results show that AI literacy is often presented as a set of skills required for individuals to be able to adapt to a future where AI is increasingly pervasive, to be responsible users of AI, and to possess skills demanded by a future labour market [6,13]. Such skills align with a functional literacy perspective [14] but leave out a critical perspective on literacy which acknowledge the socially constructed nature of meaning systems [14]. AI technology is embedded in society and thus creates issues related to power, agency, and control [5,15–17].

2.2 Eliciting Teachers AI Knowledge(gaps)

AI literacy in teacher education has to date been sparsely researched. A recent scoping review of AI literacy related and relevant to teacher education revealed that teachers' knowledge of AI needs to be understood properly and there is a need to engage in and define AI literacy in different educational contexts [13]. This identified gap is particularly urgent since previous research highlights the importance of teachers' role in informing content and methods for teaching AI literacy [9,13,18]. Understanding teachers' *current* knowledge, perceptions, and attitudes is important for informing teacher education and professional development [6]. Identifying teachers' *expected* knowledge to inform curricula and policy is equally important since this guides teacher education content and methods. Several studies aim to assess teachers knowledge and attitudes about AI [6,9,19]. Few of those studies consider the wide and varied conceptualisations of AI that their participants have. Such an issue became apparent in our previous study where both questionnaires and focus group discussions were used to elicit participants understanding, emotions and attitudes towards AI [6]. The different methods yielded different results as the focus group discussions revealed insecurities and questions about AI only after deeper discussions. The insights gained made us reflect on methods often used to elicit teachers technical, pedagogical and content knowledge related to ill defined and broad concepts such as AI and digital literacy. Based on the results of implementing these different methods we suggested an alternative methodology using scenario based problem solving

focus groups to elicit a deeper understanding of teachers AI literacy, which is informing this study [12].

In the coming section we present and discuss our study design and proposed method to explore the RQ described in Sect. 1.

3 Study Design and Proposed Method

In this study, we pilot a method for eliciting teachers' AI literacy related to different teaching contexts. This method aims at connecting with teachers' previous understanding and conceptualizations of AI and relating this to their teaching context. We draw on our previous research presented in Sect. 2.2 [6] and empirical research results from a questionnaire to teachers about generative AI in education as described below. In the following section we present the rationale for and the results of our pilot workshop.

3.1 Questionnaire Informing Design of Workshop Design

A demand emerging from the Swedish educational context prompted upper secondary high schools from a region in the north of Sweden to turn to the local University for guidance concerning the very tangible effects of Gen AI for education. More specifically, the Computer Science Department and the Software Engineering and Education Group at Mid Sweden University responded to this call by offering a presentation for teachers. To guide the design of the presentation and align with teachers' previous knowledge and specific interests in Generative AI a questionnaire was sent out to enrolled participants. The questionnaire was answered by 64 upper secondary high school teachers. The survey used open-ended questions to probe for teachers' experience of using AI, challenges they have faced related to AI and especially generative AI, their hopes and fears for using AI in education, and what knowledge they feel they would like to have to be able to teach *with* and *about* AI. Multiple choice questions were also used to gather information about teachers' experience and background. The results emerging from this effort informed the workshop conducted and as such we briefly report on the results of this questionnaire in the following sections.

Respondents. 64 active teachers returned complete responses whereof 20 had worked as teachers between 0–10 years, 20 between 11–20 years and 24 had been teaching for more than 20 years. Teachers from diverse subjects were represented with a majority (42) teaching art, language and social science subjects, 18 teaching STEM subjects and 4 teaching vocational upper secondary education.

Preliminary Results. The data from the questionnaire was analyzed using inductive content analysis. A detailed description of the methodology and analysis is beyond the scope of the present paper (in fact, an ongoing detailed analysis will lead to the creation of a second version of the questionnaire that aims

at collecting data from a larger sample; for example, topics emerging from the open-ended questions will be converted into multiple-choice questions and similar other formats to facilitate a thorough quantitative analysis). Here we focus on some preliminary results and insights that contributed to frame the workshop described in Sub-Sect. 3.2. The questionnaire revealed several challenges teachers face due to the public access to AI, especially generative AI which is briefly outlined here. A vast majority of teachers gained their AI-related knowledge from incidental sources such as social media, news and conversations whilst only 18% had some formal or intentional sources such as courses and PCD.

Many teachers (11) claimed they do not use AI at all while 12 stated that they don't use it intentionally. Those that do use it intentionally do so as they interact with smart devices (6), generate content (7) and a minority for inspiration(1). Immediate concerns such as assessment practices, cheating, and plagiarism were highlighted but also issues related to cognition and interaction with AI. The wider implications of AI for society, politics, and democracy are also highlighted as important. Teachers want to know what knowledge they should possess to be able to teach about these implications in a way relevant to students' lives.

3.2 Designing a Workshop for Educators: Promoting Critical Reflection by Incorporating LLMs Tools

To promote a deeper reflection about the actual use of Generative AI in educational contexts we designed a workshop that incorporated the use of a LLM tool in the creation of a learning scenario. In the **first part** of the workshop, we presented different AI-related discussion topics. These topics were based on the analysis from the questionnaire mentioned in the previous sub-section to ensure grounding to AI issues that teachers find challenging, urgent, and interesting to know more about.

In the **second part** of the workshop, the participants were divided into smaller groups with 2 to 3 people per group. They got 30 min to discuss the different topics and define which one(s) they felt most relevant to them, and based on this create teaching scenarios. These scenarios were to be created in collaboration with a free version of an LLM-based tool - Copilot. Participants were scaffolded in their scenario building by inspiration/scaffolding cards suggesting them to consider for example: a) educational goals present in the scenario, b) potential integration in the overall curricula, b) operational teaching goals, c) resources needed to address these educational goals, d) what forms of assessment and evaluation seem suitable to understand students' progression? The **Data** collected was: prompt history and final agreed teaching-related scenario, screen and audio recordings from each group.

In the **third part**, the groups got 30 min to analyse and criticise the scenarios they had co-created with Copilot. More specifically they were also asked to reflect on: **Epistemic issues**

- What knowledge is relevant to have in light of generative AI?
- What motivation is there for learning and understanding things?

- AI for creative practices - fostering exploration or losing creativity?
- Critical capacity - reflecting on and questioning things?
- What does it mean to understand something vs knowing something?
- Digital divide - knowledge gaps between those aware of AI and how to use it and those not.

Societal and Political

- Influence on politics and opinion formation, art, and the view of the written word.

The **Data** collected was: audio/video recording and notes from discussions (the groups were asked to keep notes from their discussions). In the **final part**, all the groups gather again for 30 min for a final discussion including a presentation of each group's scenario and their reflections from parts 2 and 3 followed by a group discussion of the findings. The Data collected was: audio and video recordings of the overall group discussion.

3.3 Participants and Data

The pilot workshop was arranged at a university in Sweden during the early Spring term 2024. Seven participants from various academic disciplines attended, all with many years of teaching experience in higher education and facing similar challenges as K-12 teachers regarding generative AI. The workshop lasted for 2 h and the participants were all physically present in the same room. Informed consent was collected from all participants and data was gathered by audio and screen recording, prompt history, and field notes.

3.4 Method for Analysis

Considering the exploratory nature of the study we remained flexible in our approach of analysis to iteratively discover what the data means and how this can be represented and supported by existing theory. As our data is qualitative, we leaned on Qualitative Content Analysis which allows some quantification of occurrences of themes and permits us to be iterative in our approach to connect with existing theory whilst expanding on and building on these. An unconstrained matrix was used to allow principles of inductive content analysis [20]. In the final step, two of the authors consulted with each other to seek agreement on how the data was labelled [21]. The method we used is described in detail in the following sections.

3.3.1 Qualitative Content Analysis

Qualitative Content Analysis (QCA) was used for analyzing the data captured during the workshops. In the first step, one of the authors transcribed the audio-recorded and screen-recorded prompt history captured during the scenario creation by all three groups, and the audio recorded from when the groups presented their reflections and discussed together at the end. In this step, the author immersed herself in the data entailing several readings. With proper familiarity with the data, we identify episodes for coding based on the cycles of interaction between participants, participants and the tool, output of the tool. Cycles of interaction here mean instances where one can see the beginning and end of an interaction episode (talking to each other, writing a prompt, analyzing a prompt). These cycles also represent the unit of analysis throughout the entire analytical process. More specifically the following categories and subcategories helped indicate the cycles of interaction:

- Teachers reflecting on topics
- Teachers writing prompts
- Teachers discussing output
  - Positive attitudes - the participants express comments/opinions that are favourable to the tool and output.
  - Negative attitudes - the participants express comments/opinions that are unfavourable to the tool and output.
  - Challenging attitude - the participants express comments/opinions that challenge the information provided by the tool.
  - Inquisitive comments - the output from the tool makes participants question their knowledge.
  - Teasing comments - the output of the tool makes participants joke and sarcastically elaborate on the information provided.
- Follow-up prompts
  - Refining prompts
  - Teasing prompts
  - Abandoning the line of inquiry
- New Topic

In the second step, a deductive approach was taken based on the perspectives of the cycles of interaction. Here we connect with the existing theory by [2] where the authors present three main perspectives related to different lenses to approach AI literacy competencies presented in AI literacy frameworks. Figure 1 is a modification of how Markauskaite et al., position these aspects that they argue are important for a holistic approach to AI literacy. AI literacy competencies (in this study represented by the framework Fig. 1) are the core of AI literacy however connecting with these competencies through different perspectives takes into consideration the context in which these competencies are relevant. We look at the perspectives and code according to the AI literacy framework by [22] to reveal related AI literacy core competencies. This final step helps highlight skills and knowledge directly attributed to AI such as How AI works - data

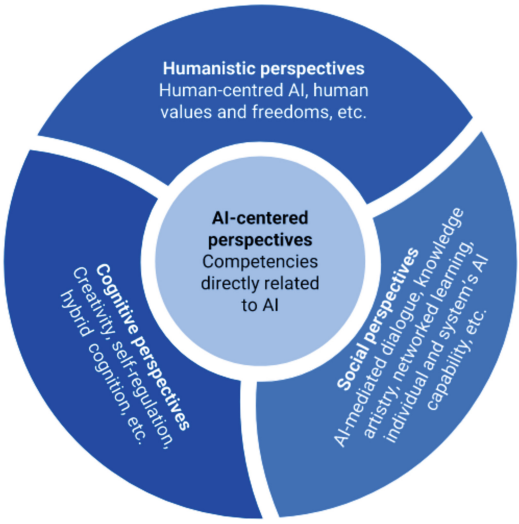

Fig. 1. Perspectives on AI literacy core competencies as presented by [2]

representation and decision-making. The framework by Long and Magerko has subcategories for each main AI literacy construct. These are also coded since different perspectives and teaching contexts such as age group, school subject, and curricular activity will alter the focus and inclusion of these specific AI literacy competencies:

- **What is AI?** - Recognising AI, Understanding Intelligence, Interdisciplinary, General vs. narrow AI
- **What can AI do?** - AI's strength and weaknesses, Imagine future AI
- **How does AI work?** - Representations, Decision making, ML steps, Human role in AI, Data literacy, Learning from data, Critically interpreting data, Action and reaction, Sensors
- **How should AI be used?** - Ethics
- **How do people perceive AI?** - Programmability

4 Results

This section describes the results starting with presenting expected vs. actual outcomes using the workshop design. We then report on the results from analyzing the data from the group's scenario creations and the final group discussion. Unfortunately, due to some technical issues one screen-recording for one of the groups (2 participants) was compromised, thus the scenario creation consists of data from two groups.

4.1 Expected vs. Actual Outcomes

Although the workshop task explicitly asked the participants to create a teaching scenario in collaboration with Copilot this outcome was not always achieved. The groups approached the task differently, as a result, the data generated revealed very varied patterns of interactions with the system. The different approaches reveal different perspectives on how to connect with and make sense of the related core AI literacy skills [22].

4.2 Results from Analyzing the Teaching Scenario of Group 1

This group consisted of two people, one professor teaching in higher education and one lecturer in history currently preparing to teach preschool teacher students about the Stone-age. The group decided to create a teaching scenario around this topic. This group displays an interactive and reflective approach using the tool throughout the scenario creation. They constantly and critically examine the output produced by Copilot. The data from their scenario creation is therefore rich in detail. However, the group did not produce a comprehensive teaching scenario as the workshop task outlined but instead got carried away in their close collaboration with the tool. The interactions presented in Table 1. are representative of their collaboration with Copilot. Where the group doesn't possess the necessary knowledge to assess the reliability of the output they follow up with a teasing prompt to see how it answers something easy to fact-check such as the last prompt in Table 1 which is available from this link: Link to Table 1. We first identify the units of analysis which are the interaction cycles between prompts. Table 1. provides a snapshot of the interaction cycle while the group created the teaching scenario. In the next step, these units of analysis are coded according to the three perspectives: humanistically, cognitively, and socially oriented perspectives [2]. This coding exercise tries to bring together the two conceptual frameworks considered, [2, 22]. At the center, we find the core AI literacy constructs from the framework by [14], and at the fringes, the perspectives from which to approach these AI literacy constructs. Figure 2 is a visual representation of the coding according to the method used. As the QCA allowed us to quantify occurrences and co-occurrences of perspectives and AI literacy construct a hierarchical structure of the data emerged and is represented as a sunburst graph Fig. 2. A link to a full size version of the image can be found here: Link to Sunburst graphic for group 1.

4.3 Results from Analysing the Teaching Scenario of Group 2

This group consisted of three participants. Two post-doctoral researchers from the arts and humanities and one doctoral student from computer science. All have teaching experience at a higher education level and one of them has teaching experience from primary school education in history. This group had a different approach to the workshop tasks and went through these step by step in sequential order with limited reflection on the outputs from the system as the teaching

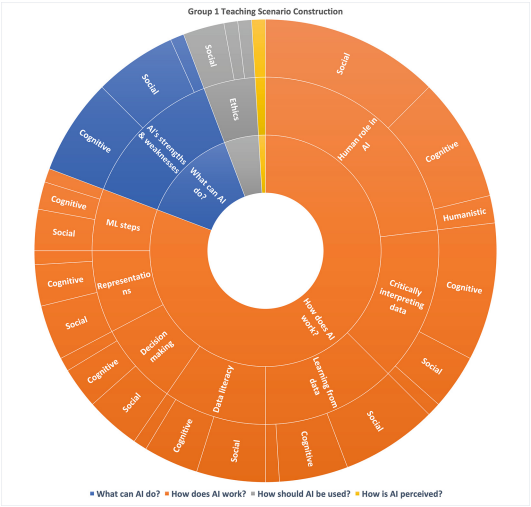

Fig. 2. A sunburst graphical representation of the hierarchical structure of the coded data from the teaching scenario created by group 1.

scenario was produced. The group did produce a teaching scenario as an outcome of the interactions with the system. This means that the workshop tasks in this sense were completed. We first identify the units of analysis which are the interaction cycles between prompts. Table 2 provides a representative snapshot of the interaction cycle while the group created the teaching scenario. Link to Table 2. This group takes a less interactive approach and limits their reflections on the system output to the time it takes Copilot to process their prompts. Once the response is presented in full the group moves on to the next task of the workshop leaving the output response unaltered and unchallenged. After prompting for all the workshop tasks of creating a teaching scenario the group finally prompts the system to summarise all previous responses to one teaching scenario. This scenario remains unchallenged as well, however the group spends the remaining time discussing the implications of using generative AI in education. This group accepts the responses from the model and based on the previous response prompts the model to answer their next workshop task. This builds on the previous response which still has not been critically evaluated. Therefore this approach provides less critical reflection and collaboration with the system. The model is in charge of the direction of the conversation, shaping and affecting the entire teaching scenario. The final prompt: "Could you summarise all the previously stated in one teaching plan I could show to others?" is the result of their completed task. Since the group "completed" the workshop task so fast they spent a lot of time reflecting on the use of generative AI in education in general and not related to the output from the model which remains largely unread. A full size version of the graph can be found here: Link to Sunburst graphic group 2.

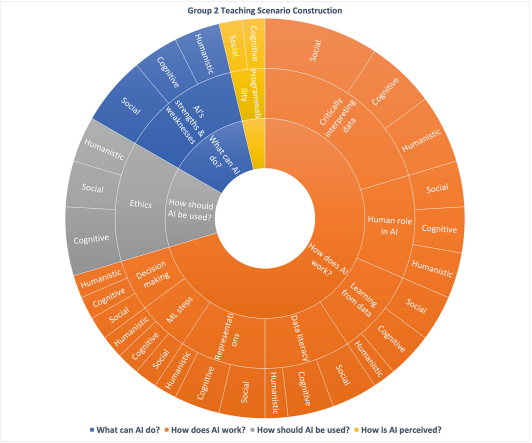

Fig. 3. Figure 3. A sunburst graphical representation of the hierarchical structure of the coded data from the teaching scenario created by group 2.

4.4 Final Group Presentations and Discussions

The last 40 min of the workshop were dedicated to the groups presenting their teaching scenarios and their critical reflections on these.

- Challenges - Accuracy especially in details, Context and concept representation, Lack of transparency related to sources, Lack of control and oversight, Time waster to analyse and assess output
- Possibilities - Ideas and inspiration, Need expert supervision, Time saver?

The hope for teachers to save time which was the most frequently expressed hope by respondents to the questionnaire as well as the participants was questioned after the workshop. The potential time saved by using the system would have to be relocated to time spent reviewing, adapting and complementing the output. The groups further discussed intelligence related to the present and future with AI. Embodied cognition was suggested as one thing that separates current human intelligence from artificial intelligence. Also highlighted was the "filler" provided by the system. This refers to the peripheral details of content such as text and images that fill out the space between what is specified in the prompts. This discussion was related to how subtle presentation of content in different contexts might shape our collective meaning-making and perceptions. Other cognitively oriented perspectives discussed were the motivation to learn considering current assessment practice in education. Participants highlighted that current assessment practices often evaluate the result rather than the process of learning. The motivation for understanding something is therefore potentially lost in favour of knowing how to prompt. One group observed that knowing how to prompt might change writing skills since it is valuable to know how to be precise and descriptive in prompting in order to receive useful responses from a system.

5 Discussion

This section concludes our paper by discussing the results presented in the previous section and its relation with the research question formulated in Sect. 1. We end our paper by briefly stating our coming research efforts and concluding remarks.

5.1 Reviewing and Discussing the Results as Indicators of Strengths and Weaknesses to Guide the Refinement of the Method

The analysis of the data collected using our workshop design suggests that the cooperation with an LLM, in this case, Copilot, helped reveal both attitudes and preconceptions about AI. Scenario creation anchored the collaboration with specific teaching practices and contexts which reveal relevant aspects of AI literacy. What is more, it revealed different perspectives of looking at AI literacy constructs present in current AI literacy frameworks. This is a positive outcome considering our previous difficulties in eliciting teachers' knowledge through questionnaires [12].

The different ways the groups approached the workshop task resulted in different interaction patterns, in turn resulting in distinct outcomes. This will be considered in the revised description of the workshop tasks which will be used in the larger study contributing to a clearer structure and expected results. We hope that these efforts will lead to other researchers being able to utilize our methodology.

5.2 Reviewing and Discussing the Method and Framework for Analyzing the Data Related to Teaching Context

Throughout the scenario creation and discussions, reflections on the implications for education of both responses from the system and the collaboration with the system were highlighted. To further make sense of this data for practical guidance for teachers these educational perspectives should be noted and considered as they could further indicate teachers' preparedness for teaching with and about AI. Estimating teachers' progression towards AI-related TPACK using an existing model relating TPACK progression to practical teaching transformation from Passive, Operational, Progressive and finally, Epistemological [23].

5.3 Eliciting Conceptual Knowledge and Guiding Teachers in Gaining Relevant AI Literacy

The overarching research goal of the project this study is part of is to explore the space between theory and practice regarding AI literacy. On the one hand, teachers express the importance of integrating AI-related competencies in their teaching as many teachers experience the implications of AI in their professional lives as well as in their pupil's lives. There is a wish to be able to connect with

pupils' experiences and concerns about AI and be able to connect these often ethical issues with more technical explanations of what makes this possible and how this might affect us [6]. Considering these practical issues for education there is a need to connect AI-related teaching in practice with relevant knowledge presented in frameworks. Analyzing the data using both perspectives and the AI literacy framework reveals a challenge for promoting critical AI literacy as the framework presents ethics only in the context of how AI should be used. Our coding reveals that ethical issues represented are broadly related and relevant to all AI literacy constructs. An example is ethical issues of values and their embeddedness in the AI literacy construct How AI works such as data representation, the human role in AI, learning from data, and data literacy. To foster critical engagement this aspect is vital as it enables questioning of current AI practices which otherwise will be left out. Confining ethical issues to how AI should be used exacerbates the already heavy burden of individuals to alter their behaviours to account for current AI practices.

5.4 Limitations and Future Research

This study reports on the outcomes of conducting a pilot workshop and is as such limited both in terms of the sample size and hence the genralisability of the results. However, it is an important step towards our planned full scale study in which teacher educators and teacher students will participate in a refined version of this workshop. The purpose of the larger study is twofold, first it aims to foster critical reflection in teachers about the use of GenAI in teaching contexts and to contribute to their AI literacy. Secondly, the data gathered will be important for research aiming to guide teachers towards relevant AI related education and support material. As teachers often feel that they need to update their skills related to AI but are not sure of what knoweledge they need this will be vital in an area such as AI which is in constant development and flux.

6 Concluding Remarks

Regarding our research question, the findings indicate that the task of creating teaching scenarios with LLMs revealed teachers' attitudes and preconceptions about AI, thereby providing valuable insights into AI literacy in the context of authentic educational settings. These findings also highlighted different interaction patterns, resulting in distinct outcomes. Reflections on the implications for education and teachers' readiness for teaching with and about AI emerged during the discussions. Furthermore, our analysis underscores the importance of aligning AI-related teaching with relevant knowledge frameworks and ethical considerations, grounded in actual pedagogical practices and specific subject matter. Embracing this holistic approach holds the promise of cultivating robust AI literacy in education, equipping both educators and students with the capacity to navigate the intricacies of AI with ethical discernment and practical comprehension.

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ChatGPT as a Learning Aid for an Introductory Programming Course

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Abstract. For the past years, technology has greatly improved—being able to perform tasks never envisioned before. Artificial intelligence (AI) provides opportunities in several fields as an assistant that could mimic human interaction. In this context, artificial intelligence offers a novel way to revolutionize educational delivery through various pedagogical methods. As a main aspect of education, practical learning could be enhanced with the intervention of AI tools. ChatGPT is a rising large language model (LLM), showing good potential for educational use, specifically in programming learning. This study aims to determine the capabilities of ChatGPT as a learning aid for secondary programming topics. The study uses qualitative and quantitative analyses of user experience and expert evaluation. The results revealed that ChatGPT performs satisfactorily as a programming learning aid with an overall weighted mean of 3.83 over 5.00 for all evaluation metrics.

Keywords: Chatbot · AI in Programming · Programming Assessment · Programming Education · Programming Learning Aid

1 Introduction

Programming has emerged as a modern skill, prompting educational institutions to equip learners with such capabilities. [4] demonstrates AI's rapid integration and growth in the education sector. Through personalized learning [3,4,9], intelligent tutoring and feedback [2,8], and administrative efficiency [1,4], AI revolutionizes learning. ChatGPT, an AI language model, has gained prominence in programming education, where it automates assessment and feedback.

Numerous studies have analyzed ChatGPT as a supplement to programming. However, its ability to provide exercises and evaluate solutions was not thoroughly assessed. As learning aids, [14] found that chatbots helped students who are hesitant to ask queries during face-to-face sessions. However, the research lacked student involvement and feedback. [5] assessed Python-bot and found that it can assist new programmers. [11] identified and categorized programming

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assessment tools, suggesting a predominance of automation tools. Moreover, [12] noted that ChatGPT performs well in debugging tasks due to its code-analyzing ability. [13] further supported this, asserting that it provides good code output for problems of varying difficulty. However, possible biases arise due to its training data being sourced from existing solutions. Furthermore, [10] evaluated the performance of large language models in generating programming exercises and code explanations, revealing that the majority were sensible, novel, and included an appropriate sample solution.

With these, [6] and [15] predict the decline of standard assessment practices considering the thoroughness of research and adequacy of ChatGPT's responses. However, this raises concerns about its potential misuse and generation of erroneous information, undermining academic integrity [7]. Thus, this study aims to explore ChatGPT's ability to create programming exercises, assess code solutions, and provide feedback using Python. Educational institutions could leverage the findings to enhance programming courses.

To evaluate the potential use of ChatGPT as a programming learning aid, the research aims:

1. to generate prompts to obtain appropriate programming exercises from ChatGPT, specifically for Empowerment Technologies 1 (ET-COTH1), an introductory programming course at De La Salle University-Integrated School (DLSU-IS) Manila, with a focus on conditional and iterative statements;
2. to determine the performance of ChatGPT in generating programming exercises for an introductory programming course. The problems generated will be categorized by difficulty, addressing different levels of complexity in both conditional and iterative statements; and
3. to evaluate ChatGPT as a learning aid through a user experience survey and an expert analysis of its responses with a determined evaluation criteria.

2 Methodology

Figure 1 illustrates a systematic approach to evaluating ChatGPT as a learning aid. Starting with prompt generation, specific prompts are formulated to obtain programming exercises. These prompts undergo verification to ensure relevance and accuracy. Subsequently, user experience surveys are conducted to gather feedback from students interacting with ChatGPT. Expert evaluation surveys are employed to assess ChatGPT's performance from an expert perspective. Moreover, a checklist is utilized to verify if the exercises comply with the restrictions given. Finally, the results of the surveys are summarized.

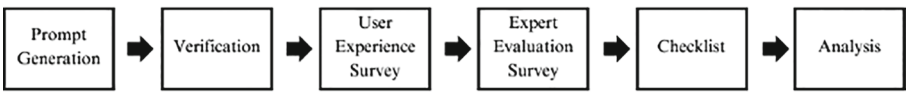

Fig. 1. Research process.

2.1 Research Instruments

In the study, two main surveys were used: a student survey and an expert survey.

The student survey aims to gather quantitative data on user experience. One survey questionnaire is created for each topic included in the study. The student respondents follow a structured process that involves sending a prompt, receiving a programming exercise, and submitting a solution, repeated for each difficulty level.

The expert survey is an evaluation survey of the conversation logs submitted in the student survey. Each evaluator is assigned to review five distinct conversation logs from the 20 survey responses. It aims to gather quantitative data through Likert scale responses and qualitative data through essay responses encompassing the predetermined evaluation criteria as summarized in Table 1.

Table 1. Evaluation criteria.

Evaluation Metric	Description
Clarity of Problem Posing	The quality of being understandable and intelligible. This is how understandable the problems (programming exercises) presented by ChatGPT are
Correctness of Evaluation	ChatGPT correctly identifies whether the submitted code generates the expected output or not
Depth and Understandability of Feedback	The feedback sufficiently explains the solution and includes suggestions for potential improvements
Correctness of Feedback	The feedback provides a correct explanation of the logical and syntactical errors found in the solutions

Finally, a basic researcher checklist consisting of yes-or-no questions is employed to check if the exercises generated are compliant with the restrictions given.

2.2 Data Analysis

This research employs both quantitative and qualitative analyses.

Quantitative data comprise ratings from 5-point Likert scale surveys completed by students and experts, as well as checklist responses. Frequency indicates the total number of responses for each survey item, while weighted mean calculates the average of all weighted values. Percentage measures the relative values of counted quantities, particularly for yes-no checklist questions.

Qualitative data consist of comments and explanations from expert analysis and user experience surveys. Comments are categorized based on their nature and similarity and are later analyzed and summarized accordingly.

3 Results

3.1 Prompt Engineering

Experiments with ChatGPT are conducted to determine the appropriate prompts. The specifications on the topic (conditional statements, iterative statements) and the difficulty level (easy, moderate, difficult) are also taken into consideration.

Table 2. Restriction 1: Topic specification.

Topic	Restriction 1
Conditional statements	The problem should be about conditional statements only
Iterative statements	The problem should be about iterative statements only

Table 3. Restriction 2: Difficulty level specification.

Topic	Difficulty	Restriction 2
Conditional statements	Easy	It should be an easy problem: Involves simple conditional statements with at most two cases
	Moderate	It should be a moderate problem: Involves simple conditional statements which have more than two cases without condition nesting.
	Difficult	It should be a difficult problem: Involves simple conditional statements which has more than two cases with incorporation of condition nesting
Iterative statements	Easy	It should be an easy problem: Involves simple iteration using a single loop
	Moderate	It should be a moderate problem: Involves single loop with incorporation of conditional statements
	Difficult	It should be a difficult problem: Involves using nested iterative statements

Table 2 presents the first restriction for every difficulty level. This restriction encompasses the topic for the programming exercise to be generated.

Table 3 presents the second restriction, which encompasses the complexity of programming exercises to be generated for each difficulty level.

Table 4 presents the third restriction. This restriction ensures that the programming exercises generated by ChatGPT are within the scope of ET-COTH1.

Finally, additional restrictions were added to ensure that ChatGPT does not reveal the solution structure. This restriction prevents ChatGPT from offering predefined algorithms, allowing students to solve the problem independently.

The general format of all prompts sent to ChatGPT is shown in Listing 1.

Table 4. Restriction 3: Scope specification.

Topic	Restriction 3
Conditional statements	The problem should be solvable using only the concepts of input-output statements (inputs must be taken in separate lines), variables, arithmetic, conditional operators, and logical operators only
Iterative statements	The problem should be solvable using only the concepts of input-output statements (inputs must be taken in separate lines), variables, arithmetic, conditional operators, and logical operators only. While loops must be used, without break or continue key words and without for-loops or recursive algorithms

Listing 1. Prompt Format

```
Make me a Python coding exercise based on the following:
1. #insert topic (Table 2)
2. #insert difficulty (Table 3)
3. #insert scope and limitation (Table 4)
4. Your problem should not provide any template,
   step-by-step process, or algorithm.
5. It should provide a sample input and output only,
   do not include instructions on what the algorithm
   flow should be.
```

3.2 Student Experience Survey

Demographics. There are 16 student respondents. All took the ET-COTH1 course at the time of the survey. They were selected as the course, which teaches computational thinking through Python, is taken in senior high school. This is to ensure that the participants have surface-level knowledge of programming.

Survey Responses. Tables 5, 6, 7, 8 present survey responses on conditional and iterative statements. Overall, students comprehended the exercises and found ChatGPT's feedback accurate and understandable. However, some feedback indicated a desire for more comprehensive explanations.

As expected, students rated exercises as more challenging with increasing difficulty levels. Iterative statements are also generally perceived as more challenging. Specifically, students noted that even easy exercises in iterative statements seemed difficult due to the topic's complexity. Nevertheless, 80% of respondents felt that the exercise difficulty aligned with their expectations, indicating ChatGPT's ability to match students' perceived difficulty levels.

Table 5. Survey results about conditional statements.

Survey questions	WM*
ChatGPT's exercises are stated in a manner that I can fully comprehend.	4.00
ChatGPT's feedbacks are stated in a manner that I can fully understand.	4.40
What do you think about the EASY problem given?	2.00
What do you think about the MODERATE problem given?	2.70
What do you think about the DIFFICULT problem given?	3.70

* Weighted mean.

Table 6. Survey results about the level of difficulty of exercises on conditional statements.

Survey questions	Response	Count	Percent
Does the difficulty level of the problems presented correspond to your perceived level of difficulty?	Yes	8	80%
	No	2	20%

Table 7. Survey results about iterative statements.

Survey questions	WM*
ChatGPT's exercises are stated in a manner that I can fully comprehend.	4.00
ChatGPT's feedbacks are stated in a manner that I can fully understand.	4.20
What do you think about the EASY problem given?	2.70
What do you think about the MODERATE problem given?	3.20
What do you think about the DIFFICULT problem given?	4.10

* Weighted mean.

Table 8. Survey results about the level of difficulty of exercises on iterative statements.

Survey questions	Response	Count	Percent
Does the difficulty level of the problems presented correspond to your perceived level of difficulty?	Yes	8	80%
	No	2	20%

3.3 Expert Evaluation Survey

The results of the expert evaluation survey are summarized in Tables 9, 10, to 12. Each table entry represents the weighted mean of expert responses with 1 as Not Evident and 5 as Highly Evident.

Table 9. Clarity of problem posing.

	Weighted Mean			
	Easy	Moderate	Difficult	AVG
Conditionals	4.50	4.20	4.10	4.27
Iteratives	3.50	4.20	3.90	3.87
AVG	4.00	4.20	4.00	**4.07**

Table 9 presents ChatGPT's clarity of problem posing. The weighted mean of 4.07 demonstrates good performance. The evaluators cited that ChatGPT could create good programming exercises as the prompts were specific and in context.

Evaluators noted that the problems provided are normally within the scope seen in online judges. Generally, it is evident from the exercises generated that the user understands them. Moreover, as [10] stated, LLMs are capable of natural language generation with programming courses, providing clear responses. [12] reinforces this, stating that natural language processing, linked with other attributes, results in effective interaction.

However, there are also several limitations. Evaluators noted that all problems did not mention restrictions, and the descriptions left room for misunderstanding. In one instance, ChatGPT provided an erroneous sample output. Considering this, ChatGPT presented good results concerning its problem generation but could improve the completeness of the problem description.

Table 10. Correctness of evaluation.

	Weighted Mean			
	Easy	Moderate	Difficult	AVG
Conditionals	4.60	4.50	4.30	4.47
Iteratives	3.80	3.70	3.80	3.77
AVG	4.20	4.10	4.05	**4.12**

Table 10 displays data on the correctness of evaluation. This metric measures whether ChatGPT correctly identifies if the user's code is correct. With a weighted mean of 4.12, the correctness of evaluation is evident. This result aligns with [10], stating that AI software provides correct code, potentially performing well with evaluation.

Evaluators cited that the evaluation of the solutions was straightforward, checking and correcting submitted codes. Here, ChatGPT can identify valid solutions to its exercises, yielding the highest weighted mean among the evaluation metrics. Areas for improvement were noted, as one solution was marked correct but was incorrect for one test case. Also, some answers were marked incorrectly due to the lack of restrictions mentioned. It was also noted that ChatGPT was unable to identify solutions that would give a "Presentation Error" or "Excess

Output Error." With a high overall rating, ChatGPT performed well in evaluating solutions, but it could be further refined.

Table 11. Depth and understandability of feedback.

	Weighted Mean			
	Easy	Moderate	Difficult	AVG
Conditionals	3.90	3.80	3.20	3.63
Iteratives	3.50	3.80	3.50	3.60
AVG	3.70	3.80	3.35	**3.62**

Table 11 presents the data on the depth and understandability of ChatGPT's feedback. This refers to being able to sufficiently explain the correct solution and provide suggestions for potential improvements. With a weighted mean of 3.62, ChatGPT's explanations are satisfactory. The evaluators stated that the feedback explained the code's nature, providing suggestions and corrections.

However, it was mentioned that ChatGPT provides generic feedback, and while mostly correct, they seem to mimic the parameters given. It was also stated that ChatGPT evaluates by looking for specific algorithmic constructs and key terms in the code. This result is less satisfactory than the first two metrics, yet the correctness of feedback remains moderately evident.

Table 12. Correctness of feedback.

	Weighted Mean			
	Easy	Moderate	Difficult	AVG
Conditionals	3.60	3.50	3.70	3.60
Iteratives	3.40	3.50	3.30	3.40
AVG	3.50	3.50	3.50	**3.50**

Table 12 shows ChatGPT's correctness of feedback. This is characterized by its ability to provide feedback that gives a correct explanation of logical and syntactical errors in the submitted code. Generally, the evaluators explained that the feedback was correct. They also noted that since most answers provided by the users were correct, there may not be enough samples to evaluate. Moreover, with the previous comment on the depth and understandability of feedback, it was mentioned that ChatGPT could be manipulated into giving inaccurate feedback as long as certain algorithmic constructs are present in the code submitted. Among the evaluation metrics, this criterion received the lowest score but remains moderately evident in the trials performed.

Generally, it could be said that ChatGPT performs better on the topic of conditional statements than iterative statements, which is attributed to the higher

ratings. This could be explained as conditional statements, which are simpler concepts usually taught before iterative statements. It is also notable that ChatGPT's performance generally declines from lower to higher difficulty levels. [13] supports this, stating that topics of greater difficulty could limit ChatGPT's capabilities.

In summary, ChatGPT performs well on problem generation and evaluating solutions, while its feedback-related aspect could be improved.

3.4 Checklist

Each programming exercise was reviewed and evaluated according to the yes-or-no criteria. The first is related to the topic, and the second and third are related to the difficulty level. The results are tallied, and the percentage of valid programming exercises is computed. Moreover, it checks for repetitions on programming exercises among all conversation logs for each difficulty level.

Table 13. Alignment to topic and difficulty level (Conditional statements).

Difficulty	Criteria	Total	% Valid
Easy	Requires a decision based on input	10	100%
	Presents at most two cases	8	80%
	Does not contain sub-cases	9	90%
Moderate	Requires a decision based on input	10	100%
	Presents more than two cases	10	100%
	Does not contain sub-cases	10	100%
Difficult	Requires a decision based on input	10	100%
	Presents more than two cases	10	100%
	Contains sub-cases	8	80%

Table 14. Alignment to topic and difficulty level (Iterative statements).

Difficulty	Criteria	Total	% Valid
Easy	Requires repetition of a certain task	9	90%
	Requires the use of a single loop	9	90%
	Does not require additional sub-cases	3	30%
Moderate	Requires repetition of a certain task	9	90%
	Requires the use of a single loop	9	90%
	Requires additional sub-cases	9	90%
Difficult	Requires repetition of a certain task	10	100%
	Requires the use of nested loops	8	80%
	Does not require additional sub-cases	7	70%

Table 13 presents the data for conditional statements. Here, most of the programming exercises were aligned with the topic and difficulty, with the percentage of valid exercises equal to or above 80%. Although ChatGPT was able to adhere to the topic restriction consistently, the difficulty restrictions were not always followed. For the easy problems, some exercises presented three cases to be considered by the programmer, which should fall under the moderate category. Likewise, some difficult problems incorrectly failed to require sub-cases.

Table 14 presents the data for iterative statements. The restriction on the topic was generally followed, with 90 to 100% valid exercises. However, inconsistency is seen in the difficulty restriction, specifically the easy level. Most problems did require the user to incorporate conditional statements to consider sub-cases within the program. This is not aligned with the requirements for easy iterative problems.

ChatGPT performs better in generating problems for conditional statements than for iterative statements. This is consistent with the expert survey result, as the topic of iterative statements is relatively complex compared to conditionals.

Table 15. Programming exercises theme.

	Easy	Moderate	Difficult
Conditionals	Eligibility to Vote Age Eligibility Check	Letter Grade Calculator	Ticket Pricing
Iteratatives	Sum of Even Numbers	Prime Number Checker	Pyramid Pattern

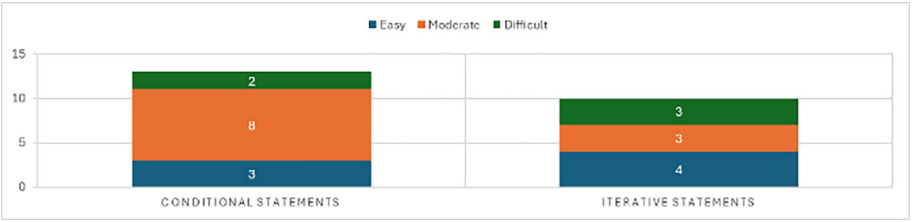

Fig. 2. Repetitions of programming exercises among trials.

Table 15 lists the themes of the different programming problems generated by ChatGPT. Figure 2 shows the number of repeated programming exercises given by ChatGPT among the topics and difficulty levels. The specific wording and parameters among these problems may be different but were counted if they had a similar task and structure.

From the data presented, it could be seen that most exercises are unique. However, the Letter Grade Calculator, a moderate conditional statement exercise, had 8 out of 10 repetitions across different trials. This could be due to it being a distinctive exercise with if-elif statements. Moreover, it could be

noted that a particular exercise appeared on different difficulties, violating its alignment to the difficulty level.

Overall, there are 43 distinct programming exercises out of the 60 user trials, with 6 exercises repeated 23 times. This result contradicts the preliminary results in the prompt generation phase. This could be due to having trials across different users, while the preliminary experiment focused only on a single conversation. Nonetheless, ChatGPT was able to generate distinct exercises for most of the trials performed.

As most of the conversation logs analyzed in the study contained a correct solution to the problem, some cases were tested ad hoc. These include incorrect code solutions submitted to ChatGPT. Likewise, incorrect solutions submitted by the participants were also reviewed. ChatGPT identified the errors in these solutions and provided corrections appropriately. ChatGPT can correctly identify syntax errors, specifically incorrect capitalizations. ChatGPT can also identify and fix logical errors, such as incorrect order of conditionals.

However, there were some instances where ChatGPT provided ambiguous problem statements or erroneous evaluations. The problem statement provided by ChatGPT did not describe how the input will be processed. It merely showed screen outputs. In another instance, the solution provided contained logical errors. ChatGPT provided the appropriate corrected code, but it does not match the sample output it provided with the problem statement.

4 Conclusion

The results have shown that ChatGPT has promising potential in programming education, with its ability to create exercises and evaluate solutions. It was found that prompts need to be specific and clear, and directly state the desired content when conversing with ChatGPT. It is crucial to state the topic, difficulty, and conceptual restrictions of the exercise. While the prompts generated were generally useful, improvements are possible. Furthermore, ChatGPT proved reliable in providing valid programming exercises, with a validity rate ranging from 80% to 100%. Student feedback indicated that the exercises were generally comprehensible and clear, with an average rating of 4.00 out of 5.00. ChatGPT also performed well in clarity of problem posing and correctness of evaluation, with weighted means of 4.07 and 4.12, respectively. Feedback depth and understandability received a moderate score of 3.62, suggesting potential improvements in this aspect. Similarly, the correctness of feedback received a moderate score of 3.50, requiring refinement through specific feedback prompts. Future research may focus on improving prompt generation by modifying restrictions and structure. More advanced topics may be explored to assess its versatility. Future works may develop on present results to expand its scope to different aspects including other programming languages and concepts.

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Exploring the Potential of Immersive Web Environments in Education: Teachers' Perspectives and Experiences

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Abstract. This research investigates teachers' perceptions of the effectiveness and feasibility of incorporating immersive web environments into their teaching practices. A total of 11 teachers from various subject areas participated, having recently attended a 30-h training course on using immersive web environments in the classroom. Following the training, these teachers implemented their created environments with their students. A survey instrument employing a 5-point Likert scale was designed to assess teachers' perceptions of the perceived benefits, challenges, and overall impact of using immersive web environments in education. The findings reveal high levels of perceived enjoyment, motivation, cognitive benefits, learning, and satisfaction among students using immersive web environments. Strong positive correlations between most variables indicate a mutually reinforcing effect, with improvements in one aspect likely leading to enhancements in others. Despite the study's limitations, including a small sample size of teachers and potential selection bias, the results emphasize the potential of immersive web environments in fostering engaging and effective learning experiences for a larger number of students. This research can inform the development of professional development initiatives and the design of innovative educational technologies that promote effective student learning experiences.

Keywords: Immersive web environments · Teachers' perception · Student engagement · Educational technology · Virtual classrooms

1 Introduction

The rapid advancement of digital technologies has significantly impacted teaching and learning practices. Emerging technologies, including immersive web environments (IWEs), offer new opportunities to enhance educational experiences [1]. IWEs, such as FrameVR.io and Spatial.io, provide users with a sense of immersion in a digital space, allowing real-time interaction and collaboration [2]. Unlike Virtual Reality (VR) and Augmented Reality (AR), these environments are accessible through standard web browsers, requiring no specialized hardware or software, making them more accessible to a wider range of learners and educators. Previous research has identified numerous

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benefits of IWEs, including increased student engagement, the development of higher order thinking skills, and enhanced social interaction and collaboration among learners [3, 4]. However, challenges such as accessibility, usability, and potential distractions remain [5]. To fully realize the potential of IWEs, it is essential to develop effective strategies for their integration into teaching and learning practices. This research investigates teachers' perceptions and experiences of using IWEs in the classroom after participating in a 30-h training course. By exploring the perspectives of trained educators, this study aims to provide insights into the benefits, challenges, and strategies for successfully integrating IWEs into education. The findings will contribute to the growing body of literature on IWEs in teaching and learning and inform future professional development and curriculum design initiatives.

To guide this investigation, the following research questions were formulated:

What are teachers' perceptions of the benefits and challenges of using IWEs in the classroom?

How do teachers perceive the impact of IWEs on student engagement, motivation, and learning outcomes?

What strategies do teachers find effective for integrating IWEs into their teaching practices?

2 Literature Review

Immersive web environments refer to online platforms that create a sense of immersion by enabling users to engage with and navigate digital spaces in real-time, often through avatars or other interactive elements [2]. These environments facilitate a wide range of educational activities, such as collaboration, communication, and problem-solving, more engagingly and interactively than traditional online learning platforms. Recent advancements in web technologies have led to the development of various immersive web environments, including FrameVR.io and Spatial.io.

2.1 Benefits of Immersive Web Environments in Education

Prior research has demonstrated that immersive web environments (IWEs) can enhance student engagement and motivation [6]. These environments provide a more interactive and dynamic learning experience, allowing students to actively participate in learning activities and collaborate with peers, leading to improved motivation and engagement. Additionally, IWEs promote the development of higher-order thinking skills, such as critical thinking, problem-solving, and creativity [7]. These cognitive benefits enhance students' mental processes and their ability to think deeply and process information effectively. Furthermore, IWEs facilitate social interaction and collaboration among learners [8], allowing students to work on projects, engage in discussions, and share ideas, thus fostering a sense of community and collaborative learning experiences. Learning effectiveness pertains to the overall impact on students' ability to understand, apply, and retain the material taught, including comprehension, application, and performance in assessments. The interactive nature of IWEs encourages students to explore, manipulate, and experiment with concepts, fostering a deeper understanding of the subject matter.

2.2 Challenges and Limitations

Despite the potential benefits of immersive web environments (IWEs) in education, several challenges and limitations must be considered [9]. Key challenges include accessibility and usability. Some IWEs require high-speed internet, powerful devices, or specific software, which may not be available to all students, particularly in low-resource settings. Additionally, complex user interfaces can pose difficulties for students and teachers with limited digital literacy, leading to frustration and hindering learning. Another concern is the potential for distractions; the abundance of interactive elements and communication tools in IWEs can lead to decreased focus and reduced learning outcomes [10].

2.3 Strategies for Effective Integration of Immersive Web Environments

Several strategies have been proposed to address the challenges of using immersive web environments (IWEs) in education. First, providing adequate training and support for teachers is essential, focusing on both technical skills and pedagogical strategies [11]. Second, selecting appropriate IWEs should prioritize accessibility, usability, and alignment with learning objectives [12]. Choosing user-friendly platforms compatible with various devices can enhance the learning experience. Lastly, designing learning activities and assessments that leverage IWEs' interactive and immersive features is crucial. Tasks should encourage collaboration, problem-solving, and exploration to maximize educational outcomes [13].

3 Methodology

3.1 Participants

This study engaged 11 teachers from diverse subjects, including Mathematics, Science, Social Studies, and Language Arts. These educators were selected after completing a 30-h training course on using immersive web environments (IWEs) in their classrooms. The two-month course aimed to enhance their understanding of IWE benefits and applications. The teachers then tested their self-created IWEs with their students, providing valuable insights into the broader educational impact of IWE adoption, despite the precise number of students being difficult to ascertain.

3.2 Instrument

A survey was designed to assess teachers' perceptions of the effectiveness and feasibility of incorporating immersive web environments (IWEs) into their teaching practices. The survey consisted of closed-ended questions structured using a 5-point Likert scale, ranging from "strongly disagree" (1) to "strongly agree" (5). Adapted from the study "Teacher Perceptions on Virtual Reality Escape Rooms for STEM Education" [14], the questionnaire measured Perceived Enjoyment, Motivation, Cognitive Benefits, Learning Effectiveness, and Satisfaction. This tool gathered data with both ordinal and interval properties, allowing for hierarchical arrangement and quantifiable differences between points, thus enhancing the rigor and interpretability of the findings.

3.3 Procedure

Before the study began, ethical approval was obtained from the relevant institutional review board. Participants were informed about the research objectives and the voluntary nature of their participation. Consent was obtained from all participants. The online survey was administered at the end of the teacher training to the 11 participant teachers, ensuring ease of access. Participants had two weeks to complete the survey, with reminder emails sent at the start of the second week to encourage timely responses.

3.4 Data Analysis

The collected data were analyzed using descriptive statistical methods and a measure of internal consistency or reliability of a scale or test. Descriptive statistics were calculated to summarize participants' responses to the Likert scale questions, including means, standard deviations, and frequencies. Internal consistency statistics, such as Cronbach's alpha (α), were used to assess how well a set of items on a scale or test measures a single underlying construct or concept.

3.5 Limitations

It is crucial to acknowledge the limitations of this study, as they could affect the generalizability of the results. First and foremost, the sample size of 11 teachers is relatively small, potentially limiting the study's statistical consistency. Furthermore, the purposive sampling method and exclusive focus on educators who completed the 30-h training course might introduce selection bias. These participants may exhibit greater motivation or interest in utilizing immersive web environments compared to the broader population of teachers. As a result, the findings may not accurately represent the wider teaching community. Additionally, while the study encompasses the perceptions of numerous students through the 11 teachers, the precise number of students involved remains uncertain, potentially affecting the interpretation of the study's impact on learners.

4 Results and Discussion

4.1 Descriptive Analysis

Perceived Enjoyment

The results in Table 1 indicate that participants generally reported high levels of perceived enjoyment across all three measures. Ratings for "I think that my students find using immersive web environments pleasant" and "I think that my students find using immersive web environments fun" were slightly higher than "I think that my students find it enjoyable to use immersive web environments." The data also show low variability in responses, indicating consistent experiences of enjoyment among participants.

Motivation

The results in Table 2 suggest that participants generally reported high motivation levels among their students while using IWEs. The data show low variability in responses, indicating consistent experiences of motivation among students.

Table 1. Perceived Enjoyment descriptive statistics.

Perceived Enjoyment	N	Mean	Median	SD	Min	Max
I think my students find it enjoyable to use immersive web environments	11	4.55	5	0.688	3	5
I think that my students find using immersive web environments pleasant	11	4.73	5	0.467	4	5
I think my students find using immersive web environments fun	11	4.73	5	0.647	3	5

Table 2. Motivation descriptive statistics.

Motivation	N	Mean	Median	SD	Min	Max
My students really enjoy working in the immersive web environment classroom	11	4.45	5	0.688	3	5
My students find it fun to do activities in the immersive web environment	11	4.64	5	0.674	3	5
My students find the immersive web environment boring (R)*	11	1.27	1	0.467	1	2
The immersive web environment did not hold my students' attention at all (R)	11	1.64	1	0.809	1	3
My students describe the immersive web environment as very interesting	11	4.45	5	0.820	3	5

(*continued*)

Table 2. (*continued*)

Motivation	N	Mean	Median	SD	Min	Max
My students find the immersive web environment quite enjoyable	11	4.55	5	0.820	3	5
My students seem to be thinking about how much they enjoy the immersive web environment while they are using it	11	4.00	4	0.775	3	5

*(R) = reversely coded item.

5 Cognitive Benefits

Participants generally reported high cognitive benefits for their students while using IWEs (Table 3). The data also show moderate variability in the responses, indicating that students' experiences and perceived benefits may vary to some extent. Overall, the findings support the notion that immersive web environments can positively impact students' cognitive abilities, such as understanding, memorization, application, and problem analysis.

Table 3. Cognitive Benefits descriptive statistics.

Cognitive Benefits	N	Mean	Median	SD	Min	Max
These type of activities in an immersive web environment make understanding easier	11	4.27	4	0.786	3	5
These type of activities in an immersive web environment make memorization easier	11	4.18	4	0.874	3	5
These type of activities in an immersive web environment help my students to better apply what they have learned	11	4.36	5	0.809	3	5

(*continued*)

Table 3. (*continued*)

Cognitive Benefits	N	Mean	Median	SD	Min	Max
These type of activities in an immersive web environment help my students to better analyze the problems	11	4.18	4	0.874	3	5

6 Perceived Learning

The analysis of the results in Table 4 suggests that participants mostly reported high levels of perceived learning for their students while using immersive web environments. The data reveal a modest response fluctuation, implying that students' experiences and satisfaction could differ somewhat. These findings support the notion that IWEs can positively impact students' interest, understanding, and application of knowledge and their ability to identify main issues and feel stimulated to learn more.

Table 4. Perceived Learning descriptive statistics.

Perceived Learning	N	Mean	Median	SD	Min	Max
My students are more interested in learning the topics	11	4.18	4	0.874	3	5
My students learned a lot of factual information about the topics	11	3.91	4	0.831	3	5
My students understood the concepts well	11	4.18	4	0.874	3	5
My students learned to identify the main and important issues of the topics covered	11	4.18	4	0.751	3	5
My students showed interest and felt stimulated to learn more	11	4.55	5	0.820	3	5

(*continued*)

Table 4. (*continued*)

Perceived Learning	N	Mean	Median	SD	Min	Max
My students were able to complete and summarize what they had learned	11	4.27	4	0.786	3	5
The learning activities were meaningful	11	4.27	4	0.786	3	5
My students are able to apply what they have learned in a real context	11	4.27	4	0.786	3	5

7 Satisfaction

These results suggest that teachers generally reported high student satisfaction levels while using immersive web environments (Table 5). The data exhibits a moderate degree of variation in the responses, suggesting that there might be some differences in students' experiences and satisfaction levels. Globally, the findings support the notion that immersive web environments can positively impact students' satisfaction with the learning experience, materials, teaching methods, and overall effectiveness, as well as their appreciation for immediate information and 3D/virtual reality-based environments.

Table 5. Satisfaction descriptive statistics.

Satisfaction	N	Mean	Median	SD	Min	Max
My students were satisfied with this kind of learning experience through an immersive web environment	11	4.64	5	0.674	3	5
My students were pleased with the wide variety of learning materials that were provided in this type of learning environment	11	4.64	5	0.674	3	5

(*continued*)

Table 5. (*continued*)

Satisfaction	N	Mean	Median	SD	Min	Max
I do not think this type of learning environment benefited my students' achievement of learning (R)	11	1.55	1	1.036	1	4
My students were satisfied with the immediate information they got in this kind of learning environment	11	4.18	4	0.751	3	5
My students were satisfied with the teaching methods in this kind of learning environment	11	4.36	5	0.809	3	5
My students were satisfied with this kind of 3D/virtual reality-based learning environment	11	4.55	5	0.820	3	5
My students were satisfied with the overall effectiveness of their learning	11	4.45	5	0.820	3	5

* *(R) = reversely coded item.*

7.1 Correlation Matrix

The correlation analysis using Spearman's correlation coefficient (Table 6) revealed strong positive correlations between most variables, suggesting a mutually reinforcing effect among them.

A very strong positive correlation was found between Perceived Enjoyment and Motivation ($\rho = 0.872$, $p < 0.001$), indicating that higher enjoyment levels in immersive web environments (IWEs) are associated with increased motivation. This aligns with Liu et al. (2020), who found that immersive environments significantly enhance student motivation and engagement [15]. Similarly, strong positive correlations were observed between Motivation and Perceived Learning ($\rho = 0.842$, $p < 0.001$), suggesting that increased motivation improves perceived learning outcomes. Makransky and Lilleholt (2018) also reported that motivated students perceive higher learning outcomes in virtual environments [16]. Additionally, a strong relationship between Cognitive Benefits and Perceived Learning ($\rho = 0.830$, $p < 0.001$) indicates that cognitive benefits, such as improved critical thinking and problem-solving skills, are linked to higher perceived

Table 6. Correlation matrix for each construct using Spearman's correlation.

	Perceived Enjoyment	Motivation	Cognitive Benefits	Perceived Learning	Satisfaction
Perceived Enjoyment	–	**0.872**	0.540	0.735	0.714
Motivation	0.872	–	0.732	**0.842**	0.660
Cognitive Benefits	0.540	0.732	–	**0.830**	0.691
Perceived Learning	0.735	0.842	0.830	–	**0.772**
Satisfaction	0.714	0.660	0.691	0.772	–

learning, consistent with Dede (2009) [17]. The correlation between Perceived Learning and Satisfaction ($\rho = 0.772$, $p = 0.004$) shows that effective learning is associated with greater student satisfaction. Freina and Ott (2015) also found that immersive environments contribute to higher satisfaction due to enhanced learning experiences [18]. Although the correlation between Perceived Enjoyment and Cognitive Benefits ($\rho = 0.540$, $p = 0.075$) was moderate and not statistically significant, it suggests that enjoyable experiences in IWEs can contribute to cognitive gains. The correlation analysis demonstrates that most relationships between the studied variables are statistically significant and positive, highlighting the potential of IWEs to enhance enjoyment, motivation, cognitive benefits, perceived learning, and satisfaction. However, correlation does not imply causation. These results underscore the importance of designing engaging and cognitively stimulating activities within IWEs to maximize their educational impact.

8 Conclusion

This research underscores the transformative potential of IWEs in education. By enhancing student engagement, motivation, cognitive skills, and overall satisfaction, IWEs can significantly improve learning experiences. However, addressing challenges related to accessibility, usability, and potential distractions is crucial for maximizing their impact. The findings provide valuable insights for educators, policymakers, and technology developers, informing the design and implementation of innovative educational technologies and professional development initiatives. Future research with larger, more diverse samples is needed to further validate these findings and explore long-term impacts on student learning and achievement. By fostering engaging and effective learning environments, IWEs have the potential to significantly enhance education, making it more interactive, immersive, and impactful for a diverse range of students. While they are not a panacea, the thoughtful integration of IWEs can address various educational challenges, providing more dynamic and inclusive learning experiences that cater to different learning styles and needs.

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Learning Swedish with AI: Exploring Multimodal Learning Analytics in Spoken Language Acquisition

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Abstract. This study investigates the application of Multimodal Learning Analytics (MMLA) in language practice, specifically within the authentic and dynamic environment of language café settings. The MMLA Model for Design and Analysis (MAMDA), a design science approach, is utilized to systematically explore the requirements for designing the MMLA system. We identify and map three elements: 1) Learning indicators, referring to spoken language learning signs, such as tone, amount and frequency of speech, and pronunciation. 2) Respective modalities and sensors, referring to the format of data to be collected and 3) Analytics models, including NLP models, that can be employed to identify and process the modalities. We propose a conceptual system that utilizes AI voice assistant while simultaneously collecting audio data for MMLA to enhance language learning experiences. The system is meant for providing insights into learning patterns, participant engagement, and the overall effectiveness of language practice strategies. While presenting a novel system showcasing the use of AI and data analytics in a unique educational setting, the study's central focus is to test and critically reflect on MAMDA as a framework for designing and analyzing MMLA systems.

Keywords: Multimodal Learning Analytics · Spoken Language Acquisition · Design Science Methodology · Generative AI

1 Introduction

Multimodal Learning Analytics (MMLA) emerges as a promising innovation in education and education science [1], suggesting a nuanced and comprehensive approach to understanding and enhancing the learning process. By leveraging multiple data sources and analytical models, MMLA provides a multi-faceted view of learners' interactions and progress. Within digital education innovations, language acquisition presents an exemplary use case for the application of MMLA. Language learning, inherently interactive and multifaceted, would benefit immensely from the depth and breadth of insights that MMLA can offer [2]. Analyzing diverse modalities, such as spoken dialogue and non-verbal cues, allows MMLA to uncover insights into language acquisition, providing

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learners and educators with valuable feedback and personalized learning pathways. In this context, language cafes, with their dynamic and authentic learning milieu, serve as an ideal setting for applying and evaluating MMLA's potential to transform language practice.

This research adopts the MMLA Model for Design and Analysis (MAMDA), a robust design science methodology, to systematically address the essential requirements for crafting an effective MMLA system. This exploration is twofold: firstly, conducting a user study in the authentic environment of language cafés to gather insights into the real-world application of MMLA, and secondly, designing a prototype system that embodies our findings and theoretical framework. Three key elements are identified: 1- learning indicators (such as tone, speech frequency, and pronunciation) that signify the nuances of spoken language learning; 2- the modalities and sensors designated to capture this multifaceted data; and 3- the analytics models, particularly those leveraging Natural Language Processing (NLP), to interpret and act upon the gathered information. We propose a conceptual integrating AI voice assistance as conversational aide, while simultaneously collecting audio data to inform MMLA, thereby enriching the language learning experience. This system aims to provide insights into learner patterns, engagement levels, and the overall effectiveness of language practice strategies within the context. While presenting a novel system, showcasing the use of AI and data analytics in a unique educational setting, the study's central focus is to test and critically reflect on MAMDA as a framework for designing and analyzing MMLA systems. Our analysis aims to evaluate MAMDA's effectiveness as a design and analysis tool and to establish it as a benchmark for developing future MMLA solutions.

The following sections provide a comprehensive exploration of the study's methodology and findings. The Background section delves into the theoretical foundations and prior research that underpin our investigation. The Methodology section outlines the design science approach and the phased structure of the MAMDA framework, detailing the user study and prototype development. In the User Study section, we present our observations and analysis of language acquisition in the dynamic setting of language cafes. The MMLA Prototype section describes the iterative development and evaluation of the system. The Results section highlights key learning indicators and their implications for language learning analytics. Finally, the Discussion and Conclusion sections reflect on the study's contributions, challenges encountered, and directions for future research. This structured approach ensures a thorough examination of the application of MMLA in enhancing spoken language acquisition.

2 Background

MMLA's analytical approach is especially pertinent to language acquisition, where multiple communication facets play crucial roles. In language cafes, which are naturalistic and immersive environments for language practice, MMLA's potential is particularly pronounced [1]. By applying MMLA in such settings, learners, educators, and researchers can gain deeper insights into language acquisition dynamics. This will enable them to tailor learning experiences that are both effective and deeply engaging. This background sets the stage for exploring how MMLA can be strategically implemented to enhance language learning in these dynamic environments.

MMLA represents a rapidly developing area in educational technology with considerable promise for enhancing language learning. By integrating diverse modalities like audio, video, and text, MMLA aspires to deliver a multifaceted understanding of learners' interactions and progress, particularly in language education [3]. Several studies highlight the potential of MMLA to offer personalized insights and support in language learning environments [4, 5]. Despite the advances in MMLA, the domain of spoken language acquisition remains underexplored. Current research and application primarily focus on written and visual modalities, with spoken language offering a rich vein of untapped potential for MMLA in language learning. This gap underscores a significant opportunity for MMLA to impact spoken language practice and assessment more profoundly. In this context, AI voice assistants emerge as promising tools for facilitating spoken language learning [2]. As conversational aids, they not only provide immediate interaction but also serve as valuable data sources for MMLA, capturing nuanced aspects of spoken language such as intonation, pacing, and pronunciation. The integration of AI voice assistants in language learning settings, particularly in immersive environments like language cafes, could substantially augment MMLA's capabilities, offering more personalized and effective learning experiences [4].

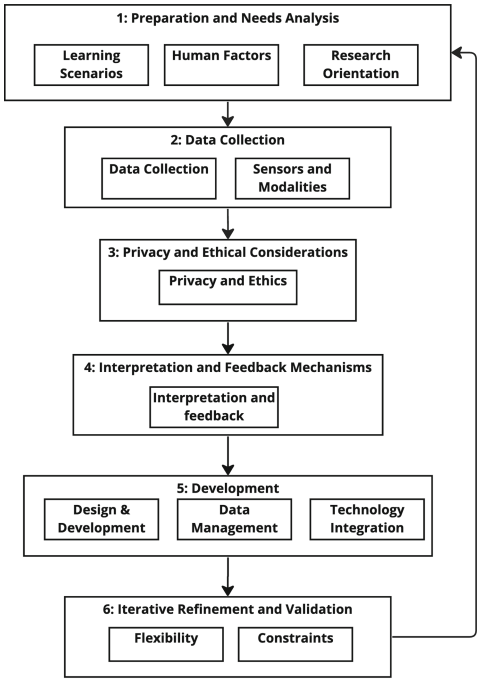

Fig. 1. Multimodal Learning Analytics Model for Design and Anaysis

The Multimodal Learning Analytics Model for Design and Analysis (MAMDA) provides a structured framework for designing MMLA systems, addressing key design

considerations [6] in integrating MMLA into educational settings. MAMDA's six-phase process includes:

- Preparation and Needs Analysis: Identifying educational goals, technological needs, and pedagogical preferences to guide the design process.
- Data Collection: Determining the types and modalities of data to be collected while considering data privacy and ethical issues.
- Privacy and Ethics: Ensuring MMLA implementation adheres to ethical standards and data privacy regulations.
- Interpretation and Feedback: Analyzing data to extract insights and providing feedback to support an adaptive learning environment.
- Development: Building the MMLA system based on earlier insights, focusing on technical development and integration.
- Refinement and Validation: Continuously testing and refining the system in real-world settings to meet the evolving needs of the educational environment.

This paper evaluates MAMDA's applicability to design an MMLA system to be used in Swedish language practice. The study follows a design science approach, detailing activities in each MAMDA phase, from conducting language cafe sessions to developing a data collection prototype and planning future system evaluations. This comprehensive approach bridges theory and practice, contributing to the discourse on MMLA's integration in language learning.

3 Methodology

A design science research methodology [7], recognized for its systematic and iterative approach to complex problems in information systems and technology, is employed. This methodology is particularly apt for our investigation into integrating MMLA within language learning contexts, which resonates with the phased structure of MAMDA. Central to our research design are the weekly language cafe sessions, serving as authentic environments for language practice. These sessions offer real-world communication settings for language learners, facilitating genuine spoken interactions in Swedish. The variability in participant numbers each week reflects the authentic diversity of language learning environments. Our study is structured into two main stages:

User Study: encompasses the first four phases of MAMDA to explore language acquisition requirements.

1. Preparation and Needs Analysis: assessing the needs and goals for MMLA integration, analyzing the language practice environment to set the foundation for effective MMLA implementation.
2. Data Selection: developing a plan to systematically collect multimodal data, focusing on spoken language interactions during the language cafe sessions.
3. Privacy and Ethics: this phase involves establishing ethical guidelines and privacy measures, emphasizing participant consent and data protection.
4. Interpretation and Feedback: Utilizing collected data, deriving insights to inform feedback mechanisms, enhancing the iterative refinement process of the MMLA system.

Prototype Development: covering the two remaining MAMDA phases.

5. Development: Insights from the earlier phase guide the design and development of the MMLA system prototype, focusing on functionality and user interface within the language cafe context.
6. Refinement and Validation: evaluating and iteratively refine the MMLA system, ensuring its effectiveness and adaptability within the language cafe settings.

To capture a comprehensive understanding of the feasibility and effectiveness of the MMLA system, qualitative data were collected through detailed notetaking and report writing at each step of the study. Notes were taken during the planning meetings to capture objectives, strategies, and anticipated challenges. During each language exchange session, observations were made regarding participant engagement, interaction patterns, and any technical issues encountered. After each session, detailed reports were written to summarize observations, participant feedback, and any immediate actions taken to address issues. Throughout the prototyping phases, both during and after development activities, notes were taken to document the design decisions, technical challenges, and iterative improvements made to the system. The integration of design science with MAMDA in our study ensures a methodical and contextualized approach to developing and analyzing MMLA systems, leveraging real-world language cafe sessions to enrich our understanding and refine our MMLA system. This methodology underscores the synergy between iterative design and phased MAMDA application, essential for advancing MMLA in language learning contexts.

4 User Study: Exploring Spoken Language Acquisition

The user study was conducted within the authentic and dynamic environment of language café settings to explore the integration of MMLA in language learning. Over four months, weekly language exchange sessions were held with four to fifteen participants, serving as touchpoints to observe dialogues and analyze interactions, thereby informing the design of the MMLA system. These consistent weekly engagements immersed both the researchers and participants in a cycle of learning and feedback, practicing Swedish. The initial four phases of MAMDA leveraged these language café sessions as platforms to develop and refine the MMLA system, focusing on the Swedish language. The varying participant numbers in these sessions provided a dynamic environment for data collection and analysis, which was essential for the application of the MAMDA framework.

Preparation and Needs Analysis: We identified learners' needs, observing interactions to determine technology's role, notably smart assistants, in language practice. Key findings included the need for a conversation moderator and targeted support for spelling and pronunciation, with speech quantity emerging as a significant learning indicator.

Data Selection: This phase involved identifying learning indicators and their corresponding digital modalities, emphasizing audio data analysis via NLP to examine syntax, semantics, and conversation dynamics. This phase aimed to capture comprehensive insights into language proficiency and interaction patterns.

Privacy and Ethics: This phase focused on ethical data handling, with an informed consent process, anonymization strategies, and a feedback loop to address participants'

privacy concerns. Challenges around data management were met with stringent privacy protocols and continuous ethical considerations.

Interpretation and Feedback: Interpretative analysis of data informed the MMLA system's feedback mechanisms, focusing on pronunciation, conversational flow, and speech quantity. This phase adapted the MMLA system to align with participant interactions and feedback, enhancing its relevance and responsiveness.

These phases underscored the necessity of a structured approach in MMLA system development, revealing insights into learner interactions and system refinement needs. The progression through these phases set a foundation for the MMLA system's including learning indicators, modalities and sensors, and analytics models.

5 MMLA Prototype

The development phase was divided into five stages to protoype the MMLA system within the MAMDA framework, focusing on the transition from concept (illustrated in Fig. 1) to functional prototype, highlighted by vision and audio modules' iterative development.

Vision: Initiated with the Arduino Nicla Vision[1] board, we explored face and object detection, shifting to fiducial markers like AprilTag[2] for efficiency and privacy. This phase yielded a "sketch" prototype, using badges and a camera base station for participant tracking in language practice, setting a foundation for further testing.

Audio: We investigated audio analysis, employing Titanet[3] for deep learning-based speaker recognition. The Audio Module, integrating voice analysis techniques, underwent evaluations to refine its functionality in capturing and differentiating speech, providing a significant potential into understanding conversational dynamics.

Evaluation of Vision and Audio Modes: This stage involved real-world testing, focusing on the Learning Analytics Platform (LAP) system's usability, privacy, and technical performance. Stakeholder feedback and technical assessments identified limitations, especially in vision and audio capture, guiding subsequent refinements.

Prototype: Addressing previous stages' insights, we developed Prototype Alpha, integrating an external webcam and Nicla Vision-equipped badges for improved detection and scalability. Enhanced communication protocols (MQTT, UDP) and data synchronization (using Redis and InfluxDB[4]) were implemented, laying the groundwork for extensive testing and empirical validation.

Evaluation of Prototype: This evaluation aimed to assess the LAP system's speaker recognition accuracy, recording delays, and overall stability. Findings highlighted an 85% accuracy rate in speaker recognition and identified the need for further enhancements, particularly in the vision module's field of view and AprilTag capture.

[1] Nicla Vision combines a M7/M4 IC processor with a 2MP color camera that supports TinyML, a smart 6-axis motion sensor, microphone and distance sensor.

[2] Fiducial marker, containing information and designed for easy recognition.

[3] TitaNet: Neural Model for speaker representation with 1D Depth-wise separable convolutions and global context.

[4] InfluxDB Time Series Data Platform.

Future development will focus on overcoming these limitations, considering RFID integration and a multi-camera setup for comprehensive coverage. This iterative process underscores our commitment to refining the LAP system, ensuring its effectiveness in collaborative learning scenarios.

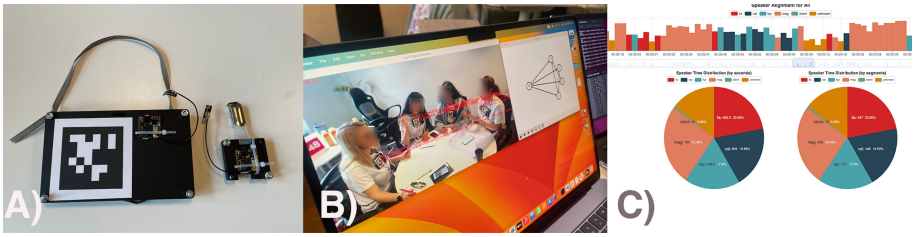

Fig. 2. LAP prototype in action.

The sixth phase is essential for the continuous enhancement of MMLA systems to better suit the dynamic requirements of educational environments. This phase should be guided by rigorous methodologies that draw on user feedback and real-world application data, allowing for targeted adjustments to improve the model's functionality and user engagement. MAMDA's adaptability is tested across different learning scenarios to evaluate its robustness and identify opportunities for enhancing its application. Challenges anticipated in this phase include addressing the scalability of the system to support a diverse range of user numbers. In addition, they include improving user interfaces for better adaptation to new technological interfaces, and integrating MMLA seamlessly with various educational technologies. Conversely, the evolving landscape of educational technology and learning analytics presents opportunities for innovative enhancements and broader applications of the model. The experiences and insights gained from MAMDA's initial phases are invaluable for this phase. They provide a repository of knowledge that informs the refinement strategy, ensuring real-world utility and effectiveness. Research questions emerging from earlier experiences will guide exploration in this phase. These questions will focus on enhancing the system's impact, usability, and educational value. The sixth phase of MAMDA is a testament to the commitment to continuous evolution of MMLA systems. It emphasizes a proactive and responsive approach to system development. This commitment ensures that MAMDA (as a design approach) will continue to evolve in alignment with the shifting paradigms of educational technology and the diverse needs of learners and educators.

6 Results

During the preliminary stages of the investigation, learning indicators in language café environments were systematically identified and analyzed, uncovering essential insights into the dynamics of spoken language acquisition. Our analysis across multiple language cafe sessions revealed significant variations and consistent patterns in learning indicators such as tone, speech rate, pronunciation accuracy, conversational flow, interaction patterns, word count, and hesitation detection. These indicators were visually represented

(see Fig. 2.) and examined through data samples to provide a detailed understanding of how spoken language is engaged with and absorbed in real-world settings. To effectively capture these learning indicators, an array of modalities and sensors was utilized. Audio data, collected via microphones, was instrumental in assessing tone, speech rate, pronunciation accuracy, word count, and hesitation detection, offering insights into verbal communication nuances. Video data, obtained through cameras, augmented this analysis by providing context through visual cues, crucial for evaluating pronunciation accuracy, conversational flow, and interaction patterns. Additionally, textual data, such as chat logs, were considered to supplement the analysis, particularly in quantifying word count, thereby enriching the dataset and offering a multi-faceted view of language learning processes. User engagement with the language cafe sessions played a pivotal role in refining the Learning Analytics Platform (LAP), with user preferences, needs, and expectations informing the platform's development. The dynamic interplay of user interactions within these settings underscored the active nature of language learning, emphasizing the importance of adapting the LAP to meet evolving user requirements and enhance its efficacy in real-world language learning contexts.

Table 1. Learning indicators and respective sensors and modalities.

Learning Indicators	Modalities and Sensors	Analytics Models	Technology
Tone	Audio: Microphones	Sentiment Analysis	Jabra
Speech Rate	Audio: Microphones	Speech Rate Analysis	Jabra
Pronunciation Accuracy	Audio: Microphones, Video: Cameras	Pronunciation Analysis	Jabra, Nicla Vision
Conversational Flow	Audio: Microphones, Video: Cameras	Discourse Analysis	Jabra, Nicla Vision
Interaction Patterns	Audio: Microphones, Video: Cameras	Interaction Analysis	Jabra, Nicla Vision
Word Count	Audio: Microphones, Text: Chat logs	Transcription Text Analysis	Jabra, Nicla Vision
Hesitation Detection	Audio: Microphones	Hesitation Analysis	Jabra
Engagement Level	Audio: Microphones, Video: Cameras	Engagement Analysis	Jabra, Nicla Vision

Through the integration of new technologies and sensors, we expanded our capabilities for data collection within language acquisition. We provide a summary of the learning indicators and their respective modalities and sensors in Table 1. We employed cutting-edge sensors to capture language practice details, ensuring a more comprehensive and nuanced understanding. As a result of technical challenges encountered during the development phase, a functional prototype has been implemented (see Fig. 3), paving the way for a more robust data collection system. Our focus extended to the deployment

of analytics models, with a particular emphasis on NLP models. These models play a pivotal role in extracting meaningful insights from the selected data. To address both syntactic and semantic aspects of spoken language learning, the analytics models demonstrated their effectiveness in analyzing language use intricacies. A visual illustration of LAP is provided in Fig. 3.

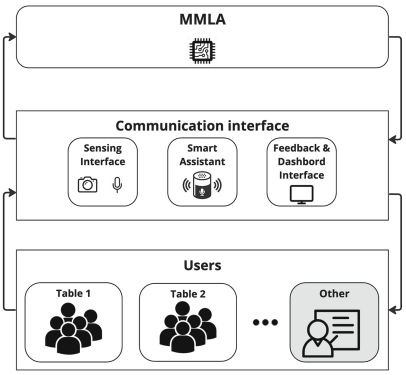

Fig. 3. MMLA system for spoken language practice in groups.

In our attempt to understand the dynamics within language acquisition, we present a visual illustration that maps the identified learning indicators across various settings. This mapping serves as a comprehensive illustration, showcasing the variations and trends observed within different language cafe sessions. By visually connecting learning indicators, modalities, and analytics models, we aim to provide a holistic understanding of their interconnectedness. The mapping not only highlights the individual nuances of tone, speech frequency, amount of speech, and pronunciation but also underscores their collective impact on shaping the landscape of language learning practices within the authentic and dynamic context of language cafes.

7 Discussion

In this study we aimed at exploring the integration of MMLA within the context of language cafes, focusing on spoken Swedish language acquisition. Through the systematic application of the Multimodal Learning Analytics Model for Design and Analysis (MAMDA), we delved into the complexities of language learning, employing a design science approach to iteratively develop an MMLA system. In the initial phases, we identified key learning indicators like tone, speech rate, pronunciation accuracy, conversational flow, interaction patterns, word count, and hesitation detection. These indicators were meticulously captured using a combination of audio and video modalities, with audio data being pivotal for most indicators and video data providing additional context, particularly for pronunciation accuracy, conversational flow, and interaction patterns. The iterative design process, underscored by the MAMDA framework, facilitated a nuanced understanding of these indicators in real-world settings. The deployment of various

sensors and modalities enabled a comprehensive data collection approach, integral for developing a robust MMLA system. The incorporation of advanced technologies, like the Arduino Nicla Vision board and deep learning techniques for audio analysis, underscored our commitment to leveraging cutting-edge tools to enhance language learning analytics.

Our study underscores the importance of a user-centered design philosophy [7], where the selection of learning indicators and modalities is informed by direct engagement with the learning context and user feedback. Moreover, the integration of various modalities and sensors necessitates a design strategy that prioritizes modularity and flexibility [8], allowing for the system to be adapted or expanded in response to evolving educational scenarios or technological advancements. The successful application of advanced technologies like the Arduino Nicla Vision board and deep learning algorithms for audio analysis in our study exemplifies the potential of integrating cutting-edge tools in MMLA design. In the broader context of educational technology design, our findings align with the growing recognition of the importance of adaptability and user-centricity in developing effective learning analytics systems. As articulated in recent literature [1, 3, 9, 10] the efficacy of learning analytics hinges on its ability to resonate with users and adapt to the unique characteristics of the learning environment. This aligns with our study's emphasis on continuous refinement and context-aware design in MMLA, highlighting a pathway toward more responsive and impactful learning analytics solutions.

The methodology employed in this study has several advantages and some limitations, aligning with existing literature on educational technology research. One significant advantage is the iterative nature of the design science approach, which allows for continuous refinement based on real-world feedback, enhancing the robustness and relevance of the developed MMLA system [11]. This approach ensures that the system is adaptable and responsive to the dynamic needs of language learners, as supported by recent studies emphasizing user-centered design in educational technologies [12]. Moreover, the use of authentic learning environments, such as language cafes, provides rich, contextualized data that can lead to more meaningful insights [13]. However, there are also notable limitations. The intensive iterative process can be time-consuming and resource-intensive, potentially limiting the scalability of the methodology. Additionally, the dependence on real-world settings may introduce variability that complicates the generalization of findings [1]. Ethical considerations, particularly related to privacy and data security, also pose challenges, requiring stringent measures to protect participant data [9]. Despite these challenges, the design science methodology remains a powerful tool for developing complex systems like MMLA, provided that these limitations are carefully managed.

Our findings demonstrate the potential of MMLA to offer rich, personalized insights into language learning, supported by the AI voice assistant's role as a conversational aide. However, the study also highlighted challenges, particularly in data collection and system scalability, necessitating further refinement of our prototype. By demonstrating the potential of MMLA in authentic settings, our study contributes to the evolving discourse on the integration of advanced analytics in language education. Moreover, the identified learning indicators and their respective modalities provide a framework for future research to build upon, particularly in exploring the scalability and adaptability

of MMLA systems in varied educational contexts. While our research has established a foundational understanding of MMLA's application in language cafes, the journey toward fully realizing its potential in language learning is ongoing. Future work will focus on addressing the current system's limitations, exploring new technological integrations, and expanding the application of MMLA to broader educational scenarios.

8 Conclusion

The aim of this study was to explore the application of MMLA in language learning, particularly within the dynamic context of language cafes. Using the MAMDA framework, we systematically designed an MMLA system integrating AI voice assistants and multimodal data collection. Our user study, conducted over several months, provided insights into language practice and interaction. The diversity in participant numbers and the dynamic nature of the sessions offered valuable insights into the real-world applicability of MMLA in language education. The development phase was iterative, with insights from the user study informing the prototype's creation, which included video and audio modalities. The vision mode utilized the Arduino Nicla Vision board for participant tracking, and advanced deep learning techniques were employed for audio analysis. This approach captured comprehensive multimodal data to support language learning analytics. Our conceptual model employs an AI voice assistant as a conversational aid, simultaneously collecting audio data for MMLA. The evaluation of our prototype in real-world settings showed promising results, with notable accuracy in speaker recognition and valuable feedback from participants. While our study established a foundational framework for MMLA in language learning, further exploration and refinement are needed. Future work will address limitations such as enhancing the vision module's field of view and improving audio data collection accuracy. Additionally, integrating RFID technology and expanding the camera setup could offer more robust tracking and analytics capabilities. Exploring MMLA's scalability and adaptability to various learning contexts beyond language cafes will also be a focus.

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SeriousGXcraft: An Innovative Framework for Multidisciplinary Serious Game Development Powered by XML

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Abstract. In the landscape of Education 4.0, Serious Games have gained traction as an engaging learning tool, blending entertainment with pedagogy. However, developing serious games poses multidisciplinary challenges, requiring a delicate balance between learning objectives and entertainment. Existing literature highlights significant gaps in addressing these challenges, with many frameworks adopting narrow perspectives that overlook the holistic nature of serious game development. This paper aims to fill these gaps by introducing SeriousGXcraft, an innovative eXtensible Markup Language (XML)-based framework for the development (design, implementation and evaluation) of serious games. Born from a collaborative effort between technology and psycho-pedagogy experts, this framework overcomes the limitations of existing ones by leveraging the flexibility of XML and providing a comprehensive and integrated solution. It promotes multidisciplinary collaboration and ensures a harmonious balance between the playful and learning aspects of serious games. By harnessing the capabilities of XML schema, SeriousGXcraft empowers the development team to effectively browse and provide visual support for the complex facets of the serious game development process.

Keywords: Serious Game · Development Framework · eXtensible Markup Language

1 Introduction

In the dynamic landscape of the digital era, Education 4.0 has emerged as a transformative paradigm, fundamentally reshaping the contours of both learning and teaching [2,13,16–18]. Within this context, Serious Games (SGs) have garnered significant attention as an innovative pedagogical tool, providing an

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engaging instrument for education [5,6,9,11,12,17,18]. These games seamlessly blend entertainment with education, leveraging the inherent power of play to facilitate effective learning experiences [6,9,11,12,17]. In fact, SGs stand out as a promising avenue for enhancing student engagement and knowledge acquisition [2,7,17,18]. Their immersive nature encourages active participation, fosters critical thinking, and promotes skill development, making them a valuable addition to modern learning practices [2,7,17,18]. By harnessing the synergy between playfulness and learning objectives, SGs pave the way for a more dynamic and impactful educational landscape.

The process of bringing SGs from ideation to realization is beset with numerous obstacles [5,9,10,12,17]. It demands a concerted, multidisciplinary endeavour, amalgamating the realms of psycho-pedagogy and computer science [9,10,12,17]. Furthermore, achieving success in this endeavour hinges upon striking a delicate balance between the learning objectives and the playful elements inherent in the game design, ensuring that the learning efficacy remains paramount amidst the entertainment value [9,10,12,17].

In this regard, an in-depth analysis of existing frameworks for developing SG reveals a significant gap in addressing these challenges [5,10,17]. Many frameworks focus on specific learning contexts [14,20], SG types [3,4,20], or development phases [4,8,10,15,19,22]. Furthermore, many frameworks tend to prioritize learning objectives over playful aspects, or vice versa, potentially overlooking the critical need to balance these two fundamental aspects in SGs development [4,10,15,19]. Lastly, these frameworks frequently lack adequate support for the inherent multidisciplinarity of SGs development; They often fail to delineate the roles of various stakeholders throughout the development process, thereby impeding effective collaboration [4,14,21,22]. Therefore, these lacks in the existing literature underscores the need for a more comprehensive approach to SG development [5,10,17].

This study aims to bridge the existing gaps in frameworks for SGs development by introducing a novel and innovative framework called **SeriousGXcraft**. This framework is the outcome of a collaborative and multidisciplinary effort involving experts with diverse backgrounds in computer science and psycho-pedagogy. By combining the expertise of these experts, SeriousGXcraft was built upon an in-depth analysis of existing frameworks (e.g., [3,4,8,10,14,15,19–22]), that allowed the identification of its main components. Then, the proposed framework was implemented as an eXtensible Markup Language (XML) schema owing to a multitude of factors. Firstly, XML's flexibility, adaptability, and scalability align seamlessly with the dynamic nature of SGs, enabling the framework to evolve by accommodating new elements and attributes as the SG landscape progresses. Moreover, XML's built-in data validation mechanisms ensure that SGs developed following the SeriousGXcraft framework adhere to predefined standards, thereby bolstering their quality and effectiveness. Additionally, the widespread adoption and accessibility of XML, coupled with the availability of diverse tools for XML document manipulation, render SeriousGXcraft user-friendly and accessible to a broad spectrum of users, regardless

of their programming proficiency. Finally, the utilization of an XML Schema enhances visual comprehensibility and facilitates data browsing, thereby aiding in understanding the underlying structure and relationships inherent in the SG development process.

The paper follows this structure: Sect. 2 presents the key features of the proposed framework, pointing out its practical implications and an example of its practical usage; Sect. 3 concludes and suggests future directions.

2 SeriousGXcraft Framework

This section presents the key characteristics of the SeriousGXcraft framework, beginning with the approach chosen for its inception. The approach entailed a comprehensive study and in-depth analysis of existing frameworks for SG development, along with a collaborative, multidisciplinary effort. Specifically, this approach has facilitated the identification of:

1. Phases that must be undertaken in the development of SGs;
2. Aspects and elements that must be considered in the development of SGs;
3. Relationships between identified aspects and elements;
4. Professional profiles that must be involved in the development of specific aspects and elements of SGs.

As a first result, the SeriousGXcraft framework is composed of four sequential primary elements, each referring to a specific aspect/phase of SG development: **Target Audience**, which refers to the group of people who will interact with the SG; **Abstract Level**, which refers to the conceptualization of the *Serious Aspects*, followed by the *Gaming Aspects* of the SG; **Concrete Level** which refers to the *Technological Aspects* and *Presentation Aspects* of the SG; and **Actual Level**, which refers to the practicalities related to *Implementation Aspects* and *Evaluation Aspects* of SGs. In Fig. 1, a conceptual view of the SeriousGXcraft framework is shown, emphasizing the levels and corresponding aspects in SG development phases and the iterative nature of the process. Specifically, through the examination of existing frameworks and leveraging multidisciplinary expertise, two types of forward temporal and functional relationships have been discerned (that is, mandatory and suggested), and three types of backward temporal and functional relationships. These relationships are identified considering the effort and time required for their realization, as well as the frequency of their occurrence. For the sake of example, revisiting Serious Aspects after defining Gaming Aspects is a scenario that can frequently occur at a low cost, aiming to strike the right balance between learning and playful facets. Conversely, returning to the Abstract Level of SG development (i.e., Serious and Gaming Aspects) after the evaluation of the developed SG is a situation that is ideally infrequent. This is due to the substantial investment required in terms of effort and time, necessitating a redesign of the foundational aspects of the SG's development.

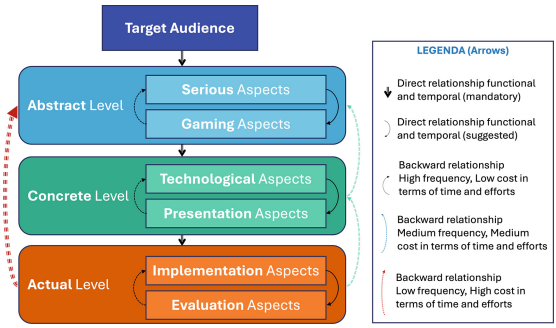

Fig. 1. A conceptual view of SeriousGXcraft highlighting Target Audience and the main aspects at three levels.

The framework's intrinsic value would remain incomplete if its development were to conclude at this stage. However, by translating the conceptual framework into an XML schema, we improved its functionality by adding a detailed layer. This enhancement involves integrating a comprehensive set of elements related to each identified aspect, specifying details like constraints, data types, and cardinality for each element. Additionally, the XML schema explicitly states the professional profiles required for developing specific aspects and elements of the SG through the `Actors Involved` element. For instance, experts in psychopedagogy are primarily responsible for designing elements related to *Serious Aspects*. Figure 2 illustrates an overview of the XML schema, depicting its first two hierarchical levels. Notably, the Serious Aspects section displays all its subordinate elements. For a comprehensive understanding, the complete XML schema can be downloaded here [1]. Delving deeper into the proposed SeriousGXcraft framework, here below is an overview of the main components of the XML schema:

1. **Target Audience** refers to the group of people who will interact with the SG. This includes their demographics (i.e., *Age*, *Biological Sex*, *Gender*, *Nationality*), needs, and behaviours (i.e., the presence or not of *Disabilities or Disease* and their *Baseline* knowledge. Understanding the target audience is the first phase for developing a SG that is learning-effective, engaging, and user-friendly.
2. **Abstract Level** refers to the design aspects to be considered in the second phase of the development process. This phase primarily involves conceptualizing the ***Serious Aspects***, followed by the ***Gaming Aspects*** of the SG:
   - Serious aspects include the *Purpose* of the SG (e.g., education, training, healthcare, rehabilitation), the *Stimulation Plan* (i.e., the plan governing the SG's usage), the *Learning Strategy* (i.e., the approach or method used to achieve the purpose), and the *Knowledge Content Model* (i.e., the specific knowledge or skills that the SG is intended to impart). It is

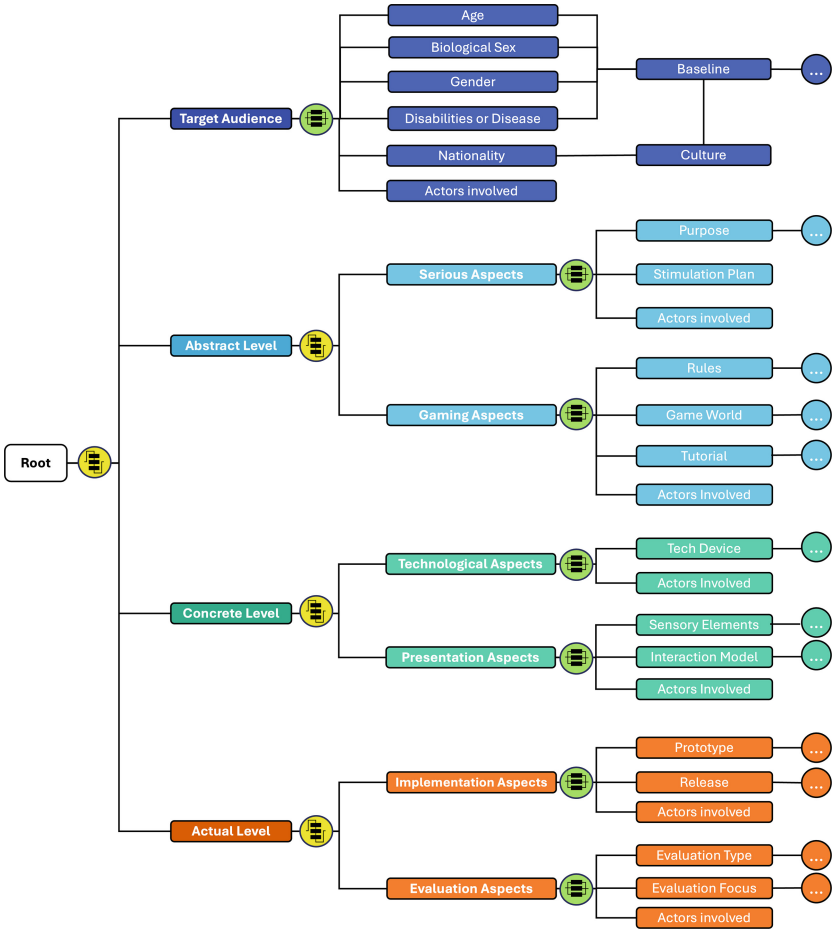

Fig. 2. SeriousGXcraft: An overview of a portion of the XML schema.

important to highlight that the definition of the *Learning Strategy* and the *Knowledge Content Model* are intrinsically linked to the Purpose, as they must align with and support the overall goal of the SG.

– Gaming aspects include the *Rules* and *Core Mechanics* that make up the gameplay of the SG (e.g., the *progression mechanics* and the *social interaction mechanics*); the *Game World* that encompass the *Scenario*, the *Characters* (both playable and not), and the *Storyline*; and the *Tutorial* (both explicit and implicit).

By considering both the Serious Aspects and Gaming Aspects at the abstract level, the team involved in the SG development strive to create a successful SG by balancing serious content and engaging gameplay. This approach establishes a solid foundation for a learning-effective and playful experience for the target audience.

3. **Concrete Level** refers to the design aspects to be considered in the third stage of the SG development, where the focus is on more concrete aspects of the SG, such as the technology used to deliver the SG (***Technological Aspects***) as well as how to present and interact with the SG (***Presentation Aspects***). Specifically:
   - Technological Aspects include the *Input devices* (e.g., mouse, keyboards, controllers) and *Output devices* (e.g., computers, head-mounted displays), and associated *Technological Drawbacks* (e.g., cybersickness that may arise from the use of head-mounted displays).
   - Presentation Aspects include: *Sensory Elements* that make up the User Interface (e.g., *visual* and *auditory elements*); the *Interaction Model* that determines how the play interacts with the SG's game world and characters (e.g., Avatar-based or Party-based); the *Camera Model* that represents how the player views the game world (e.g., first-person view or third-person view); the *Navigation Mechanism* that players use to navigate the game world (e.g., Rail Movement or Point-and-Click).
4. **Actual Level** refers to the last critical juncture in the SG development process, where the focus shifts from the creative phases of conceptualization and design to the practicalities of implementation (***Implementation aspects***) and evaluation (***Evaluation aspects***). Specifically:
   - Implementation Aspects encompass elements that are related to two key stages of implementation. The first stage is the *Prototype* deployment, which involves considerations such as the *Level of Interaction*, which could be a high or low-fidelity prototype, and the *Type of Prototype*, which could, for example, be a paper-and-pencil prototype. The second stage is the final *Release* deployment, which involves considerations such as the choice of *Programming Language*.
   - Evaluation Aspects encompass elements such as the *Evaluation Type* (i.e., expert-based or user-based evaluation) and the *Evaluation Focus* (e.g., evaluation of the learning effectiveness or of the engagement of the SG).

2.1 Practical Implications of SeriousGXcraft

The SeriousGXcraft framework aims to offer valuable and comprehensive support throughout various phases of SG development, providing enhanced browsing capabilities and visual assistance in diverse scenarios. The framework goes beyond its traditional role of assisting the team solely in initiating SG development from scratch and exhibits versatility in several other crucial situations.

One notable situation is its usefulness in evaluating existing SGs. By assessing the appropriate instantiation of all elements within the XML schema, the framework can identify any potential oversights in SG development and specific elements. This evaluation process ensures that the SG is built with a comprehensive understanding of the required elements, enhancing its overall quality and effectiveness.

Moreover, the framework proves beneficial for development teams already engaged in projects dealing with SG development. It enables seamless navigation

through the XML schema, allowing verification of the proper instantiation of preceding elements and ensuring the correct instantiation of subsequent elements during ongoing development phases. This capability streamlines the development process, minimizes errors, and promotes the consistent implementation of all necessary elements.

Furthermore, SeriousGXcraft is especially valuable for development teams commissioned to create SGs with specific features or constraints. For instance, consider a project involving the development of an SG targeted at children under the age of 6, accessible exclusively through a head-mounted display for immersive virtual reality. In this scenario, the proposed XML schema prompts careful considerations regarding the downstream decision-making process of ***Technological Aspects*** in relation to defining the **Target Audience**. This ensures the selection of the most suitable technology for the specified demographic, taking into account factors such as safety, user experience, and compliance with established guidelines. In the given example, where immersive virtual reality headsets are not recommended for children under 10 years old, two potential approaches emerge: revising the target audience to an older age group or reevaluating the choice of reference technology. Both options necessitate consensus and agreement with the client to align the SG development with the intended audience and ensure a safe and appropriate user experience. This highlights the framework's ability to guide development decisions and address potential challenges or constraints that may arise during the SG development process.

2.2 How to Use SeriousGXcraft: An Example

Suppose a multidisciplinary team, that includes two experts in psycho-pedagogy and two experts in computer science, wants to develop a serious game for the rehabilitation of social skills in individuals with Autism Spectrum Disorder using the SeriousGXcraft framework.

The first step is to create an XML document based on the SeriousGXCraft XML schema. The development process can then begin using this XML document as a guide, support, and record of all key information about the SG under development. To facilitate reader comprehension while adhering to page limitations, we will provide a practical example of using the SeriousGXcraft framework. This example will concentrate solely on the first two elements of the XML schema: **Target Audience** and ***Serious Aspects***. Specifically, the first aspect that needs to be addressed and described is the **Target Audience**, including all its child elements (i.e., *Age*, *Biological Sex*, *Gender*, *Nationality*, *Disabilities or Diseases*, and *Baseline* knowledge). As defined in the SeriousGXCraft XML schema, defining all this information requires a collaborative and multidisciplinary effort among all team members. Therefore, the *Actors Involved* child element should be filled out accordingly. Below is a possible completion of this section of the XML document:

```
<Target_Audience>
    <Age Age_value="18-30"/>
    <Biological_Sex sex_value="Not Relevant"/>
    <Gender gender_value="Not Relevant"/>
    <Nationality nationality_value="Various">
        <Culture culture_value="Various"/>
    </Nationality>
    <Disabilities_or_Disease disabilities_value="Autism Spectrum Disorder (
        High Functioning)"/>
    <Baseline>
        <Technological_Skills>Intermediate</Technological_Skills>
        <Other_Skills>High School Education or Higher</Other_Skills>
    </Baseline>
    <Actors_involved>two computer science experts and two psycho-
        pedagogical experts</Actors_involved>
</Target_Audience>
```

Next, following the structure outlined by the XML schema, the team focuses on the **Abstract Level** element, starting with its first child element, **Serious Aspects**, and its related sub-elements (i.e., *Purpose*, *Stimulation Plan*, *Learning Strategy*, and *Knowledge Content Model*).

Filling out this portion of the XML document involves a team discussion, similar to the Target Audience element. However, because the child elements under **Serious Aspects** pertain more to the psycho-pedagogical aspects of the SG, the experts in this field will have greater influence over the decisions. Below is a possible example of the XML document:

```
<Serious_Aspects>
    <Purpose purpose_value="Rehabilitation">
        <Learning_strategy>Experiential Learning</Learning_strategy>
        <Content_Knowledge_model> Key Areas= Social Skills Training,
            Communication Techniques, Emotional Recognition, Interactive
            Role-Playing, Conflict Resolution, Empathy Development</
            Content_Knowledge_model>
    </Purpose>
    <Stimulation_plan>
        The serious game should be played three times a week, in a
            supervised setting such as a therapy center or at home with the
             guidance of a therapist. Each session should last
            approximately 30-45 minutes.
    </Stimulation_plan>
    <Actors_involved>Mainly psycho-pedagogical experts</Actors_involved>
</Serious_Aspects>
```

The filling out of **Target Audience** and ***Serious Aspects*** elements exemplifies the activity that will be done for all other elements included in the XML document, following the structure proposed by the SeriousGXcraft XML schema. The adoption of the SeriousGXcraft XML-based framework may significantly benefit the team by providing a structured way for organizing the development process of SG, ensuring comprehensive consideration of all key elements, balancing the learning and playful aspects, and maintaining consistent and clear documentation. This structured approach can enhance communication among team members, clarify the project scope, and assist in tracking progress and making informed decisions throughout the development of the serious game.

While these advantages are still theoretical, the SeriousGXcraft framework shows considerable promise in supporting the development process of SG.

3 Conclusions and Future Works

SGs have emerged as a transformative tool within the landscape of Education 4.0, blending entertainment with education to create engaging and impactful learning experiences. Leveraging the inherent power of play, SGs have proven to enhance student engagement, foster critical thinking, and promote skill development. However, the journey from conceptualization to implementation of SGs is riddled with challenges, primarily due to the absence of comprehensive development frameworks capable of accommodating the multidimensional nature inherent in balancing learning objectives with playful elements. The proposed SeriousGXcraft framework aims to address the identified gaps in SG development leveraging from the capabilities of XML schema: It not only facilitates multidisciplinary collaboration and balances learning and playful aspects but also provides comprehensive guidance for all phases of SG development in several crucial situations.

To validate and refine the SeriousGXcraft framework, we plan to conduct a comprehensive evaluation. This will include an expert-based evaluation using a Delphi panel and a user-based evaluation through hands-on workshops. Furthermore, we aim to conduct four case studies to test the applicability of SeriousGXcraft in different contexts: (1) ASD rehabilitation interventions, (2) Music education, (3) Cybersecurity education, and (4) Physical rehabilitation. These case studies will provide valuable insights into the effectiveness and versatility of the SeriousGXcraft framework in various real-world scenarios.

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Alignment of Learning Design with Learning Analytics: Case of MOODLE Blended Learning Course in Higher Education

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Abstract. This paper studies the effects of the design for an Engineering Education introductory course in a brick-and-mortar university through the alignment of learning design with learning analytics. It looks at how an analysis of students' data (logs in the Learning Management System (LMS) and assessment grades) can be interpreted in the context of educators' pedagogical intentions and the influence in students' performance. The findings show that even in this case of blended learning it is possible to align learning analytics with learning design, leading to interesting conclusions nurturing course redesign. Learning analytics, which is the way data about students can be represented and analyzed can be used to understand the impact of learning design. In blended learning settings, it is possible to collect data about students' interactions with the LMS which are configured, at least partially, according to educators' designs. This study assumed a censors approach with a cross sectional design. Learner actions in the Moodle LMS for a course offered in a blended learning mode for engineering first-year students in a Spanish university were analyzed. The learning design employed in the course had a mix of collaborative task and individual tasks. The result indicates that; there is an alignment of learning design and learning analytics, there are observed behavior patterns in learners, there is a relationship between observed patterns and performance, and there is significant difference between the students in different clusters in individual tasks as opposed to group tasks.

Keywords: Learning design · Blended learning · learning analytics · learner behavior · Learning Management System

1 Introduction

Learning analytics (LA) is an important field that is useful in understanding the impact of learning design. The rise of e-learning and the availability of data generated from e-learning systems has led to the growth of the learning design (LD) and LA research. E-learning systems generate a lot of data that can be used to inform the instructors and learning designers on the student progress and effectiveness of the learning design on learner behavior [1]. With integration of technology in education, researchers have

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recognized that it is important to ensure that learning happens effectively in digital learning spaces. This has led to the effort of developing learning designs that will foster effective learning in e-learning environments. One of the capabilities of technology in education is the ability to collect and understand the learners' behaviors and learning progress from logs using learning analytics. Learning analytics can be used to understand the effectiveness of the learning design and to understand the learner behavior and characteristics. The use of learning analytics provides an opportunity for the educators to make changes on the learning design on the go as deep insights are revealed through the interactions of the learner with the learning design [2].

In their attempt to look ahead to the future of LA, [3]provide a conclusion that it is necessary for learning analytics experts to consider pedagogy. They argued that LA should start with considering the approaches to learning and teaching outlined in the theories. These approaches will provide guidance on the kind of learning analytics to be carried out. Researchers have observed that there is a very close relationship between learning analytics and learning design as the insights from learning analytics can be used to improve learning design and learning design can provide a framework for interpreting the insights from learning analytics [4, 5].

Even though there is a growing interest in both the LD and LA, there is little attention given to the relationship between LA that measures learner engagement on one hand, and LD on the other [6]. In addition, there is not enough empirical evidence to support that LD is an important driver for learning [7]. Furthermore, the potential of the data generated from learning management systems (LMS), has not been fully explored [8]. Also the focus of past research has been on the conceptualization of the learning design without considering their impact [10].

1.1 Research Questions

This study is an effort to provide evidence that there is an alignment between learning design and learning analytics which can be seen through the learner behavior and performance. Specifically, the study seeks to answer the following questions:

1. Does learning design align with learning analytics?
2. What patterns of behavior are witnessed when learning design aligns with learning analytics?
3. Is there a relationship between the observed learner behavior patterns with performance?

This work lies in the learning analytics layer of the learning design framework and its interaction with the design analytics layer [5].

2 Relevant Literature

LD influences learner engagement in online learning. The decisions that instructors make on the changes in learning design influence the behavior of learners as they learn in virtual learning environments [11]. Depending on the type of resource and its place in the learning design, different learner behavior patterns are seen. The behavior patterns of

learners reflect specific aspects of the learning design [4]. In addition, the use of different LDs result in different learner behavior patterns [12]. Research has also established that learner behavior patterns and performance are influenced by learning design [13] as high performing learners tend to be more self-regulated in online learning environments as compared to low performing learners [14].

Time allocation to tasks in the learning design tend to influence the behavior patterns of learners. For example, the number ideas generated by students was found to be influenced by the amount of time allocated to the task [15]. In addition, the time the learner spends in learning activities is dependent on the individual learner characteristic as high performing students tend to spend more time interacting with learning activities as compared to low performing students [6]. There is a relationship between the learning activities and the student behavior, satisfaction, and performance [16]. Even though there is some work in this area, there is still a need for more evidence on the alignment of LD and LA and how this alignment generates patterns of behavior that influence the learning performance.

3 Methodology

3.1 Participants

The data used in the study was drawn from first year engineering students undertaking an ICT Course in their first trimester. The course had two offerings using three languages: Spanish/Catalan and English. The course had a total of 85 students who were enrolled for the English undergraduate course and 128 for the Spanish/Catalan version. The study assumed a censors approach with a cross sectional design. Data was generated from students' interactions with the Moodle Learning Management System (LMS) where the course was hosted. The data consisted of the log data that had records of the time stamped views of each student to particular learning components, along with a record of students' grades for all the assigned tasks. The data was anonymized before any analysis was carried out.

The course focused on an introduction to transversal skills relevant in engineering education (with a focus on teamwork, oral and written communication skills, self-regulation) and an introduction to ICT engineering sectors [17]. The course was offered in a blended approach in that learners were expected to devote 150 h to engage in its proposed tasks, of which only 50 were in the classroom. The learning materials (files, web pages) and tasks (upload assignments, quizzes) were provided using the institutional LMS (Moodle).

3.2 Data Analysis Methods

Data was analyzed using descriptive statistics, correlational analysis, cluster analysis, and t-test. Graphs and charts were used to visualize the analysis. Python's pandas, matplotlib, seaborn and scipy libraries together with Microsoft excel were used to carry out the analysis. Before formal analysis of the data began, the data was anonymized. Log files were aggregated to indicate the total views of students on the learning materials. The

aggregation was based on the Moodle components. The components used included; Number of system views, forum views, page views, file views, URL view, files submitted, submission comments, video views, quiz views, assignment views.

4 Findings

4.1 Research Question 1: Is There an Alignment Between the Learning Design and the Learning Analytics?

To show how the learning design was aligned to the learning behavior, total views for each component was calculated and visualized as shown in Fig. 1. Each learner was expected to access a learning design component at least once. Each component had more than one view. The most frequently accessed component was system followed by file, quiz and URL components, while the least accessed components were user report, overview reports, submission comments and logs.

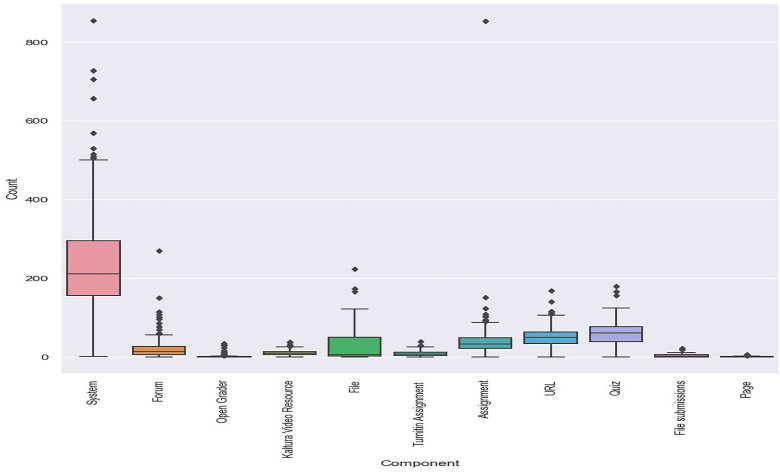

Fig. 1. Cumulative frequency of access to different learning components

The students seem to be active in the first three months that is September, October and November as shown in Fig. 2. December has minimal interactions in both the Spanish/Catalan and English course offering. This trend is consistent with the design of the course because many learning tasks were given in the months of September, October and November.

4.2 RQ 2. What Behavior Patterns Exist with the Alignment of Learning Design and Learning Analytics?

To see if the behavior pattern that were emerging from the alignment of LD and LA, correlation analysis and clustering was carried out. The correlation analysis was carried

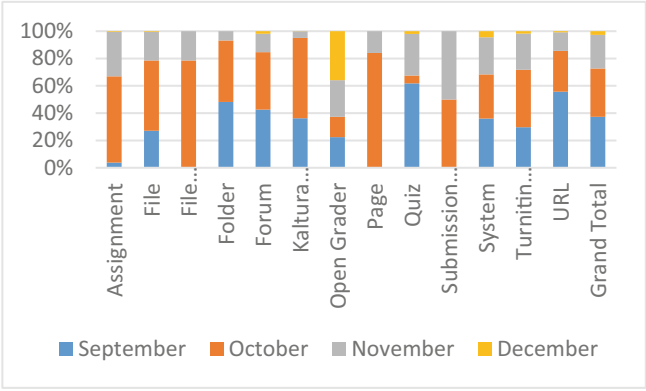

Fig. 2. Course activities over the period of the course

out in order to determine if there was a correlation between students access behavior of different components.

Correlation of the Various View Counts

All the components were positively correlated with each other. The system views had positive correlation with all the other components with the highest correlation being with URL and Quiz at 0.75 and 0.7. This shows that most of the times when students logged into the learning platform, they accessed the URL and the quiz. According to the learning design, the learning materials were offered using URLs and the assessments were majorly through the quizes, this explains why there is a high positive correlation between the URL, Quiz and the system.

Clustering

Clustering was used to find out the emerging behavior patterns of students based on the number of times they accessed the different components. To do so, we first experimented with three clustering algorithms in order to determine the best clustering algorithm and best number of clusters. We measured the performance of the algorithms using the Silhouette score. We found that K-Means had the highest score of 0.45 for 2 clusters followed by agglomerative clustering at 0.34 and spectral clustering being the poorest with −0.15. Guided by the scores, KMeans with two clusters was selected for clustering the students based on the number of views for the different learning components. KMeans produced two distinct clusters as shown in Fig. 3.

The students seemed to be generally active, with those in cluster 1 seeming to be more active than those in cluster 0. The average access behavior being higher in all the components for the active class. The total average access for all the components for cluster 1 is 745.5 and the total average access for cluster 0 is 365.5.

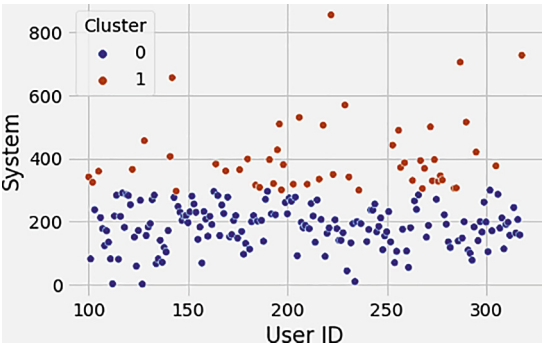

Fig. 3. Clusters using the KMeans algorithm

4.3 Q3. What is the Relationship Between the Observed Behavior Patterns with Performance?

Correlation, and t-test were carried out to establish if there was a significant a relationship between the observed behavior patterns and grades on the different assigned tasks. It was hypothesized that there was no correlation between students' access behavior and performance and that there was no difference between the performances of students in the different clusters. The average performance of the students in the most active cluster is higher than the average performance in the less active cluster in both group work and individual work. Cluster 0 had an average of 25.9 in group project, 15.9 in debates, 41.7 in class tasks and activities and quiz at 38.2. In cluster 1 the average for group 1 was 26.2, debates 16.0, class tasks and activities 44.9 and quiz 87.1.

Correlation of Access Behavior with Performance

To get the relationship between learning components and student performance, Pearson correlation analysis was carried out to see if there is any correlation between the access behavior to the different components and performance. The result indicates that performance in the quiz has a positive correlation with all the components except for files. The highest correlation between performance quiz with access to quiz and URL at 0.5 followed by system at 0.4. This is understandable given the fact that most times the students accessed the system they accessed either quiz or URL. Performance in class tasks activities had a positive correlation with all the components except for the open grader. This is majorly because open grader was not part of the components the students were required to access. The highest correlation of performance in in class activities is with system URL and Quiz with all of them being at 0.4. Group project and debates have low correlation with all the component views, with the highest correlation being 0.2 with file submissions. This can be explained by the fact that group work and debates happened outside the learning management systems and students just submitted reports of those activities using files in the learning system.

Checking for Statistical Difference Between the Generated Clusters

A t-test was used to find out if there is any statistical difference between the two generated clusters of students in terms of performance in the different assigned tasks. It

was observed that there was no statistical significance difference between the two clusters in the students' performance in both the group work and debates with a p-value of 0.47512 and 0.69705 respectively. There was a statistical significance difference in the performance in the quiz and in class activities with p-values of 0.0002 and 0.0025.

Discussion of the Results

As seen from the results, students access the learning system more where there are tasks assigned to them. We see diminished activities when the courses come to an end and the learners have fewer or no tasks to accomplish in the LMS. This is consistent with the results of other studies that investigated the relationship between learner behavior and learning design. Studies such as those reported in [18]. Other studies reported in [4, 11] found out that the access behavior of learners followed the path that the learning designers had intended.

The results indicate a relationship between the learner's actions in the system with the performance. All the interactions reported some form of correlation between the views and the learner performance. There are some similarities and differences between these results and those in a study in [19]. It is similar in that they both report on the existence of a relationship between the access behavior of learners and performance, but different in the sense that the study [19], found significant relationship in online courses but not in cases of technology supported face to face learning [19]. The study reported in this paper, in contrast found that the learner access behavior affects performance in individual tasks and not in collaborative tasks. In addition a study by [20] also found that success in learning is related to students interactions; students with higher interactions were likely to succeed in online learning.

From the t-Tests, there is no significant difference between the different access behaviors in group work and debates. This can be explained by the fact that those tasks were done outside the system and therefore cannot be affected by the interactions of an individual on the learning platform. In contrast, we see statistically significant differences between the quizzes and in class activities. Because these are tasks that were done individually and therefore the individual learner interactions affect their outcomes. These results corroborate the results of [21], who found out that learners who are active in the learning systems tend to have higher learning achievement as compared to those who are less active. Similar results were reported by [13, 22] in their study of learner behavior where they found out that students who were more active tend to outperform those who are less active. Earlier on [23], found out that even though online activities did not significantly affect performance, poor learning performance was experienced by learners who were not active in their online learning activities. The lack of statistical significance in the learner performance in the study reported in [23] and to some extent in this study in group work and the debates can be explained by the fact that there might be other factors that influence performance of learners in group work given the dynamics of the people involved in the work. It can be noted that the individual learner engagement in the learning management system only affects the tasks that were carried out individually and not the group work.

Conclusion and Recommendations

The first conclusion we can draw from this study is that the individual access behavior of

learners has a significant relationship with learner performance in individual activities especially the quizzes. Teachers should therefore consider developing learning design that encourage learner interaction. The second conclusion we can make is that based on the access to learning design components, learners can be clustered, and these clusters can be a basis for providing adaptive support and encouragement to learners to engage with learning design elements to increase their chances of being successful in the online learning process. Having adaptive support is known to help in managing learners cognitive load [24] especially in complex subjects such those in engineering education. Thirdly, the study has revealed that, even in blended leaning of engineering courses, it is possible to have an alignment of learning design and learning analytics hence leading to improvement of learning design. Having the alignment will help the engineering learning designers to be able to pinpoint points of correction. Through the insights gained from learning analytics, engineering education learning designs can be evaluated for their betterment. Fourthly, we can see that online interaction with the LMS components did not have an influence in the group work and debates. In this study context, group work and debates were tasks carried outside the LMS hence the individual interactions could not affect their performance. Further study should be carried out to investigate the influence of group interactions on the learner performance in group tasks.

Limitations of the Study and Future Research

This study had some limitations. First, the study was based on data from one institution and only one course with two offerings from one field. This might affect the generalizability of the results. We recommend that future studies consider having more courses from different fields and observe if the same conclusions can be drawn. Secondly, the data is limited to conclusively draw distinct clusters. Therefore, future studies can consider having more data from different courses with more students. Thirdly, the study only analyzed the engagement of individual learners. Further studies can analyze group engagement and evaluate how it effects group work.

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Teaching Dataset Creation in a Classroom Environment

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Abstract. Datasets are crucial for data-driven decision-making in businesses and organizations, allowing for the optimization of processes and the identification of improvement opportunities. They are foundational for training machine learning models to recognize patterns and make predictions, encompassing various forms of data from simple text and numbers to complex images and graphs. Furthermore, datasets are vital for educating future data scientists and machine learning engineers, offering hands-on experience that enhances analytical skills and practical knowledge. However, a gap in university curricula regarding dataset creation skills is noted, which is essential for ensuring the quality of datasets and the robustness of decision processes. This paper proposes to address this educational gap by developing methodologies for teaching dataset creation within university settings, mindful of the distinct skill levels of students compared to professionals.

Keywords: Datasets · Teaching · Teamwork

1 Introduction

Datasets are an essential part of any data-based decision process. They serve as the foundation for training machine learning models to make predictions or decisions. Datasets provide a common ground for comparing the performance of different algorithms or models, i.e., establishing a benchmarking baseline. Standardized datasets allow researchers to objectively measure the effectiveness of their approaches and compare them with existing methods. Getting hands-on experience with real datasets helps students understand data analysis techniques, develop critical thinking skills, and gain practical knowledge so as to prepare the next generation of data scientists, machine learning engineers, and researchers.

However, we must notice that dataset building is typically not a skill included in university curricula or at least not adequately reported in the literature. Such skills are anyway strongly needed to avoid neglecting the importance of adequate quality datasets and founding decision processes on shaky ground.

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In this paper, we aim to provide a starting ground to fill that gap, by proposing a methodology for teaching dataset creation in a classroom environment at the university level. Though guidelines for dataset management exist, introducing those skills in a classroom requires special care, at least because students do not possess the same skills and experiences as the professionals to whom those guidelines refer. Here, we first draw a sequence of steps that should be part of every dataset creation teaching and provide indications for each step (Sect. 3). Finally, we report our experience in a classroom where we applied those indications (Sect. 4).

2 Literature Review

As hinted in the Introduction, there is no literature specifically devoted to teaching how to create datasets in a classroom environment. However, there is a fair amount of literature devoted to the general theme of data lifecycle management (DLM) and its phases in a professional environment.

There is no general agreement on the number of phases involved in DLM. Harvard Business School identified a sequence of eight steps [25]. Rahul and Banyal adopt a sequence of six phases which starts with creating a dataset and finally destroying it [22]. Ofner et al. focus on master data that serve a strategic purpose and include deactivation and/or archiving as the final phase [16]. The topic has become particularly crucial in the Big Data era, with contributions ranging from embedding DLM into an ISO9001 framework [12] to considering security issues [27] or focusing on the public administration context [24].

The task we are dealing with here focuses on creating a dataset, which is what most sources refer to as the generation and collection phases in DLM.

3 Methodology Framework

In this section, we review the methodological framework we have devised to build a dataset from students' joint work. We aim to build a dataset of significant size (to be used for further research or for more complex teaching assignments).

We have subdivided the framework into the following sequence of phases, described in greater detail in the following subsections: 1) Defining objectives; 2) Assigning tasks; 3) Collecting data; 4) Formatting data; 5) Releasing data. This sequence intersects with that identified in [25], though we are mainly concerned with getting the class involved in steps 1 through 4 of [25].

3.1 Defining Objectives

The first step is to clearly define which data we are looking for and the purpose of the collection.

Since data are typically not available in the exact form and coverage that we expect, we must warn students that they must first collect all the data somewhat related to the research scope and then process it, since that data may

include additional information or fail to cover the whole scope (e.g., missing some attributes or exhibiting a limited time coverage). They must also be advised to take note of where they retrieved their data (for later access and proper referencing) and when (again for proper referencing). The purpose of data collection must also be very clear to the students to help them be aware of the significance of their effort and of the embedding of their data in a wider activity.

3.2 Assigning Tasks

Assigning a task to the students in the classroom involves some choices concerning the individual vs group assignment and the degree of overlapping.

First, we have to decide whether to assign data collection tasks to individual students or groups. In both cases, their tasks are part of the collective task of creating a dataset. The dilemma between individual and group assignments is a long-standing one. Either choice has pros and cons. Individual assignments allow the instructor to properly evaluate the effort of all students, as group assignments are marred by free-riding [17]. Though the latter issue is a major one, working in a group helps improve team-building skills. In addition, several positive recommendations can be made to make teamwork better, e.g., encouraging students to adopt an integrative workload approach rather than a divided one, or suggesting holding student meetings to resolve conflict via grievance procedures [23]. Also, some imbalance of skills may find a sort of compensation inside the group, since students lacking computing skills may seek help from their teammates; this may be a significant issue in classes outside computer science degrees [14]. In the following, we refer to both individuals and groups as project work units.

A second choice concerns the allotment of the overall task to work units. Adopting a partitioning approach, where all units are assigned disjoint tasks, allows us to build a larger dataset. However, this may result in a poorer-quality dataset. In fact, assigning tasks in an overlapping way so that at least two units share tasks makes room for error reduction. Though two units that have been assigned the same task (portion) may have obtained their data from different sources, their outputs should be identical or at least easily comparable. Detecting discrepancies between the datasets obtained by the two units helps trigger a warning signal, which should call for a deeper analysis of the dataset. Here, we must consider separately the cases where we are collecting data subject to randomness or not. An example of the former case is the collection of data concerning traffic at a roadway junction; if the measurements by two units are taken on different days (though on the same weekday at the same time so as to hope for statistical homogeneity), then there is not a *correct* dataset, as both represent two samples of the same phenomenon and can be safely considered jointly (statistical tests or other approaches may be employed to testify their consistency [18]). Instead, the case we are mostly interested in is when we expect to find the same data whatever the source, e.g., when we are trying to retrieve the dialogues in a TV show as in the case study of Sect. 4. In this case, the presence of inconsistencies may be healed if we adopt a majority vote (with each task being assigned to three work units at least) or resort to more sophisticated

schemes for handling disagreement between annotators [3,9]. The amount of redundancy introduced during task assignments will reduce the overall amount of data collected.

3.3 Collecting Data

When it comes to collecting data, students need to know where to search, how to format data, and how to provide information about the data.

Data Repositories. We must provide students with directions about the websites to search, looking first into well-known data repositories such as Kaggle (www.kaggle.com) or Google Dataset Search [6] (https://datasetsearch.research.google.com). However, it is better to introduce students to a wider panorama. An excellent guide in the search for data is [5], which lists both general resources and separate lists for specific application domains. If those ready-made repositories prove to be insufficient, students should be introduced to the advanced use of general search engines, e.g., Google Dorks [1].

Annotation and Labelling. Data annotation may involve adding relevant information (e.g., metadata) and labelling the area of interest in images and videos. In supervised machine learning, labelling should also include assigning a class to each input instance for training and testing purposes. Consistency in annotation should be guaranteed across work units in a classroom environment, where the annotation effort is distributed. Many systems and toolkits for annotation can serve as standardized platforms for many different tasks in many different forms, e.g., LabelStudio[1], which can be used to annotate several types of data. Using annotation software guarantees also the availability of integration and export methods. In fact, storing the data to be annotated could rely on a cloud storage service, which could make it difficult to provide data to many different users while keeping the database coordinated. Annotation software can show data directly from the cloud provider, eliminating any difficulty in coordination. The same advantage is also guaranteed for exporting annotated data: indeed, leaving each user the responsibility to export annotation (especially in non-computer science degrees) may lead to incompatible formats or structures.

Quality Control and Validation. In the case of data extracted from the web, students should be called to look for two sources of errors: 1) unreliable data sources; 2) errors in retrieving. The first issue is particularly significant as anyone can post material on the web. Assessing the reliability of the source is then the first issue to tackle [20]. Students should be pushed to critically assess the historical reliability of the source they are using. They should ask themselves the following questions: "Is this source a well-known and respected source of data? Is it an institutional source? Is it an individual?" Diversity in sources should anyway be sought [11]: students should be encouraged to look for data

[1] https://github.com/HumanSignal/label-studio/.

from different sources and maybe see if different secondary sources point to the same primary source of data. In the case of merging contents from different sources, with possible inconsistencies, a methodological approach to deal with information conflicts has been proposed in [10]. In the simple case of the very same piece of information collected through different sources, a simple majority voting may be applied if some redundancy is introduced in assigning tasks to students, as described in Sect. 3.2.

3.4 Formatting Data

The data provided by all work units must be put in the same format, so that the aggregation of data coming from different work units is easier. A tabular format is probably the most suited for later use. The work units should then employ the same labelling and ordering of rows and columns. Recommended formats are comma-separated values (.csv), JSON, XML, or Excel sheets. In the case of numerical data, care must be exercised in having the same decimal separators.

3.5 Releasing Data

In our educational context, we are mainly interested in having students document the data so that they can be handed over containing all the needed information for later use. A notable approach is based on Data Cards [21]. Data Cards employ a box-style rendering of the most relevant information about the dataset and allow for 31 content themes. In addition to including credits for the authors and ownership and reuse information, Data Cards include a snapshot view of the data (e.g., size, number of instances, number of classes), the nature of the data (mode, format), possible use cases and intended application, the evolution (the data cards allows for dataset modifications throughout its lifecycle), some descriptive statistics, and cleaning, parsing and processing details. Other proposals included Model Cards, which reported both qualitative information (ethical considerations, target users, and use cases) and quantitative information about the data [19]; Datasheets for Datasets, made of a set of questions designed to describe the motivation, composition, and collection process [15]; FactSheets, which includes a large set of items to provide information about the creation and deployment of an AI service, grouped around the categories of statement of purpose, basic performance, safety, security, and lineage [2]; and Data Statements, which were originally developed for documenting natural language processing systems and aim to minimize critical scientific and ethical issues [4].

4 Case Study: Creating a Dataset from TV Series

In this section, we report a real-world case of application of the framework described in Sect. 3. The class tasked with building the dataset was *Automatic Language Processing* in the degree course in Linguistic and Cultural Mediation at LUMSA University.

The objective was to build a dataset including the dialogues in the South Korean TV series Doctor Cha[2]. The series is a medical drama set in a university hospital, where the main character is Cha Jeong-suk, a twenty-year-old housewife who becomes a first-year medical resident. The expected coverage was the whole set of seasons and episodes in both English and Korean. This task was conceived in a larger framework aiming to build a set of datasets extracted from TV series [7,8,13]. Students were asked to find some information about the series and to retrieve subtitles, including the speaker's indication, as a proxy for dialogues.

Since the series has 16 episodes, students were grouped so that each episode would be retrieved by three concurring groups. In such assignments, we can use some basic results from graph theory to compute the number n of groups needed for that requirement. We can form a graph, where nodes represent groups, and two nodes are connected if they share an episode. Hence, the number of edges A equals the number of episodes E times the required redundancy R (three in our case). We can use the degree-sum formula for undirected graphs (see Chap. 2 of [26]) to derive the number of groups

$$2A = 2ER = \sum_{i=1}^{n} G_i = nG \rightarrow n = \frac{2ER}{G} \tag{1}$$

where G_i is the degree of node i, i.e., the number of episodes assigned to group i, and the load is shared uniformly across the groups ($G_i = G$).

In our case, we had a constraint on the number of groups (due to the number of students in the class), so we derived the number of episodes G to assign to each group. Since we formed $n = 12$ groups, and we required a redundancy $R = 3$, we needed to assign $G = 2ER/n = 96/12 = 8$ episodes to each group. The average number of episodes s shared between two groups can be obtained as the number of edges divided by the number of group (i.e., node) pairs:

$$s = \frac{A}{n(n-1)/2} = \frac{2A}{n(n-1)} = \frac{nG}{n(n-1)} = \frac{G}{n-1} \tag{2}$$

The resulting degree of sharing was $s = 8/11 \simeq 0.73$, with any two groups sharing less than one episode on average. In general, if we wish to allocate the load uniformly across the groups and have some degree of freedom to define the number of groups, the choices for the number of groups and the number of episodes assigned to each group will be the integer solutions of the equation $xy = ER$, where the overall number of episodes and the required redundancy are the known parameters.

Students could not find the subtitles on either Google Dataset Search or Kaggle and resorted instead to https://www.opensubtitles.com/it, where they retrieved the subtitles in plain text format. The speakers were added by manual annotation. Each group was required to form a separate csv file for each episode and either language (English and Korean), with two columns reporting respectively the speaker and its line. In order to document the dataset, we asked

[2] https://en.wikipedia.org/wiki/Doctor_Cha.

Movie Dialogues

The dataset consists of the subtitles of all the episodes of the South Korean TV series Doctor Cha, in both English and Korean. There are 16 episodes, each consisting of a number of speech lines between 664 and 1460.

PUBLISHER Lumsa University	**INDUSTRY TYPE** Academic	**INTENDED USE CASE(S), SUITABLE USE CASE(S)** This dataset was collected to carry out a sentiment analysis on the main characters of the series and to identify the dynamics of dialogues in the framework of a network analysis of the relationship between characters. It is intended for the following use cases: – Computing the sentiment of main characters – Extracting the character network
DATA SUBJECT Fictional	**VALIDATION METHOD** Triple redundancy Majority voting Manual inspection	**UNSUITABLE USE CASE(S)** This dataset is not intended for the following use cases: – Commercial exploitation
LAT UPDATE February 15, 2024	**KEY APPLICATIONS** Sentiment analysis Dialogue network structure	**SOURCE DATASET(S)** List of all the web sources that this dataset was derived from: – www.opensubtitles.com
PRIMARY DATA FORMAT(S) Textual data	**BREAKDOWN** Total Instances 15,292 Number of episodes 16	**NATURE OF CONTENT** The dataset consists of 16 files. Each file contains the speech lines for the whole episode. An instance is a single speech line (in English and Korean) accompanied by the indication of the speaker.
PRIMARY DATA TYPE Text data	**EXAMPLE OF ACTUAL DATA POINT** – Speaker: Doctor Cha – English line: Give that to me. I'll wear it myself. – Korean line: 이리 줘, 내가 입을게	**DATA COLLECTION METHODS** Retrieving data from the web Manual annotation

Fig. 1. Excerpt of the Data Card for the TV series *Doctor Cha*

students to adopt the Data Cards approach described in Sect. 3.5 [21] (see the excerpt of the data card in Fig. 1).

Since redundancy was introduced in the assignment process, the data retrieved by the groups were merged. Merging the datasets created by the groups is a recursive task, because merging two datasets leads to a new dataset in the same form as the original one, but made consistent. So, it is enough to analyse

the process of merging two datasets. In this case, we need to consider all the possible cases:

1. a line was assigned by both groups to the same character;
2. a line was assigned by the two groups to different characters;
3. two lines were merged by a group but not by the other, which left both lines assigned to the same character;
4. two lines were merged by a group but not by the other, which assigned one to the same character as the former group and the other line to another character;
5. two lines were merged by one group but not by the other, which assigned both lines to a character different from the former group;
6. two lines were merged by a group but not by the other, which assigned the two lines to different characters;
7. two lines were merged by both groups but assigned to different characters.

We can easily see that: Case 3 leads us to Case 1; in Case 4, we can split again the lines, considering the lines after splitting as assigned to the same character, and so we can consider any instance of Case 1 as an instance of Case 2; in Case 5, we can merge the lines and assign the merged line to the same character, and end up with Case 2; in Case 6, we can split again the lines (just like in Case 4) and go back to two instances of Case 2; Case 7 leads us to Case 2. This leaves us with just instances of Cases 1 and 2.

In Case 1, the line and character attribution will be reported as in the contributing datasets because of their consistency. Instead, Case 2 cannot be solved just by using the two contributing datasets. We need to exploit the other groups too, but we may face two situations: either the same line is present in another dataset, or the only instances are the ones we are analysing. Even though it should always be preferable to assign each sample to an odd number of groups (at least three), this is not always possible. So, if the same sample is assigned to an odd number of groups, we just adopt a majority vote. If this is not possible, we need to reconcile data through a different strategy. In this case, the strategy we used exploited the fact that many samples (even if not all of them) were assigned to at least three groups, so the strategy consisted in assigning each group a score given by the proportion of the times they were part of the majority in a decision. The group with the highest score was considered the rightful one. In the rare cases when the groups had the same score, we considered the number of students in each group, favouring the largest one. In the very rare cases when the groups have the same size, we let the students discuss to reach a final decision by consensus. Each time two datasets were merged, the groups were considered merged for those episodes. This form of gamification of the merging stimulated the students to be part of all the data annotation processes.

5 Discussion and Conclusions

We have drafted a methodology for teaching students how to create datasets, letting them develop skills for searching for the data, collecting and processing

it to remove errors, and documenting it for later use. We have reported our experience with a non-computer science major classroom, which adds to the complexity of the task. While the students were quite skilled in searching the data and collecting them, and they were also at ease with moving away from the Excel-oriented approach to use csv formats, the most critical phase turned out to be processing and documenting the data. Some processing was needed to merge data coming from different sources (students had to couple each speech line with the corresponding speaker). Documenting the data was a bit hard since the beginning, as students were not used to keeping track of the searching task and reporting all the information useful for later use of the data.

A major source of problems was teamwork. Some free-riding issues emerged as well as difficulties in smoothly merging the individual contributions. As indications for the future, we realized that some formal instruction is especially needed for the processing and documenting phases.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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Counting Triangles with 3D Printed Didactic Materials for People with Visual Impairment

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Abstract. Digital accessibility includes producing learning materials for various sensor needs such as objects supporting tactile learning in visually impaired people. Here are presented 3D printing models related to triangle-counting tasks suitable for use by people with visual impairment. 3D printing files and the corresponding models made from them are given, as well as the system of triangle-counting tasks that help solve with links to 3D triangle model files, an example methodology for their use, as well as some typical errors and options for overcoming them. Some results of work with them by people with visual impairment and attendees of an interdisciplinary seminar are described. The developed scenarios for working with the 3D models are suitable for use in the classroom and in STEM centers with different consumer groups.

Keywords: People with Visual Loss · Mathematics Education · 3D Printing · Manipulatives · Didactic Resources · Analogy

1 Introduction

In the National External Assessment, in Republic of Bulgaria, in the 4th grade in 2022, among the tasks with a selectable answer, the result of a task related to counting isosceles triangles was the lowest [1]. For the counting of squares and rectangles in geometric compositions, students are prepared by working in a square grid, as well as by the presence of similar tasks in recreational mathematics and in mathematics textbooks. In research, competitions and extracurricular work in mathematics, there are examples related to isometric grid tasks, as well as triangle counting, but they are not yet widely used [2, 3]. Single tasks of the considered type are also found in textbooks and study aids [4, 5]. It is interesting to see if visually impaired people meet the same difficulties in solving the triangle counting task, as is observed in the assessment exercise above.

Some of the activities and typical errors when solving such tasks are analogous to the activities and typical errors when counting rectangles and, in particular, squares. Most often, the "big" or "larger" triangles, or the differently positioned ones, are omitted. When compiling a system of tasks related to counting triangles, it is appropriate: to increase

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the number of elements, to complicate their mutual arrangement, to use isometric mesh nodes for triangle vertices, to use manipulatives, for example, matchsticks [6], through which to build or modify a composition, etc. In the Virtual Mathematics Laboratory [7], developed at the Institute of Mathematics and Informatics of the Bulgarian Academy of Sciences, there are virtual models that can be used to easily set such tasks, to facilitate the relevant research, to visualize the solution, to compose similar tasks. Such tasks are set in some editions of the online competition "VIVA Mathematics with a computer" [8] and computer models can be used to compose new examples.

Accessibility to digital resources in the country for people with visual impairment continues to develop and update according to the latest technological and organizational policies and standards [9–12].

Here we will present 3D printing files and the corresponding models made from them, as well as the system of tasks for counting triangles, the solution of which they help, the links to 3D triangle model files, an example methodology for their use, as well as some typical errors and options for overcoming them. For printing the 3D models, the free XYZMaker3DKit software was used, allowing direct construction of geometric figures from the ready-made shapes in the library (Figs. 1 and 2).

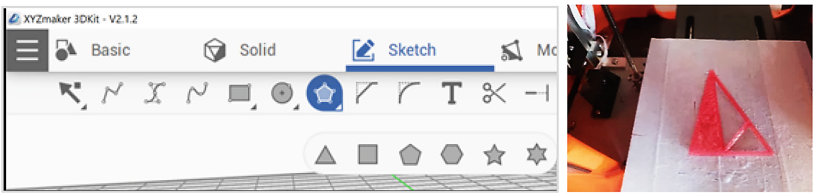

Fig. 1. Ready-made shapes of geometric figures in the XYZMaker3DKit software and model in the printer.

2 A System of Tasks for Counting Triangles

Task 1. Count the triangles in the figure.

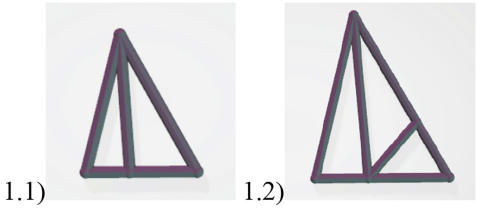

Fig. 2. Virtual 3D task models 1.1[1]) and 1.2[2]) for triangle counting.

[1] Stl file, https://cabinet.bg/storage/N4.1.stl last accessed 2024/03/01.
[2] Stl file, https://cabinet.bg/storage/N4.2.stl last accessed 2024/03/01.

If necessary, a 3D printed model of a single triangle is provided and a description of the figure is given. In case of an incomplete answer, the models with filled specific triangles are used (Fig. 3).

When solving Task 1.2, solvers should focus on an analogous activity. In case of an incomplete answer, the models with filled concrete triangles are used. The triangle from the fourth pattern below, which is a union of two smaller triangles (Fig. 4), is most often omitted.

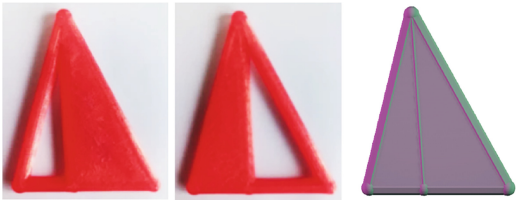

Fig. 3. Help 3D models representing the solutions of Task 1.1.

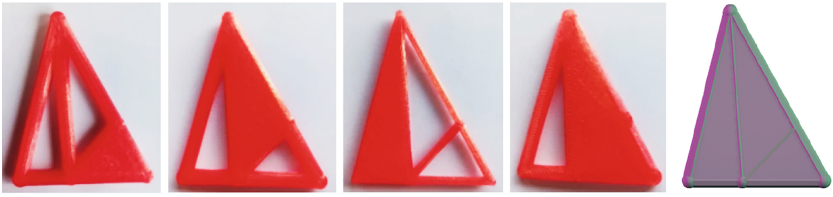

Fig. 4. Help models for the solution of Task 1.2.

And one can use the solution of the previous Task 1.1 and add the number of triangles connected to the added segment, which are two (the first two in Fig. 4).

Task 2. Count the triangles in the figure (Fig. 5).

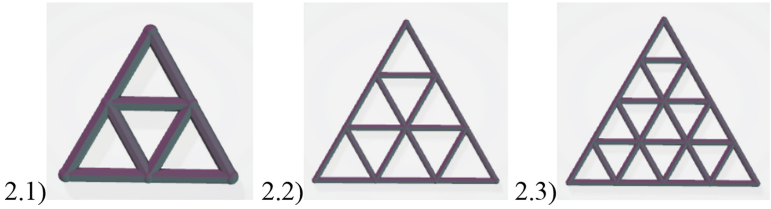

Fig. 5. Virtual 3D models of Tasks 2.1[3], 2.2[4] and 2.3[5] for counting triangles.

In case of an incomplete response, models with filled concrete triangles are used (Fig. 6).

[3] Stl file, https://cabinet.bg/storage/N5.1.stl last accessed 2024/03/01.
[4] Stl file, https://cabinet.bg/storage/N5.2.stl last accessed 2024/03/01.
[5] Stl file, https://cabinet.bg/storage/N5.3.stl last accessed 2024/03/01.

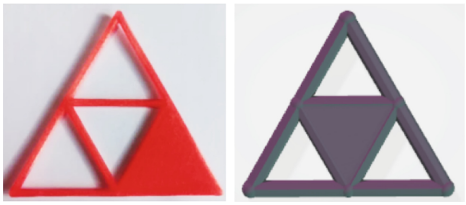

Fig. 6. Models of help files for Task 2.1.

It is appropriate to discuss that all triangles are equilateral and to refer to their systematized description. One way of describing it is: 4 equilateral triangles of side length 1 and one equilateral triangle of side length 2 (Fig. 7).

Fig. 7. Solution of Task 2.1.

It is appropriate to monitor the way solvers take notes during their independent work, bearing in mind that some tasks can also be solved orally.

When counting the triangles in the figure of Task 2.2, the solvers can turn to an analogous activity, as when solving Task 2.1. If necessary, the different sized triangles can be discussed first and then the counting done (Fig. 8).

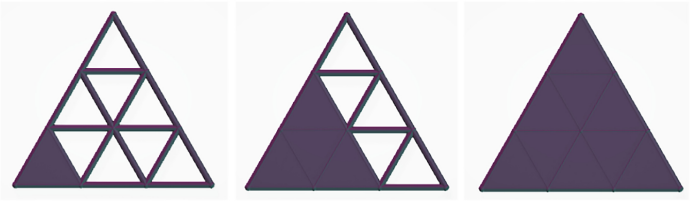

Fig. 8. Virtual 3D models of equilateral triangles with side lengths 1, 2 and 3 from Task 2.2.

In this case, there are 9 equilateral triangles with side length 1, three equilateral triangles with side length 2, and one equilateral triangle with side length 3 (Fig. 9).

In case of an incomplete answer, during the analogous solution of Task 2.3, models with filled concrete triangles are used, or help models cut from cardboard are pre-pared (Fig. 10).

Analogous to the previous two tasks for counting equilateral triangles, here it is al-so recommended to discuss rational options for solving, as well as ways to record results, for example, according to the length of the side of the types of triangles.

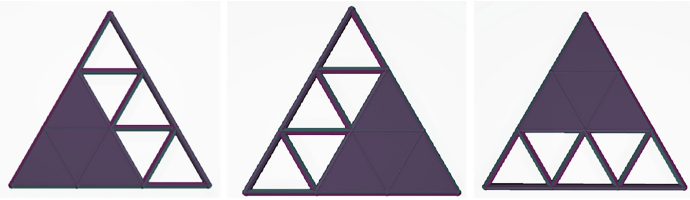

Fig. 9. The three equilateral triangles with side length 2 from Task 2.2.

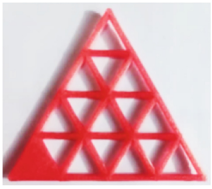

Fig. 10. Printed model of the help file for Task 2.3, with a filled triangle of side length 1.

In this case, there are 16 equilateral triangles with side length 1, 7 equilateral triangles with side length 2, 3 equilateral triangles with side length 3, and one equilateral triangle with side length 4 (Fig. 11).

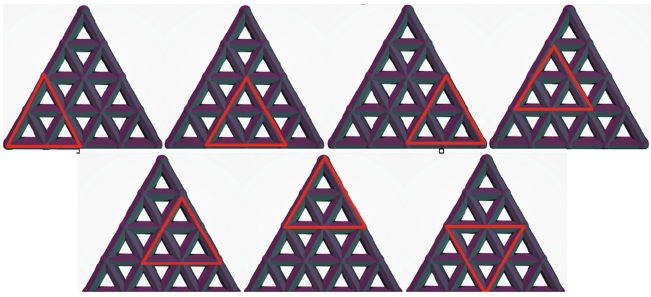

Fig. 11. The seven equilateral triangles with side length 2 from Task 2.3.

Here, it is appropriate for solvers to be directed to compose a triangle-counting task.

Task 3. Count the triangles in the composition (Fig. 12).

In case of an incomplete answer, models with filled concrete triangles are used, or help models cut out of cardboard are prepared (Fig. 13).

It is appropriate for solvers to aim for a rational calculation when counting the triangles of Task 3.2., using the answer of Task 2.1. They must consider that each of the walls of the body represents the figure from task 2.1.

To the $4 \times 5 = 20$ triangles (5 triangles each on the four sides of the large tetrahedron) the four "inner" triangles must be added (Fig. 14). Another option is to discuss the different size triangles and then do the counting (Fig. 15).

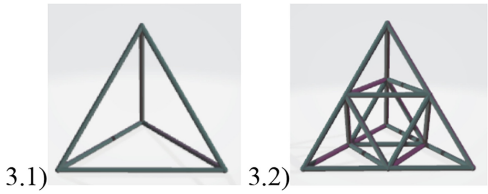

Fig. 12. Virtual 3D models of Task 3.1[6] and 3.2[7] for counting triangles.

Fig. 13. Virtual 3D Help file model for Task 3.1.

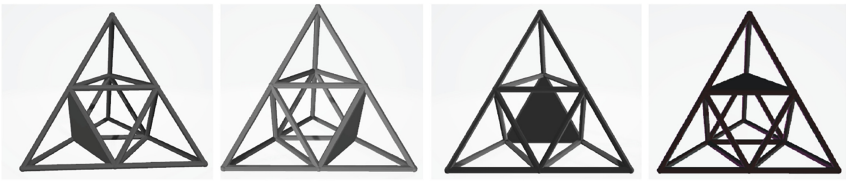

Fig. 14. Virtual 3D model of help file for Task 3.2, showing the filled "inner" triangles.

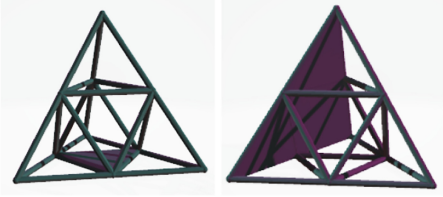

Fig. 15. Variant for solving Task 3.2 according to the size of the triangles in the composition.

In case of an incomplete answer, models with filled concrete triangles are used, or help models cut out of cardboard are prepared.

3 A Pilot Experiment

During the experiment was monitored:

[6] Stl file, https://cabinet.bg/storage/N6.1.stl last accessed 2024/03/01.
[7] Stl file, https://cabinet.bg/storage/N6.2.stl last accessed 2024/03/01.

- the way of solving tasks, with the help of the printed 3D models, by people with visual impairment;
- the time taken to solve an individual task;
- the type and number of mistakes made;
- the degree of independent work by each solver are recorded.

A solver evaluation of the 3D printed models and a self-rating of satisfaction were requested. Since it was a pilot study, quantitative indicators were not recorded. The descriptive statistics will be recorded and analyzed in future studies.

Within the framework of the first part of the spring seminar "Information Society", held in March 2024 in the city of Veliko Tarnovo, Bulgaria (Scientific network INIS and interdisciplinary seminar "Information Society" [13]) participated 10 people in place and 15 online. The audience included researchers in the professional field of mathematics and informatics, between the ages of 20 and 65, who voluntarily attended the public invitation to the free event. Those present on the spot were given the task system on paper, on which anyone could take notes if they wished. The online participants were able to observe the dialog in place, while seeing the tasks in presentation.

The solvers had the opportunity to work independently before moving on to working with the supporting 3D printed models and dialogue about the correct answer and mistakes made. No one from the participants solved this system of counting tasks completely, i.e. every solver got it wrong at least once. In examining the model from Task 2.1, all solvers listed the correct number of triangles within a few seconds, but spent an additional 1 min discussing with each other, looking for any "hidden" triangles.

The visually impaired person in the audience (male, aged 46, reported a complete loss of visual acuity during his twenties) in the audience also rotated the figure several times, counting the triangles (Fig. 16).

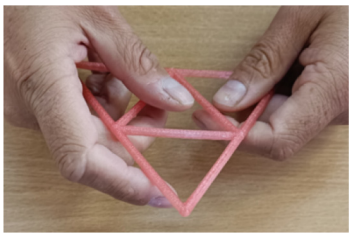

Fig. 16. Counting triangles in a plane figure for Task 2.1 by a visually impaired person during an "Information Society workshop".

When working with the models in Tasks 2.2 and 2.3, all solvers used counting based on the length of a side of a figure. Several solvers asked if there was a formula to solve the task without having to count all the triangles. They had a short discussion among themselves, looking for formulas. All solvers supplemented and/or edited their answer to Task 2.3, after the presentation of all solutions by the lecturer.

Moving on to the tasks with solids, on Task 3.1 everyone directly recorded an answer, which they then repeated out loud. The final Task 3.2 was a challenge for all solvers. Those present in the hall directly asked for some time to work with the printed model.

By analogy with the tasks of counting squares in solids, here too the solvers did not take into account that the sides of the pyramid are formed by an already known figure whose number of triangles is known to them. The visually impaired participant indicated that he liked this task the most (Fig. 17).

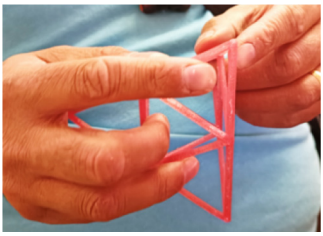

Fig. 17. Counting triangles in a composition for Task 3.2, by a visually impaired person, during an "Information Society workshop".

During the second part of the study, the audience included members from the Pensioners' Club and the Union of the Blind, Veliko Tarnovo branch, whose meetings are housed in the same building. The two participants were – a visually impaired person, 50 years old, male; and pensioner, 84 years old, female.

From the model in Task 1.1, both solvers indicated the correct answer without doubt and directly put the printed figure aside to wait for the next one. Obtaining the next pattern from Task 1.2, the pensioner counted all 5 triangles, including the intermediate one, formed by two smaller ones. The visually impaired person did not recognize this intermediate triangle and indicated "1 large and 3 smaller" as the answer. In contrast, when solving Task 2.1, the visually impaired person with one study indicated 5 triangles, while the pensioner counted 1 large outer triangle and 1 smaller "inverted" one in the middle. Only after presenting the help model was the pensioner able to separate all the triangles from the model. When working with the next two models from Tasks 2.2 and 2.3, the visually impaired person counted only the large outer and the identical smallest squares with side length 1. In contrast, the pensioner also completed the intermediate ones with side length 2 and 2, paying special attention and looking for "inverted" triangles by analogy with Task 2.1. In the first seconds of examining the model from Task 3.1, the visually impaired person said that it was a pyramid and indicated the correct answer. The pensioner also indicated the correct answer, but after several rotations of the printed composition (Fig. 18).

Both failed to realize that several internal built-in connections were also available in the model from Task 3.2. Accordingly, both of their initial answers were 16, multiplying the number of triangles on one wall by 4. After the lecturer directed them to explore the interior of the figure as well, they still failed to count them in full. Both shared that, in general, nothing from the system of tasks was difficult for them. The pensioner elaborated that perhaps the last Task 3.2 had many elements and he should have spent more time on it. The pensioner shared that all the tasks were interesting. The visually impaired man singled out the pyramid and commented that "the triangle is the magic figure for me, and the pyramid is a combination of them. A figure with a stable foundation".

Fig. 18. Individual work with a pensioner in the Union of the Blind, Veliko Tarnovo branch.

In overall, they enjoyed solving this type of tasks and found them fascinating and enlightening.

4 Conclusion

The present work is most similar to the study of applying 3D printed models to teach statistics concepts to people with visual impairment in using help models, additionally printed by 3D printers to help in understanding the notion of a probability distribution and the elements it consists of [14]. Working with the printed 3D resources on the triangle counting tasks confirmed the expectations regarding common errors; finding rationality in counting triangles, from different figures and solids; ways to record intermediate results; separation of figures according to side length and by species.

The solvers went through the processes of using rationality in counting; separation of triangles by types; recording intermediate results according to the length of the side of a figure; reducing work time. The usefulness and attractiveness of this type of didactic materials irrespective of age and condition was clearly evident in this study.

Future studies may help understand better the underlying cognitive mechanisms of shape perception in different groups of solvers for better accessibility design of games, developing intellectual abilities. Based on the pilot study with the two blind people, the feedback from the audience during the two days of the seminar, work will be done on the public provision of the 3D printing files on the project site for their wider dissemination to interested parties.

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Turning the Gaze Inwards: Exploring Simulator Instructors' Professional Vision Through Eye-Tracking and Stimulated Recall Interviews

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Abstract. In the domain of Maritime Education and Training (MET), simulator training is a key component in imparting the requisite skillsets to prospective seafarers. The maritime instructors lead the process of simulator training for the trainees by setting up diverse scenarios, monitoring and scaffolding the execution, and facilitating post-simulation discussion and reflection in the debriefing phase. In this pilot study, eye-tracking was used to record gaze videos of experienced instructors (n = 2) as they conducted simulator examinations for maritime students in a Scandinavian university. More than 2.5 h of gaze recording associated eye-tracking parameters were extracted through processing of these videos. In the next step, we performed stimulated recall interviews with the instructors, showing them their gaze behavior during examinations and asking them to reflect. The interviews helped in articulating their decision making and thought processes during simulation scenarios. Findings from the study provide insights into the professional vision of maritime instructors, showing how they direct attention and reason about student performance during examinations.

Keywords: Maritime Education and Training · Simulators · Professional Vision · Eye-tracking · Stimulated recall interviews

1 General Background and Introduction

The concept of teacher professional vision is increasingly being utilized in the learning sciences to understand the situated pedagogical practices in various educational arenas (Keskin et al., 2023). As a distinct field of research, studies on teachers' professional vision further our understanding of how teachers or instructors in a learning setting notice and interpret the events occurring during learning activities and engage in "knowledge-based reasoning about the noticed information" (Gegenfurtner et al., 2020, p. 59). Gegenfurtner et al. (2020) outlines how teachers' knowledge-based reasoning draws upon three processes that are closely interrelated: description, explanation, and

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prediction. Hence, a teacher's professional vision is based on selective visual attention and skilled professional reasoning about classroom dynamics, that is, a process mostly hidden from direct observation (Huang et al., 2021). The use of eye-tracking technology enables the collection of detailed records of teachers' gaze patterns, creating stable representations of how teachers direct their attention during teaching activities (Keskin et al., 2023). When paired with stimulated recall interviews, eye-tracking data can be used to reveal insights into the teachers' interpretation of classroom dynamics (Wyss et al., 2021).

The use of simulators for training has a long tradition in maritime education and training (MET), preparing students to work in the safety-critical environment on board a ship (Homlong et al., 2016). Since simulators offer a virtual, but work-relevant, context for learning, simulator training is seen as a risk-free, economical, and time-efficient way to train for professional work (Hontvedt and Øvergård et al., 2020). In the last two decades, maritime education has, at least partly, moved from paper-based tests, such as written exams, to competency tests based on practical scenarios in simulators (Sellberg et al., 2023). To conduct assessments that both examine courses in higher education and, at the same time, form the basis for issuing maritime certificates involves carrying out assessments where values such as validity, reliability and fairness are put at the forefront and come into conflict with the higher goals of education, i.e. to educate competent mariners for a professional life at sea (Sampson et al., 2011; Ernstsen and Nazir et al., 2020; Ghosh and Bowles, 2020). As a result, to conduct examinations in simulated environments puts the instructors' professional knowledge as well as their pedagogical skills to the test, requiring them to navigate complexities involved in balancing the scientific rationality of measuring skills in a fair and reliable way, and the art of skillful professional practice (Sellberg, 2020).

This pilot study explores the professional vision of maritime instructors through an exploration of the instructors' noticing and professional reasoning about student performance during examinations. Eye-tracking technology was utilized to record gaze videos of the instructors during the simulation scenario involving examination. Through stimulated recall interviews based on eye-tracking data, we aim to explore the professional vision of simulator instructors in MET. Exploring the simulator instructors' professional vision in this setting moves the applied method from classroom contexts to the technology intense control room environment of the instructors' room. As per the standard layout usually present in the instructors' room, some screens are mirroring the navigational instruments and technologies at use in the bridge simulators, such as radars, conning displays, and electronic charts (ECDIS). There are also screens showing the students' visual outlook into the surrounding marine environment and audio-visual recordings of their teamwork interactions on the simulated bridge. There are also possibilities to gain an overview of the scenario, monitoring the actions of the whole student group and their interactions with surrounding traffic. Through their work in the instructors' room, maritime instructors orchestrate simulator-based competence tests by setting up diverse navigational scenarios, monitoring the students' performance, and gather information for facilitating debriefings where assessments can be delivered (Sellberg et al., 2023).

Although there have been previous studies in the context of maritime settings utilizing eye-tracking technology to evaluate navigators' performance or associated cognitive

parameters (Hareide and Ostnes, 2017; Lützhöft and Dukic, 2007, Atik and arslan, 2019), the method has not been used to evaluate the professional vision of maritime instructors. Our study not only explores the use of eye-tracking data to assist the instructors to articulate their professional vision and reflect on the pedagogical activities occurring during examinations, but also serves as a starting point to explore methods from the learning sciences that can provide novel insight into maritime simulator-based training and assessment processes at large. This can contribute to improving assessment and, therefore, safety in this high-risk profession.

The research questions investigated by this pilot study are as follows:

RQ1: How can eye-tracking and stimulated recall interviews provide insight into the professional vision of maritime instructors?

RQ2: What future research directions and limitations can be identified by a small pilot study?

2 Methods

This section describes the general methodology and the demographic characteristics of the participants, the details about the eye-tracking equipment and the data analysis process in the study.

2.1 Participants and Setting

The topic's complexity called for participants known to possess the required skills and knowledge to facilitate collection of relevant and meaningful data (Campbell et al., 2020). Consequently, the participants were recruited through purposive sampling of professional contacts. The recruited participants were Nautical Science Instructors (n = 2) who led simulator sessions in the university. Both instructors have a background as navigators in the merchant fleet, an average age of 54 years and average experience as simulator instructors of 16.5 years. Local ethics procedures were adhered to during data collection and analysis. Prior to data collection, the instructors were presented with informed consent forms, describing in detail the purpose of the study, its voluntary nature, and their rights during the complete process such as privacy, and the ability to withdraw at any stage. They were further briefed regarding the subsequent data processing steps involved in the study.

Data was collected during pre-scheduled maneuvering and navigation examinations for third year students in a Kongsberg bridge operations simulator, and for final-year students in a Transas full-mission bridge simulator. These settings were part of the participants' standard work schedule, and due care was taken to ensure data collection did not interfere with their everyday work practice. Since the simulation scenarios form part of the regular examination, instructors conducted formal communications, briefing, and debriefing of students throughout the recording session. The students were tasked with navigating from Point A to Point B in a selected coastal region and were observed for various performance parameters such as navigation according to the passage plan, adherence to the collision regulations, communication to nearby traffic and reporting to vessel traffic services (Fig. 1).

Fig. 1. One of the instructor rooms where eye-tracking was conducted.

2.2 Equipment and Analysis

Wearable Tobii© Pro Glasses 3 were used to record eye-tracking data from two instructors at a Scandinavian university (see Table 1). The device has four integrated cameras attached in the scratch proof lens, affording 106-degree field of vision. In addition to capturing sound, it is also possible to record gaze metrics with 50–100 Hz sampling rate. The Tobii Glasses 3 application (Version 1.19.1) installed in a laptop computer was used during data collection. Before beginning the recording sessions, calibration of the eye-tracking device was performed to ensure accuracy. 160 min of eye-tracking and audio data was recorded over three simulator training sessions, while the instructors monitored and observed students from a nearby instructor room. Instructor 1's data was collected during two training sessions (50 min + 52 min) where groups of third-year students were being monitored in collaboration with a second instructor. These exercises ran consecutively on the same morning, during one examination session. Instructor 2 (58 min) was recorded separately while monitoring fourth year students' final examinations, where each student operated independently. Following collection, the data was prepared for analysis by Tobii Pro Lab software (Version 1.232.52758).

Table 1. Data collection during simulator sessions

Participant	Simulator	Students	Course	Time
Instructor 1	Kongsberg bridge	3rd Year	Maneuvering and navigation	102 min
Instructor 2	Transas Navi-Trainer Professional 500	4th Year	Maneuvering and navigation	58 min

2.3 Stimulated Recall Interviews

During the week following the data collection, video footage captured from the eye-tracking device was used as a basis for conducting stimulated recall interviews (Wyss et al., 2021) with both participants. These interviews were conducted by two researchers, and participants were interviewed separately, with a semi-structured interview guide used to prompt discussion as they viewed their own gaze patterns (see Figs. 2 and 3) and reflected upon recorded actions and events (Minarikova et al., 2021). The interview guide focused on how attention was directed, and how participants reasoned about student performance, aiming to capture the interpretation behind the gaze, and allowing the flexibility to do so (Brinkmann, 2014). During each 48-min interview, the recordings showing gaze patterns were projected onto a large screen (see Fig. 4), where participants could also hear the audio captured by the eye-tracking devices, thus using their own voices as further prompts in recalling instances of general interest. The interviews focused on decision making demonstrated by the instructors while monitoring the examination exercises and assessing student performance. Instructors engaged in a process of thinking out loud as they recounted their thought processes and professional reasoning behind making specific assessment decisions. The interviews were video recorded on a dedicated portable video camera. Audio from the interviews was later transcribed using OpenAI© automatic speech recognition system, Whisper, with resulting text checked by a member of the research team. The transcripts were then analyzed by the research team utilizing the Gioia method (Gioia et al., 2013). This approach focuses on sensemaking of elicited expert knowledge by the researchers, taking an inductive approach to building theory to be presented as a narrative (e.g., Magnani and Gioia, 2023). Transcripts were analyzed separately by two researchers, before discussing and narrowing down findings to themes which reflected those points emphasized by participants, building upon the research team's prior knowledge of instructional practices within MET. This reflects the context-specific nature of teachers' professional vision (e.g. Minarikova et al., 2021) and facilitates the co-creation of research. In keeping with the research questions and objectives, preliminary findings have been presented as (1) gaze behavior, (2) protocol and assessment, and (3) suggestions for future research. The approach to data analysis will be further developed as the research progresses.

3 Preliminary Findings

During the stimulated recall interviews, numerous instances of general interest selected by two researchers through discussion and mutual agreement were used as triggers for gaining insights into how the instructors viewed their own gaze behavior. These instances allowed the instructors to reason about how they position themselves in their role during examinations. Due to the exploratory nature of the pilot study, and small sample size, a qualitative approach was taken. Quantitative eye-tracking data will feature in future research.

3.1 Gaze Behavior

Our preliminary findings show that the gaze behavior identified through eye-tracking is broadly in line with the participants' expectations about how information about student

performance is accumulated. For example, Instructor 1 stated that the recorded gaze patterns (see Fig. 2) were in keeping with what they would have expected to see, saying:

> "*That is... I think this is quite representative for what I would, without the eye-tracking evidence, try to tell you. That I'm doing this. I have this row with this data. I look at this in this way. So, if you had asked me in advance, I think you could see that my... Answer would have followed what I actually look at on the eye-tracking*"

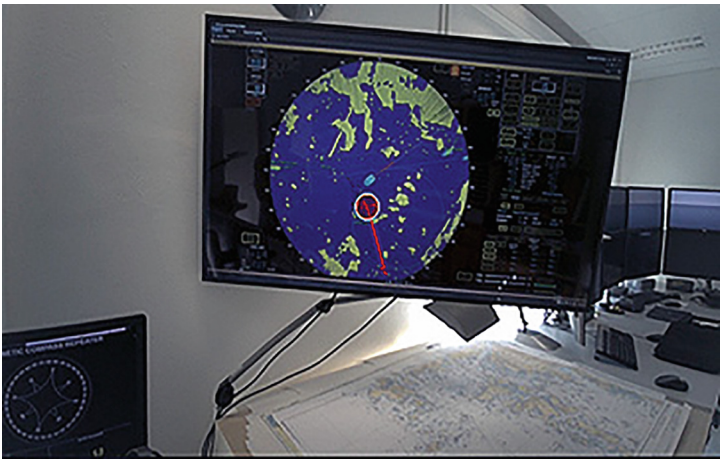

Fig. 2. An example of gaze recording from Instructor 1

Different data sources available to the instructors during the examination are combined to create a comprehensive picture of the state of performance. For example, instructors can view mirrored versions of the displays on the bridge, showing various settings and information used by the students as the exercise progresses. They may also refer to databases on their own computers, or paper charts in the instructor room. Instructors can listen to audio from each bridge, which is particularly relevant in team-based exercises. The speed at which this information is collected can vary depending on the task at hand and other demands on the instructor.

While we have mentioned that the eye-tracking data broadly reflected the participants' expectations, there were some particularly interesting events to be explored. Instructor 2, when shown the initial minutes of their gaze recording (see Fig. 3), expressed some element of surprise, primarily around the speed of their own actions. In response to this discovery, they attempted to justify what appeared to them as relatively high saccade velocity, saying:

> *"In my view, this, all of this takes more time. I look more time, I mean. I think that red dot just flies all over, I go and go like this. But in in real, when this happened, I was calm as…."*

This alludes to the need for the instructor to merge all the available information to create an overall picture of performance during the examination, a process that cannot be captured by eye-tracking alone. As information is gathered at different speeds and paces, it may result in a variety of behaviors, both conscious (looking for something e.g. closeness to objects) and unconscious (e.g. speed recorded by the eye-tracking, in comparison with their own impression of being calm).

Having highlighted the juxtaposition of the movements viewed on-screen with their awareness of their actions, Instructor 2 went on to confirm that *"it looks quite systematic"*, mirroring their expectations of where their attention was directed. "*Another thought when I see this is that honestly, it feels like I'm looking at what I want to look at. I was afraid that I will be looking at completely obsolete stuff*" (see Fig. 4). That is, while the speed was surprising, the content was not.

Fig. 3. An example of gaze recording from Instructor 2

3.2 Protocol and Assessment

Both instructors confirmed that the gaze patterns and actions recorded during the examination sessions reflected "protocol". The term "protocol" appeared repeatedly in the transcripts, referring to a set of semi-formal parameters against which the instructors assessed the students. Instructors had assessment sheets in front of them during the examinations, where notes were made against pre-defined grading criteria.

> *"I think it's interesting to know that we examine in the picture what we have in the protocol that I and that's interesting to see that that this is consistent, so that we are actually following the protocol and those things that in there that is actually what we are looking at"*

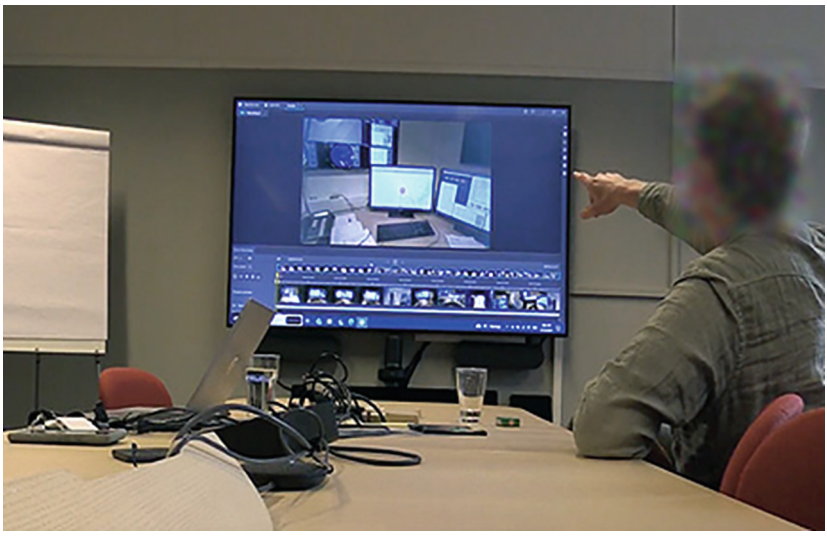

Fig. 4. Image from stimulated recall interview with one instructor

With the above statement, Instructor 1 noted the focused nature of their own gaze and attention, before continuing to note:

> *"That's interesting to see, especially when I zoom in on a picture like this. That I can see what I'm following actually. That I can see, I look for the P-I and where the P-I is touching. I'm almost certain that what I'm looking for now"*

Only one event was surprising to a participant, and this was quickly recalled as an unrelated administrative task conducted as the student wrapped up the examination.

The discussions around protocol and assessment, where instructors recounted what they saw and how they as experts packaged this verbally, were very much in line with Goodwin's seminal work on professional vision (Goodwin, 1994).

Both instructors spoke of their intention to give students every chance of performing well during examinations through taking measures to induce certain outcomes. In doing so, instructors strive to be fair and consistent, while also ensuring that students face challenges suitable for demonstrating successfully attaining intended learning outcomes. For example, altering the course of vessels to ensure that the student has to make a decision on how to handle approaching traffic, but not making any further amendments once the student has noticed the vessel, as this moves away from fair assessment.

3.3 Suggestions for Future Research

Both participants were aware this was a pilot study and considered the feasibility and usefulness of a larger follow-on study. Both supported the use of eye-tracking in further studies, drawing our attention to aspects of their teaching practice where such a study could result in valuable insights. However, attention was also drawn to the fact that

both recorded sessions were examinations, and therefore different to general teaching practice.

Instructor 2 recommended that the larger study focus on one element of training, perhaps how two experienced instructors work together in the instructor room, while Instructor 1 suggested eye-tracking in the instructor room could be helpful for confirming and updating assessment protocol. The underlined text in the following quotes indicates points that Instructor 1 suggests may be raised by a researcher in such a study, as they act out a dialogue while engaging in self-conversation.

> *"It could also be tool to see if we are looking at other things and that we don't have in the protocol to..." ""You actually examine this, but you don't write that down," which could be interesting if it has a value that we have missed"*
>
> *""You always look at this, why don't you examine that?" We didn't think about we do, we do think, we do examine it, but we don't write it, we think about it, so that could also be interesting and…".*

In doing so, they acknowledge the importance of instructors' tacit knowledge during training and assessment, and the complexity of the instructors' work. As instructors have extensive experience, assessment is conducted using professional intuition, meaning instructors know exactly what to do and where to look, but are not necessarily always aware of their own actions. While the eye-tracking in this pilot study showed actions that were broadly in line with the instructors' expectations, the sample size was small, and this may differ amongst a larger number of participants. However, a small sample size is not unusual in eye-tracking studies of teachers' professional vision (e.g., Minarikova et al., 2021; Kaminskiene et al., 2023).

4 Discussion

Eye-tracking gaze recordings obtained during simulator examinations and subsequent stimulated recall interviews shed light on the decision-making processes of maritime instructors in specific contexts. While viewing their own gaze, the instructors tried to articulate the reasoning behind their assessment of students, and how they strive to induce certain actions from students to maximize opportunity for good performance during examination. For the most part, this could be articulated easily, as the instructors recognized their actions and immediately recalled what they were doing and why. The focused nature of the gaze patterns reflected their high level of expertise and experience, as well as familiarity with the tasks being performed while evaluating student examinations. This underscores the need for careful selection of participants known to have the required skills and knowledge.

As outlined in the preliminary findings, the eye-tracking data recorded during this pilot study appears to illustrate professional reasoning, and the instructors' ability to recall and explain their actions and decision-making process provided rich insights. Although the pilot study was small, it is clear that instructors' "knowledge-based reasoning" (Gegenfurtner et al., 2020 p. 59) can be elicited using eye-tracking and stimulated recall interviews, where instructors' noticing and understanding of their own thought

processes enable the co-creation of meaningful insights into assessment processes. In terms of co-creation, the level of expertise of the instructors also meant that they could suggest further ways in which an expanded study could help to both understand and inform assessment practices in MET. These suggestions will be used along with the findings from this short pilot study to inform a larger eye-tracking study. The addition of other eye-tracking metrics such as heat map analysis and fixation duration on Areas of Interest (AOIs) is expected to add a further level of insight, offering increased prompts for discussion during the interview stage.

4.1 Limitations and Lessons Learned

Eye-tracking glasses, to some extent, can be inconvenient to use, particularly when the participant wears vision correction glasses while working. Both participants in our pilot study wore such glasses, and both easily found workarounds which made eye-tracking possible. However, other potential participants were unable to take part due to the incompatibility between equipment and their own glasses. Use of contact lenses may counteract this limitation for some potential participants. Ongoing work will be designed with this limitation of usability in mind.

Participants may find that awareness of wearing eye-tracking glasses affects how they work. Neither participant in this pilot study reported this as an overarching issue, but it will be considered in future research design, as it cannot be ruled out completely. Similarly, awareness of the researchers' presence may affect participants' behavior. As in the pilot study, the research team will leave the instructor room during the simulator session to minimize this impact and ensure that the participants' focus is on the students as it would be in normal circumstances.

One further limitation is that imposed by scheduling of regular simulator sessions where data can be collected. As our research aim relates to instructor's professional vision, it is essential that data is collected during genuine simulator training sessions and not in an experimental setting. As suggested by a participant, a copy of the relevant protocol will be requested in advance of each eye-tracking session, to aid with preparation for the subsequent stimulated recall interview. Due care will be taken to ensure this does not influence interpretation of the findings.

4.2 Informing Future Research

In addition to our research contributing towards understanding the concept of professional vision in the context of MET, the preliminary findings from this pilot study have potential to inform similar studies in alternative simulator-based training settings such as process industry, healthcare, driving safety and similar. Researchers considering the use of this method should consider the physical limitations, such as participants' (instructors') own glasses, and the scheduling of simulator sessions where meaningful data can be collected. It is also recommended that particular attention is paid to those moments during stimulated recall interviews where participants encounter unexpected events, such as our participant discussing the difference between their perception of their work during monitoring of simulator scenarios, and the speed of their gaze as viewed on-screen during the interview.

5 Conclusion

The preliminary findings from this short pilot study have confirmed that an eye-tracking study focused on instructors' professional vision is both appropriate and welcomed in the context of MET. Core concepts of professional vision can be seen in the recorded eye-tracking data and confirmed by stimulated recall interviews with participants. Instructor comments on how such a study could benefit them and inform teaching practices will be considered as the research team embarks on a larger study. Analysis of the findings from the pilot study is ongoing and will be used to inform the larger study involving nautical instructors from a range of maritime universities across Europe.

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Storytelling with Technology-Enhanced Artefacts: A Literature Review of Toolkits for Children

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Abstract. This paper explores how technology-enhanced artefacts can augment storytelling experiences for pre-school and school-aged children, particularly when utilised with puppets or doll-like figures. It presents a systematic literature review focusing on this specific application of technology in storytelling. The analysis employs a four-lens framework to evaluate the integration of technology in children's storytelling, assess the alignment with STEM learning goals, and finally, investigate the facilitation of open-ended and collaborative storytelling by children.

Keywords: Storytelling · child · physical artefact · interaction · tangible · making · physical computing · puppet · doll

1 Introduction

As Cassel and Ryokai wrote in their 2001 seminal paper, there is abundant evidence in the literature that storytelling is "an extremely important developmental arena for children" [1]. In particular, storytelling with physical artefacts for them allows children to experiment with diverse interaction possibilities in a safe manner. Using dolls and puppets as physical artefacts can make learning more engaging and enjoyable. The analysis by Råde critically assessed and sustained different benefits of using these in play, e.g., besides engaging different children in open-ended play, they can make learning less intimidating, in relation to both personal and social development [2].

Research on technology-enhanced learning, child computer interaction and education shows that stories with interactive elements, using digital tools or game-like elements, can help further engage children and promote different forms of learning, e.g., [3,4]. However, storytelling technology has not been often designed to "support child-initiated and child-driven play" [1].

Recently, the availability of physical-computing boards for children, such as Adafruit Circuit Playground Express and BBC Micro:Bit, or of educational robots, such as Pepper and NAO, has broadened the panorama of children's expressive potential, allowing for the construction of various technology-enhanced physical artefacts for their storytelling [5–8]. Technology-enhanced

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physical artefacts open doors to multi-sensory experiences, e.g., that the ground trembles as a giant approaches can be achieved by attaching a vibration motor to a cardboard storytelling surface.

In other words, technology-enhanced physical artefacts enable children to tap into touch, sound, and even movement, and create engaging storytelling experiences [9]. In line with constructionism, which sustains that children can learn better through the creation and sharing of objects meaningful for them, children can practise not only their storytelling skills by playing with such artefacts, but also Science, Technology, Engineering, and Mathematics (STEM) skills [10].

Moreover, due to their physical nature and the manipulative possibilities they offer, physical artefacts lend themselves to social play. In particular, if endowed with screenless technology, they enable head-up play and can foster several social learning gains [11–14].

2 Methodology

To conduct this literature review, the PRISMA guidelines were considered [15]. The leading research question of the review was as follows: *what is the status of research about storytelling for pre-school or school-age children, with technology-enhanced physical artefacts?* The specific focus of the review was related to artefacts in the form of puppets or doll-like elements, enhanced with making, tangible, digital, electronic, or other interactive technologies. Figure 1 recaps the workflow adopted in the literature review.

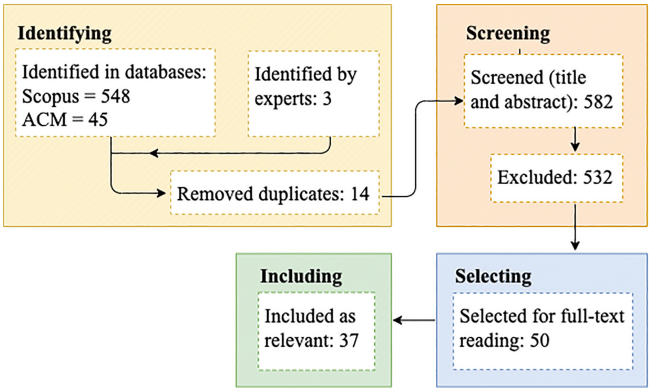

Fig. 1. The literature-review workflow

2.1 Search Strategy

To address the leading research question, and maintain the focus of the review, three researchers independently analysed search queries for online databases.

They revised queries and discussed outcomes iteratively. The final query was the conjunction of the following four expressions:

1: storytelling OR "story telling" OR story-telling OR (puppet AND play) OR (puppet AND theatre) OR (doll AND play) OR (doll AND theatre)
2: digital OR electronic* OR interact* OR tangible OR mak*
3: *school
4: child* OR kid*

The query was used to search for papers with those terms occurring in the title, abstract, and keywords. Scopus and ACM were used for this literature review. The reason for selecting ACM and Scopus was their broader coverage of journals and conferences that publish work on children, interaction, and education. The search on Scopus was restricted to Computer Science, Engineering, Arts and Humanities to find relevant papers suitable for our study, that is, with making, tangible, digital, electronic, or other interactive technologies for storytelling. The number of records identified from Scopus was n = 548 and ACM was $n = 45$. In total, the number of records identified from both databases was $n = 593$. After the removal of 14 duplicates, it became $n = 579$.

Three further papers were identified through experts and hand searches in major journals and conferences related to human computer interaction, child computer interaction, and education. In total, $n = 582$ papers were identified after removing duplicates.

2.2 Screening and Selection Criteria

The screening and selection was then iteratively discussed by the team of three researchers and the following criterion emerged: a paper is to be retained as relevant if it is related to the leading research question, and reports on artefacts, in the form of puppets or doll-like elements, that are enhanced with making, tangible, digital, electronic, or other interactive technologies.

All the identified papers were thus added to a spreadsheet for coding them as relevant or irrelevant. One of the researchers initially read the title and abstracts, and selected 47 papers as potentially relevant for full-text study. The other two researchers firstly cross-checked all the papers that were excluded as irrelevant and found that 3 papers could instead be relevant. Those papers were brought into discussion. Secondly, the other two researchers cross-checked the potentially relevant papers, and considered that 13 papers could be irrelevant and brought them into discussion. In the next iteration, all papers to discuss were jointly assessed by reading firstly abstracts and secondly the papers themselves in full. After several discussion iterations, the total number of papers jointly selected as relevant for the analysis was $n = 37$.

2.3 Analysis Process

The leading research question was iteratively elaborated by the three researchers. Working initially independently and then comparing their results, they derived

the following specific research questions and lenses to address with the analysis of the selected relevant literature:

R1. Does the adopted storytelling technology promote **STEM** learning?
R2. Does it enable open-ended storytelling (**open**)?
R3. Does it enable social play (**social**)?

For answering the **R1**–**R3** questions, the three researchers analysed the literature by adopting an iterative, inductive and deductive process along the following steps: familiarising with the research articles, coding independently, generating and reviewing analyses jointly, and lastly writing up results in relation to the three research questions [16]. The related results are presented in the following.

3 Results of the Literature Review

This section is divided into subsections, presenting the results of the analysis, divided into research concerning toolkits without screens, with screens, and with educational robots. Each subsection comes with recap tables that frame toolkits along the **R1**–**R3** research questions.

3.1 Toolkits Without Screens

Toolkits in this category do not use screens as the main interaction media for storytelling. They use instead artefacts with physical-computing or purely electronic devices, besides paper or similar materials. They are sub-divided into toolkits with making materials for artefacts, or pre-made artefacts for storytelling.

Making Materials for Artefacts. Several toolkits offer making elements, like Micro:Bit boards and conductive material, besides paper or cardboard, to tell stories and learn STEM concepts. Table 1 lists them.

For instance, Raposo and Vairinhos report that children animated a story by animating characters, "indirectly, apprehending the fundamental concepts of input, output and processing of physical computing" [17]. Children together followed a scripted story, so the nature of storytelling was only partly open-ended and it is unclear whether storytelling was a social activity.

In the work by Minuto et al., children learnt what smart materials are, created origami creatures and told their own stories with them in an open-ended collaborative manner, and also programmed their behaviours [18]. The paper by Mattews et al. reports on a co-design experience with middle-school children of puppets with embedded digital devices, which encouraged collaboration, exploration, and persistence "with inevitably 'glitchy' technology" in making their own stories [19]. The making kit, called the Maker Theater, invited children to make theatre artefacts and tell a story through a recorded puppetry show [20].

Table 1. Phygital toolkits with making materials for artefacts

Ref.	Age	Technology	STEM	Open	Social
[17]	7–9	Arduino, Scratch, sensors, speakers, LEDs	Yes	No	Unclear
[18]	9	Arduino, resistive wires, LEDs, thermochromic paints, NiTiNOL	Yes	Yes	Yes
[19]	7–12	Micro:Bit, Makey Makey, Scratch, servos, LEDs	Yes	Yes	Yes
[20]	8–11	Paper switch, resistors, LEDs, battery	Yes	Yes	Yes

Pre-made Artefacts. In other cases, toolkits came with pre-made puppets endowed with computing or electronic devices (e.g., servos, tactile motors, mics, speakers). Children narrated stories with them and improved, e.g., emotional literacy and not, in general, STEM related skills, e.g., [21]. Table 2 lists them.

Perhaps the best-known for open-ended play is StoryMat: it was a soft, quilt-like mat, designed to support children's open-ended storytelling. Through an infrared transmitter in the toy and receiver in the computer, StoryMat records their storytelling voices and movements with the toy on the mat, so that stories can be later shared with other children on the mat using a projector [1].

Lately, a puppet, with an embedded mobile phone, was used to scaffold children's reflective habits on storytelling [22]. Though not meant for children, Non-myth prioritises, "an integrated physical storyworld rather than a mapped one", used physical-computing materials, puppets, and other physical elements for enabling purely tangible narratives, open-ended and collaborative storytelling [23].

A multi-sensory tangible system was also prototyped and tested with visually impaired children to enable them to narrate stories with puppet-like characters [24]. An interactive book was also designed for them, integrating multi-sensory interactions (touch, sound, and smell) [25]. Children with mixed visual abilities instead participated as co-designers in the creation of a multi-sensory story mapping tool [26]. Although the authors did not formally assess children's learning gains, their findings indicated that children could learn about storytelling, design, and science concepts as well.

Table 2. Phygital toolkits with pre-made artefacts

Ref.	Age	Technology	STEM	Open	Social
[21]	5–9	Interactive puppet, servos, tactile motors, potentiometer, microphone, speaker, buttons	No	Partly	Unlcear
[1]	5–8	Infrared, computer, projector	No	Yes	Partly
[22]	4–5	Stuffed puppet, with embedded mobile phone	No	Yes	Yes
[23]	unclear	ESP32, RFID, diverse sensors, conductive tape, audio components	No	Yes	Yes
[24]	5–12	Arduino, interactive mat, NFC, microphone, speakers	No	Unclear	No
[25]	7–11	Conductive threads, speakers, olfactory device	No	No	No
[26]	7–10	Linux-based board, sampler, playback, RFID	No	Yes	Yes

3.2 Toolkits with Screens

Toolkits, listed in Table 3, mainly use screens for storytelling. Though diverse learning gains are reported (e.g., linguistic and cultural diversity), the related

studies seem to have no specific STEM learning aims, and in general to enable for open-ended and social storytelling. Such toolkits also use other devices, such as NFC tags or computer vision devices, to enable the interaction with children: a toy is identified with a tag, or sensor or via a computer vision device, and then a virtual character is replicated on a screen for storytelling [27–29].

MAB, for instance, came with a box recognising objects to then display on a screen [30]. More recently, Mobeybou used physical blocks tagged with an ID and an electronic board to manipulate a digital story on a screen, also collaboratively [31–35]. I-Theatre used a tabletop, microphone, and two RFID readers to create and narrate a story, with children in the last year of preschool [36]. This and other similar solutions, with large screen surfaces, tended to facilitate collaborative storytelling about diverse topics [37,38].

StoryMakAR was instead an augmented reality and making toolkit for older children (14–18 y.o.), for teaching programming or electronics via storytelling, e.g., by creating virtual characters controlled via electro-mechanical devices [39].

Table 3. Toolkits with screens

Ref.	Age	Technology	STEM	Open	Social
[27]	5–9	PC, magic board, RFID, NFC, STELT	No	No	No
[28]	5–7	Tablet, touch sensors, conductive materials	No	Yes	Yes
[29]	3–7	Laptop, electronic tablet, playmate, infrared, light sensors	No	Yes	Yes
[30]	6–7	Screen, STELT, controller module, RFID, speaker, battery	No	Partly	Unclear
[31]	4–5	Laptop, USB, electronic board, picture-blocks	No	Yes	Yes
[32]	4–5	*As above*	No	Yes	Yes
[33]	4–5	*As above*	No	Yes	Yes
[35]	8–9	Laptop, Bluetooth, peripheral interface controller, magnets, battery	No	Yes	Yes
[34]	3–11	*As above*	No	Yes	Yes
[36]	4–5	Tabletop, RFID readers, mic	Unclear	Yes	Yes
[39]	4–5	PC, projector, RFID, webcam, audio recorders, speakers	No	Yes	Yes
[38]	5–7	Computer, screen, projector, USB hub, pressure sensors, video tracking, barcoding	No	Yes	Yes
[39]	14–18	Laptop, mobile phone, MakAR board, motors, buzzer, sensors	Yes	Yes	Yes

3.3 Toolkits with Educational Robots

Educational robots are smart devices for children with sensing and motor capabilities. Sophisticated ones have also conversational abilities. They tend to engage

children in open-ended and social storytelling, often between the robot and children, only partly addressing STEM learning. Moreover, either robots interact with puppets or puppet-like elements, used by children for storytelling, or robots act as puppets and directly interact with children in storytelling. Table 4 lists them, including a co-design work of robots for storytelling [40].

An example of the first type of interaction, less frequent, is reported by Stal et al., with the Cozmo robot that reads RFID tags, associated with emotions, that children place on puppets [41]. There are instead diverse investigations on the usage of robots as puppets that interact directly with children. In particular, robots like Ozobot, and Bluebot have been used to help young children of different age groups tell stories, and practise coding by instructing robots on how to move [42–44] or to enhance their creativity and collaborative storytelling [45].

In other cases, instead, robots like Tega, NAO or Dragon Bot were employed to create dialogues with children or orally narrate stories on cues, to promote language learning mainly, and in some cases emotional learning or physical activities besides storytelling [46–50]. Finally, collaborative storytelling with children has been practised with robots and projectors in one case, and with multiple robots in another, thereby promoting children's embodied participation in one case, and emotional interpretation of the narrated stories in the other [51,52].

Table 4. Toolkits with educational robots

Ref.	Age	Technology	STEM	Open	Social
[40]	4–6	ROBIN	No	Partly	Yes
[41]	5–10	Cozmo	No	Partly	Yes
[42]	3–10	Ozobot	Yes	Yes	Yes
[43]	8–9	Ozobot	Yes	Yes	Yes
[44]	4–5	Blue bot	Yes	No	Yes
[45]	3–6	Ozobot	Unclear	Yes	Yes
[46]	4–6	Tega	No	No	Yes
[47]	3–3.6	Nao	No	No	Yes
[48]	4–6	Dragon bot	No	Yes	Yes
[49]	4–6	Dragon bot	No	Yes	Yes
[51]	11–12	GENTORO	No	Yes	Yes
[50]	8–10	Nao	No	Partly	Yes
[52]	6–8	MyKeepon	No	Partly	Yes

4 Discussion

Novel technology, like physical computing kits or educational robots, can help create technology-enhanced artefacts and enhance the storytelling experiences of pre-school and school age children. This paper presents a systematic literature

review on storytelling with different interactive technology-enhanced artefacts, focusing on puppets or doll-like elements. A three-lens analytical framework was employed to evaluate the relevant literature. It examined the role of technology in children's storytelling experiences, and it analysed papers considering if there were any (**R1**) STEM learning goals or gains, and whether the technology-enabled interactions promoted (**R2**) open-ended or (**R3**) social storytelling.

According to the reported analysis, the majority of the 37 toolkits use screens (13) or educational robots for storytelling (13). The remaining toolkits use making materials: they invite children to make their own storytelling physical artefacts with technology (4) or to use pre-made ones for their storytelling activities (7). Probably due to their making and manipulative nature, screenless toolkits with making materials turn out to more frequently promote STEM learning gains and invite children into open-ended, social storytelling than toolkits with pre-made artefacts seem to do (see Tables 1 and 2). Though toolkits with screens as interaction media tend not to promote STEM (e.g., only 1 out of 13 does so clearly), they address in general open-ended, social storytelling (see Table 3). Similarly, toolkits with educational robots seem not to aim for STEM learning but they tend to enable open-ended and social storytelling, mainly through conversations between a robot and a child (10 fully or partly do so, see Table 4).

5 Conclusions, Limitations and Future Work

This paper offers a literature review of recent work in the evolving research field of storytelling with technology-enhanced physical artefacts [53]. The review focuses on storytelling for pre-school or school age children. It excludes narrative technology without technology-enhanced physical artefacts, like purely virtual or purely embodied narrative toolkits. Moreover, it focuses on toolkits with technology-enhanced physical artefacts like puppets or dolls. It analyses paper reporting such technology, considering (1) if STEM learning, (2) social, (3) or open-ended storytelling are promoted. The paper starts explaining the methodological approach to the review. The reviewed papers are then analysed with the aforementioned lenses, and discussed.

The review referred to the PRISMA methodology. The selected database sources may limit the generalizability of the reported findings. To address this, information from additional sources was also considered, such as expert consultations. Future work may expand on it, by considering different physical artefacts and integrating further lenses, e.g., from the framework by Harley et al. [53].

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Herding Cats: Assessing Complex Procedural and Tacit Knowledge with Multi-agent Simulations

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Abstract. Complex procedural knowledge and tacit knowledge are hard to assess and practice constructs. In this paper, we demonstrate the application of autonomous multi-agent simulations to assess teacher's classroom management skills. We developed a visual virtual environment within which virtual agents mimic students. These agents act and interact autonomously with each other. Human teachers are requested to interfere on the basis of a pre-defined set of actions to optimize the simulated study time and happiness of the agents. This paper presents analyses of systematically configured simulation runs to elucidate the simulation characteristics and to evaluate the rationality and credibility of the simulations.

Keywords: multi-agent simulations · teacher training · tacit knowledge · classroom management · professional training

1 Introduction

Agent-based Simulations (ABS) [1] are systems within which autonomous artificial entities (agents) act and interact with each other with a virtual environment, characterized by rules and constraints. ABS have been used in various domains to study complex systems that result from the interaction of many individual entities. Examples of such systems are the stock market, crowds, beehives, social networks and many more. Social Simulations are a type of ABS that focus on modeling social dynamics of humans or animals [2]. Early examples of type of such social simulation come from [3], who studied the dynamics of social segregation, or [4], who studied the spread of contagious diseases. The main idea of ABS is to elucidate phenomena emerging in large, complex (social) systems but which are absent in the individual agents [1]. A series of open source as well as commercial distributed Agent Software [5–8] exists.

A novel field of application for ABS is teachers' classroom management (CM) skills (CMS), which are composed of certain declarative and procedural knowledge. CMS are amongst the key determinants of successful teaching and encompasses a multitude of abilities and techniques to interact with students, to organize lessons and to keep students

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focused and attentive. Effective strategies to classroom management minimize student behaviors that hinder and distract other students from learning, while maximizing beneficial and constructive behaviors. Theoretical underpinnings offer models of teaching quality [9]. [10] describe CM as the activities set by a teacher to provide students with an optimal learning environment. Three dimensions of CM are considered essential [11, 12], that is, lessen planning, fostering social relationships in a class as well as behavioral control to enable a constructive handling of disruptions as well as preventive and proactive action. As important adequate CM is, as complex and challenging is the assessment of (pre-service) teachers' CMS and an effective training of those skills. Typically, methods for the assessment are interviews of students and teachers [13], observations in the classroom [14] as well as the evaluation of key situations presented as text descriptions or videos [15, 16].

These methods for the assessment of CMS have advantages and disadvantages; partly based on the relation to real classroom situations and a strong behavioral component, partly it is a strong standardization of assessment items, and partly is possibility to elucidate motives and rationales. The key challenge occurs from the fact that the assessment of beneficial or helpful actions needs to be based on a valid theoretical foundation. This foundation, however, hardly exists; there is no ground truth of what is right or wrong in a particular situation.

2 The EKIS Project

ABS may potentially mimic the dynamics of an entire class and model classroom activities in relation to the elementary components such as multi-dimensionality, simultaneity, immediacy, unpredictability, and the history of events [17]. When teachers interact with virtual, simulated classes, the effect of certain actions and interventions can be represented and investigated. With reference to an integrative understanding of teachers' professional competencies [18] and the assessment of CM related perception, interpretation, and decision processes, it might be possible to observe authentic sequences of action-reaction-reflection processes between teachers and students. CM, thereby, is not limited to a specific generalized or abstracted action or activity; instead, the effect of the same action on different students and in different situations can be investigated. Concerning the training of CMS, this means that different definable yet complex situations and scenarios can be practiced repeatedly and a number of different actions and activity can be tested – without risking serious real-life consequences.

With the repaid technological evolution, specifically in the field of AI, vivid and realistic simulations become doable, even in small-scale projects. ABS [19], in essence, consist of a number of virtual agents that act and interact autonomously within a specific environment and on the basis of a given set of rules. The agents are defined by a set of properties and parameters (their personality, so to speak) and they typically purse a certain goal. ABS have been applied in a variety of settings, for example in modelling the spread of infectious diseases.

The origin of the EKIS project ("assessment of CM skills using intelligent, agent-based simulations") is a different view on the dimension of social dynamics in CM. In reference to the research on implicit knowledge in psychology, we consider this CM

dimension as a teacher's tacit capability to regulate a complex system with unknown rules. Tacit knowledge, in essence, is knowledge that is hard or even impossible to extract, express, or verbalize. It means that one can do something without being able to explain exactly how. The famous sugar factory experiment of Berry and Broadbent [20] investigated the tacit knowledge. The participants' task was to optimize the output of a sugar factory by altering certain production parameters, without knowing the underlying relationships and rules. The results showed that participants could reliably improve their ability to optimize the factory output, while being unable to explain the underlying mechanisms. They just 'knew somehow'. We argue that the same principle may apply to CM. A class is composed of a number of students, all having individual characteristics and personalities, all pursuing a certain goal, and all act and interact with each other and the teacher. The dynamics are characterized by various class composition effects, stable traits but also by oscillating states and random effects. We cannot fully understand all the dynamics, relationships, and rules; at best, we can distill a limited number of general assumptions. Thus, what may work today in terms of CM might not work tomorrow.

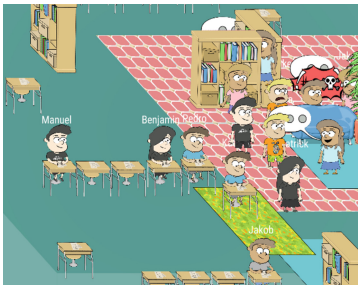

Fig. 1. "Breakfast Club" – A multi-agent classroom simulation

Therefore, we suggest defining CM as the ability to learn to regulate and to interact with any complex system without knowing (all) the underlying relationships, rules, and parameters. For the assessment and perhaps the training of this ability, we explore the utility of ABS. By our definition, it is actually not relevant whether an ABS behaves like a 'real' class – especially when we cannot fully capture how a real class functions.

In a first step, we developed an ABS including a visual representation of a classroom and virtual students using Unity3D, which is a programming environment for digital games and simulations. In our case, the environment is a virtual classroom (Fig. 1) that contains a number of desks for students to study individually and in small groups. In addition, the environment is modeling the noise that is accumulating in the classroom resulting from the different actions performed by the agents. The noise model implemented is accumulating the noise produced by the different actions performed by all agents in the classroom, whereby different actions produce different amounts of noise depending on the simulation configuration. Agents are moving within the classroom as part of the various actions they perform. The Unity3D Navigation Agent Infrastructure is used to control the movement of agents, including path finding and collision control.

Agents: Virtual Students. The agents are defined by general characteristic, internal states, and behaviors. In the present version of the ABS the agents/students are characterized by (i) personality parameters based on the so-called OCEAN personality traits (often referred to as "Big 5"), that is, openness, conscientiousness, extraversion, agreeableness, neuroticism [21], (ii) by internal states (happiness, attention, activation/motivation), and (iii) by a set of behaviors (move, talk to other agents, study alone, study with other agents, quarrel, and rest). In addition, the agents may transition (e.g., when moving towards a table to study), wait (e.g., for other agents to study in groups), or execute a certain behavior. The agents' intrinsic goal is to maximize their happiness. Happiness, in turn, is determined by the personality setup of agents. Also, an element of peer pressure is included in the simulation, meaning that the action a particular agent selects to perform is influenced by the average action preference of the entire class.

Dynamic System Control. The agents' actions are determined by their specific configuration and by an element of randomness. The simulation follows a set of basic rules (e.g., 'happiness increases when studying successfully', 'studying reduces the activation level', 'resting increases the motivation to study') and can be configured with any number and types of agents (defined by the personality parameters) a runs for a given time. By this means, we can realize a broad variety of scenarios and trace class composition effects (e.g., observing the effects of the percentage of aggressive/disruptive agents). In the visual representation, one can directly observe the activities in the virtual classroom. In addition, the simulation steadily outputs the agents' parameters and those for the entire class (including noise level, average happiness, and average productivity).

Technically, for each of an agent's behavior options a score is calculated, depending on the personality characteristics and momentary internal states. In addition, an action score bias is added to the score of the current action and subtracted from the score of the previous action. This mechanism is used to keep the agent from switching between actions too quickly. In addition, the added bias models the tendency to continue with an ongoing task and switching costs between tasks (similar to sustaining attention). Reducing the score of the previous action keeps the agent from looping between two of the possible actions, and cause a more diverse action selection. The action score bias depends on the conscientiousness of the agent and is following an exponential decay curve, where time is the number of ticks the agent is performing the current action. The number of ticks is only taken into account while the action is executed, not in its other states, making sure that transitions or waiting do not affect the action score bias. The action score is computed as follows:

$$action - score - bias(a_i) = \alpha * e^{-(1.0-C)\beta * t} \tag{1}$$

where a_i refers to the set of behaviors, α is a simulation parameter defining the maximum bias, β is a simulation parameter defining a dampening factor scaling exponential decline, C is the agents conscientiousness, and t is the time (number of ticks) the current task is executed.

As mentioned, the action selection is not only depended on the agents own focus on maximizing individual happiness, but is modulated by peer pressure. This aspect was included in the model by building the action score vector that is used to select an action,

as a weighted sum of the action score vector of the agent plus the mean action score vector of the classroom (i.e., all agents in the class). The weight given to the classroom average is defined by a simulation parameter. The probability for an action to be selected is defined by the square of the normalized action scores. Taking the squared action score makes sure that the highest rated score has a clear advantage over the other actions, but still gives other actions a chance to be selected. The probability is given by

$$p(a_i) = \frac{s_i^3}{\sum s_i^3} \tag{2}$$

where $p(a_i)$ is the probability of action i to be selected and s is the score for action a_i. This multi-level dynamic system can express very sophisticated behaviors and dynamics, making it one of the main reasons agent-based models are such a powerfully tool to study real world social phenomena. One of the most interesting aspect of complex dynamic systems are emergent phenomena which describe aspects of the entire system (the classroom), absent in the individual (i.e., students) components. None of those behaviors is obviously deducible from the initial basic interaction rules. Instead, those properties emerge in the interaction between the components or agents following basic rules.

3 Exemplary Simulation Studies

We ran a number of systemic studies with the simulation to investigate the simulation characteristics and outcomes. The main goal was to gain insights into emerging phenomena and the credibility and rationality of the outcomes. Although we argue, that it is not necessarily key that a virtual environment for the assessment and training of CMS is "realistic" in the sense of being equivalent or comparable to real "human" classes, it may be important that the simulation acts reasonably und meaningfully for a human observer.

In addition, we carried out systematic studies with theoretically and empirically well described personality types such as aggressive children [22] attention-deficit hyperactivity disorder (ADHD) [23]. The literature provides insights into the relationship between personality characteristics and behaviors. The goal is to compare the behavior of agents that are configured in such a way with the behavior described in the literature.

The simulation also features a basic set of actions enabling a real person to intervene and to regulate the simulation. This 'virtual teacher' may exert influence on the class level (ordering to study alone or in groups, urging the agents to be quiet, allowing free time) and on the level of individual agents (praising, admonishing, ordering to study, removing an agent from the classroom). We ran various simulation trials and systematically varied and applied external interventions, at this stage on at class level. In this paper we want to illustrate the overall approach by provide some exemplary results.

3.1 General Simulation Characteristics

In essence, simulations are determined by the composition of classes with differently configured agents and the emerging interplay of these agents. To study the class composition effects, for each agent the likelihood of certain actions as well as the transition

probabilities from one action to another are studied. This provides insights about the way individual agents act and interact with the class. In order to have a statistical description of the simulation results, we ran multiple instances of the same simulation (identical simulation and classroom configurations) but different seed values for the random number generator used during the simulation. There are several random elements in the simulation. Depending on the classroom configuration, agent personality can be partially randomized. The actions selection process has a random element and agent interactions are effected by the output of a random number generator.

To investigate the simulation characteristics, we ran several studies, which are composed of a number of trials (i.e., simulation runs), a number of turns (i.e., fictitious time intervals), and a number of individually configured agents. The exemplary study presented in Figs. 2 and 3 includes 100 trials with 100 turns each based on a class of 20 agents with a wide range of characteristics. Figure 2 shows the transition likelihood matrix in this study, meaning the likelihoods with which an agents switches from a particular action (x-axis) to another (y-axis). Figure 3 shows the percentage of time (measured in simulation turns) an agent spent on certain actions.

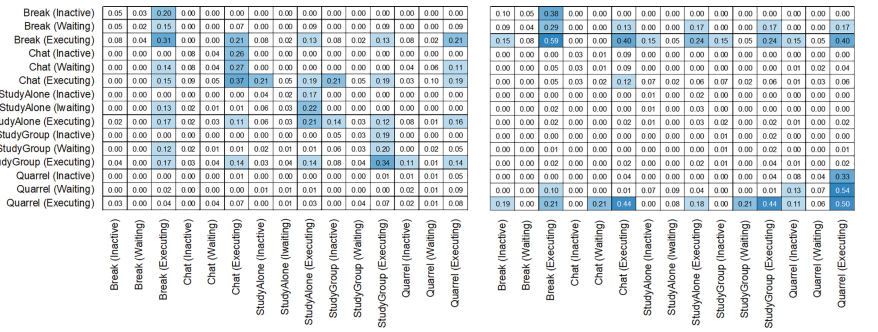

Fig. 2. Transition likelihood matrices for two distinct agents. The left matrix shows the study results for a quiet and dilligent student, the right matrix that of a disturbing student.

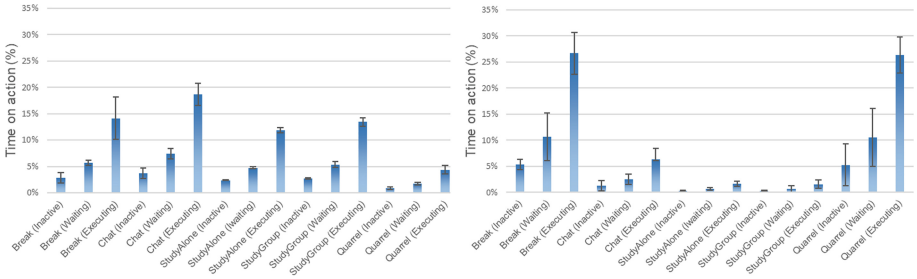

Fig. 3. Average time (measured in simulation turns) spent on possible actions/behaviors. The top chart shows the study results for a quiet and dilligent student, the bottom chart that of a disturbing student.

What we can learn from these statics is that agents behave reasonable according to the fictitious personality profile. For example, an agent characterized by [Openness: .80;

Conscientiousness: .75; Extraversion: .45; Agreeableness: .80; Neuroticism: .00], who could be thought of as a quiet, open, and diligent student, produced the results shown in Fig. 2, top and Fig. 3, top. In comparison, an agent characterized by [Openness: .10; Conscientiousness: .20; Extraversion: .75; Agreeableness: -.50; Neuroticism: .80], who could be thought of as a disturbing, aggressive, and is interested student, produced the results shown in Fig. 2, bottom and Fig. 3, bottom. As opposed to the former agent, this agent predominantly rests or quarrels.

These results indicate that the simulation produced outcomes that appear reasonable and realistic. A quiet and diligent agent switches between a broader range of actions (rests, chats with other agents, studies alone and studies in groups), however, quarrels with a very low probability. In turn, the aggressive and disinterested agent predominantly quarrels or rests.

On the class level, we observed the overall noise level and the total levels of happiness, motivation, attention, study time, and quarrel time. Exemplary results are shown in Fig. 4. These results show a quite oscillating classroom climate. However, the patterns of mutual dependencies of variables one would expect can be largely found in the data. For example, an occurring high intensity of quarreling results in a high noise level, which in turn leads to a decrease in happiness, motivation, attention, and study time. We could observe these patterns through all simulation trials.

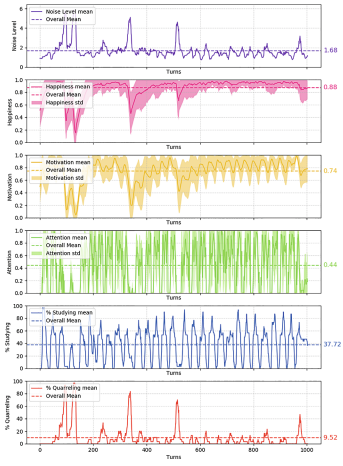

Fig. 4. Changes of simulation variables over time

3.2 ADHD Simulations

Attention-deficit hyperactivity disorder (ADHD) is amongst the most common childhood behavioral disorder, with a prevalence between 5% and 10%. Because of the availability of studies correlating ADHD with personality traits, we decided to study the effects of children with personality profiles that match ADHD children on the classroom climate.

These authors suggest that the overall ADHD symptoms are related to low conscientiousness, low agreeableness, and high neuroticism. For the study, we composed classes of various combinations of normal, ambitious, DHS students as well as additional randomly defined agents. Table 1 provides an overview. Configurations covered a very high (50%), (high (33%), medium (17%), and low (7%) percentage of ADHD agents in the class and no ADHD agents (0%). These classes are completed with different sets of highly ambitious (diligent, interested, motivated), normal and random agents.

Table 1. Agent Configuration in the ADHD Study

Student Type	O	C	E	A	N
ADHD	RND	0.20	RND	0.20	0.80
Normal	0.75	0.60	0.55	0.65	0.50
Ambitious	0.80	0.80	RND	0.80	0.20
Random	RND	RND	RND	RND	RND

Figure 5 (left panel) shows a summary of 100 simulation trials with 100 turns each for each class. The Breakfast Club simulations resulted in a clear tendency that the more ADHD agents are in a class the lower is the average happiness of agents and the average attention level. Interesting is also the comparison of class compositions with high and low ambitious agents. The former produce higher attention while the latter produce higher happiness. These results, again, indicate that the simulations procedures reasonable and meaningful outcomes.

Figure 5 (right panel) reveals an interesting and unexpected emergent phenomenon. In the context of the ADHD studies, we configured highly disturbing ADHD students and added none to three of those agents to an average class. The simulation trials indicate that having one highly disturbing agent in a class of 20 students produces higher happiness while resulting in a similar attention level as a class without a highly disturbing agent. This is a phenomenon, which cannot be (easily) explained by the simulation logic, however, which could be rationally explained for human classes. The regular class can accommodate a low number (one) of disturbing students quite well. The behaviors of such disturbing student might sometimes be funny or amusing, which could increase the total happiness of the students in the class.

4 Conclusions and Outlook

Overall, the results of numerous simulation studies run with a wide range of configurations and class compositions revealed reasonable and credibly results, which by all means could likely be overserved in real human classes or other social groups. This gives rise to the hope that the main goal of the project, which is developing an innovative approach to the assessment and training of complex procedural skills and tacit professional knowledge, can be achieved. This approach could be a role model for new immersive and intelligent training environments in many professions. However, we are

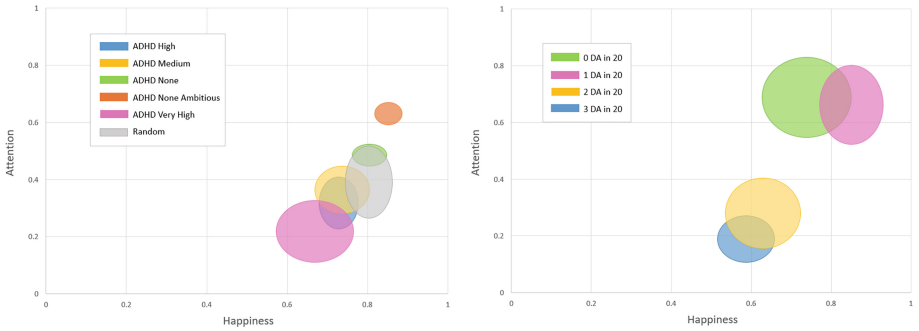

Fig. 5. Left panel: Hapiness-attention comparision for ADHD related simulations; right panel Happiness-attention comparision for simulations with 0 – 3 highly disturbing agents (DA) in a class of 20 agents.

still at an early stage in which we primarily study the general simulation characteristics. In a second step, we implemented a basic set of actions enabling a real person to intervene and to regulate the simulation. This 'virtual teacher' may exert influence on the class level (ordering to study alone or in groups, urging the agents to be quiet, allowing free time) and on the level of individual agents (praising, admonishing, ordering to study, removing an agent from the classroom). We ran various simulation trails and systematically varied and applied external interventions. Also, these simulations result in reasonable outcomes. For example, removing a disturbing agent from a class immediately increases the attentional level and study time and decreases the overall noise level. As a next step, we intend to compare the activities and behaviors in real classes with simulations of those classes, trying to reproduce the student personalities in the virtual environment. By this means, we aim to look into the validity of the ABS. In addition, we intend to compare the achievements of real teachers in controlling the ABS with their achievements in traditional methods of assessing CMS. In the future, we will revise and advance the simulation based on the results of the carried out trials and analyses and intend to integrate Machine Learning features to enable the virtual agents to learn from their interactions with their peer agents and the external teacher.

In conclusion, we strongly believe that AI technologies and in particular agent-based simulations open a new dimension for researching and assessing classroom management skills as well as establishing new teacher training methods in the future. Even if the simulations address only one dimension of the complex construct CMS, and even if the approach will be discussed controversially, we see a great potential in utilizing AI technologies in the field of professional raining.

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Intelligent Tutoring Systems in Smart Learning Environments for Higher Education: A Systematic Literature Review

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Abstract. E-learning integrates information and communication technologies on technological platforms to facilitate access to learning, as well as make it interactive and personalized. Within e-learning, intelligent tutoring systems (ITS) use artificial intelligence to adapt content and activities to the needs of each student, perform academic monitoring and facilitate interaction with virtual learning objectives, allowing for interaction between teachers, students, and intelligent tutors. Smart learning environments (SLE) integrate ITS to create innovative, motivating, and effective learning experiences, based on educational theories and models for the teaching-learning process under the online modality. This article presents a systematic literature review (SLR) on these topics, with the objective of identifying the trends, challenges, and research opportunities in this field. Educational theories that support e-learning and promote student interaction with virtual platforms are analyzed. Student-centered ITS models focused on learning styles, and the integration of data mining and learning analytics, as well as strategies for the implementation and validation of intelligent systems in SLEs. The design of a self-adaptive architecture in the Moodle platform is proposed, for the academic monitoring of students in the online modality with the use of autonomous agents. Finally, it is concluded that the research should be geared towards adaptive learning, to perform academic monitoring focused on the student and their learning styles within the SLE.

Keywords: Adaptative Learning · E-Learning · Higher Education · Intelligent Tutoring · Smart Learning Environment · ICT

1 Introduction

Virtual education has revolutionized the teaching and learning processes, with the use of virtual environments and technological platforms that offer unprecedented access to knowledge and flexible learning opportunities in different study modalities [1, 2]. A myriad of organizations and academic institutions promote the integration of ICT in

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higher education, founded on educational models and theories, which have allowed to establish learning pathways for students [3]. In the process of knowledge construction, contents are designed according to learning achievements. Methodological strategies and instructional design are used to guide students in the use of resources and in the fulfillment of activities [3, 4]. Ultimately, all the educational planning is integrated in smart learning environments (SLE) [5, 6]. However, on many occasions, intelligent tutoring systems (ITS) do not succeed in adapting to the students' needs, since they are designed for tasks such as content generation, automatic answers, and participation, leaving aside the personalized academic follow-up performed by a human tutor. To maximize the effectiveness of these technological platforms, it is necessary to identify the learning needs of each student and their personal context, since in many cases there is no personalized follow-up, causing little interest in learning under the online modality, which is less accepted among students seeking training in different areas.

Intelligent tutoring systems (ITS) offer high quality learning experiences, as they can adapt to the diverse needs of students and keep pace with digital advances, taking advantage of personalized adaptive learning (PAL) [7], to adapt the learning experience to the unique cognitive style and pace of each student. Virtual tutors developed from artificial intelligence (AI) algorithms, are designed to provide personalized support, ranging from content delivery to the assessment of student progress, oriented towards achieving a balance between the use of technology and human interaction for effective learning, ensuring smart learning [5]. These systems can identify learning patterns and provide personalized recommendations to improve academic performance [2, 8].

A significant trend in virtual education is the growing interest in the impact of educational models and potential AI techniques used to research personalized adaptive learning platforms (AAP) [2, 4, 9, 10]. The advancing development of deep learning (DL) is impacting on the online learning process by providing intuitive algorithms and automated content delivery in virtual learning environments [9].

Generative artificial intelligence and deep learning (DL) have revolutionized the world of e-learning, allowing for the adaptation of traditional strategies for the teaching–learning process so as to improve the virtual learning experience, considering instructional design, accessibility, adaptive learning experiences based on learning style models (LSM) and theories, as well as the automated adaptation of the content to the different languages and formats. However, there is a need for further research on educational models oriented to the online modality and intelligent academic tracking, it is necessary to develop more advanced DL algorithms, such as deep neural networks, conduct comparative studies of DL architectures with the aim of improving the adaptability and recommendation capability of platforms in online learning contexts [10–14].

The present study analyzes theories and models for online education, considering smart learning environments (SLE) [13, 15], intelligent educational systems (IES), intelligent tutoring systems (ITS) [2, 18], to facilitate academic monitoring in the teaching–learning process in higher education. The contributions of this article are evidenced by the following research objectives (RO): i) RO1 - conducting a systematic review of the literature to highlight educational theories, didactic strategies, and the use of smart learning environments (SLE), which are adjusted to e-learning in higher education. ii) RO2 – examining trends in intelligent tutoring systems (ITS), considering adaptive and

personalized learning in higher education and iii) RO3 - proposals for the adoption of academic monitoring strategies in higher education with the use of ITS in SLE.

2 Methodology

The systematic literature review (SLR) aims to identify, evaluate, and combine evidence from primary studies available in scientific databases and digital repositories, through research questions [16–18]. The following is the protocol for a literature review conducted in 4 phases: i) research questions (RQ), ii) document search, iii) scientific article selection and iv) extraction, see (Fig. 1):

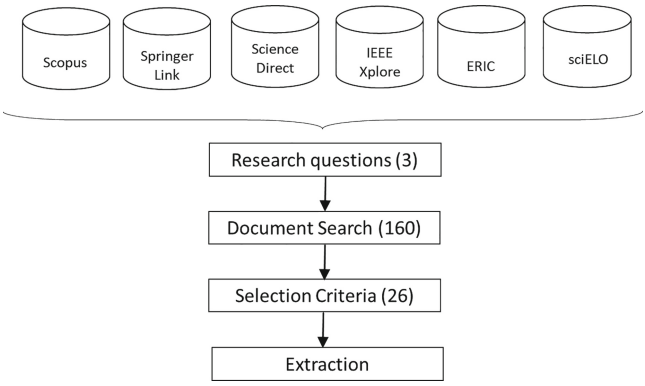

Fig. 1. Diagram of the protocol used in the SLR on e-learning.

2.1 Research Questions

Research questions allow to formulate the problem, extend the knowledge of findings, and provide new results, for scientific and societal knowledge. Table 1 shows 6 scientific databases: Scopus, SpringerLink, Science Direct, IEEE Xplore, ERIC and SciELO.

2.2 Article Search

The article search was performed on February 03, 2024. Boolean operators ("AND", "OR") and keywords were used: ("intelligent tutoring system in higher education" OR "virtual tutoring in higher education" AND "online education" AND "virtual learning environment") (Fig. 2).

In addition, education and technology categories were selected in the database filters (Table 2).

Table 1. Research questions (RQ)

Number	Research Question	Justification
RQ1	What are the educational theories that have been recently reported in higher education for e-learning?	Provide context and analyze current trends in the subject area
RQ2	What models of intelligent tutoring systems (ITS) are used in intelligent education systems (IES)?	Identify and evaluate common intelligent technology system (ITS) models used in e-learning
RQ3	What are the didactic strategies that support the development of an intelligent tutoring system in the virtual learning environments of higher education institutions?	Determine didactic strategies that relate educational models and adaptive learning with artificial intelligence

Fig. 2. The applied SLR methodology.

Table 2. Articles reported in scientific databases

Source	Source Type	Document
Scopus	scientific databases	8
SpringerLink	scientific databases	33
Science Direct	scientific databases	63
IEEE Xplore	scientific databases	23
ERIC	scientific databases	31
Scielo	scientific databases	2
Total		160

2.3 Selection Criteria

This section presents the document selection process and the specific criteria applied to the search results obtained in the scientific databases and digital repositories. In total, 3 phases had been defined with exclusion and inclusion criteria (Table 3). In the first phase, the inclusion criteria were scientific articles and open access studies with abstracts and keywords related to the topics of interest, reviews and articles published between 2020 and 2024, to understand the development of ITS during the period of the COVID-19 pandemic. In the second phase, exclusion criteria were established, discarding technical

papers, reports, book chapters, redundant studies by the same author, secondary studies related to the search words, since the aim was to obtain original studies from the databases. In the final phase, inclusion criteria related to information on methodologies and ITS models that respond to the research questions and contribute to the development of this research were considered.

Table 3. SLR document selection

Source	Phase I	Phase II	Phase III
Scopus	8	7	3
SpringerLink	30	25	11
Science Direct	63	33	9
IEEE Xplore	8	1	1
ERIC	10	1	1
Scielo	2	1	1
Total	121	68	26

2.4 Extraction

Table 4 shows the 26 scientific articles and documents selected for the research. The code assigned to each document corresponds to the relevance, pertinence and validity identified in contrast to the research questions in Table 4.

3 Results

In phase 1, 126 scientific articles from 6 databases were reviewed, after applying the inclusion and exclusion criteria, keywords, and database search filters. In phase 2, we discovered the limited applications of an intelligent tutoring system in higher education teaching. Nevertheless, after applying the exclusion criteria, we identified the studies related to the adaptation of generative artificial intelligence strategies and techniques in virtual learning environments for tutoring in primary and secondary education. In phase 3, the articles that respond to the research questions posed were reviewed. These articles provide key information for the development of a proposal framed within an intelligent tutoring system in a virtual learning environment, for an optimized process of academic monitoring in higher education. There is limited research on the topic of the development of intelligent tutoring systems (ITS), which can be applied in virtual learning environments, for this reason, the established publication timeframe, of the selected studies, was from 2020 to 2024. In this period, a series of studies have been published on generative AI, chatbots and machine learning applications that work independently in virtual learning environments. (See Fig. 3).

Table 4. SLR document selection

Code	Title	Source	Year	Reference
A1	Application of Artificial Intelligence Techniques in Operating Mode of Professors' Academic Governance in American Research Universities	SCOPUS	2021	[19]
A2	Intelligent and adaptive tutoring through a social network for higher education	SCOPUS	2020	[20]
A3	STUART: An intelligent tutoring system for increasing scalability of distance education courses	SCOPUS	2020	[21]
A4	Artificial intelligence in intelligent tutoring systems toward sustainable education: a systematic review	Springer Link	2023	[13]
A5	Smart education framework	Springer Link	2021	[15]
A6	A student-centered approach using modern technologies in distance learning: a systematic review of the literature	Springer Link	2023	[12]
A7	A comprehensive analysis of personalized learning components	Springer Link	2021	[22]
A8	A systematic literature review of personalized learning terms	Springer Link	2020	[23]
A9	A critical evaluation, challenges, and future perspectives of using artificial intelligence and emerging technologies in smart classrooms	Springer Link	2023	[11]
A10	The AI generation gap: Are Gen Z students more interested in adopting generative AI such as ChatGPT in teaching and learning than their Gen X and millennial generation teachers?	Springer Link	2023	[24]
A11	A low complexity heuristic to solve a learning objects recommendation problem	Springer Link	2020	[10]
A12	Adaptive learning: toward an intentional model for learning process guidance based on learner's motivation	Springer Link	2022	[21]
A13	Gaze awareness and metacognitive suggestions by a pedagogical conversational agent: an experimental investigation on interventions to support collaborative learning process and performance	Springer Link	2020	[26]
A14	Automatic engagement estimation in smart education/learning settings: a systematic review of engagement definitions, datasets, and methods	Springer Link	2022	[7]
A15	Smart MOOC integrated with intelligent tutoring: A system architecture and framework model proposal	Science Direct	2023	[9]

(*continued*)

Table 4. (*continued*)

Code	Title	Source	Year	Reference
A16	Developing an intelligent virtual reality interactive system based on the ADDIE model for learning pour-over coffee brewing	Science Direct	2021	[27]
A17	Enhancement of online education system by using a multi-agent approach	Science Direct	2022	[28]
A18	Systematic literature review on opportunities, challenges, and future research recommendations of artificial intelligence in education	Science Direct	2023	[29]
A19	Automatic feedback in online learning environments: A systematic literature review	Science Direct	2021	[8]
A20	Automatic assessment of text-based responses in post-secondary education: A systematic review	Science Direct	2024	[30]
A21	A systematic review of ontology use in E-Learning recommender system	Science Direct	2022	[4]
A22	Understanding the role of study strategies and learning disabilities on student academic performance to enhance educational approaches: A proposal using artificial intelligence	Science Direct	2024	[31]
A23	A multi-perspective study on Artificial Intelligence in Education: grants, conferences, journals, software tools, institutions, and researchers	Science Direct	2020	[32]
A24	Personalized Adaptive Learning Technologies Based on Machine Learning Techniques to Identify Learning Styles: A Systematic Literature Review	IEEE Xplore	2023	[2]
A25	Technology Enhanced Learning Analytics Dashboard in Higher Education	ERIC	2022	[1]
A26	Intelligent Tutoring Systems and their application in higher education	sciELO	2021	[33]

3.1 RQ1- Educational Theories Applied to E-Learning

UNESCO-IESALC [3], among other authors [4, 7, 12, 15, 18], report traditional educational theories and models for the accessible, adaptive and quality development of e-learning in higher education, such as: behaviorism, cognitivism, constructivism and connectivism, are presented as classical theories, which start from the epistemological foundation, and improve the online educational experience in a personalized, flexible, user-centered way, analyzing the context and interactions of students. Educational theories presented by EduLabs [5], which consider the holistic approach in the integral formation of the individual, are: i) constructivist learning: promotes the active construction of knowledge by the student through interaction with the content in the SLE. ii) collaborative learning: development of the collaborative activities of study teams, to

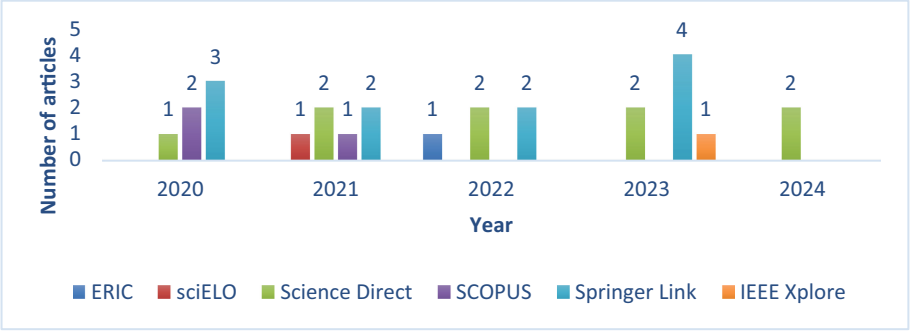

Fig. 3. Number of articles published since 2020 related to this scope of study.

achieve common learning objectives. iii) self-directed learning: considering the student's ability to direct his or her own learning process, establishing study goals and objectives. iv) experiential learning: considering practical experience and reflection through ICT-mediated activities. v) social learning. Current educational theories that are the result of the evolution of classical and holistic theories in education are oriented to the integration of AI in the teaching and learning process, such as: student-centered learning, adaptive learning and intelligent learning [12, 21, 23], which consider the incorporation of personalized adaptive learning platforms (PAL) and ITS [2, 8, 32].

3.2 RQ2 Intelligent Tutoring Systems (ITS)

Models for Classifying Learning Styles or Mastery Drivers. Intelligent tutoring systems (ITS) are categorized according to models that make up their architecture: teacher, student, and tutor, which correspond to traditional models based on classical educational theories, oriented to the development of thinking [15]. The different study modalities and the use of technological platforms allowed for the development of ITS models oriented to identifying learning styles, which focus on the student and the virtual tutor, leaving aside the teacher's direct participation in the teaching process. These intelligent systems include the VARK model, the Kolb model, the Felder and Silverman model (FSLSM), the Myers-Briggs Type Indicator (MBTI) theory, Gardner's Theory of Multiple Intelligences (GTMI) and Honey and Mumford's model [13, 21]. These models consider sensory aspects such as: visual, auditory, textual, and kinesthetic in virtual learning objects [9, 31], for the identification of students' learning styles, oriented to improve academic performance and encourage active participation through didactic strategies, adapted to the new technological platforms [4, 18].

Student-Centered Learning Models. These models involve teachers, students, and intelligent virtual tutors, which have the ability to become an integral part of the teacher's teaching process, through the use of AI techniques [2, 5]. In these models, the virtual tutor becomes a dynamic instructor, simulating the interaction with a human tutor who identifies possible problems in the collaborative learning process [26]. The models that fit these new study modalities are: Malvi system [32], STUART [21], which simulate the mentioned interaction, integrating the training focused on the behavior of each student.

Other models are oriented to the use of read aloud techniques to facilitate accessibility to information for students with visual disabilities, voice recognition during activities and the automatic translation of texts in the development of activities [13, 22]. The ARCS model for motivation is based on four key elements that influence students' motivation: attention, relevance, and satisfaction with great acceptance for adaptive learning in online learning [25].

Models Integrating Data Mining and Learning Analytics. These models use algorithms that evaluate learning methods based on data generated in the intelligent learning environments of variables such as: percentage of completed activities, obtained grades, connection time, amount of interactions with the contents, among others [1, 27], to predict the academic performance of the student within the virtual learning objects, generally decision trees are used to estimate the predictions (LVO) [7, 22].

3.3 RQ3 - Teaching Strategies for ITS in VLE

Strategies for the implementation and validation of ITSs involve domain and student-centered learning, the use of autonomous agents, chatbots, learning analytics (LA), educational data mining, to support students in problem solving and learning processes [2, 22, 23]. These systems use Bayesian approaches and sequential likelihood ratio testing (PPPS), to improve the adaptability and personalization of the educational experience for students through the proposed intelligent tutors, starting from the selection of the appropriate model and technology, content personalization, integration of learning algorithms, continuous assessment, personalized feedback. These didactic strategies support the development of ITS in the LE of higher education institutions and are adjusted to the new trends of the learning process within e-learning [4, 18, 33].

4 Limitations and Future Work

Future research should integrate AI developments into ITS, to personalize the learning experience, offering effective feedback and improving interaction with students within VLEs. These innovations in ITS should consider the interaction between the tutor and the student [27]. In addition, alerts and recommendations that allow for the evaluation of academic follow-up should be developed. These interactions should be developed based on didactic strategies and a legal framework, which allows for the achievement of learning goals under this modality of study. The integration of autonomous agents in smart learning environments (SLE) is a strategy used in the implementation of intelligent tutoring systems focused on the user and learning styles. According to the results obtained for RQ1, traditional and holistic theories are considered for ITS model development, which integrates personalized student monitoring on technological platforms [4, 7, 12, 15, 18]. This integration implies a digital transformation for the achievement of processes, activities and fulfillment of student learning objectives based on educational theories and instructional designs within the e-learning framework. We propose the design of a self-adaptive architecture in the Moodle platform, based on the following models: FSLSM, Malvi system [32], STUART [21], which could facilitate adaptive

and intelligent learning by means of a virtual tutor, simulating human interaction and enabling the academic monitoring of students in the online modality, with the use of autonomous agents [12, 21, 23].

5 Conclusions

This study presented a systematic review of the literature on intelligent tutoring systems (ITS) in higher education and virtual learning environment (SLE) studies. This review highlights the focus of recent research on the presentation of ITS models focused on student-centered adaptive learning, oriented towards academic monitoring and the students' individual learning styles. Educational theories on e-learning in higher education have been underscored, starting from classical theories such as: behaviorism, cognitivism, constructivism and connectivism, to new trends oriented to the application of AI in the teaching–learning process such as: adaptive learning and intelligent learning. There has been limited research on the development of intelligent tutoring systems (ITS) in virtual learning for higher education, within the articles analyzed in the period from 2020 to 2024. Nevertheless, there are other proposals oriented to different contexts of the education sector, such as the use of generative AI, chatbots, ML algorithms and autonomous agents, among others. This systematic review has highlighted the importance of the development of ITS models for the generation of accessible content. ITS models should respond to different learning styles and adapt to students' abilities. Future lines of research are oriented towards the integration of AI to personalize the learning experience, provide effective feedback, and improve interaction with students, ensuring accessibility, and usability, as well as simulating the personalized assistance provided by a human tutor, within an intelligent learning environment.

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Cybot: A Chatbot for Teaching and Testing Cybersecurity Courses

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Abstract. As technology advances and public awareness of cybersecurity grows, protecting data and promoting cybersecurity best practices is essential. Educational chatbots were developed to address this need, but current systems respond artificially and lack user-friendliness, resulting in a bad user experience. This paper presents Cybot, a chatbot specifically designed to tackle this gap in cybersecurity education for university students. Cybot can answer cybersecurity questions, offer human-like interactions, provide assessments to improve students' understanding and performance and provide emotional support with its motivational personality. Additionally, it decreases the workload of academic staff due to its constant availability and instant replies. An experiment was conducted on 30 university students to evaluate Cybot's performance. The results showed positive feedback as users reported that Cybot was effective and a supportive study partner who always responded to their inquiries at any time.

Keywords: Chatbots · Conversational agents · Cybersecurity · Education

1 Introduction

Human-Computer Interaction (HCI) experts believe that technology would influence society and that a more socially connected world would arise from the increased use of smart devices [19]. Conversational agents, also known as chatbots, are examples of these technologies. Although most people have used personal assistants such as Siri or Alexa, they are still unaware of what a chatbot is. Chatbots can be designed for voice or text interaction, providing information, and completing tasks [4]. Nowadays, one of the popular conversational agents is ChatGPT which OpenAI developed. It helps users generate human-like text based on input. It can be used for various tasks such as conversation generation and language translation [16]. However, the current chatbots still lack a balance between providing important functionalities and simulating human contact [19].

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In the end, the efficacy of chatbots is determined by their capacity to deliver precise responses, involve users through suitable psychological techniques, and satisfy their requirements in important domains such as healthcare and education [5,7,8].

Chatbots are improving the education sector by automating administrative tasks and helping students obtain information for academic papers or exams. Both higher education institutions and students can benefit from chatbots. They can handle admission processes [9], provide course information readily [1], and even act as personal assistants reminding students of deadlines [6]. It frees up staff time, potentially reducing costs in the long run. Most educational chatbots use a rigid question-and-answer format, with some lacking the ability to understand complex queries or provide a human-like interaction.

The topic of cybersecurity education chatbots has gained importance these days because cybersecurity is becoming extremely popular, which is encouraging more individuals to want to learn about it. Before answering simple, often requested inquiries about cybersecurity, chatbots that provide security education first instruct users about current security trends [12]. This raises a vital question: "Is it possible for an educational chatbot focused on cybersecurity to act as your at-home tutor?" In this paper, we aim to address this question by developing an instructional chatbot for university cybersecurity classes called "Cybot". This tool aims to answer the following research question: "Can a chatbot effectively supplement traditional cybersecurity education?"

Regarding instructing and testing students in cybersecurity courses, Cybot aims to maintain the ideal mix between being a helpful teacher and a compassionate, human-like assistant. It is designed to answer any questions from students and provide clarifications on cybersecurity courses by utilizing data gathered from the university content management system (CMS). Moreover, it can assess students on the tutorials for the course to make them feel ready for tests. It is also trained to converse with the students in a friendly manner, providing them with emotional support and encouragement.

The paper is structured as follows: Sect. 2 presents the related work. Section 3 describes the features and capabilities of Cybot. Moreover, Sect. 4 explores the architecture of the system and the tools used. Section 5 discusses testing, data analysis, and statistical findings. Finally, Sect. 6 concludes the paper and presents future work.

2 Related Work

Chatbots are computer programs that answer questions and engage in conversations similar to those between people. Chatbots are created using various methods, including pattern matching and scripting languages [18]. The authors of [18] reviewed a few chatbot designs and discussed the applications of chatbots in various industries, including business, healthcare, and education. However, a well-designed chatbot should provide a positive user experience [3]. A study investigated the discoverability and characteristics of educational chatbots on

Facebook Messenger, focusing on language, subject matter, and developer platform [21]. The reported findings showed that educational chatbots on Facebook Messenger range from sending personalized messages to recommending resources, suggesting the field is still evolving [21].

Chatbots have emerged as a tool for cybersecurity awareness. A chatbot was created that educated users on password management, privacy, and safe browsing practices with motivational interviewing techniques[13]. Another model focused on a knowledge base, a structured database of information. The chatbot extracted keywords from user messages and searched the database to provide answers and explanations for cybersecurity questions [14]. Moreover, the authors in [12] explored the Dialogflow platform, a cloud service for natural language processing by Google. The chatbot used a vocabulary of cybersecurity terms to address user queries, test knowledge, and help identify phishing emails. Apart from chatbots, a security education training and awareness program incorporating games and surveys was proposed by [2]. This interactive approach fosters a more engaging learning experience than traditional training methods.

The authors in [20] designed SecuBot which focuses on user education. First, the chatbot expresses gratitude to the user. Then, it evaluates the user's security awareness across five key aspects: password management, data storage, phishing scams, and ransomware threats. Finally, it highlights the area requiring the most improvement and provides guidance accordingly. SecuBot uses a combination of buttons, text, and videos to enhance user experience [20]. Another chatbot, SecBot, was presented as a conversational agent for managing cybersecurity from a business perspective [11]. This chatbot goes beyond education, aiming to assist businesses in planning and incident response. SecBot can analyze potential cyberattacks based on user-reported reports and business risks. It then helps users understand the impact of such incidents and recommends appropriate actions at various levels. Additionally, SecBot can guide users through modifying or acquiring security solutions [11].

Despite advancements in cybersecurity chatbots, many fail to provide a comprehensive learning experience for students. They often lack motivational elements and do not assess their understanding. Although it provides users with information in a conversational manner, it neglects users with different preferences, needs, and degrees of digital literacy [10]. Cybot addresses these issues by acting as a personalized teaching assistant for cybersecurity courses. It motivates students to further explore the course subject by providing positive and engaging interactions, resulting in a richer learning experience.

3 Cybot Design and Features

University semesters can be overwhelming, with constant deadlines, assignments, and exams causing stress and anxiety for students. Cybot aims to decrease this burden by acting as a friendly, always-available study partner. It uses the university's CMS-posted material to answer student inquiries and guide them toward the correct learning path. It is helpful for students who study at night and

encounter topics that they do not understand at odd hours when professors and teaching assistants are not available. Additionally, Cybot offers a more efficient way to navigate the CMS than manually searching through all of the posted documents. Before its development, students would ask the academic staff for clarification, search through all the worksheets for solutions, and constantly check their emails for their new deadlines.

Effective educational chatbots should be approachable and use natural language to minimize student stress and improve learning. Cybot balances friendliness and professionalism, offering motivational messages to anxious or overwhelmed students. Cybot uses a wide range of responses including emojis and is trained for small talk as shown in Fig. 1 to create a more human-like experience. Moreover, Cybot serves as a valuable study tool. Students can request questions to practice a specific topic after studying it, or it can initiate questioning to assess their understanding and adjust its support accordingly. Cybot is also designed to adapt its responses to the user's mood, further personalizing the learning experience.

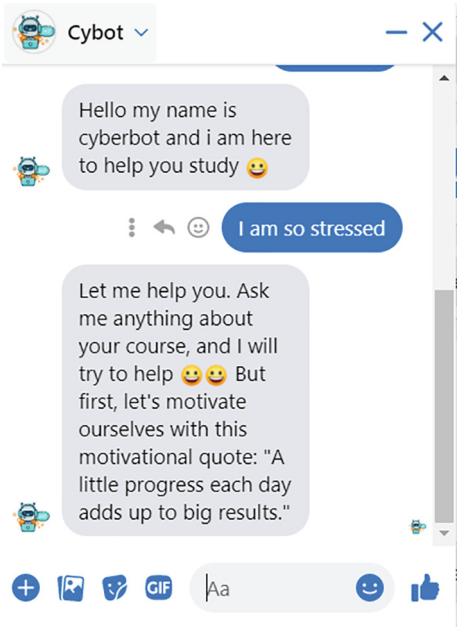

Fig. 1. A small talk dialogue example of Cybot interacting with students

Upon interacting with Cybot for the first time, the "Get Started" button appears, prompting the chatbot to introduce itself when the user clicks it. Afterward, Cybot presents a carousel of topics to ease navigation. As shown in Fig. 2, each carousel entry presents the topic title, definition, and subtitle for a quick

overview. Students can browse and choose any topic to access detailed explanations or ask related questions. Following the initial greeting, Cybot's guided approach prompts students to select a specific topic. Once chosen, Cybot provides a clear explanation. Students are then free to ask any questions about the course material or the chosen topic. Cybot uses a comprehensive knowledge base containing all course material, allowing it to answer student inquiries confidently. It promotes trust, as students can rely on the accuracy of the responses.

Fig. 2. A user choosing a topic from the carousel

Once students have finished with questions and knowledge review, Cybot's "Quiz Me" feature transforms learning into a game-like experience, making assessment more engaging. Students simply type "Quiz Me" to activate the quiz. Afterward, it prompts them to choose a specific topic or tutorial for the assessment. The quiz uses cards which is another sort of Facebook Messenger response. Cybot presents the students with a set of randomized questions on the chosen topic and asks them to attempt to respond on an external paper without any aid. After attempting the questions, students can use the "Get Answer" button to reveal the answer, allowing for self-review and knowledge reinforcement.

4 Cybot Implementation

Cybot is an educational chatbot designed based on a modular framework for future expansion [5]. It allows Cybot to manage all university courses due to its flexibility. Currently, Cybot focuses on the Business Continuity and Risk

Management course, providing students with relevant information and answering FAQs. The majority of chatbots follow a common four-step cycle. First, users interact with the chatbot through a user interface. Their input is then processed and converted into a format the system can understand. This triggers intent matching in the back-end, where Cybot identifies the user's intent. Finally, Cybot formulates a response, retrieves any additional data needed from external sources, and delivers the answer back to the user.

Cybot's architecture consists of four key modules as shown in Fig. 3 working together to deliver information and answer user queries. The database is the first module which acts as a knowledge repository that stores FAQs and relevant content retrieved from the university's CMS. The second module which is the processing module uses Google's Dialogflow knowledge base to understand user intent. It analyzes user messages to identify the main goal or question. The third module, the chatbot module, contains all the intents and entities defined within Dialogflow. Intents represent user goals (e.g., "greetings," "ask a question"), while entities are specific pieces of information extracted from user queries (e.g., "topic name," "worksheet number"). Finally, the user interface module is the fourth module which serves as the communication channel. In Cybot's case, this is Facebook Messenger, where users interact with the chatbot. When a user submits a message, it is forwarded to Dialogflow where intent matching and entity extraction occur. Dialogflow then retrieves the most relevant response from Cybot's knowledge base and responds in the messaging platform.

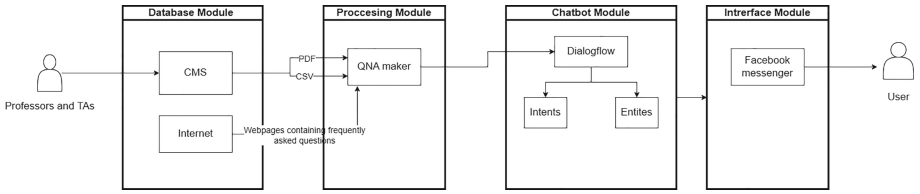

Fig. 3. The architecture used for the chatbot implementation

The base of Cybot's knowledge is in the database module. The academic staff uploads lectures and worksheets to the university's CMS. Cybot then gathers all of the uploaded material and integrates them into its knowledge base. Additionally, the database module stores frequently asked questions compiled from reliable sources, allowing Cybot to address common cybersecurity inquiries. The processing module takes the raw materials from the database and transforms them into a readily usable format. It utilizes a Natural Language Processing (NLP) tool called QnA Maker to automatically convert information in lectures and worksheets into question-and-answer pairs. This structured format allows Cybot to locate the most relevant response to any student query efficiently.

The chatbot module acts as the engine that analyzes user input. It uses Dialogflow to define intents and entities. The previously identified questions

and answers by the processing module are mapped to corresponding intents in Dialogflow. The intention of a user for one conversational turn is called an intent. Dialogflow matches the user message to the best intent. Moreover, the entity type of each intent parameter specifies how data from a user message should be extracted. For example, system entities can compare dates, times, colors, and other basic data. Developers can create their customized entities to match specific data. Each intent pairs a user question with its appropriate answer. When a user interacts with Cybot, the chatbot module analyzes the input, extracts any relevant entities, and then retrieves a personalized response from the database module based on the matched intent.

Finally, the user interface module provides a suitable platform for student interaction. Cybot is integrated into Facebook Messenger based on the findings from the literature review. This choice allows students to discover Cybot through their existing social media habits. Students are more likely to message Cybot on Facebook Messenger than visit a dedicated website or application, making Cybot readily accessible for any academic assistance they may need. To better enhance the user experience, Cybot employs a friendly avatar that visually represents its supportive and informative nature. Additionally, Facebook Messenger offers various response formats besides text, including buttons, lists, GIFs, and images, enhancing the system's usability. Deployment on Facebook Messenger ensures Cybot's constant accessibility. The process involved creating a dedicated Cybot Facebook page. To connect the Facebook Messenger interface with the chatbot engine, we integrated Dialogflow with Meta for Developers. This integration involved using callback URLs, Dialogflow tokens, and Meta's verification key. This seamless connection allows Facebook Messenger to send user inputs to Dialogflow for processing and response generation. Cybot can then use Facebook Messenger's diverse response formats to deliver the most appropriate and engaging answer.

Cybot uses multiple Facebook Messenger user interface elements to deliver information in a user-friendly way. When responding to user queries, it can present a carousel of a list of topics. Carousels display each topic alongside a brief definition and buttons for discovering subtopics. Students can easily browse the carousel (Fig. 2) to find relevant information or choose to view a specific topic's definition. The carousel also features a "Quiz Me" card, inviting students to test their knowledge. Selecting this card prompts Cybot to present quick reply options, allowing students to choose the specific tutorial they wish to be quizzed on. Instead of typing out answers, Cybot offers quick replies which are suggestions that guide users on the type of input it expects. Simply a student can choose a reply, and Cybot will adjust its response accordingly. Once they are finished, they can ask any questions or request the list of topics again by typing "list". Cybot also uses buttons in some responses, allowing students to access links or submit text for an immediate answer easily.

5 User Evaluation

To assess Cybot's impact on student learning, an evaluation was conducted by the end of 2022 and the beginning of 2023 with 30 university students aged 19–22. All participants were familiar with their university's CMS as they were recruited from the same university while taking their consent. Two key aspects were investigated: user preference (Cybot vs. traditional CMS) and the effectiveness of Cybot in raising cybersecurity awareness. The evaluation was divided into two experiments, each involving 15 students. The first experiment included 10 males and 5 females, while the second experiment included 8 females and 7 males.

5.1 Experiment 1

To assess Cybot's usability and learning impact, participants were asked to fill out two questionnaires after interacting with the chatbot and an assessment before and after interacting with it. The first questionnaire was the System Usability Scale (SUS) which measures the overall usability and learnability through a 10-item, 5-point Likert scale questionnaire (1 = Strongly Disagree, 5 = Strongly Agree) [17]. The second questionnaire was the Chatbot Usability Questionnaire (CUQ) which had 16 questions on a similar scale, evaluating aspects like response clarity, error handling, and perceived intelligence [15]. Finally, a background assessment tested the participants' cybersecurity knowledge before and after interacting with Cybot. This included a self-reported knowledge level question and four course-specific questions worth a total of four points. The experiment was conducted by all years of university students with various specialties, including software, security, and data science. The experiment was conducted in the following order:

1. Students received an assessment on cybersecurity courses provided to assess their background knowledge and familiarity with the topics. The assessment consisted of five questions with a 4-point score
2. Students were free to use the chatbot and attempt to obtain more information about the themes covered in the assessment after the assessment was completed. The following suggestions were made to the users to guide them:
   - Choose a topic from the Cybot carousel menu
   - Try typing "Quiz me" and receive a quiz from Cybot
   - Ask Cybot about its courses and functions
3. Students were given another chance to retake the assessment to see whether or not Cybot had improved their understanding. Some questions were added about whether Cybot was useful on these topics or not.
4. Finally, the participants were asked to fill out the SUS and CUQ forms

5.2 Experiment 2

The second experiment was conducted to demonstrate Cybot's effectiveness on students' grades and knowledge gain. All participants completed a background

assessment with five cybersecurity course-related questions to measure their prior knowledge. They then received access to all course materials and had 24 h to study independently. After this period, they repeated the background assessment to assess their knowledge improvement. Comparing the pre- and post-test scores allowed us to measure how effectively the course materials facilitated learning.

1. Students took the first assessment on the cybersecurity course given to assess their background knowledge
2. Students were then given access to all course materials from the CMS and 24 h to study
3. Students retook the assessment to identify how much their studying affected their performance
4. Results from both assessments were gathered to evaluate their progress

5.3 Results

The two experiments were designed to compare learning effectiveness with Cybot against studying independently. In the first experiment, students used Cybot to learn the course material. In the second experiment, they examined the materials independently. To assess the knowledge gain in each approach, all participants completed the same background assessment before and after their designated learning period. This allowed us to determine which approach led to better learning outcomes—the first assessment aimed to measure the students' prior knowledge of cybersecurity. As expected for students from different majors or without prior coursework, the mean score was low (1.73 out of 4) and the median score was 2 as shown in Fig. 4, indicating limited initial knowledge in this subject area.

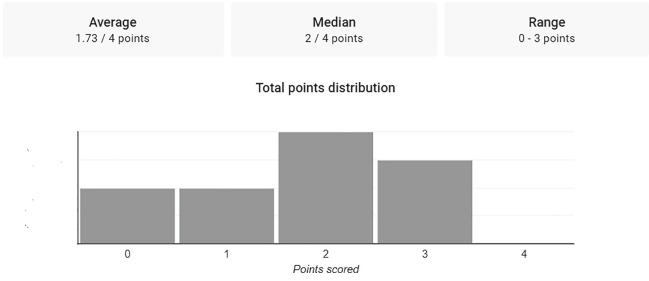

Fig. 4. The first assessment results for both groups

The second assessment in the first experiment revealed positive outcomes. Students showed a significant increase in knowledge after interacting with Cybot. The mean score jumped to 3.33 out of 4, and the median score remained steady at 3 as shown in Fig. 5. Usability evaluations were also promising. The SUS score

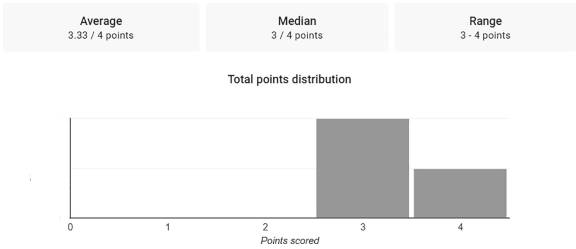

Fig. 5. The results of the second assessment for the first group

yielded a mean of 95, indicating a high level of perceived ease of use. The CUQ score, with a mean of 85.9, further confirms Cybot's user-friendliness.

Meanwhile, the second assessment in the second experiment, where students studied the course materials independently, resulted in lower knowledge gains than the Cybot group. The mean score mean was 1.56, and the median score was 1 as shown in Fig. 6. These results suggest that Cybot may be a more effective learning tool than independent study for this particular subject matter.

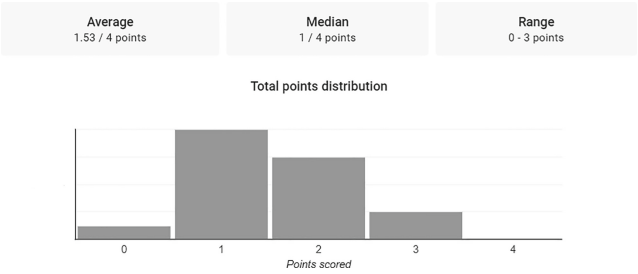

Fig. 6. The results of the second assessment for the second group

5.4 Discussion

This study explored the use of chatbots to replace traditional content management systems and act as teaching assistants, supporting academic staff and reducing their workload. The SUS and CUQ results were positive, with average scores higher than the minimum acceptance score of 71.1%. However, some students suggested Cybot could benefit from initial user guidance and improved error handling to user queries that it did not understand. Still, the majority found it easy to use, informative, and approachable. Cybot's clear and helpful responses motivated students to explore its different functionalities and ask more questions. The significant improvement in knowledge gain between the pre- and post-assessments (Experiment 1) is particularly encouraging. Students reported

feeling more confident in understanding various cybersecurity topics after interacting with Cybot. Moreover, the second experiment (independent study) suggests that Cybot-assisted learning may lead to better outcomes than traditional self-study. Overall, the study highlights the effectiveness and efficiency of educational chatbots like Cybot. It also reflects the importance of user-friendly design elements like clarity, approachability, and friendliness in chatbot development.

6 Conclusion and Future Work

A major challenge in educational chatbots is maintaining a balance between effectiveness and user experience. Current chatbots tend to fall into two extremes: efficient but robotic, and amiable ones that lead to time-consuming interactions. This study aimed to tackle this gap by creating a chatbot that acts as a teaching assistant at home by improving student knowledge without overwhelming them. The results were positive. Cybot helped improve the students' cybersecurity knowledge and reduced some workload for the academic staff. Moreover, students found Cybot's friendly and engaging character motivating, encouraging them to learn and discover its features. This highlights the importance of user-friendliness and efficiency in chatbot design.

Multiple challenges were faced during the study. Extracting information from lectures and worksheets was difficult due to the different formats and designs used by the academic staff. As a workaround, standardized worksheets were used instead. Several advancements can further enhance Cybot's capabilities. Firstly, addressing the challenge of extracting information from different lecture formats is important. Secondly, offering Cybot access to a wider range of courses would allow students to select specific topics for learning. Additionally, integrating a scheduling function would enable students to set reminders and plan their studies, creating a better learning experience. Analyzing user queries and responses could allow Cybot to expand its knowledge base continuously. Finally, increased language support and improved error tolerance would be significant improvements, enabling students to interact with Cybot in their preferred language and creating a more natural learning environment.

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CyberChallenge.IT@UnivAQ: A Cybersecurity Educational Training Path for High School and University Students

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Abstract. We are experiencing the core of the Fourth Industrial Revolution, living in a digitally interconnected world. This digital interconnectedness has improved social interactions and work efficiency but has also led to an increase in the frequency and complexity of cybersecurity threats. Consequently, the need for cybersecurity professionals has exponentially increased. However, there is a shortage of cybersecurity professionals, and their training is a constant battle due to the lack of specific cybersecurity training. Thus, became imperative to create cybersecurity programs aimed at producing cybersecurity professionals. The present paper goes in this direction by presenting the CyberChallenge.IT, a pioneering cybersecurity program for high school and university students aged 16–24, in the University of L'Aquila (UnivAQ) venue: CyberChallenge.IT@UnivAQ. Moreover, we analyzed the registration number of high school and university students at the UnivAQ program during the editions from 2020 to 2023. The preliminary results are worrying, with a considerable decrease in registration numbers in 2023. The awareness of this decrease has led us to propose and implement strategic activities to improve the success of such program. Concluding, this paper paves the way for a reflection on how to effectively address the cybersecurity challenges of the near future, which are expected to be even more interconnected than the current ones.

Keywords: Cybersecurity · Education · Students · Training Program

1 Introduction

At the heart of the Fourth Industrial Revolution, individuals find themselves in a highly interconnected world with many opportunities for innovation and progress. In recent years, the digital boom has affected every sector, from the

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health sector to the world of industry and education. Consequentially, individuals are interconnected in all aspects of daily life, from social relationships to everyday work [5,11,12,18,19].

Although such interconnectedness in cyberspace produces benefits for social welfare, it comes at a high cost: a heightened vulnerability to cyber threats [13, 20]. Nowadays, the number, sophistication, and impact of cyber-attacks (such as phishing, malware, meeting invasions, and website and social media defacement) have increased greatly, violating the security and privacy of citizens [6,21]. In this troubling and evolving field, the human element remains a critical line of defence, and in fact, research paints a troubling picture [13,22]. A significant portion of security breaches is due to human error, which highlights a crucial gap in user awareness and the urgent need for solid cybersecurity training programs to create cybersecurity professionals [13,22]. To understand the importance of cybersecurity, just think that, among governmental and non-governmental institutions, there is a growing concern that cyberspace may become the next battleground. Despite the increasing significance of cybersecurity, the existing workforce in this field is unable to meet the rising demand for qualified professionals [6,25].

Indeed, there are few cybersecurity professionals; recruiting and training enough cybersecurity professionals is a constant battle, not only because of the technical aspect of cybersecurity but also because of the lack of specific, high-level training [9,17]. Effectively designed cybersecurity training equips individuals with the knowledge and skills to recognize and combat cyber threats [22]. By educating users on prevalent attack vectors like phishing scams and malware, these programs empower them to make informed decisions online and safeguard sensitive information [22]. Moreover, training fosters a culture of security within organizations, fostering a shared responsibility for protecting digital assets. Creating a high-level cybersecurity training program is not a simple challenge, as cybersecurity encompasses numerous specialized sectors and job roles. For this reason, no single educational program can cover all the specialized skills and sector-specific knowledge desired by each employer [6,9]. However, there are general guidelines and standard programs that should always be followed regardless of the context. For example, an understanding of basic computer architectures, data, cryptography, networks, secure coding principles, and operating system internals. Additionally, other competencies, such as proficiency with operating systems, fluency in low-level programming languages, and familiarity with common exploitation methods and mitigation techniques, are considered equally important [5].

Concerning the types of educational activities, cybersecurity training programs generally include a combination of various types of educational activities, such as: (1) *Theoretical lessons*, where instructors deliver lectures to impart foundational knowledge and concepts related to cybersecurity; (2) *Practical lessons*, where participants apply their theoretical understanding in simulated or real-world scenarios through hands-on exercises; (3) *Workshops*, where participants engage in group activities, problem-solving exercises, and discussions to deepen their understanding of specific cybersecurity topics; (4) *Gamified Learning Expe-*

riences, such as Serious Games and Capture the Flag (CTF) competitions, which simulate real-world cybersecurity challenges in a competitive game-based learning environment [2–4,7].

Several institutions worldwide have begun to define and offer cybersecurity programs to address the shortage of cybersecurity professionals [5,9,23]. Following this trend, in 2017, the Italian Cybersecurity National Laboratory (CINI) launched the CyberChallenge.IT project, a groundbreaking initiative standing out by specifically targeting young people and nurturing a pipeline of future cybersecurity professionals [10]. Recognized by the Italian Ministry of Education, CyberChallenge.IT bridges the gap between theoretical knowledge and practical application by operating within the framework of high schools and universities. CyberChallenge.IT provides an educational training path that blends theoretical/practical lessons with gamified learning experiences.

The present paper goes in this direction by presenting the CyberChallenge.IT implemented at the University of L'Aquila (UnivAQ) venue. Moreover, we reported the preliminary results related to the registration numbers of high school and university students during the last editions of the UnivAQ program, i.e., from 2020 to 2023.

The paper is structured as follows. Section 2 provides an overview of the CyberChallenge.IT project, aiming to present a clear description of the initiative. In Sect. 3, we present how this project is implemented at UnivAQ, summarizing the data analysis and results. Finally, in Sect. 4, conclusions are drawn.

2 CyberChallenge.IT

The CyberChallenge.IT project is a pioneering cybersecurity training program specifically designed for high school and university students aged between 16 and 24 years old. It is organized annually by the National Cybersecurity Laboratory (CINI), and its eighth edition (2024) is currently running.

One of CyberChallenge.IT's key strength is its strong partnerships with Italian universities and industry leaders. These collaborations keep the program current with the evolving cybersecurity landscape and introduce participants to potential career opportunities. Cybersecurity experts from these institutions provide valuable mentorship and contribute to developing the program's curriculum.

The second key strength of CyberChallenge.IT is its gamified learning environment. Participants engage in interactive activities that use gamification techniques to introduce essential cybersecurity concepts like cryptography, web application security, incident response, and digital forensics. These engaging and challenging activities allow participants to apply their knowledge in simulated scenarios. This approach attracts young people through gamification while providing substantial educational content by blending technical, scientific, and ethical aspects of cybersecurity. The primary gamified strategy employed is the Capture The Flag (CTF) competition. CTFs are exercises designed to test and develop cybersecurity skills in a safe, engaging environment. Participants, individually or

in teams, compete to find hidden "flags" within simulated systems or challenges, requiring various cybersecurity techniques to discover. There are two main types of CTFs: (1) *Jeopardy-style CTFs* are organized as a set of stand-alone challenges focusing on a particular topic (e.g., network security, web application security etc.). Each challenge usually contains a single flag, and it is worth a fixed or dynamic amount of points; (2) *Attack & Defence* are a CTFs format in which each team has to protect a set of provided network services which contains, by default, a set of security vulnerabilities. To gain points, teams have to keep services running, locate the vulnerabilities and patch them while using the acquired knowledge to attack other teams.

CTFs, whether Jeopardy-style or Attack & Defense, typically end with participants sharing write-ups of their solutions. These write-ups benefit everyone: successful participants solidify their understanding, while unsuccessful ones learn new techniques and thought processes. CTFs offer a safe space for aspiring cybersecurity professionals to apply knowledge, learn new tools, develop critical thinking and problem-solving skills, foster teamwork, and assess and improve their cybersecurity knowledge.

The CyberChallenge.IT program is structured to provide a comprehensive and multidisciplinary educational training path in cybersecurity, from the initial pretest to the final national competition. Here below is presented an overview of the CyberChallenge.IT program's structure:

- **Online Registrations Phase:** Interested participants can register online.
- **Pretest Phase:** An initial test is conducted to determine the participants' current knowledge of logic, mathematics, algorithmics and basic computer science.
- **Programming Test Phase:** Following the pretest, an admission test is conducted consisting of four algorithmic or implementation problems to be solved using a programming language (e.g., C++ or Python).
- **Admission Phase:** At the end of the programming test, a weighted average of the Pretest and Programming Test scores will be calculated. The final ranking will be sorted by score in descending order, with ties broken by age in ascending order. Each venue will select up to 20 participants from this ranking for their team, along with up to 5 reserves who can attend lessons and replace any dropouts.
- **Training Path Phase:** It consists of 12 weeks of training, independently managed by each venue regarding activities, dates, and times, focusing primarily on cybersecurity topics such as cryptography, network security, and software security. All participants will have access to the project's official training portal, featuring slides, video lectures, and exercises prepared by national leaders, along with past tests from previous local competitions.
- **Local Competitions Phase:** At the end of the Training Path phase, a local competition is held at each participating venue to assess the skills of the trainees. This competition consists of a Jeopardy-style CTFs competition.
- **Final National Competition Phase:** The top performers from local competitions gather for a national-level contest that consists of an *Attack &*

Defence CTFs competition organized each year in a different location. During this event, all the teams, their instructors, and the sponsors meet to celebrate and conclude the entire project.

The well-structured program of CyberChallenge.IT, along with its key strengths, has led to its widespread popularity and success. This initiative has notably boosted interest in cybersecurity among Italian youth, as evidenced by a steady increase in participation, with over 4700 students registered across 43 locations in the latest edition (see Table 1).

Table 1. CyberChallenge.IT participants over the last seven editions (2017–2023)

Year	Venues	Participants												
		Total	Gender						Origin					
			Male		Female		Other		High School		University		Other	
		#	#	%	#	%	#	%	#	%	#	%	#	%
2017	1	683	603	88.3	80	11.7	–	–	57	8.3	626	91.7	–	–
2018	8	1864	1697	91.0	167	9.0	–	–	583	31.3	1250	67.1	31	1.6
2019	18	3203	2829	88.3	374	11.7	–	–	1314	41.9	1810	56.5	52	1.6
2020	28	4449	3854	86.4	604	13.6	–	–	1959	44.0	2414	54.2	76	1.8
2021	33	4902	4258	86.9	644	13.1	–	–	2255	46.0	2583	52.7	64	1.3
2022	34	5336	4629	86.7	660	12.4	47	0.9	2452	46.0	2782	52.1	102	1.9
2023	43	4720	4101	86.9	567	12.0	52	1.1	2066	43.8	2585	54.8	69	1.4

In the following section, we examine the implementation of the CyberChallenge.IT initiative by UnivAQ, specifically focusing on the local structuring of the training phase. We place particular emphasis on analyzing registration data, specifically during the previous editions spanning from 2020 to 2023.

3 CyberChallenge.IT@UnivAQ

The University of L'Aquila is involved as a local venue of the CyberChallenge.IT project since 2020. As mentioned before, each edition of the CyberChallenge.IT involves a well-defined national-level structure wherein all local venues adhere to the same phases, with the exception of the Training Path phase. In fact, each venue has freedom over the organization and scheduling of topics and lessons. In particular, given the 12-week time frame, each venue is free to select the topics to teach from the list of main topics nationally selected. In this regard, the organising committee of the CyberChallenge.IT initiative at UnivAQ venue adopted the following criteria:

1. Avoid topics that teach little or no practical skills (in terms of CTF challenges);

2. Condense similar topics into one week to increase the number of different topics taught during the Training Path phase;
3. Simplify or partially omit complex topics to enable all participants to participate actively while providing additional material to more motivated and skilled participants who wish to learn more.

Following those criteria, UnivAQ selected the following topics, organized into distinct teaching modules:

- **Introduction** (6 h/1 module)—It aims to present the CyberChallenge.IT project and CTF competitions in general. In addition, it addresses legal and ethical considerations pertaining to cybersecurity, privacy, and data protection. Notably, it emphasizes the boundaries that a cyber-defender must never exceed.
- **Network Security** (12 h/4 modules)—It aims to present an in-depth overview of network security principles, protocols, and practices. In this way, participants acquire the skills necessary to design, implement, and maintain security measures for networked devices and data.
- **Web Security** (15 h/3 modules)—It aims to present the fundamental security principles specific to web applications. In this way, participants learn about concrete threats that target web applications and explore common attack vectors.
- **Software Security** (18 h/4 modules)—It aims to present principles, mechanisms, and tools for creating software resilient against vulnerabilities and threats. Additionally, it covers common causes of vulnerabilities and techniques to prevent and defend against them. This approach helps participants understand that poorly written software can be exploited for malicious purposes, causing it to behave in unintended ways.
- **Cryptography** (18 h/4 modules)—It aims to present the practices, protocols, mechanisms, and techniques used to secure information by transforming it into a format unintelligible to unauthorized individuals. Participants learn the basics of cryptography, essential for protecting data and communications from unauthorized access, interception, tampering, and forgery.
- **Other Topics** (3 h/1 module)—It aims to present other topics relevant to cybersecurity education.

For the Training Path phase, the organizing committee of CyberChallenge.IT at UnivAQ has implemented a 50% theoretical/practical ratio. This ensures that practical sessions are integrated into each theoretical lesson, enhancing the learning experience. Regarding the instructors for the Training Path phase, UnivAQ has consistently maintained a faculty of six permanent instructors, with additional instructors varying each edition. Over the years, UnivAQ has included a diverse group of CyberChallenge.IT instructors, including researchers, technicians, students, and former participants, all contributing their time and expertise voluntarily to enhance the CyberChallenge.IT experience at UnivAQ.

Finally, during the Training Path phase, UnivAQ hosts talks and seminars by companies involved in security research and product manufacturing, such

as Leonardo S.p.A., Cisco Systems Inc., and ZTE Corporation. These sessions provide participants with insights into the real-world application of their security skills and offer companies opportunities to identify and recruit talented individuals. Notably, many UnivAQ ex-students who participated in previous CyberChallenge.IT editions were contacted by these companies for job positions and internships.

3.1 Data Analysis and Synthesis of the Results

A frequency analysis was carried out to provide an overview of registration numbers in the latest editions (2020–2023) of the CyberChallenge.IT initiative at UnivAQ venue. From this overview, we can draw valuable insights, providing suggestions for improving participation in the program.

In detail, Table 2 presents the number of students who registered for CyberChallenge.IT program at UnivAQ venue over the last editions, i.e., from 2020 to 2023. All students summarised in Table 2 have an age range of 17–24.

Table 2. CyberChallenge.IT@UNIVAQ: participants data over last editions (2020–2023)

Year	Registered Participants												
	Total	Gender						Origin					
		Male		Female		Other		High School		University		Other	
	#	#	%	#	%	#	%	#	%	#	%	#	%
2020	137	111	81.0	26	19	–	–	69	50.3	67	49	1	0.7
2021	87	74	85.0	13	15.0	–	–	56	64.4	30	34.5	1	1.1
2022	78	69	88.5	8	10.2	1	1.3	55	70.5	23	29.5	–	–
2023	38	32	84.2	6	15.8	–	–	11	28.9	27	71.1	–	–

The frequency analysis performed on the data summarised inTable 2 shows three main results:

1. There has been a reduction in the number of students registering at the UNIVAQ venue over the years, with a particularly pronounced decrease in 2023 (as highlighted in the graph in Fig. 1). The decline in registration can be attributed to multiple factors, including the CyberChallenge.IT program's demanding nature. In contrast to the initial years, students are now acutely aware of the program's requirements, resulting in a more selective enrollment of motivated individuals. Additionally, unlike previous years when UnivAQ was the sole venue in the Abruzzo region, in 2023, D'Annunzio University of Chieti-Pescara joined the project. This university is located in an urban centre with better accessibility, utilizing both private and public transportation options.

2. The number of high school students, initially larger than that of university students (during the 2020 edition), is smaller than that of university students in the 2023 edition. A reversal of the trend has therefore occurred, as shown in the frequency plot graph in Fig. 2. This reversal of the trend in 2023 could be due to the fact that until 2022 a large part of the CyberChallenge.IT program was provided online due to restrictions related to the COVID-19 pandemic. On one side, the remote classes made it easier for high school students to follow CyberChallenge.IT Training Path phase after school time while avoiding transportation costs/time to reach the university site. On the other side, university students have lessons and courses all day and they have to follow university courses from home. This situation caused them to avoid making commitments to follow extra activities e.g., CyberChallenge.IT.
3. A considerable gap emerges between male and female students' involvement in CyberSecurity.IT program at UnivAQ venue, as shown in the frequency plot graph in Fig. 3. The low number of female students can be traced back to a well-known issue in STEM fields, where females tend to show less interest and engagement compared to their male counterparts in these specific areas, and cybersecurity is no exception [1,14,26]. In the realm of cybersecurity, despite its association with innovation and progress, enduring stereotypes have a profound impact, particularly on females [24,25]. Prevailing stereotypes, often reinforced by popular media, depict cybersecurity professionals as male *nerds* toiling away in dimly lit, isolated spaces such as windowless rooms or cramped cubicles. This portrayal perpetuates the perception that cybersecurity work is solitary, intense, and lacking in social interaction. Unfortunately, such stereotypes discourage women from pursuing careers in the field, contributing to their underrepresentation. When the prevailing image is one of male experts working in isolation, it becomes challenging for women to envision themselves thriving in such an environment. Consequently, fewer women choose to pursue cybersecurity careers, thereby perpetuating the gender gap.

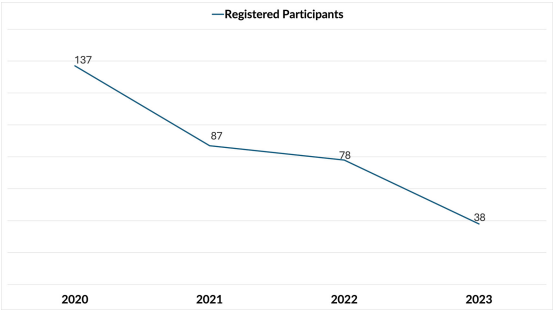

Fig. 1. Frequency plot graph that shows the registered participants at CyberChallenge.IT@UnivAQ during the last editions: 2020–2023

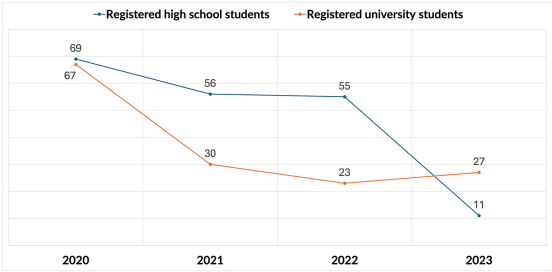

Fig. 2. Frequency plot graph that shows the High school and University students' registration over the last editions (2020–2023) of CyberChallenge.IT@UnivAQ

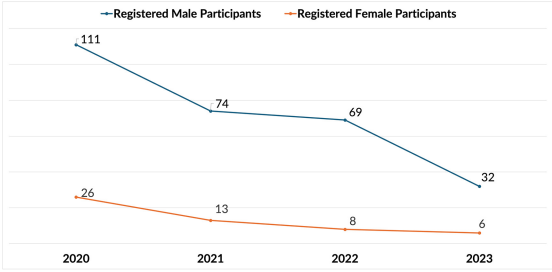

Fig. 3. Frequency plot graph that shows the gap between male and female registered students over the last editions (2020–2023) of CyberChallenge.IT@UnivAQ

3.2 Strategic Activities: A Results-Driven Proposal

Based on these results, the local organizing committee of CyberChallenge.IT@UnivAQ are considering strategic activities to motivate students to enrol in the program, thus ensuring the program's success in future editions.

Focusing on the identified results, to overcome the decline in registrations at the UnivAQ venue, we could consider enhancing the visibility and marketing of the CyberChallenge.IT program. For example, we could:

- Change the CyberChallenge.IT program at UnivAQ to highlight its unique features, success stories, and career prospects;
- Invest in targeted digital marketing campaigns, such as utilising social media, webinars, and online platforms to showcase program benefits, alumni achievements, and exciting projects;
- Improve the collaboration with local, national, and international industrial tech companies by organizing additional guest lectures, workshops, and internships;
- Integrate the CyberChallenge.IT@UnivAQ program with other similar initiatives to make UnivAQ a major reference point for cybersecurity training.

UnivAQ is participating in the industrial initiative CyberX:Mind4Future [15], led by the Italian Cyber4.0 consortium and Leonardo S.p.A., targeting univer-

sity students from eight universities, including UnivAQ. The first edition was launched in 2023, and a second edition is planned for Autumn 2024. This initiative provides training on advanced cybersecurity topics (e.g., cryptography, industrial network and system security, malware and binary analysis) followed by an in-site experience and lessons at Leonardo S.p.A. After the training phase, a final competition awards prizes and study grants to the best students. In the 2023 edition, UnivAQ students excelled, securing 6th and 2nd place [16].

Furthermore, we guess, the decline in the number of students (both high school and University) registered for the CyberChallenge.IT@UnivAQ program can also be countered through strategic activities with the primary objective of boosting student motivation. For example, to involve high school students, we could:

- Promote the high school students' engagement by collaborating with local high schools to raise awareness about CyberChallenge.IT program by conducting workshops, presentations, and career talks to showcase the program's benefits;
- Appoint enthusiastic ex-participants as ambassadors to visit high schools, share their experiences, and encourage registrations.

Likewise, to involve university students, we could:

- Engage university faculty to promote the program during regular classes, and the professors could highlight its relevance to coursework and career prospects;
- Share success stories of university students who excelled in CyberChallenge.IT program, highlighting their achievements and career growth.

Then, to address the gender imbalance, a potential approach is to partner with high schools to promote CyberChallenge.IT program by organizing workshops and guest lectures specifically tailored for young girls. These initiatives aim to spark interest in cybersecurity and demonstrate that it is a field open to everyone, dispelling the notion that it is exclusively for males. Additionally, involving accomplished female cybersecurity professionals to share their personal experiences can further inspire and motivate young girls to cybersecurity training paths. In this regard, UnivAQ is already moving and is participating in many parallel initiatives (e.g., [8]) to address the gender unbalance in future editions of CyberChallenge.IT.

Finally, as a general strategic initiative, we have planned to implement interviews and focus groups with participants of the ongoing CyberChallenge.IT at the UnivAQ venue. The primary objective of these data collection methods is to gather valuable feedback and insights to effectively refine and enhance CyberChallenge.IT@UnivAQ.

4 Conclusion

Nowadays, the increasing complexity and frequency of cyber threats in our digitally connected world require a greater effort in the qualified training of cybersecurity professionals. In the wake of national and European programs that are

currently trying to promote the qualified training of cybersecurity professionals, the CyberChallenge.IT project in the UnivAQ venue specifically targets young talents, cultivating a pipeline of future cybersecurity professionals.

In this paper, we provide an overview of high school and university students' enrollment at **CyberChallenge.IT@UnivAQ** in the last editions, i.e., from 2020 to 2023. Unfortunately, the preliminary analysis of the data showed worrying results characterized mainly by a considerable decline in the number of registered students (both high school and university) from 2020 to 2023. These results have a dual value: (i) they lead us to implement strategic activities to promote the program at UnivAQ and, similarly, to improve the motivation of students to participate (as discussed in the previous section); (ii) they make us reflect on the fact that in a future that will be even more digitally interconnected than the present one, it becomes a categorical imperative to implement campaigns to promote and recruit young students in order to incentivize them to train as cybersecurity professionals.

Overall, the present paper lays the first foundation for building a stronger cybersecurity engagement in the future that can incentivise organisations and educational institutions, at least at the national level, to strengthen their commitment to developing a competent workforce capable of dealing with the ever-changing cybersecurity landscape.

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Study on the Influence of Different Spatial Ability Students in AR Geography Teaching

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Abstract. In the geography class where the spatial concept is relatively abstract, the application of AR technology can improve students' learning performance and stimulate their learning interest. However, existing studies have not fully considered whether AR teaching will have different effects on students with different spatial abilities. Therefore, this study focuses on answering two questions: 1. What are the advantages of AR based geography teaching compared with the traditional teaching? 2. What is the impact of AR-based teaching on students with different spatial abilities? This study involved the design, development, and empirical testing of the AR education software 'Explore contour lines'. A total of 69 students from grade 7–8 in a middle school participated in this experiment and a two-group pre-and post-test design was adopted. The experimental group used AR technology-supported learning mode, while the control group used traditional teaching methods. It is found that AR based geography teaching can significantly improve student achievement and positively impact learning outcomes, and students with higher spatial ability are more effective in facilitating learning when using AR tools.

Keywords: Augmented Reality · Spatial ability · Geography learning

1 Introduction

With the development of multi-disciplinary integration and technology-supported teaching, augmented reality (AR) technology has gradually occupied a unique position in the field of education. As a technology that skillfully integrates virtual information with the real world, AR is increasingly widely used in teaching, its functions are constantly enhanced, and its application scope is gradually expanded to different disciplines and education stages [1–3] AR technology has three significant characteristics: virtual-real combination, real-time interaction and 3D perspective [3], which not only provides students with a richer and more intuitive learning experience, but also effectively enhances students' understanding and memory of knowledge. Especially in the geography, physics and other spatial concepts are more abstract courses, the application of AR technology is particularly important.

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Although the application of AR technology in teaching has been widely concerned, the research on the effect of AR teaching is still in the exploratory stage. The research on the influencing factors of AR teaching effect, especially the spatial ability of learners, still lacks sufficient practice test and research. Spatial ability, as an individual's key ability in understanding and manipulating spatial information, has a significant impact on the learning effect in AR teaching [4]. However, most of the existing researches focus on the theoretical discussion and technical implementation of AR technology, and there is still a lack of in-depth and systematic research on its practical application effect in specific subject teaching, especially for the differentiated impact of learners with different spatial abilities.

In this study, the content of "contours" in the geography course of junior high school is selected as the research object. As an important tool to describe terrain relief changes, the understanding and application of contours require students to have good spatial imagination and operation ability, which is highly correlated with spatial ability. Therefore, through empirical research, this study aims to explore the differences in the performance of students with different spatial ability levels in the learning of geography "contours" with the support of AR, including the changes in their geographical knowledge test scores, flow experience and cognitive load. Through comparative analysis of the learning data of students with different spatial abilities, we can further understand the relationship between spatial ability and AR teaching effect, and provide scientific basis for further optimization of AR teaching.

2 Literature Review

2.1 The Application of Augmented Reality Technology in Geography Teaching

In geography teaching, special attention is paid to the core elements such as man-land coordination, comprehensive thinking, regional cognition and geographical practice. In recent years, educators and researchers have tried to use AR experimental environment to solve the difficult problems in geography education. At present, the research of AR in geography education mainly focuses on investigating its different effects from traditional teaching, exploring its promoting effect on teaching efficiency, and its impact on students' cognitive load, spatial ability and emotional experience. Academic achievement, motivation, and classroom engagement are key variables for research in this area [5]. A number of studies have shown that geography teaching tools based on AR technology have significant advantages in improving students' learning performance and stimulating their learning interest. For example, Halik & Wielebski [6], through experimental research and user testing, found that geography applications based on AR technology can help students understand related concepts, principles and applications of volcanology more effectively than traditional teaching methods [7]. Adedokun-Shittu [7] from the University of Ilorin used augmented reality teaching tools (ARIT) to study the performance and knowledge retention of geography learners in Osun State, Nigeria, and the results showed that AR tools can significantly improve learners' academic performance and knowledge retention. Its advantages in geography education are significant and not affected by gender [8]. However, the role of existing AR devices in improving academic performance is still limited. Some studies have found that although AR application has

a positive effect on students' attitudes, it does not show a significant advantage in terms of performance improvement [9].

Although some studies have explored the application of AR in geography teaching and its influence on students' learning motivation and effect, it is still unclear how AR teaching affects students' cognitive load and flow experience. In addition, existing studies have not fully considered the impact of students' geospatial ability differences on AR learning outcomes. Future studies need to explore these issues in depth to more fully understand the role and potential advantages of AR in geography teaching.

2.2 Spatial Ability

Spatial ability, as an individual's ability to search the field of vision, analyze the shape, location and mental representation of objects, plays a fundamental role in scientific learning [10]. A number of studies have pointed out that spatial visualization ability is significantly positively correlated with the achievement of STEM subjects, but in-depth research on this ability in the field of education is still insufficient, which leads to some students facing challenges in science learning due to the lack of corresponding spatial ability. In particular, students are often troubled by understanding scientific concepts that require three-dimensional thinking [11].

Although the traditional teaching environment has limitations in cultivating students' spatial ability, the introduction of computer technology provides a new way to enhance the spatial visualization ability. In recent years, AR technology has attracted a lot of attention as one of the emerging technologies and has shown positive impact on spatial visualization capabilities [11] Some studies have found that AR technology can significantly improve students' spatial ability and have a positive impact on learning results [12]. Moreover, there is evidence that the spatial ability level of students also has a significant impact on the effect of AR application, and learners with high spatial ability are usually more efficient in using AR tools to promote learning [4]. However, research by Elinda Ai-Lim Lee et al. provides a different perspective, and they found that students with weak spatial ability improve their academic performance more significantly in virtual learning environments [8].

Meanwhile, there is a certain relationship between spatial ability and cognitive load, that is, it is usually negatively correlated with cognitive load [13]. In addition, the level of spatial ability is closely related to learning motivation and academic performance [12]. It is worth noting that some studies have confirmed that AR technology can make it easier for students to achieve the state of flow experience, but the current in-depth research on the relationship between flow experience and spatial ability is insufficient. Based on students' different spatial ability levels, this study aims to deeply explore the relationship between AR teaching and students' cognitive load, flow experience and academic performance, and explore the impact of AR technology on students' learning results.

2.3 Research Questions

To sum up, AR technology has great potential in promoting geography learning because it can make abstract things concrete. Spatial ability also has an important impact on

students' learning effect and flow experience. However, few studies have focused on whether AR based instruction has different effects on students with different spatial abilities. Therefore, this study attempts to answer the following two questions:

(1) What are the advantages of AR-based geography teaching compared with the traditional teaching?
(2) What impact does AR-based teaching have on students with different spatial abilities?

3 Method

3.1 Participants

A total of 69 students aged 13 to 14 participated in this experiment, all of whom were from a rural middle school in Yunnan, China, and had never been exposed to the knowledge of "contour lines". In order to further explore the practical effects of AR technology in geography teaching, this study adopted a two-group pre - and post-test experimental design, in which the experimental group (EG) included 33 students and the control group (CG) 36 students. After the intervention, a total of 30 valid data were collected in the experimental group and 35 valid data were collected in the control group.

3.2 Research Design

This study adopts quasi-experimental research method. In this study, 69 beginning students were divided into two groups. The experimental group (Class A, $n = 33$) used AR education software for teaching activities, that is, the teacher provided Huawei tablets to the students in the appropriate exploration process, and a group of three people operated a tablet to conduct cooperative exploration together. In the process of exploring the concept of contour lines, students scan the marker, observe the three-dimensional mountain, and draw contour lines every 100 m on the learning sheet; In the process of exploring the contour lines of different parts, students continued to scan the corresponding cards with flat plates, observe the contour characteristics of different parts of the mountain, and fill in the learning sheet according to the observed situation. The control group (Class B, $n = 36$) adopted the traditional form of learning, that is, teachers explained the definition, characteristics, drawing methods and practical applications of contour lines through slide presentation and blackboard writing. Students then consolidate what they have learned by listening in class, working in groups to observe maps, and completing exercises.

The class situation is shown in Fig. 1. The control group (Class B, $n = 36$) was taught in the traditional form. The experimental group and the control group were taught by the same teacher with rich teaching experience. Except for the intervention of AR interactive equipment, the course tasks and contents were completely consistent. After the experiment, we conducted a post-test on the students and collected the geographical knowledge test, cognitive load questionnaire and flow experience questionnaire of the students to measure the influence of learning style supported by AR on the academic performance, cognitive burden and learning experience of the middle school students. The specific experimental process is shown in Fig. 2.

Fig. 1. Class situation

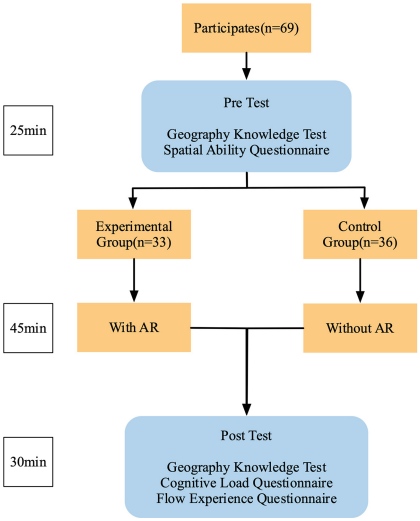

Fig. 2. Experiment procedure

3.3 Teaching Material

According to the actual needs of teachers and literature research, this study finds that "contour lines" is an important and difficult subject in junior middle school geography, and it is also a concept often misunderstood by students [14]. Therefore, this study developed the AR case of " Explore contour lines" with the theme of "contour lines", which mainly includes the following knowledge points: 1. The basic concept of contour lines; 2. Contour lines can judge the steepness of peaks, ridges, valleys, cliffs, saddles and slopes on topographic maps; 3. Contour maps can be used for site selection of different building lands.

The AR software is developed on Windows10 platform using Unity3D and Vuforia. The development process involves creating tags, uploading them to the Vuforia database to generate database files, which are then imported into Unity3D along with 3D models. Interactive features are programmed in C# to increase user engagement with the model. The final software is packaged into an APK file using the Android SDK, which can be installed and run on Android devices. After launching the software, students use the device's camera to scan the marker, triggering the display of the corresponding 3D model. The software is divided into two scenarios. The corresponding marker is placed

in the camera's field of view. When mountain appears on the screen, you can click the button below the flat screen by hand, click "Show contour lines" to control whether there are contour lines or not, click "contour surfaces" to control whether there are contour surfaces or not, click "layered colored terrain" button to control the three-dimensional mountain displayed on the layered colored topographic map, and click the triangle on the left side of the screen to display 100–800 m of different buttons, click each button can control the display of each level of elevation plane.

Click the triangle on the right side of the screen to display the buttons on different parts of the mountain. Click on different buttons such as "ridge" and "valley" to display the specific position of the summit, valley, ridge, saddle and cliff on the three-dimensional mountain. Click the arrow in the upper right corner to enter the next scene. Place the corresponding markers in the field of view of the camera, and the screen will display the three-dimensional terrain of the marker, as shown in Fig. 3. In this way, the selected places are arranged for reservoirs, factories, residential areas and other buildings, so that students can practice the application of contour lines in life. If the matching is wrong, the screen will show wrong feedback, give tips, and guide to and summarize the application law of contour lines in life.

Fig. 3. Game screen

3.4 Study Tools

- Geographical knowledge test

The geographical knowledge test questions of this experiment come from the textbook after class, which were divided into pre-test and post-test. The content of the questions in the pre-test and post-test is different, but the type and amount of questions are the same, there are choice questions, judgment questions and fill in the blank questions, a total of 14 questions.

- Questionnaire

The questionnaire involved learners' spatial ability level, flow experience assessment and cognitive load. The Spatial Ability questionnaire, from Purdue University Spatial Test [15], is a 21-question questionnaire designed to see how well students can imagine rotating three-dimensional objects. The flow experience questionnaire is from Pearce [12], with a total of 8 questions. The cognitive load questionnaire comes from the cognitive load questionnaire of Jimmie [16]. It is divided into three dimensions, internal load, external load and closely related load, with a total of 10 questions. After a number of rigorous research tests, the above questionnaires have good reliability and validity.

4 Data Analysis and Findings

4.1 Covariance Analysis of Learning Achievement

In this study, we first conducted a covariance analysis of the academic performance of the experimental group and the control group, using the geographical pre-test scores as the covariable. The basic hypotheses were tested before covariance analysis. The homogeneity test of variance met the homogeneity of variance ($F = 0.606$, $p > 0.05$), and the interaction effect was not significant ($F = 0.018$, $p > 0.05$). The data were suitable for covariate analysis.

Table 1. ANCOVA of posttest geography scores with pretest scores as a covariate

Source	Type III Sum of Squares	df	Mean Square	*F*	Sig.	Partial η^2
Corrected Model	10649.685a	2	5324.842	22.464	0.000*	0.42
Intercept	6301.433	1	6301.433	26.584	0.000*	0.3
Geography Pre-Test Total	8778.769	1	8778.769	37.035	0.000*	0.374
Group	2222.266	1	2222.266	9.375	0.003*	0.131
Error	14696.469	62	237.04			
Total	247500	65				
Corrected Total	25346.154	64				

[a] $R^2 = 0.420$ (Adjusted $R^2 = 0.401$).

The results (as shown in Table 1) of covariance analysis showed that the experimental group (Mean = 63.43, SD = 3.046) had a significant advantage in geographical posttest scores compared with the control group (Mean = 52.67 SD = 3.781) ($F = 9.375$, $p < 0.05$). This indicates that AR based geography teaching can effectively improve students' academic performance.

4.2 The Differentiated Impact of AR on Students with Different Spatial Abilities

In order to further explore whether AR teaching has different effects on different spatial abilities, students in the experimental group were divided into high spatial ability group (EH) and low spatial ability group (EL) according to spatial abilities. There are many ways to distinguish high and low ability students. The median score [17] and the average score [18] are the two common dividing points. In this study, we choose the average of students' spatial ability test pretest score as the cut-off point. Students whose score is less than or equal to 6.2 will be marked as low spatial ability students, and students whose pre-test score is higher than 6.2 will be marked as high spatial ability students. Through the independent sample t test, we compared the achievement improvement of students with different spatial abilities in the experimental group. The results showed that the difference in geography post-test scores of the high spatial ability group was more significant ($F = 17.313$, $p < 0.01$), indicating that the improvement of high spatial ability students in AR teaching was more significant. This result further validates the importance of spatial ability in AR teaching (as shown in Table 2).

Table 2. T-test of Pre- and Post-Test Geography Scores for Experimental Group Learners with High and Low Spatial Abilities

Variable	EH (n = 15)		EL (n = 15)		*F*	Sig.
	M	SD	M	SD		
Pretest	57.85	21.18	53.75	18.90	.249	.621
Posttest	73.92	10.59	56.50	19.20	8.553**	.006

*$p < .05$ **$p < .01$.

4.3 Comparison of Flow Experience and Cognitive Load

Finally, we compared the differences in flow experience and cognitive load between the experimental and control groups using the independent sample T-test. The results showed that there was no significant difference in flow experience between the two groups ($F = 0.316$, $p > 0.05$). However, in terms of cognitive load, the mean value of the experimental group was significantly lower than that of the control group ($F = 7.430$, $p < 0.01$), indicating that AR-based geography teaching can reduce the cognitive load of students and provide a more relaxed and pleasant learning experience (as shown in Table 3).

5 Conclusion and Discussion

The purpose of this study is to deeply explore the impact of AR based geography teaching on students with different spatial abilities. Through the design and development of the AR education software "Explore contour lines" and an empirical study of 65 students in grades 7–8, we aim to reveal the advantages of AR technology in geography teaching and its differentiated impact on students with different spatial abilities.

Table 3. T-test of Flow Experience and Cognitive Load between Experimental and Control Groups

Variable	EG (n = 30)		CG (n = 35)		*F*	Sig.
	M	SD	M	SD		
Flow	31.37	5.006	28.50	5.507	.316	.576
Cognitive Load	28.50	9.508	37.93	14.61	7.430**	.008

*$p < .05$ **$p < .01$.

First of all, from the experimental results, AR based geography teaching has significant advantages compared with the traditional teaching. This is mainly reflected in the improvement of students' academic performance and the reduction of cognitive load. AR technology makes abstract geographical concepts intuitive and easy to understand by providing a three-dimensional, interactive learning environment. Students can deepen their understanding of geography knowledge such as contour lines through hands-on operation and exploration, thus significantly improving learning results and reducing cognitive load.

Secondly, this study found that students with higher spatial ability were more effective in promoting learning when using AR tools. This may be because these students have stronger spatial imagination and operation ability, and can make better use of the three-dimensional perspective and real-time interaction function provided by AR technology, so as to achieve better learning results.

However, there are some limitations in this study. First, the sample size is relatively small and may not fully reflect the actual situation of students with different spatial abilities. Future studies can expand the sample size to improve the reliability and universality of the study. Secondly, this study mainly focuses on the short-term effect of AR technology in geography teaching, and the investigation of long-term impact is still insufficient. Future research could track changes in students' learning outcomes and spatial abilities to more fully assess the application value of AR technology in education.

Based on the findings of this study, educators can consider the following suggestions to optimize the use of AR technology in geography teaching:

1. AR teaching activities are designed according to the individual differences of students, and appropriate learning materials are provided for students with different spatial abilities to meet their learning needs [19].
2. Using the advantages of AR technology, the abstract geographical concepts are visualized to help students better understand and remember, while stimulating their interest in learning and desire to explore.
3. Pay attention to students' cognitive load, reduce students' learning pressure and improve learning efficiency by providing intuitive and interactive learning experiences.
4. Continuously evaluate and reflect on the application effect of AR technology in teaching, combine the feedback and learning results of students, and constantly adjust and improve teaching strategies and methods.

In short, AR based geography teaching brings new possibilities to the traditional education teaching, which helps to improve the learning effect of students and reduce the cognitive load. Through in-depth research and practical exploration, we can further develop the potential of AR technology and contribute more to the development of education.

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Advancing Occupational Safety and Health Training: A Safety-II Integration of the ADDIE Model for Virtual Reality

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Abstract. This paper examines the enhancement of occupational safety and health (OSH) training in manufacturing through the integration of Safety-II principles within a Virtual Reality (VR) training framework, applying the Analysis, Design, Development, Implementation, and Evaluation (ADDIE) model. Amidst the digital transformation in manufacturing, innovative training methods such as VR have become instrumental in improving operational safety and efficiency. However, the incorporation of Safety-II, a philosophy emphasizing the complexity of organizational systems and the need for resilience, has not been systematically applied to VR training designs. Through a series of focus group sessions, this study presents a new method for creating VR training materials. Designed to foster systemic thinking, resilience, proactivity, learning, flexibility, and leverage human performance variability, these courses are in line with Safety-II's transition from merely preventing negative outcomes to understanding and increasing positive capacities. The results lead to a newintegration of the ADDIE model within a Safety-II framework for VR-based OSH training, enhanced by Skill-Rules-Knowledge (SRK) specific guiding principles.

Keywords: VR training · ADDIE · Safety-II · SRK · occupational safety and health

1 Introduction

Manufacturing organizations are using cutting-edge digital solutions to improve their operational effectiveness and efficiency as they relentlessly pursue digital transformation. Here, digital solutions are being integrated into several areas, most notably Occupational Safety and Health (OSH) training. Virtual reality (VR), e-learning platforms,

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and gamification stand out among the many cutting-edge digital training techniques for OSH in manufacturing for their increasing popularity and efficacy [1–3]. Despite the existence of studies on the use of VR for OSH training, there is a lack of a systematic approach to designing these interventions with consideration for the principles of Safety-II [4]. Safety-II's philosophy, which views organizations as complex systems, suggests a review of the traditional approach to enhancing VR training. Using the Analysis, Design, Development, Implementation and Evaluation (ADDIE) model, this paper explores the integration of Safety-II principles within VR training framework to improve OSH manufacturing. The approach can be easily extended to other industries which are showing interest in the application of VR-based training solutions, e.g. construction and building sectors, mining, and healthcare [5, 6].

This paper, in Sect. 2, presents the research methods made of a four-step process. Sections 2.1 and 2.2 introduce the first two steps, Sect. 2.3 reveals the results from the third step, and Sect. 2.4 outlines the validation plan activities planned for the upcoming months. Section 3 shares insights and reflections from the research team on their process and findings.

2 Methods

The research methodology is a structured and iterative process centered around the utilization of a focus group. This approach is inherently exploratory and iterative in nature, aiming to harness the collective insights and expertise of the group to define, analyze, and validate the research objectives and findings [7]. The method unfolds through a series of planned steps, each designed to progressively refine and enhance the research's scope, methodology, and outcomes.

The preliminary step involves the creation of the focus group, selected to include a diverse range of expertise and perspectives relevant to the research domain. This foundational stage is essential for setting the mode and direction of the subsequent inquiry [8]. The first meeting establishes the research's objectives and scope, and it sets the collaborative framework, specifying the tools for communication and coordination. This phase ensures that all members are aligned on the goals and understand the research's boundaries.

In the second meeting, the reference models to be adopted are selected. This involves a critical examination of existing theories and frameworks to identify those most applicable.

The third meeting is dedicated to a collective review of the analysis made by every participant of the focus group between the second and third meetings. This session allows for the preliminary exploratory analyses to be shared, discussed, and critiqued, fostering a consensus among the group's members who come from varied scientific backgrounds. This step integrates diverse perspectives and it ensures that the analysis is comprehensive and robust.

Finally, the fourth meeting focuses on outlining a validation plan. This phase involves determining the criteria and methods for assessing the reliability and validity of the research findings, ensuring that the conclusions drawn are reliable and can withstand scrutiny. So, the research method is a dynamic and collaborative process that leverages

the collective intelligence of the focus group to explore complex research questions [9]. It facilitates a deep dive into the subject matter, encouraging the synthesis of ideas and the convergence of diverse scientific viewpoints. This approach enriches the research process and enhances the validity and applicability of the findings.

2.1 Meeting I – Definitions and Workplan

A harmonisation of the definitions of the research aspects was carried out by the focus group. The participants' varied backgrounds is a well-known prerequisite for focus groups to be effective. The focus group consisted of 10 participants with diverse backgrounds, as detailed in Table 1. The group included experts from academic fields, spanning multiple scientific research topics, as well as researchers from a public institution specializing in OSH. This scientifically oriented composition aligns with the study's methodology assessment objective. However, all participants also collaborate with multiple organizations that will be involved in the validation phase. The research provides insights into the application of VR for OSH in manufacturing, focusing on Safety-II principles. To achieve this, it is necessary to know about VR, industrial processes, smart factories, training, and education, as well as OSH.

Table 1. Focus group composition.

Background	Main area of expertise	Main focus area in the research	Years of experience
Public research	Training and education	Occupational safety and health	25
Public research	Innovation and communication	Occupational safety and health	16
Public research	Psychology	Occupational safety and health	28
Public research	Legal	Occupational safety and health	18
Academic/Industrial	Operations management	Manufacturing systems	20
Academic	Smart factories	Manufacturing systems	5
Academic	Advanced industrial applications	Manufacturing systems	3
Academic	Telecommunications	Virtual Reality	20
Academic	Digital signal and image processing	Virtual Reality	3
Academic	Immersive multimedia	Virtual Reality	3

The focus group worked on defining the requirements of VR, OSH, and Safety II. VR refers to a computer-generated environment that simulates three-dimensional spaces, typically experienced through specialized electronic devices such as headsets, aiming to provide high levels of immersion to the users [10]. Safety-II [11] is a contemporary approach to safety management that shifts the focus from preventing accidents and incidents by reducing things that go wrong (as emphasized in Safety-I) to enhancing the capacity of an organization to succeed under varying conditions, thereby ensuring high levels of safety performance. Here's an outline and brief explanation of the core principles of Safety-II, as these are used in the following research steps.

1. **Systems thinking**. Safety-II advocates for understanding the complexities of systems and their interdependencies. It emphasizes that safety emerges from the interactions within the system rather than from the components themselves. Systems thinking encourages looking at the big picture and understanding how changes in one part of the system can affect the whole [12].
2. **Resilience engineering**. This principle focuses on the system's ability to function effectively during expected and unexpected conditions. Resilience engineering aims to enhance the system's capacity to adapt, learn, and bounce back from failures or unexpected events, thereby maintaining safety [13].
3. **Proactivity**. Safety-II requires a proactive stance towards safety management, as opposed to the reactive nature of Safety-I. It involves anticipating potential issues and variations and preparing for them before they lead to failures or accidents [14].
4. **Learning and adaptability.** A core tenet of Safety-II is continuous learning from both what goes wrong and what goes right. It encourages organizations to adapt their safety practices based on lessons learned from day-to-day operations and from the industry at large [15].
5. **Human performance variability as an asset**. Unlike Safety-I, which often views human variability as a source of errors, Safety-II sees it as a valuable asset. It recognizes that human adaptability and variability in performance can enhance system resilience and safety under varying conditions [16].

2.2 Meeting II - Reference Models

The focus group considered a variety of models, to select the most appropriate considering the research's objectives. The group aligned on the opportunity to consider the ADDIE model for training [17]. The motivation relies on the necessity for a systematic approach to develop a VR solution for OSH that is both flexible and rigorous. The ADDIE model, with its structured phases of Analysis, Design, Development, Implementation, and Evaluation, offers an ideal framework for creating training programs effective and adaptable to the evolving needs of learners and the dynamic nature of workplace safety. The steps of the ADDIE model are the following.

Analysis. It involves an assessment of learners' needs, learning objectives, and the context in which the training will be applied. For the VR solution, this means understanding the specific health and safety challenges faced by the target audience, identifying the skills and knowledge required to address these challenges, and recognizing the unique affordances of VR as a training medium. This step ensures the training program is grounded in real-world needs and leverages VR technology to its fullest potential.

Design. Building on the insights gained from the Analysis phase, the Design phase focuses on planning the instructional strategy, content, and interaction design of the VR training program. This includes outlining the training scenarios that will be developed, the learning objectives for each scenario, and the instructional methods that will be employed. By integrating principles from educational theory and cognitive psychology, this phase aims to create a blueprint for a VR training experience that is engaging, immersive, and pedagogically sound.

Development. With a detailed design, the Development phase involves the actual creation of the VR training materials. This includes programming the VR environments, developing the interactive elements, and integrating the content into a cohesive training experience. Here, iterative testing with end-users and subject matter experts is crucial to ensure that the VR solution is intuitive, effective, and aligned with the learning objectives.

Implementation. The Implementation phase addresses the delivery of the VR training program to the intended audience. This involves preparing the learners, facilitating the training sessions, and providing the necessary support to ensure a smooth and effective learning experience. For VR training, this might also include technical setup and orientation to familiarize learners with the VR equipment and interfaces.

Evaluation. The final phase of the model focuses on assessing the effectiveness of the training program, gathering feedback from learners, measuring learning outcomes, and evaluating the impact of the training on workplace safety practices. The insights gained from this phase are invaluable for refining the VR training solution and informing future training initiatives.

In conclusion, the ADDIE model provides a robust framework for the design and development of a VR solution for OSH training. Its systematic, iterative approach ensures that the training program is tailored to the needs of learners and to the goals of the organization, resulting in an engaging, effective, and impactful learning experience. Few applications of ADDIE with VR are available [18, 19], with no specific work on how to integrate Safety-II principles in it.

The focus group identified a further model as a reference for the research, i.e., the Rasmussen's Skills, Rules, and Knowledge (SRK) framework [20]. In the traditional management of OSH, the prevention of accidents relies on the identification and mitigation of risks, a perspective that overlooks the complexity of human behavior and the adaptive capacities that individuals and organizations employ to navigate and manage risks in dynamic environments. In response, the Safety-II philosophy emerges as a complementary paradigm, emphasizing the importance of understanding and enhancing the capabilities that enable successful performance under varying conditions. The focus group identified the SRK framework as a reference to consider the human capabilities cited above. Indeed, SRK categorizes human behavior into three distinct levels: skill-based, rule-based, and knowledge-based. Each level represents a different level of cognitive processing and decision-making, from automatic responses and heuristic shortcuts to analytical reasoning and problem-solving. More in details the three levels are the following.

1. **Skill-based level**: This is the most automatic level of performance, requiring minimal conscious thought. It involves well-practiced routines or actions performed

almost automatically, such as typing on a keyboard or driving a familiar route. Skills are developed through repetitive practice and are executed without much conscious awareness.
2. **Rule-based level**: At this level, individuals apply rules or procedures that they have learned to handle situations that are somewhat familiar but not automatic. This involves if-then type reasoning where specific situations trigger predefined responses or actions. Rule-based behavior is more conscious than skill-based behavior and is often employed in situations that are routine but require the application of learned rules to navigate.
3. **Knowledge-based level**: This is the highest and most complex level of cognitive performance, engaged when individuals face new or unfamiliar situations that cannot be addressed by existing skills or rules. At this level, problem-solving and decision-making are based on explicit reasoning, analysis, and the application of theoretical knowledge. Individuals must rely on their understanding and knowledge to formulate a response to novel situations.

By integrating the SRK framework into VR solution, the training helps to set and reach learning objectives on every of the three distinct modes. To clarify, the VR solution can set a situation where the learner spontaneously activates responding actions (skill), the trainee acts with safety procedures (rule), the learner creates novel solution to the safety issue (knowledge). Thus, the application of the SRK framework is a novel strategy for augmenting VR-based training initiatives from a Safety-II perspective. SRK was rarely used to design VR training [21], even if it is a proven categorization useful to design training in complex environments. To the authors' knowledge there is no application of SRK framework and ADDIE model, despite the evidence of the need to consider skills, rules, and knowledge capabilities of trainee in all the ADDIE steps of a training.

2.3 Meeting III - Design by Consensus

The consensus process unfolded in two phases. Initially, participants shared their insights and findings, presenting how each step of the ADDIE model could be enhanced or reinterpreted through the lens of Safety-II principles. This stage was characterized by open dialogue, analytical thinking, and cooperative interpretation. Following this, the group engaged in a more focused discussion, aiming to reconcile different positions and forge a unified, interdisciplinary perspective. The conclusion of this process was a consensus-driven Table 2 that mapped the synergies between the ADDIE model and Safety-II principles, reflecting a shared understanding that was comprehensive and nuanced. This Table 2 is as a valuable tool for guiding future educational design projects with an emphasis on resilience and adaptability.

A similar consensus-building procedure investigates how SRK affected each stage of the ADDIE model, leading to a cooperative comprehension of the ways in which these cognitive aspects may improve instructional design (Table 3).

Table 2. ADDIE evolution in VR with Safety-II.

ADDIE step	Systems thinking	Resilience engineering	Proactivity	Learning and adaptability	Human performance variability as an asset
ANALYZE	Identify instructional goals considering the coexistence of human and technologies, such as the complex interrelations within work environments, people, and systems to simulate these in VR	Analyze resilience needs by identifying unusual scenarios and critical incidents that workers have faced, analyzing positive and negative elements to be put in VR training	Identify specific instructional goals in VR training to increase the capacity of recognize paths to OSH incident/accident/risks in work environments and encourage proactive risk management	Assess the adaptability of the workforce to VR training tools and methods, identifying learning styles and needs. OSH objectives should be experienced in VR in a iterative process, to address an evolutionary learning process	Recognize individual differences in perceiving and reacting to OSH risks, tailoring VR scenarios to accommodate diverse learning curves. Integrate work-as-prescribed into instructional goals, but analyze work-as-done and permit it in the VR environment
DESIGN	Design VR scenarios that mirror the complex dynamics of real-world systems, encouraging learners to think systemically about OSH	Design VR scenarios that challenge learners to maintain operational effectiveness under adverse conditions, fostering resilience	Design proactive safety management scenarios in VR, such as identifying early warning signs of hazards	Design VR scenarios with adjustable difficulty levels and branching paths to adapt to learner feedback and progression	Utilize VR to create diverse scenarios that reflect the variability in human responses to safety challenges, encouraging innovative solutions
DEVELOP	Develop VR content that realistically simulates system interdependencies and their impact on safety, stimulating different human's sensory aspects by using realistic feedback and physics engines	Develop VR modules that simulate stressful conditions, requiring learners to implement non-prescribed responses by practicing resilience strategies such as emergency response and adaptation	Develop VR training modules that allow learners to explore and intervene in unfolding scenarios that could lead to safety incidents, fostering a proactive mindset	Incorporate real-time feedback and adaptability into VR training, allowing learners to experiment with different approaches and learn from their actions	Develop VR training that capitalizes on human performance variability, offering scenarios where diverse strategies can lead to successful outcomes

(*continued*)

Table 2. (*continued*)

ADDIE step	Systems thinking	Resilience engineering	Proactivity	Learning and adaptability	Human performance variability as an asset
IMPLEMENT	Implement VR training sessions that emphasize understanding and managing complex systems for health and safety, using interactive and immersive scenarios. Debrief session should emphasize how elements apart simultaneously contributed to the events experienced in VR	Implement VR training that allows for repeated practice in dealing with unexpected events, building psychological resilience	Implement VR scenarios that enable learners to actively identify, assess, and mitigate risks, emphasizing the importance of proactive health and safety measures	Implement adaptive VR training sessions that can evolve based on learner progress, feedback, and emerging health and safety challenges. Iterative implementation of training is expected, to stimulate learning by doing in VR	Implement VR training that values diverse problem-solving approaches, encouraging learners to share and reflect on different strategies for safety management. Provide debrief sessions to underline any positive behaviors that contributed positively to OSH
EVALUATE	Evaluate the effectiveness of VR training in improving systemic health and safety awareness and understanding among learners. Evaluate the capacity to consider connection among activities, controls, input, output, resources, etc	Assess how well VR training prepares individuals to respond non-prescribed events, adapt and recover from safety incidents, focusing on resilience outcomes. Assess problem solving capacity, critical thinking, collaboration and communication for health and safety through measures in the VR environment	Evaluate the impact of VR training on learners' ability to anticipate safety issues and take preventive actions. Evaluate the ability to identify risks, to anticipate increasing risks, and take actions to prevent future health and safety problems	Evaluate the adaptability of the VR training program, including how well it meets learning objectives and accommodates learner feedback for continuous improvement. Assess VR by considering sickness, immersivity, usability, etc	Assess how effectively the VR training utilized and valued human performance variability, including the range of successful strategies employed by learners. Assesses the understanding of how alternative strategies or unexpected contexts impact OSH

2.4 Meeting IV - Validation Plan

The focus group embarked on defining a validation plan for the VR training program, with an emphasis on integrating the ADDIE model and the insights from Tables 2 and 3 to ensure a comprehensive evaluation of the training content and its delivery. The validation plan commenced with the identification of a specific risk scenario that embodies critical OSH challenges within a manufacturing environment. The chosen scenario is a maintenance process where lock-out tag-out (LOTO) procedures should ensure OSH.

Table 3. Ta ADDIE evolution in VR with SRK.

ADDIE Step	Implementation through SRK framework
ANALYZE	**Skill:** Identify specific physical skills and reflexes required in health and safety scenarios. Assess VR's capability to simulate these accurately. Analyze the trainees (competences, roles, experience, etc.) to identify the specific skills **Rule:** Determine the set of OSH regulations and procedures pertinent to the target audience. Analyze gaps in current training regarding rule application **Knowledge:** Understand the theoretical underpinnings of OSH practices. Identify knowledge gaps and misconceptions in the target audience. Define acceptable misalignments from rules
DESIGN	**Skill:** Plan VR scenarios that enable hands-on practice of identified skills in a controlled, simulated environment, enhancing muscle memory and reaction times. Design risky situations with no specific rules is available because it is expected that trainees do not need it **Rule:** Design interactive modules that guide learners through decision-making processes based on OSH procedures. Use branching scenarios to illustrate the consequences of different choices **Knowledge:** Structure content to build unexpected situations in VR, where complexity emerges, and no rules directly apply. Insert strong elements of variability, e.g. missing tools, environment that diverge from usual, unexpected behavior of people, etc
DEVELOP	**Skill:** Create immersive, realistic VR simulations that require learners to perform tasks or respond to situations as they would in real life. Incorporate feedback mechanisms for skill refinement **Rule:** Develop VR experiences that prompt learners to apply rules in varied contexts, offering immediate feedback to reinforce correct procedures and correct mistakes **Knowledge:** Integrate interactive lessons that allow exploration and application of OSH knowledge within the VR environment, encouraging active learning. The scenarios present many unforeseen variables which the learner can manage by integrating skills and procedures acquired in the basic scenarios
IMPLEMENT	**Skill:** Roll out VR training sessions focused on skill practice, ensuring accessibility and ease of use for all learners. Monitor and support learners in using VR to practice specific skills **Rule:** Implement structured VR training pathways that sequentially introduce rules and their application in increasing complexity, ensuring understanding and retention **Knowledge:** Facilitate VR-based workshops or sessions that engage learners in applying theoretical knowledge in simulated real-world scenarios, promoting knowledge transfer. Also, encourage learners to integrate the management of unforeseen variables and situations with skills and procedures acquired in the basic scenarios
EVALUATE	**Skill:** Assess learner performance in VR simulations to gauge skill acquisition and improvement. Use metrics like speed, accuracy, and decision-making time **Rule:** Evaluate the ability of learners to correctly apply rules in varied scenarios within VR, using quizzes and decision-making simulations **Knowledge:** Measure the depth of understanding and retention of OSH principles through pre- and post-tests, scenario analyses, and applied problem-solving in VR

The group emphasized the need to understand how individuals interact within the system and how these interactions could lead to safety incidents. It has been created a VR simulation that accurately reflects these dynamics, offering learners the opportunity to navigate the scenario with an emphasis on resilience, adaptability, and proactive risk

management. Unexpected situations will be in the VR environment, e.g. missing tools, and time shorter than expected to conduct all the activities. Following the development of the VR simulation, the validation plan includes a testing phase. Here, participants will be assessed on their comprehension of the OSH protocols illustrated within the VR scenario and also on their overall experience and usability of the VR training tool. The usability is defined as the perceptions of the tool's user-friendliness and engagement, and it is commonly assessed by System Usability Scale (SUS) and Technology Acceptance Model (TAM) [22]. Moreover, when employing a VR solution, it is also necessary to balance theoretical knowledge and practical application opportunities, assessing the so-called Quality of Experience (QoE) by the Absolute Category Ranking (ACR) or Group Environment Questionnaire (GEQ) [23].

This approach ensures that the training program is effective in conveying critical information but also engaging and accessible for users, facilitating a deeper understanding and retention of the safety procedures. The evaluation will utilize a combination of quantitative and qualitative methods to gather comprehensive feedback, enabling further refinement of the VR training program.

3 Conclusions and Future Research

This paper presents a comprehensive exploration into the design, development, and preliminary validation of a VR training program for OSH in the manufacturing sector, grounded in the principles of Safety-II and the ADDIE model, and enriched by the SRK framework. The approach can be easily extended to other industries which are showing interest in the application of VR-based training solutions, simply modifying the training contents, and exploiting the remainder of the method's structure.

The validation plan resulting from an iterative focus group-driven approach can improve learning outcomes and also has positive implications for practitioners in the field of VR training. By embedding systemic thinking, resilience, proactivity, learning and adaptability, and human performance variability into the fabric of VR training scenarios, we anticipate a paradigm shift in how OSH training is conceived and delivered. For practitioners, our work offers a blueprint for designing VR training programs that are not only immersive and engaging but also deeply rooted in contemporary safety management philosophies.

The implementation of the experiment for validation is a critical next step. Future research will focus on developing a VR tool for a real industrial scenario, following the ADDIE-VR model proposed. Currently, the team is working on a VR tool for maintenance procedures, specifically the LOTO scenario. According to EU-OSHA, most workplace accidents occur during maintenance activities, and LOTO procedures are often underestimated by operators. The team is developing multiple storytelling approaches: one that covers general skills and procedures, and another that incorporates elements of variability to teach resilience engineering principles and manage unexpected situations. Future activities will also focus on refining the VR training modules based on feedback and performance, exploring scalability options for wider deployment, and investigating the integration of emerging technologies to enhance the realism and effectiveness of VR training in the OSH field.

After testing the focus group approach, some methodology's strengths and weaknesses emerged, such as the possibility for participants to develop ideas jointly, bringing forward their own priorities and standpoints, which at the same time may led to the tendency for socially acceptable opinion to emerge, and for certain participants to dominate the process. Nevertheless, the approach was generally beneficial since the range of argumentative behaviours exhibited by participants results in a depth of dialogue not often found in individual interviews.

The proposed contribution highlights the need to bridge the gap between advanced VR capabilities and the practical requirements for effective and immersive training programs. To this end, many aspects must be considered simultaneously from the early design stages, such as identifying the learning outcomes, the target audience, the appropriate technology, and the user engagement, Then, a technical setup, and user onboarding are required to address potential barriers to adoption. Finally, it is imperative to devise effective methodologies for validating the efficacy of the VR solution. This entails designing experiments to assess the effectiveness of VR training in comparison to traditional methods, analyzing data to assess learning outcomes, and gathering user feedback to identify areas for improvement.

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Using the Repertory Grid Technique in a Co-design Process for Learning Analytics:
Conceptualisation and Connection to 21st Century Skills

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Abstract. This paper introduces an innovative approach to designing data requirements for the development of learning analytics tailored to diverse, constructionist, open learning tools. Leveraging the Repertory Grid Technique, we systematically elicited and analyzed stakeholders' personal constructs to inform evidence-based user-centered design specifications. Through requirements elicitation workshops with educators, we uncovered essential information pertaining to teacher perceptions about student interactions with exploratory learning tools and their alignment with 21st-century skills. The process yielded concepts in relation to tool-specific indicators of learner interactions, forming a comprehensive design specification for a customizable learning analytics tool. Initial feedback from stakeholders and learning analytics experts indicates strong alignment with implementation needs and a coherent synthesis of diverse perspectives. Future steps include evaluating the effectiveness of the designed tools through real-world implementation and practitioner feedback, further validating the efficacy of this design approach.

Keywords: Learning Analytics · Repertory Grid Technique · User-centered Design · 21st-century Skills · Exploratory Learning Environments

1 Introduction

EU and international agendas for education highlight the need for digital transformation in education and encourage the shift in educational paradigms towards

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a trans-disciplinary and inclusive skill-oriented approach, focusing on the development of digital competences and the emerging so-called 21st century skills [1].

Modern educational approaches aiming to support the development of 21st century skills, are characterised by the integration of interactive and constructionist learning media, offering dynamic learning opportunities. This complicates instructional design, turning it into a multifaceted and challenging task. In this context, the role of Learning Analytics (LA) is pivotal in enhancing instructional design and optimising teaching and learning tools, environments and processes.

However, a top-down implementation of LA would fail to address the intricate needs of such environments if it does not incorporate sufficient early-stage involvement from educators [7]. Despite existing approaches that involve stakeholders in the development of Learning Analytics (LA) systems [10], challenges remain particularly during the critical early stages. One significant issue is the varying levels of awareness around the potential of LA among stakeholders, which can impact their ability to effectively contribute to and engage with the development process. To address these challenges, in the unique context of constructionist environments and 21st-century skills, the study reported here proposes the co-design, development, and evaluation of a customisable LA dashboard with input from teachers with regard to the collection, analysis, and visualisation of data generated from students' engagement with various tools.

The aim of the paper is to present a methodology that involves teachers in the first steps of the design of a customisable visualisation dashboard and to derive implications about 21st century skills conceptual models that render LA meaningful for end-users. The paper presents the implementation of a methodology based on the Repertory Grid Technique that is deployed through a series of requirements elicitation workshops with teachers, offering a structured context for stakeholders early involvement.

2 Background

2.1 Co-designing Learning Analytics

The notion of learning analytics as a means to facilitate evidence-based decision making in educational practice is well established in the research community. Yet there are cases where LA tools are not being adopted, used and valued by teachers. One of the pain points identified, as indicated in [7], is that teachers' perspectives are not taken into account when designing LA and the natural consequence of this is that LA fail to address crucial issues in the learning process.

Co-design - as part of the field of Human-Centred Design - is a well established method for incorporating stakeholders' needs into the design process [8]. Participatory design is defined as a series of actions aimed at mutual learning among diverse participants, facilitating the inclusion of new voices into the design process. By incorporating teachers' experiences and perspectives, the design process identifies the key role of teachers and is more likely to produce practical and useful tools that will gain their trust. Despite the recognition of co-design as an approach that democratises the use of LA, studies on co-design projects underpin

the complexities involved in incorporating diverse stakeholders into the co-design process for LA tools [2]. These challenges span from practicalities such as time and space constraints to factors that relate to self-efficacy and self-confidence of teachers with low background in IT and data literacy. Other difficulties have to do with genuine involvement of the teachers throughout the design stages instead of having them act as testers of a nearly-completed design. Furthermore, involving teachers in early design stages presents a challenge for designers in terms of effectively empowering participants to voice their opinions and comprehend the specific feedback being sought from them [9].

2.2 Context

This work took place in the context of a large EU co-funded project - ExtenDT2[1] - and aims to investigate how the fusion of emerging technologies with existing constructionist learning environments combined with the methodology of design thinking can have a transformative impact in educational practice. To accommodate experimentation with technologies an innovative learning platform was designed and developed. This platform can be thought of as a web based learning ecosystem that allows the seamless integration and interoperability of heterogeneous learning environments as well as third party web components that carry educational value. These components are diverse in nature as they follow different architectures, they expose dissimilar application programming interfaces, they use different communication protocols and exchange information in different data formats. In the context of this platform these components operate as a single, unified and coherent whole and can be used to dynamically synthesise compelling learning activities that address a quite diversified spectrum of 21st century skills. The basic building blocks are highly interactive constructionist learning environments that allow learning and discovery of knowledge in exploratory settings (examples are given at the end of this section). These components can also be dynamically enhanced with automated support as well as adaptability by authoring tools embedded in the system. As instances of these learning activities are being used by learners, the system can intercept and store interactivity indicators that may then be analysed to offer valuable insights to learners and educators. These outputs may be used by learners as formative and summative feedback that can improve awareness of their learning progression and adjust their learning strategies. They can also be used by educators to assess and verify the effectiveness of their designs and to potentially revisit them in subsequent iterations. It is self-evident that this offers opportunities for learning but at the same time reveals the immense difficulty to generate those outputs. This diversity as well as the exploratory nature of the learning process

[1] ExtenD.T.2: Extending design thinking with emerging digital technologies, https://extendt2.eu/, co-funded by the European Union (Grant agreement 101060231). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union. Neither the European Union nor the granting authority can be held responsible for them.

are the major challenges this work is trying to address. The learning tools integrated with the platform are MaLT2 (programming environment for creation and tinkering of 3D dynamic graphical models), SorBET (authoring classification tool), ChoiCo (choice driven simulation games) and GearsBot (educational robotics).

3 Research Approach

The current study adopts a mixed-methods approach with emphasis on the qualitative strand. The method of choice to address the main objective of the project is the Repertory Grid Technique (RGT) [4]. Widely used in human-computer interaction studies (e.g., [3]), RGT offers a systematic and structured approach to understanding stakeholder's perceptions, preferences, and constructs, leading to user-centred design decisions minimising designer and researcher bias. The method is based on Personal Construct Theory (PCT) which is a psychological framework that views people as proactive agents who continuously interpret their experiences through their unique construct systems. This theory highlights the dynamic and personal nature of cognition emphasising on how individuals perceive the world through a unique set of cognitive constructs [5].

We adapted the method to accommodate the collection of feedback regarding information deemed useful by educators for exploratory activities in open learning environments. This technique allows educators to associate events generated by digital tools with educational concepts and related 21st century skills. The methodology is deployed through a requirements elicitation workshop. The workshop was designed to achieve authentic engagement and bias-free feedback from teachers as end-users. The process is open and prioritises teacher needs and wants over any other design constraints following an open conceptualisation process. This way the researcher has access to in-depth qualitative data coupled with numerical rating. The quantitative component in this data is given in the form of Likert scale ratings presented in the grid cells. This uncovers patterns and creates groups of concepts. Each group is converted into a theme, providing the framework for the subsequent thematic analysis. Analysis takes into account qualitative data, namely texts written by the participants that explain concepts and the rationale behind them, cards with 21st century skills matched to the concepts as well as observation notes made by the researcher/facilitator.

3.1 Participants

A purposive sampling strategy ensured participants met the following criteria a) to be familiar with at least one of the project technologies and b) to have used the educational tool(s) in the classroom. As the sampling method is non-random, representativeness was achieved by selecting teachers with different characteristics in terms of educational background, IT and data literacy skills, teaching subject, gender, age, experience. All the participants were in-service primary and secondary school teachers with various specialisations ranging from maths and

ICT to language and classics. Participants provided informed consent for their involvement in the study and were given the option to withdraw at any time. During the sessions the researcher and one facilitator were present.

3.2 The Workshop

As discussed above, the purpose of this workshop is to reveal the learner's interactions with the learning tool that can help educators assess their progression as well as their achievements in the learning process and to identify how these interactions are related to 21st century skills. The workshop is designed to uncover this knowledge through the educators' perceptions and utilise these findings to co-design LA to satisfy those requirements.

Phase 1: As a preamble, the participants are given a simplified scenario as teachers who use a specific tool for an activity in their classroom. The objective in the first component of the workshop is to establish that the participants have the same level of awareness in terms of the data that is produced by the learning tool during the learning process, to ensure that they understand the same semantics about this data and to allow them to introduce new and potentially useful data items that are currently missing. The latter is especially useful because it reveals more information about how the participants perceive the learning process themselves and their own expectations. In this phase the participants are introduced to a complete collection of events potentially generated by the chosen tool during a learning session. Events are given as individual cards on a board. The participants are asked to read them carefully and see if there are things that are not known, understood and require clarifications. The facilitators are responsible to resolve possible misconceptions and make sure the same level of understanding is achieved at the end of this session. At the bottom of the board there is a designated space for the participants to propose events they would like the tool to generate regardless of technical limitations. This step is included to give the researcher the chance to communicate from the beginning to the participants that they have to focus on their needs and that IT and data literacy skills are not relevant and not required. Proposed events are shared with the other participants, each one of the contributors presents and explains the rationale behind it, and upon agreement the new event is added to the board (Fig. 1).

Phase 2: The objective in the second component of the workshop is to reveal and highlight the differences in terms of how the participants perceive the learning process. Different perceptions reveal the variability of different expectations, needs and requirements. This is essential in design as it enhances diversification of initial requirements and leads to more comprehensive technical and functional specifications for the thing to be built.

This phase starts with the prompt "Pick three random cards and create a group based on your experience". The aim here is to identify similarities and

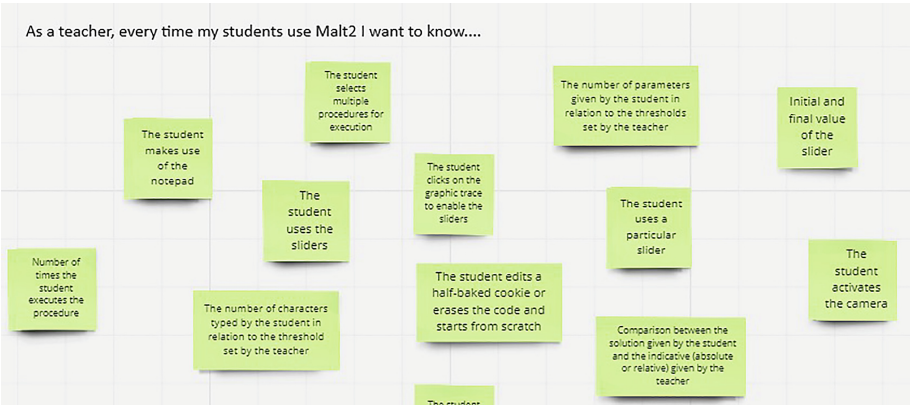

Fig. 1. Cards representing tool-specific events for phase 1

differences among the selected cards in relation to an underpinning concept. A written example is also given to help the participants understand the process. This, as a thought process, is expected to be very subjective as it relates to what each of those cards mean to the teacher in the context of a learning activity. Driven by this process, the teacher is expected to create a concept that these cards, as a group, may be related to. This concept represents the one thing that this teacher expects to learn about the process by using the related events on those cards. The teacher gives a title to the group that reflects the concept of interest and then is asked to explain their thought process in writing.

Initially the participants are asked to consider this process and do nothing. To avoid misunderstandings, an example not relevant to the tool or the context of this exercise is presented and explained by the facilitator. The example considers three cards titled basketball, orange fruit, and pencil, and suggests concepts like colour or shape as possible connections between those cards while at the same time highlights the differentiator. That allows the positioning of the cards on the two edges of the concept continuum. Allowing a few minutes to discuss the example encourages creativity, sets aside right/wrong answers, and puts stress on subjectivity as the key factor of this activity. After this example is discussed among the participants and a consensus is agreed as to how the activity should be performed, the task begins. The participants are encouraged to think creatively. They select three random cards from the board, consider them, and identify the way the two cards are different from the third exploring the bipolar concept hidden. After this, the participants are given at least thirty minutes to create at least five constructs out of the given triads. Creating and grouping triads allows participants to identify, give names to, and communicate with the facilitator the most important concepts advocating their perspective (Fig. 2).

Phase 3: Once the second phase is completed there are concepts identified that need to be associated with 21st century skills. The objective of the third phase

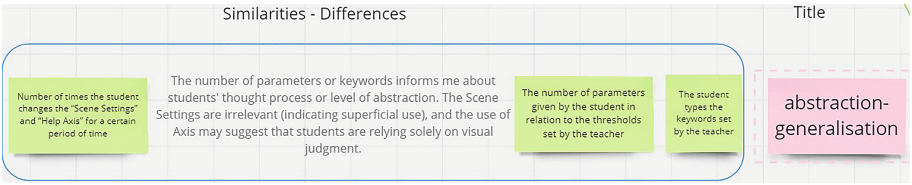

Fig. 2. Triad formulation and naming for phase 2

is exactly that. A number of cards depicting 21st century skills drawn from the current literature are presented to the participants in a designated area of the board. The participants are asked to consider these skills, and match them up with relevant concepts. The groupings are supposed to contain one concept combined with three skills at the most. The participants are then encouraged to discuss the reasoning behind the connections they identified.

Phase 4: The last phase is an overall reflection of the process. This is quantified and translated into a matrix of values. The concepts are given as columns in this matrix and the events generated by the tool are given as rows. The intersection of a concept and an event in the matrix shows the perceived relevance between them by the teacher. The matrix is populated with numerical values ranging in a 1–7 Likert scale. The participants are asked to reflect on the whole process, consider the tool affordances in relation to the concepts and rate the relevance between affordance and concept (Fig. 3).

	random use	experiment in depth	iteration	towards solution	struggling	generalisation	experimentation
The student makes use of the notepad	3	3	7	6	4	3	3
Initial and final value of the slider	7	7	1	5	3	1	7

Fig. 3. Concept - Event perceived relevance matrix for phase 4

The workshop is structured as a generic process that may be utilised for any digital educational tool participating in the project learning ecosystem presented in Sect. 2.2. As presented above this structure comprises four consecutive phases and its designated duration is 120 min. Every implementation of the workshop is inextricably bound to a single educational tool as it is tailored upon the affordances of the particular technology. The general rules of engagement are that the participants are allowed to work independently and interact with each other. Interaction with the facilitator is allowed only if they have a question or they need clarification. Discussion and interaction is in general encouraged but

is not mandatory. This is to respect introverted participants that may not feel comfortable sharing their outputs.

4 Implementation

The workshop is designed to be implemented either online or in a physical space. As a preparatory phase, a pilot study was implemented prior to the deployment of the planned workshops. The feedback received led to modifications and the finalised version of the workshop template. The workshop was implemented three times, one session for each of the available educational tools in the platform. One workshop was delivered online and the other two face-to-face. The number of participants were 4, 3, and 4 for the sessions, 11 in total. In every session there were two facilitators, one of which was the researcher. The role of the facilitator is to organise the process according to the designated structure of the session, give clarifications where needed, keep track of time and make sure the process flows as planned. They also observe participants' reactions, interactions, while keeping notes. In the actual sessions special care was given to participants' questions. The answers given were intentionally not direct in order to avoid guiding their thought process. The idea was to give them cues as to how they have to think in the context of the methodology used. Face-to-face implementation seemed to be more engaging as participants were more active and the interactions observed were richer in terms of quantity and quality. Participants that were more verbose and extroverted created opportunities for discussions by asking questions or just thinking aloud and others followed. Participation in those sessions in general was more enthusiastic and vibrant. That also affected the level of feedback received. During the online session the participants worked more as individuals as they preferred to turn off their microphones or lower the sound on their devices, in order to focus on the tasks and not get distracted by other's questions. They expressed discomfort using the online board as they often needed to zoom in and out to perform the required tasks. All the participants found it especially hard to do the part with the triads.

5 Results

As explained in the previous section, the quantitative output of each session is a grid that shows how events generated by the learning tools are related to concepts, or aspects educators consider worth monitoring. Each session was dedicated to a specific learning tool. That means that the outputs of each session are processed separately and only in relation to the tool they are associated with.

5.1 Analysis Approach

The analysis of this data looked both at the measurable and qualitative aspects and aimed to identify events rated quite high and the raters are in a general

agreement about the rating. The method used is the Intraclass Correlation Coefficient (ICC) that calculates the proportion of the total variance in the ratings that is attributable to differences between the ratings of individual raters. The analysis in detail was carried out as follows:

Individual Raters

1. Rename synonymous concepts: If the same name is given to more than one concept and the ratings differ then we examine the descriptions given in the triads and more suitable names are given.
2. Find commonalities between concepts: Consider the ratings given in each individual matrix separately and try to find common patterns between concepts identified by individual participants. In this case a correlation matrix was used and combinations exhibiting a correlation coefficient greater than 0.5 were considered.
3. Verify similarity of concepts: Consider the concepts identified as similar in step 1 using the annotations from the triad definitions combined with notes taken by the facilitators. This is to verify that the patterns reveal a meaningful resemblance in a qualitative way.
4. Merge similar concepts: Merge concepts identified in step 3 as semantically identical. Give a single common name, and replace the ratings with the mean value.

Group of Raters

5. Give generic concept names: Consider all the individual grids for all the educational tools and give common names to semantically identical concepts. Commonality is achieved here due to the fact that the concepts were determined by tool affordances through the available events.
6. Identify raters' agreement: Identify common concepts in all matrices related to a single learning tool and use the ICC method to verify raters agreement. This creates a separate matrix for each set of common concepts and after the ICC method is applied an additional column is appended for the ICC indices.
7. Find average rating for each event: Compute the average rating values for all events in all the matrices for a particular concept. This is all the matrices derived from step 6.
8. Re-examine events: The average rating calculated in step 7 is valid only if the ICC index for the event is relatively high. If it is lower than a given threshold and the dispersion of the ratings is high, the participants that give high ratings are considered in a qualitative manner. The concept explanations given in triad definitions are used to re-evaluate those ratings. If the high ratings are validated they replace the average rating given in step 7.
9. Select the most important events for each concept: The criterion used as a threshold is the median value of the Likert scale, in this case 4. We consider important all the events rated above the median. The rest of the events are excluded.

10 Generate output: Create a report for each educational tool that gives a list of the important concepts, along with the related 21st century skills, and the list of events to inform these concepts.

The results of the three tools were then compared to each other to highlight context-independent similarities.

5.2 Emerging Categories

The main concepts identified in step 5 of the analysis represent the common and agreed understanding of all the participants about what is worth monitoring when cultivating 21st century skills using the project technologies. These are: motivation, experimentation, understanding, interaction with technology and originality. As shown in the respective figures, these concepts are common for all three tools used in the study. Each concept is associated with one or more 21st century skills and a list of tool-specific events that show how the tool is used by the learner during a learning activity. We can think of the concepts, in this respect, as an extra variable, or dimension in the generated data, that can provide a sort of a category, as this is perceived by the educators. We can also map these categories with skills and the respective learner actions. This is what is communicated to the team responsible for implementing a learning analytics tool for the project (Fig. 4).

Fig. 4. Data Design for SorBET. (A) Key common concepts identified through RGT associated with (B) specific 21st-century skills and (C) tool-specific events

6 Discussion

This section is a discussion and a reflection on the overall experience from this study with a focus on what can be learnt from the process and proposed as good practice for future implementations.

As mentioned in previous sections, before the commencement of the study, a pilot session was conducted with a single participant in order to verify the soundness and the applicability of the approach and give us feedback as to what adjustments need to be made before the first run. The pilot in general went smoothly as a process as there was quite a bit of discussion between the participant and the researcher, so there was nothing major identified that would lead to a big adjustment in the structure. All the results yielded were close to the expected ones, and the only useful and worth noting feedback was the expected duration for the session that was set to 90 min.

When the first session took place, it became apparent that 90 min were not enough. The different dynamics of bigger groups revealed that this was an overly optimistic estimate. There were people handling things in different ways. Some people wanted to work alone and others wanted to discuss things with the team. Some people were expressing doubts and queries about the process and the tasks whereas others felt completely lost and wanted long explanations and discussions. After this experience the duration was extended to 120 min. After this session, there was enough feedback to suggest that the most difficult part of the process is the formation of the triads. This seems to require the most time and explanation as it was the part that confused most of the participants. What was observed is that whoever did not understand the triads properly, then struggled with completing the final matrix as well. While initially, it had been decided to provide an example related to the tool in the triads section, we observed that some tried to "copy" the thinking process of this particular example and they got confused. Therefore, it was decided to provide an unrelated example (the one with the orange and the ball), which helped a lot. Thus, in the following sessions, this became a formal part of the methodology.

Another worth noting observation is that people in online sessions tend to work individually, whereas in face to face ones there is a tendency to work in groups and communicate and collaborate a lot.

After completing all the sessions we concluded that what needs careful consideration in this approach is a very good and detailed explanation with examples and discussion about how to approach the triad formation task. Also, as a general recommendation, we think that more time is needed for the tasks as well as emphasis on interaction among participants. Experience also shows that the process is better conducted with three to five people; more are really difficult to handle, and fewer cannot not develop enough interaction in the team.

7 Conclusion, Future Work

In this paper we presented an innovative approach to design the data requirements for the development of learning analytics for diverse, constructionist, open learning tools. For this we used a methodology based on the Repertory Grid Technique which is a psychological technique developed as part of the Personal Construct Theory. This method is used to elicit and analyze an individual's personal constructs, which are the mental frameworks or ways of understanding the

world that people use to interpret and make sense of their experiences. We used this methodology to provide a systematic and structured approach to understanding stakeholder perceptions, preferences, and constructs, in order to derive evidence-based, well-informed and meaningful user-centred design specifications for a learning analytics tool suitable for this kind of learning. We implemented a series of requirements elicitation workshops with teachers to explore what type of information they find important when students interact with exploratory learning tools with regard to 21st century skills. The process revealed a number of concepts, associated with 21st century skills translated into tool-specific indicators of learner interactions. This forms a design specification that feeds into the design of a customisable learning analytics tool for open learning environments.

The feedback we have received so far from the stakeholders that participated in the design is more than encouraging. A preliminary review of the outputs by the learning analytics experts that will be using this design specification for the implementation of the tool is that the outputs articulate well what needs to be implemented and they are quite comprehensive in technical terms as they include direct references to the underlying data formats used by the learning tools. Another positive remark is that this work managed to aggregate into a coherent and meaningful framework, information coming from very diverse and heterogeneous environments and that helps a lot in terms of the effort required to put all of those seemingly dissimilar things under a single perspective. Such a framework has the potential to facilitate the development of a learning analytics platform like the one presented in [6]. What we envisage to see in the future is an implementation of the learning tools based in this specification being evaluated by teachers and practitioners in the field as an additional step to verify the value of this design approach.

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Automated Math Word Problem Knowledge Component Labeling and Recommendation

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Abstract. The accurate annotation of math exercises and word problems according to a diverse set of knowledge components is an important task for many education applications. It is an extremely complex and resource-intensive process, and has traditionally been done manually by experienced educators. There has been research work in recent years to apply machine learning to automate the process, however, due to the varied datasets used by different researchers and the private nature of these datasets, there is no good benchmark for which model performs the best for such a task. Moreover, the datasets used in literature typically comprise math exercises that follow similar templates. In this paper, we benchmark some of the best reported models on a math word problem dataset with fine-grained knowledge component annotations, and highlight the challenges in making accurate predictions. We propose models to improve the prediction accuracy, and demonstrate how they can be used to extract similar problems based on knowledge component from an unlabelled pool of questions, thus empowering educators to better identify the knowledge gaps of their students and tailor suitable practice problems for them.

Keywords: Math Word Problems · Knowledge component · Annotation · Recommendation · BERT

1 Introduction

Accurate annotation of math exercises and problems with relevant knowledge components from the curriculum is extremely important to online tutoring platforms such as ASSISTments and Cognitive Tutors for the purposes of personalized recommendation [1,2] and knowledge tracing [3,4] to help tailor the best learning experience for students. The task of annotating new math questions is laborious as the range of knowledge components can go up to the hundreds. For instance, the Common Core State Standards used in the US Curriculum has over 200 knowledge components for K-12 mathematics [5]. It is typically performed by experienced domain experts, usually teachers, but even then, there can often

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be disagreements with the labeling due to how challenging it is [6], thus more cross-checks are required to ensure consistency, resulting in even more work.

In recent years, many researchers have proposed applying machine learning methods to solve this problem using both unsupervised learning [7] and supervised learning techniques [8,9], with some promising results. Usually this task is framed as a multi-class classification problem [9] where the content of a math problem is analyzed by a model, which predicts the correct knowledge component for it. However, these works train and evaluate their models on different data sources that are not publicly released, making them difficult to benchmark and analyze. Moreover, there have been reports that the math exercises used in these works follow certain templates that lead the model to overfit to the templates, thus being non-generalizable to different data sets [6]. In this paper, we benchmark different models on challenging math word problem datasets with fine-grained knowledge component annotations that do not follow fixed question templates, to better compare the models. We also highlight some challenges with knowledge component annotation of math questions for practical usages and propose some solutions to address them.

2 Related Work

Early work on automated knowledge component annotation operated on the assumption that problems with the same tags share regular keywords [8], leading to models that learn to label math problems based on the similarity of their word distributions to tagged content. One of the earliest reported work was by Cetintas et al. [10], who trained a support vector machine (SVM) classifier to differentiate Multiplicative Compare and Equal Group problems from other types of math problems from a fourth grade math textbook. Later, Karlovcec et al. [11] employed a text mining approach with SVM and K-nearest neighbour on mathematical text with up to 106 knowledge components taken from the ASSISTments Platform between 2005 to 2006, and they were able to identify the correct tag in 62.1% of cases. This value goes up to 88.9% when 10 suggestions are provided. Working on ASSISTment problems from 2012–2023, Pardos et al. [8] presented a model that can accurately match problems with their associated knowledge components out of a pool of 198 in total, achieving an accuracy of 90%, using a combination of skip-gram and bag-of-words.

More recently, Shen et al. [9,12] demonstrated improved performance with models based on BERT (Bidirectional Encoder Representations from Transformers) [13]. They introduced Task-Adaptive Pretrained (TAPT) BERT, that is pretrained on various math data, as well as MathBERT, a model based on BERT that is further pretrained on a large mathematical corpus ranging from prekindergarden all the way to college level. Tested on an ASSISTments dataset with 13722 problems and 213 possible labels, MathBERT was able to achieve an F1 score as high as 92.67% and an accuracy as high as 93.92%. There have also been other works on knowledge component labelling for different languages such as Japanese [14] and Chinese [15], but they use relatively small and inaccessible datasets.

There has been some controversy over the dataset used in some of these works. While replicating the methodology used by Pardos et al., Patikorn et al. identified an issue with using certified Skill Builder problems from ASSISTments as many of the questions are generated based on templates and are nearly identical with only differing numerical values in most cases [6]. Patikorn purported that this caused models to remember specific words in the templates instead of learning the terminology of the skills, inflating the prediction performance. As evidence, Patikorn replicated the results of Pardo et al. on ASSISTments data, but showed that the performance dropped drastically, from over 90% to less than 15% when tested on a dataset from Illustrative Mathematics, a different data source also based on the Common Core standard. This showed that the models developed lacked generalizability and were not ready for real-world application. Additionally, some of the works report a reliance on key phrases in the math exercises to make predictions, suggesting that their models do not actually understand the underlying math concepts in the problems [14].

In this work we conduct experiments to answer the following questions:

1. Do the models and methods reported by recent literature and the latest large language models (LLMs) perform well on more complex math word problem datasets that contain more realistic problems which do not follow templates?
2. What are some of the key features that affect model accuracy and how can the automated knowledge component annotation be further improved?
3. Can we leverage on these models to make recommendations of similar math word problems based on the knowledge component?

3 Methodology

3.1 Data

We use two publicly-available math word problem datasets in English language that were introduced by TAL Education Group for the purpose of training and evaluating math word problem solvers. The first dataset is TAL-SCQ5K-EN, a high quality mathematical competition dataset comprising a total of 5000 questions that span topics from primary, junior high, and high school levels. The other dataset, TAL-SAQ6K-EN, was later introduced as the test set for the AAAI2024 Global Challenge on Math Word Problem Solving, and consists of 5927 questions. Questions from both datasets are annotated with detailed knowledge point route(s) that can go up to 5 levels, thus presenting valuable fine-grained annotations for model development and evaluation. All the mathematical expressions in the datasets are presented in standard text-mode Latex.

To evaluate knowledge component annotation models, we use word problems from TAL-SCQ5K-EN as the training set and those from TAL-SAQ6K-EN as the evaluation set, and use the knowledge point routes as knowledge components. We remove the problems with multiple knowledge components to train and evaluate our model on only a single knowledge component per problem. To ensure that the annotations are fine-grained, we remove problems with knowledge components

that have only one or two levels. As majority of the problems' knowledge components do not extend beyond three levels, we terminate the longer ones at level 3. From the training set, we also remove the knowledge components with fewer than 5 examples, as there will be too few examples for the model to learn from. We end up with 2247 problems with 109 unique knowledge components in the training set. For the evaluation set, we have 3234 problems with 206 knowledge components, of which 103 were seen in the training set and 103 are completely unseen. This deviation from more conventional 70:20 or 80:20 train:test splits is intentional as we want the evaluation set to be as vast as possible relative to the training set to better resemble realistic scenarios. We keep the questions with unseen knowledge components in the evaluation set as it is common to encounter unseen labels in real-life application, and we will compare the model's confidence on seen versus unseen labels in the Results section. Overall, the evaluation questions that have knowledge components found in the training data comprise 78.8% of the evaluation set.

To demonstrate that the math questions in our dataset are truly diverse, we employ sentence vectorization to obtain the sentence embeddings and compare them using cosine similarity. We use two different well-established models for vectorization, namely bag-of-words vectorization [16] and Sentence-BERT [17] vectorization and compare the similarities between math word problems of the same and different knowledge components for both the train and evaluation datasets. The cosine similarities obtained are summarized in Table 1. We observe negligible difference between the sentence embeddings for both datasets across the knowledge components with both models, confirming that the TAL dataset word problems are indeed well-diversified and do not follow templates.

Table 1. Cosine similarity between the sentence embeddings

	Train set cosine similarity		Evaluation set cosine similarity	
	Same KC	Different KC	Same KC	Different KC
Bag-of-words	0.1591	0.1592	0.1771	0.1762
Sentence-BERT	0.2237	0.2237	0.1657	0.1642

3.2 Models

Benchmarking. For benchmarking purposes, we pose the knowledge component classification task as a multi-class classification problem, in line with the recently reported literature on this topic [9,12]. Models are given the problem text as input and trained to predict the correct knowledge component out of all possible 'seen' knowledge components. We benchmark the following models/methods:

1. Fasttext vectorization of problem text [18] + XGBoost classification model

2. Multi-class classification with fine-tuned open-source large language models. Besides MathBERT, we also compare with BERT-base, BERT-large, and RoBERTa-large [19] to study the effect of different model sizes and pretraining. We train all models with a batch size of 16 and a learning rate of 1e-5 on an NVIDIA A40 GPU.

Since being introduced to the public in November 2022, ChatGPT [20] has become widely used for personal as well as commercial use for tasks such as question and answering and code generation. To study its usefulness for knowledge component annotations without any finetuning, we use the GPT-3.5 Turbo model from OpenAI, and prompt it with the following system role: *"You are an experienced math educator, and highly skilled with annotating math word problems with the correct knowledge label from a pool of candidates."*, this is important to set the behavior of the model for our desired task. Next we prompt the model to pick the right knowledge component label for each question as follows: *"What is the correct label for the question: <question>, please select the most suitable knowledge point route from the following options: <all kcs>"*, where *<question>* is the text for each question and *<all kcs>* represents a string of all the knowledge point routes in the training set, separated by a comma. The option selected by the model is recorded and compared to the ground truth.

4 Results

4.1 Benchmarking

To obtain the benchmarking results, we perform the training of each model 5 times and the results on the test set are averaged and shown in Table 2. As we have over 100 class labels, we report top-N accuracy, which measures the accuracy of the true class being within the N most probable classes predicted by the model. We use values of 1, 3, 5, and 10 for N.

We see that the results for Fasttext + XGBoost method performs the worst, with a top-1 accuracy of only 0.124. Meanwhile, the BERT-based models show much improved performance, with a top-1 accuracy ranging from 0.309 for BERT-large to 0.359 for RoBERTa-large. For top-3, top-5, and top-10 accuracy, we also observe that the best results are obtained by RoBERTa-large, while MathBERT, BERT-base, and BERT-large perform very similarly. These results suggest that neither the larger parameter size of BERT-large over BERT-base, nor the pretraining on mathematical corpus of MathBERT over BERT-base led to much improvement. Instead, more optimized training methods in the case of RoBERTa-large, which was trained for longer, with larger batch sizes and longer sequences of data, and used dynamic masking compared to the BERT models, can yield better performing models for math problem annotations [19].

We can see that when applied to the challenging TAL dataset, there is a significant drop in prediction accuracy by models that reported over 90% accuracy on their own private datasets, suggesting that those results could be due to the ease of those datasets and possible overfitting to repeated key phrases [8,12]. One

reason for the relatively lower accuracy scores on our evaluation dataset is the presence of unseen labels, which comprise about 21.2% of the data. If we consider only the accuracy metrics on problems with knowledge point routes that exist in the training set, we can get a top-1 accuracy of 0.455 and top-10 accuracy of 0.815 for the best-performing RoBERTa-large model. The "in-train"-only results are summarized in the right columns of Table 2.

Table 2. Benchmarking results

Model	Accuracy				Accuracy (in-train only)			
	Top-1	Top-3	Top-5	Top-10	Top-1	Top-3	Top-5	Top-10
Fasttext + XGB	0.124	–	–	–	0.158	–	–	–
Multi-class (MathBERT)	0.323	0.479	0.535	0.597	0.409	0.608	0.6789	0.757
Multi-class (BERT-base)	0.327	0.477	0.531	0.600	0.415	0.605	0.673	0.761
Multi-class (BERT-large)	0.309	0.471	0.530	0.602	0.391	0.598	0.672	0.763
Multi-class (RoBERTa-large)	0.359	0.535	0.589	0.642	0.455	0.678	0.747	0.815

For knowledge component annotation using GPT-3.5 Turbo, we obtained an overall top-1 accuracy of 0.053, which is worse than even the Fasttext + XGB method. The low performance suggests that the terminology and complexity of the knowledge components of the TAL datasets is very different from the data exposed to the commercial LLMs during pretraining, and using these commercial LLMs off-the-shelf is insufficient without performing further finetuning on dedicated task data.

4.2 Analysis

How Many Training Examples Are Required for Each Unseen Label? As manual annotation of training examples is costly, we should determine the minimum number of training examples per unseen label to achieve good performance. For this purpose, we examine how the prediction accuracy for each label class for the top-performing RoBERTa multi-class classification model varies with the number of examples present in the training set. We plot scatter plots for all the top-N accuracies in Fig. 1. Across the different values for N, we observe that at small training examples (less than 30), there is a wide spread in the accuracy, spanning from 0 to 1.0. This reveals that while some knowledge components can achieve high accuracy with relatively few examples, others perform poorly. As the number of training examples increases, we observe that the accuracy stabilizes. In short, there is a tendency that more training examples lead to higher annotation performance, up till a certain point beyond which the performance plateaus. For the TAL dataset, this occurs at around 50 training examples.

Can Machine Learning Models Produce Reliable Confidence Scores of Predicted Knowledge Components for Individual Questions? Besides simply looking at the top-N predictions provided by a language model, we can also consider the confidence of each prediction, which is typically calculated by the difference between the softmax prediction probability of the top two candidates. Figure 2(a) plots the top-1 accuracy of the RoBERTa-large classification model vs. the confidence of the prediction, and we can observe that the accuracy generally increases with prediction confidence. This means that by setting a confidence threshold for predictions, we can ensure that the model only makes annotation suggestions for high confidence predictions with correspondingly higher accuracy. For instance, at 98% confidence and above, the top-1 accuracy is 0.673, a large improvement over the overall accuracy of 0.366. We further investigate the model's confidence for problems with knowledge components that are found in the training set versus those that are not, and found that the mean prediction confidence for the former is 72.1% compared to 58.8% for the latter, indicating that the model is more confident about the predictions for previously-seen knowledge components.

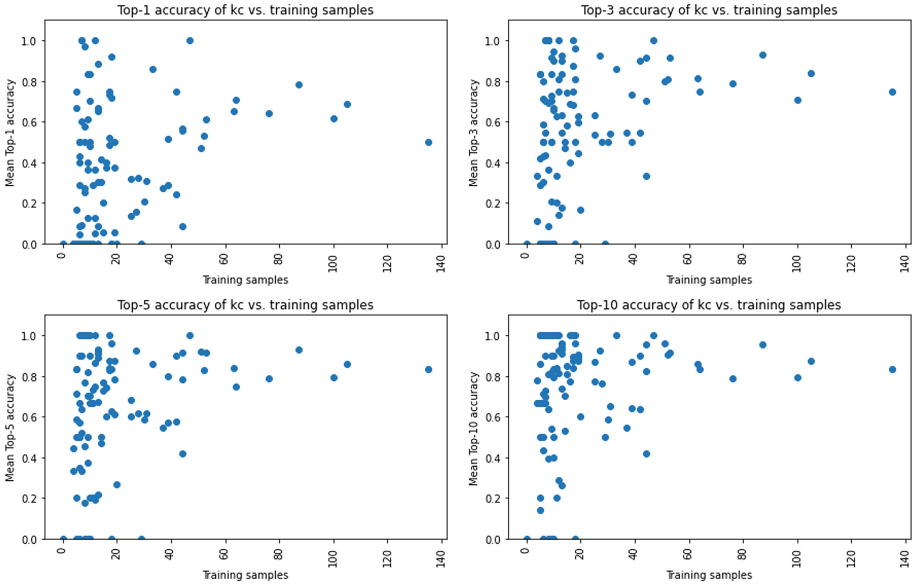
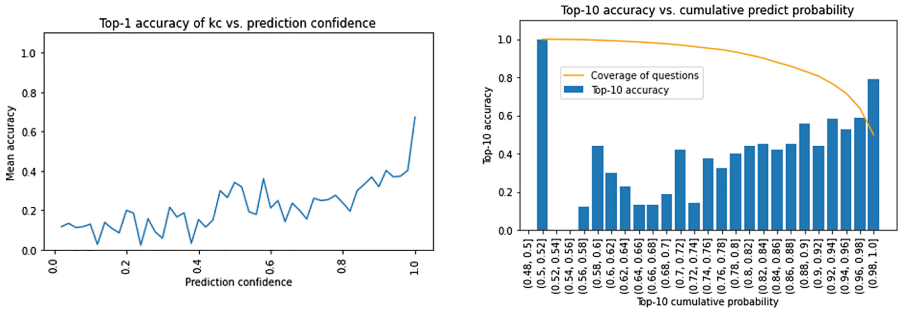


Fig. 1. Number of training examples vs. accuracy

We also study the relationship between the top-10 accuracy and the prediction probabilities, using the cumulative probabilities of the top 10 classes as a measure of confidence. In Fig. 2(b) we divide the cumulative probabilities into bins of width 0.02 and plot the top-10 accuracy against it. We also plot the coverage of the problems above a certain cumulative probability, denoted by the



(a) Top-1 accuracy vs. confidence (b) Top-10 accuracy vs. cum probability

Fig. 2. Plots for prediction accuracy vs. prediction confidence and cumulative probability

orange curve in the figure. We observe a similar trend of accuracy increasing with cumulative probability. There is a strange peak with accuracy of 1.0 for the confidence bin between 0.50 and 0.52, but there is only one problem in that bin so it is disregarded as an anomaly. In the highest bin with cumulative probability above 0.98, the accuracy is 0.79 with about 50% of the problems, indicating that by setting a high cumulative probability threshold, we can filter for the higher accuracy predictions whilst maintaining a good coverage of all questions.

4.3 Problem Recommendation Based on Similar Knowledge Components

A key benefit of having math problems annotated with knowledge components is the ability to recommend other questions with the same knowledge components. One way to do this is by considering the class (knowledge component) probabilities for each question and finding other questions with the most similar probabilities. In this section we investigate how many of the other questions of the same knowledge component are retrieved (or recommended) for the given question with high ranks. For this analysis, we only consider the problems with ground truth labels that were in the training set. For each math problem in the evaluation dataset, we computed and ranked the cosine similarity of its knowledge component class probabilities as predicted using RoBERTa-large against all other questions. We tabulated the mean cosine similarity of problems with the same knowledge component and found it to be 0.458, significantly higher than the overall mean of 0.027, proving that problems with the same knowledge component have more similar class probabilities.

To evaluate how well this approach performs for making problem recommendations we use hits@k, which considers the top k recommendations for each problem based on similarity and tabulates the number that have the same knowledge component. We use k = 1, 3, 5, and 10, and compare the hits@k for all the knowledge components as well as for only the knowledge components with

at least 30 and 50 training examples. The hit@1 was 0.53, 0.61, and 0.66 respectively, demonstrating that using cosine similarity of prediction probabilities is an effective way to retrieve similar problems. We also observe improvements in hits@3, hits@5, and hits@10 with more training examples, proving that more training examples is beneficial to making better problem recommendations. The results are consolidated in Table 3.

We compare the results to using Sentence-BERT to obtain the sentence embeddings of the question texts and computing the cosine similarity based on that to make recommendations of similar problems. Interestingly, the hits@1 for Sentence-BERT is slightly better than using the RoBERTa-large prediction probabilities. A possible reason for this is that for each knowledge component there are problems that are semantically very similar, so finding the one with the highest similarity yields good results. As k increases, we observe that the RoBERTa-large model performs significantly better, for example, hits@10 for RoBERTa-large with at least 50 training examples is 5.99 versus. 5.03 for Sentence-BERT. This shows that for making recommendations of a larger number of similar problems, finetuning a RoBERTa-large model produces superior results.

Table 3. Hits@k versus number of training examples

Hits@k	RoBERTa-large			Sentence-BERT		
	All	Min. 30 examples	Min. 50 examples	All	Min 30. examples	Min 50. examples
hits@1	0.53	0.61	0.66	0.57	0.62	0.67
hits@3	1.43	1.60	1.88	1.46	1.64	1.84
hits@5	2.28	2.69	3.04	2.18	2.48	2.83
hits@10	4.3	5.23	5.99	3.76	4.25	5.03

5 Conclusion

From our results, we conclude that accurately labeling the knowledge components of math word problems is a challenging task that even commercial LLMs such as GPT-3.5 Turbo are unable to solve without finetuning. We believe that human input is still necessary for annotation tasks, but finetuned models such as RoBERTa-large can aid in the process by providing suggestions for a manual annotator to select from. We show that with confidence thresholding, it is possible to filter for more accurate predictions, and also show that limitations on training data still constrain model performance at the top-1 accuracy level. It is possible to design an LLM-powered annotation system which makes suggestions for high-confidence predictions, which can be verified by human annotators, and added to the training data pool to improve the model further.

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